The CMB temperature angular power spectrum through fourth order in a radiation-only universe: exact observable, initial data, perturbative maps and one-loop assembly

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Abstract

We compute the observed temperature angular power spectrum $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ of a spatially flat, radiation-only universe in general relativity ($\Lambda = 0$, perfect fluid $\rho = \alpha T_{\rm phys}^4$), for a Gaussian non-linear primordial curvature perturbation $\zeta_{\rm nl} = \epsilon\zeta_1$, photons emitted isotropically in the fluid rest frame on the hypersurface $T_{\rm phys} = T_*$, and an observer comoving with the fluid who observes at fixed proper time after the crossing of $T_{\rm phys} = T_i$. Three structural results organise the calculation. (i) The measurement is exactly $T_{\rm obs}(\hat{\mathbf{n}}) = T_o\, e^{-S(\hat{\mathbf{n}})}$, with S the fluid shear integrated along the ray; it contains all Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing, time-delay, emission-surface and observer effects, and the normalised anisotropy is $1 + \Theta = e^{-S}/\langle e^{-S}\rangle_{\rm sky}$. (ii) Gaussianity of $\zeta_{\rm nl}$, regularity at $\eta = 0$ and the absence of primordial vector and tensor modes fix the initial data at every order; they are non-zero, non-local and satisfy the constraints. (iii) The normalisation and the order-by-order maps $\Theta^{(n)}$ follow by exact algebra from $\Lambda = -S$. We give the first-order transfer function and $C_\ell^{(11)}$ in closed form; all second-order fields in closed form (elementary functions, Si and Ci); the contracted third-order fields with closed-form initial amplitude and one declared time integral; the complete line-of-sight maps L_1, L_2, L_3 (all terms are listed); the harmonic machinery; the kernel K_2, $C_\ell^{(22)}$ and its monochromatic evaluation; the contracted third-order kernel, $C_\ell^{(13)}$ and the monochromatic $O(\epsilon^4)$ coefficient (an evaluation by quadrature at one parameter point). Checks: the exact formula, the field equations, the second-order closed forms and the harmonic rules are verified symbolically or to high precision; the maps $\Theta^{(1,2,3)}$ are verified end to end against a non-perturbative solver in a different gauge; the contracted third-order kernel is checked by form equivalence, residual-gauge invariance and its squeezed limit; the cancellation of the collinear poles of the (13) integrand and the UV behaviour of the loop are established numerically only, and the formulas (E2), (E3), $T^{(\rm ii)}$ and an analytic zero-mode proof are open (Sec. 14).

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1 Introduction

The problem. We compute the observed angular power spectrum of the CMB temperature through fourth order in the primordial amplitude, $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}]$, in the simplest universe in which every ingredient of a one-loop calculation is present: spatially flat, general relativity with $\Lambda = 0$, a single perfect radiation fluid, Gaussian non-linear primordial curvature perturbation, last scattering on a surface of constant physical temperature, and an observer who is a fluid element measuring at fixed proper time. No piece of the observable (emission surface, propagation, observer motion and clock, normalisation by the measured mean) is approximated or dropped; the result is expressed through named, analytic objects, and every integral that is not evaluated in closed form is declared with its complete integrand, limits and nesting.

Why the calculation is tractable: the exact observable. For a radiation fluid, energy–momentum conservation alone implies that the expansion and acceleration of the fluid are gradients of $\ln T_{\rm phys}$ (Lemma 4.2). Combined with the kinematic decomposition of the redshift along a null geodesic (Lemma 4.1), this gives the exact, non-perturbative identity (Theorem 4.3)

$$T_{\rm obs}(\hat{\mathbf{n}}) = T_o\, e^{-S(\hat{\mathbf{n}})}, \qquad S = \int_e^o \sigma_{\mu\nu} e^\mu e^\nu \,\mathrm{d}l,$$

with T_o the fluid temperature at the observer and $\sigma_{\mu\nu}$ the fluid shear. What this buys: (a) there are no separate Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing, time-delay, "intrinsic δT" or observer terms to be added and reconciled – all of them are contained in S, the only inputs being the shear along the actual ray and the two endpoints; (b) T_o is a scalar at the observer and drops out of the normalised anisotropy Θ identically, so no background temperature or clock bookkeeping enters the anisotropy except through the location of the endpoints; (c) the background shear vanishes, so $S = O(\epsilon)$ and the third-order map needs the ray only to second order. The normalisation by the measured sky mean then becomes exact algebra on $\Lambda \equiv -S$ (Sec. 4.7). The remaining work is to expand S about the background ray (Secs. 8–9), to supply the fields from the regular solution of the Einstein–fluid equations with Gaussian initial data (Secs. 5–6), and to perform the harmonic transforms and Wick contractions (Secs. 10–12).

Relation to existing work. No complete $O(\epsilon^4)$ (one-loop) computation of the temperature power spectrum exists in the literature: the only (22)-type temperature calculation [23] is for dust in synchronous gauge (ISW terms only) and explicitly leaves out the (13) term for lack of third-order metric perturbations; third-order temperature anisotropies have been studied on large scales in matter domination [24, 25]; second-order Boltzmann codes [20, 22] and the (22)-type B-mode spectrum [21] exist, as do the second-order Sachs–Wolfe analyses [18, 19, 8, 9, 10] which, "following standard practice", drop the observer terms; only [6, 7] keep them, for long-wavelength modes. The lensed C_ℓ at $O(C^\phi)$ [16, 17] is the (22) + (13) structure restricted to lensing and serves as

our control case. The radiation-era second-order vector and tensor sectors are treated in [15] and [12, 13, 14]; general second-order perturbation theory is reviewed in [11].

Organisation. Section 2 fixes the model and all conventions; Sec. 3 is a two-page summary (target, structural results, pipeline, where each requirement is met, status, delicate points). Sections 4–6 contain the exact observable, the initial data and the field solution. Section 7 runs the whole pipeline at first order as a worked example; Secs. 8 and 9 are the same steps at second and third order. The harmonic machinery and loop formulas (Sec. 10) are then applied in the assembly (Sec. 11) and the monochromatic reduction (Sec. 12). Section 13 summarises the checks; the ledger of App. G records each of them, and App. F lists the code that reproduces them.

2 Setup and conventions

Model. Spatially flat FRW background, general relativity with $\Lambda = 0$, signature $(-+++)$, and a single perfect radiation fluid

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g^{\mu\nu}, \qquad P = \rho/3, \qquad \rho = \alpha\, T_{\rm phys}^4, \tag{1}$$

with α constant and $u^\mu u_\mu = -1$. The second relation *defines* the physical temperature $T_{\rm phys}$; no chemical potential or particle-number assumption is made. Test photons propagate collisionlessly without backreaction after emission.

Assumptions defining the observable. The problem statement leaves three choices implicit; we fix them here, and they are part of the definition of the observable.

(O1) *Emission frame.* Photons are emitted with an isotropic Planck spectrum of temperature T_* *in the fluid rest frame* (the frame of u^μ), at the last crossing of the hypersurface $T_{\rm phys} = T_*$ by their past light ray.

(O2) *Observer.* The observer is the fluid element whose position at the initial event is the label $\mathbf{x}^{\rm lab}$: its worldline is $\mathbf{x}_o(\eta) = \mathbf{x}^{\rm lab} + \int_{\eta_{\rm init}}^{\eta} \mathbf{v}(\eta', \mathbf{x}_o(\eta'))\, \mathrm{d}\eta'$, where the *initial event* is the crossing of $T_{\rm phys} = T_i$ by this worldline (Poisson time $\eta_{\rm init}(\mathbf{x}^{\rm lab})$). The observation is made at fixed proper time $\Delta\tau$ after the initial event, and the sky directions are measured in a Fermi–Walker transported orthonormal frame (Sec. 4.4).

(O3) *Normalisation.* $\Theta(\hat{\mathbf{n}}) \equiv T_{\rm obs}(\hat{\mathbf{n}})/\langle T_{\rm obs}\rangle_{\rm sky} - 1$ with the sky mean $\langle\cdot\rangle_{\rm sky}$ measured by the observer; the dipole is removed after normalising, i.e. only $\ell \geq 2$ is reported.

Background and units. Conformal time η, $\mathrm{d}s^2 = a^2(\eta)\, \hat{g}_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$ ($\hat{g}$ is the *conformal metric*). In radiation domination $a \propto \eta$; we fix the normalisation $a(\eta) = \eta$, so that $\mathcal{H} \equiv a'/a = 1/\eta$ and the Friedmann equation $3\mathcal{H}^2 = 8\pi G a^2 \bar{\rho}$ reads

$$8\pi G\, a^4 \bar{\rho} = 3, \qquad \text{i.e.} \qquad 8\pi G\, \bar{\rho} = 3/\eta^4. \tag{2}$$

Background temperature $\bar{T}(\eta) = T_i\, \eta_i/\eta$. The *conformal* fluid variables

$$\hat{\rho} \equiv a^4 \rho, \qquad \hat{P} \equiv a^4 P, \qquad \hat{T} \equiv a\, T_{\rm phys}, \qquad \hat{u}^\mu \equiv a\, u^\mu \quad (\hat{g}_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1), \tag{3}$$

are constant in the background ($\hat{\bar{\rho}} = 3/8\pi G$, $\hat{\bar{T}} = \text{const}$) and are the natural variables of Sec. 4.

Fourier conventions, statistics and the connected average.

$$f(\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}} f_{\mathbf{k}}, \qquad \langle \zeta_{\mathbf{k}} \zeta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k}+\mathbf{k}')\, P(k), \qquad \langle \zeta^2(\mathbf{x}) \rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k) = \frac{1}{2\pi^2} \int k^2 \mathrm{d}k\, P(k). \tag{4}$$

Products in position space become convolutions, $(fg)_{\mathbf{k}} = \int \mathrm{d}^3q/(2\pi)^3\, f_{\mathbf{q}}\, g_{\mathbf{k}-\mathbf{q}}$, and $\partial_i \to ik_i$, $\nabla^2 \to -k^2$, $\nabla^{-2} \to -1/k^2$. $\langle\cdot\rangle$ denotes the ensemble average over the Gaussian field ζ_1 (Sec. 5); $\langle\cdot\rangle_{\mathrm{sky}}$ denotes the solid-angle average over the observer's sky. The *connected average* of two multipoles is

$$\langle a^{(r)}_{\ell m} a^{(s)*}_{\ell m} \rangle_c \equiv \langle a^{(r)}_{\ell m} a^{(s)*}_{\ell m} \rangle - \langle a^{(r)}_{\ell m} \rangle \langle a^{(s)*}_{\ell m} \rangle, \tag{5}$$

i.e. connected with respect to the $a_{\ell m}$, *not* with respect to ζ_1. By statistical isotropy $\langle a_{\ell m} \rangle = 0$ for $\ell \geq 1$, so for the multipoles reported here $\langle\cdot\rangle_c = \langle\cdot\rangle$; in particular products of two ensemble averages that are disconnected in ζ_1 but connected in the $a_{\ell m}$, such as $\langle \langle m_1 \rangle^2_{\mathrm{sky}} \rangle\, C^{(11)}_\ell$ in $C^{(22)}_\ell$, are part of C_ℓ.

Perturbative expansion. Every perturbation $X \in \{\Psi, \Phi, \omega_i, h_{ij}, \delta, v^i, \ldots\}$ is expanded as

$$X = \epsilon X_1 + \frac{\epsilon^2}{2} X_2 + \frac{\epsilon^3}{6} X_3 + O(\epsilon^4), \tag{6}$$

and the same convention is used for every derived quantity (S, Λ, m, Θ, ζ_{nl}, $\delta\eta_o$, ...). For an equation $E[X] = 0$ we write $E^{(n)} \equiv n!\, [\epsilon^n] E$, so that X_n enters $E^{(n)}$ linearly with unit weight. The bookkeeping parameter ϵ is the amplitude of ζ_1; the final expansion parameter is $\epsilon^2 \langle \zeta_1^2 \rangle$. Products of lower-order quantities are never dropped.

Poisson gauge. The conformal metric is parametrised *linearly* (no exponentials),

$$\hat{g}_{00} = -(1+2\Psi), \qquad \hat{g}_{0i} = \omega_i, \qquad \hat{g}_{ij} = (1-2\Phi)\delta_{ij} + h_{ij}, \qquad \partial_i \omega_i = 0, \quad h_{ii} = 0 = \partial_i h_{ij}, \tag{7}$$

with ω_i and h_{ij} starting at order ϵ^2: there are no primordial vector or tensor modes at any order (Definition 5.2). Indices on ω_i, h_{ij}, v^i are raised and lowered with δ_{ij}. The fluid is parametrised by

$$\rho = \bar{\rho}(\eta)\,(1+\delta), \qquad 1 + \delta_T \equiv (1+\delta)^{1/4} = \hat{T}/\hat{\bar{T}}, \qquad \hat{u}^\mu = \hat{u}^0\,(1, v^i), \tag{8}$$

with $\hat{u}^0$ fixed by the normalisation $\hat{g}_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1$ as a series in ϵ. Irrotationality of v^i is *not* imposed (Sec. 6.2). The Einstein equations are $E_{\mu\nu} \equiv G_{\mu\nu}[g] - 8\pi G\, T_{\mu\nu} = 0$ with $g = \eta^2 \hat{g}$ and $8\pi G\, T_{\mu\nu} = (3/\eta^4)(1+\delta)[\frac{4}{3} u_\mu u_\nu + \frac{1}{3} g_{\mu\nu}]$; the conservation laws $\nabla_\mu T^{\mu\nu} = 0$ are implied and are used as checks. Dictionary to the exponential parametrisation of [8, 9, 6] ($\hat{g}_{00} = -e^{2\Phi_{\mathrm{e}}}$, $\hat{g}_{ij} = e^{-2\Psi_{\mathrm{e}}}\delta_{ij} + \ldots$, note the swapped names): $\Phi^{(2)}_{\mathrm{e}} = \Psi_2 - 2\Psi_1^2$, $\Psi^{(2)}_{\mathrm{e}} = \Phi_2 + 2\Phi_1^2$ at second order.

Directions and rays (one convention throughout). $\hat{\mathbf{n}}$ is the direction in which the observer looks (a unit vector with respect to the observer's orthonormal triad, Sec. 4.4); the photon propagation direction at the observer is $-\hat{\mathbf{n}}$. The background ray through the observer event $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$ is

$$\bar{\mathbf{x}}(\eta) = \bar{\mathbf{x}}_o + \hat{\mathbf{n}}\, \chi(\eta), \qquad \chi(\eta) \equiv \bar{\eta}_o - \eta, \qquad \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}\eta} = -\hat{\mathbf{n}}, \tag{9}$$

so a Fourier mode on the ray carries the phase $e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}$ (observer at $\bar{\mathbf{x}}_o = 0$), and we use the Rayleigh expansion

$$e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = 4\pi \sum_{\ell m} i^\ell j_\ell(k\chi)\, Y_{\ell m}(\hat{\mathbf{n}})\, Y^*_{\ell m}(\hat{\mathbf{k}}) \tag{10}$$

everywhere. The total derivative along the ray acting on a field is $D \equiv \partial_\eta - \hat{\mathbf{n}} \cdot \nabla$. (Writing maps as functions of the propagation direction $-\hat{\mathbf{n}}$ instead, with $e^{-i\mathbf{k}\cdot(-\hat{\mathbf{n}})\chi}$ and $(-i)^\ell$, multiplies every $a_{\ell m}$ by $(-1)^\ell$ at every order; all transfer functions Δ_ℓ and all $C_\ell^{(rs)}$ are unchanged.)

Times. Four times are kept distinct: the *initial* time η_i at which initial data are posed, $\bar{T}(\eta_i) = T_i$ (the initial event of a given fluid element is at the perturbed time $\eta_{\rm init}$, $1 + \delta_T(\eta_{\rm init}) = \eta_{\rm init}/\eta_i$); the background *emission* time η_*, $\bar{T}(\eta_*) = T_*$, and the actual emission time $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$ along each ray; the *observation* time $\eta_o = \bar{\eta}_o + \delta\eta_o$ (direction independent, fixed by proper time, with $\bar{\eta}_o$ its background value, $\Delta\tau = (\bar{\eta}_o^2 - \eta_i^2)/2$); and the dummy time η along the ray. The ordering is $0 < \eta_i < \eta_* < \bar{\eta}_o$. Initial data are *defined* at finite η_i (Sec. 5.2); the limit $\eta_i \to 0$ is taken only where stated, with its $O(\eta_i^2)$ remainder displayed (Sec. 8.2).

Angular quantities and target.

$$a_{\ell m}^{(r)} = \int \mathrm{d}\Omega_{\hat{\mathbf{n}}}\, Y_{\ell m}^*(\hat{\mathbf{n}})\, \Theta^{(r)}(\hat{\mathbf{n}}), \qquad C_\ell^{(rs)} = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} \langle a_{\ell m}^{(r)} a_{\ell m}^{(s)*} \rangle_c, \tag{11}$$

with $Y_{\ell m}$ carrying the Condon–Shortley phase, and the target is

$$C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4 \Big[\frac{1}{4} C_\ell^{(22)} + \frac{1}{3} \operatorname{Re} C_\ell^{(13)} \Big] + O(\epsilon^6), \qquad \ell \geq 2, \tag{12}$$

where the factors $\frac{1}{4}$, $\frac{1}{3}$ come from the coefficient convention (6), $\Theta = \epsilon\Theta^{(1)} + \frac{\epsilon^2}{2}\Theta^{(2)} + \frac{\epsilon^3}{6}\Theta^{(3)}$.

Notation. Symbols that could be confused are kept distinct as follows. $S(\hat{\mathbf{n}})$ is the shear integral; order-n sources of the field equations always carry indices ($S_{\mu\nu}^{(n)}$, S_Φ, $S_{\rm TL}$, …). $\Lambda \equiv -S$ and $M \equiv e^\Lambda$; the transverse-traceless projector is $\Pi_{ij,kl}$. The fluid vorticity tensor is $\varpi_{\mu\nu}$ and the spatial projector $\perp_{\mu\nu}$; ω_i is the metric vector and h_{ij} the metric tensor mode. The affine parameter is ς; λ alone denotes a helicity in Sec. 10.3, and $\lambda(\epsilon)$, $\beta(\epsilon)$ are the time and space rescalings of the dilation family (Sec. 5.4). $\mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{q}}$; $K \equiv |\mathbf{k}_1 + \mathbf{k}_2|$ in the (22) loop, while kernels are K_X, K_r; $U(x) \equiv 3u_1(x)$ is the regular homogeneous solution of the scalar equation; $\sigma_{\mu\nu}$ is the fluid shear and $\sigma^2 \equiv \langle \zeta_1^2 \rangle$ the primordial variance of the monochromatic spectrum.

3 Roadmap: target, structure, status

Target and conventions. $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4 [\frac{1}{4} C_\ell^{(22)} + \frac{1}{3} \operatorname{Re} C_\ell^{(13)}] + O(\epsilon^6)$, $\ell \geq 2$ (12), with $C_\ell^{(rs)} = \frac{1}{2\ell+1} \sum_m \langle a_{\ell m}^{(r)} a_{\ell m}^{(s)*} \rangle_c$, $\langle\cdot\rangle_c$ connected in the $a_{\ell m}$ (5). Fourier $f = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} f_{\mathbf{k}}$, $\langle \zeta_{\mathbf{k}} \zeta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') P(k)$ (4); every quantity expanded as $X = \epsilon X_1 + \frac{\epsilon^2}{2} X_2 + \frac{\epsilon^3}{6} X_3$ (6); Poisson gauge with linear metric functions (7); $a = \eta$. $\hat{\mathbf{n}}$ is the observation direction, the ray is $\mathbf{x} = \mathbf{x}_o + \hat{\mathbf{n}}\chi$, and $e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = 4\pi \sum i^\ell j_\ell(k\chi) Y_{\ell m}(\hat{\mathbf{n}}) Y_{\ell m}^*(\hat{\mathbf{k}})$ (10) ($Y_{\ell m}$ with Condon–Shortley phase). Observable: emission isotropic in the fluid rest frame on $T_{\rm phys} = T_*$; observer = fluid element labelled at its $T_{\rm phys} = T_i$ event, fixed proper time, FW frame (replaceable by the canonical triad for C_ℓ); $\Theta = T_{\rm obs}/\langle T_{\rm obs} \rangle_{\rm sky} - 1$, dipole removed after normalising (assumptions (O1)–(O3), Sec. 2). Monochromatic: $P = 2\pi^2 \sigma^2 \delta(k - k_0)/k_0^2$ with $\sigma^2 = \langle \zeta_1^2 \rangle$ held fixed, applied after assembly (127).

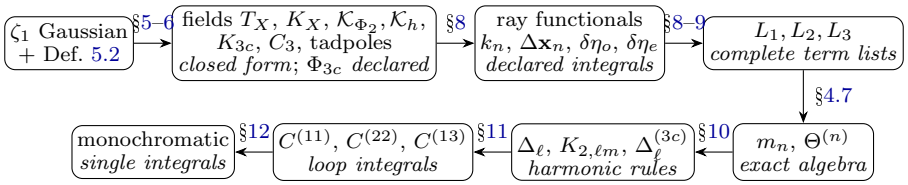
Three structural results.

(a) Exact observable (Thm. 4.3, Cor. 4.5): $T_{\rm obs}(\hat{\mathbf{n}}) = T_o\, e^{-S(\hat{\mathbf{n}})}$, $\quad 1+\Theta = e^{-S}/\langle e^{-S}\rangle_{\rm sky}$.

(b) Initial data (Def. 5.2): $\zeta_{\rm nl} = \epsilon\zeta_1$ Gaussian, $h_n(0) = 0$, regular $\Rightarrow$ $X_n(\eta_i)$ fixed, non-local, constraint-satisfying.

(c) Log-normalisation algebra (32), with m_n the coefficients of e^{-S}: $\Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1\rangle_{\rm sky} m_{2,\ell m} - 3\langle m_2\rangle_{\rm sky} m_{1,\ell m} + 6\langle m_1\rangle^2_{\rm sky} m_{1,\ell m}$.

Pipeline. Each arrow is labelled by the section that performs it; boxes give the type of object.



At first order every box is one line (Sec. 7); Secs. 8–9 repeat the same steps at higher order.

Where each requirement is met.

requirement	where
model; Gaussian non-linear variable	Sec. 2; Defs. 5.1–5.2 ($\zeta_{\rm nl}$ of Salopek–Bond/Maldacena)
no independent higher-order modes	Def. 5.2 (scalars by $\zeta^{(n)}(0) = 0$; tensors and vectors by (36)); Sec. 5.2
constraints at every order, data not set to zero	Sec. 5.2 (Bianchi propagation, (37)); $X_2(0)$, $v_2(0)$, C_3 in Sec. 5.3
emission on $T_{\rm phys} = T_*$; fluid-frame emission	Thm. 4.3 ($T_e = T_*$ exactly), (21), (93); (O1)
observer: fluid worldline, proper time from initial event	(O2), (19)–(20); finite η_i in (90)–(92)
FW tetrad, initial orientation	Sec. 4.5 (rotation drops out of C_ℓ, not of $C^{(22)}$, $C^{(13)}$ separately)
E_o/E_e; normalise by sky mean; dipole after normalising	Thm. 4.3, Cor. 4.5, Sec. 4.7
all terms of $\Theta^{(1,2,3)}$ incl. vectors, tensors, products	(30); Secs. 8–9; App. C (every term of L_1, L_2, L_3)
$a_{\ell m}$, $C^{(rs)}_\ell$, interference terms	Secs. 10, 11; (E0)–(E3)
analytic; evaluated vs unevaluated marked	status table below; [unevaluated integral: ...] markers; analytic ℓ-cut-off (130)
finite times $\eta_i < \eta_* < \eta_o$, boundary terms	Sec. 2 (times), (94), Sec. 8.2
constraints, normalisation, null condition, gauge invariance	Sec. 13: EF-5, R-2, R-3; dilation ($k = 0$) and plane wave vs comoving-gauge solver (finite k)
monochromatic only after the full observable	Sec. 10.8
exchanges of limits; unresolved steps	delicate points below; Sec. 14

Status.

object	closed form	declared integral	verified by	open
T_X, Δ_ℓ, $C_\ell^{(11)}$	yes (ISW: Ci, Si, all ℓ)	—	EF-3, FO-1–4, PW-2	—
K_X (2nd order), tadpoles	yes	—	EF-17, EF-19	—
$\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$	yes (App. B)	—	CF-1–4	—
$K_{3c,X}$, C_3	yes	—	EF-20, EF-22	— (pole (46) cancels in κ, L-1)
Φ_{3c}	—	one time integral	PW-1 (plane wave)	—
L_1, L_2, L_3	complete term lists	ray functionals	X-9, R-4, R-6, PW-2	—
clock, label at finite η_i	yes (91)–(92)	worldline integrals	R-8, PW-5	—
harmonic rules, (B1), (C1)	yes	—	H-1–9, H-11	—
K_2, $K_{2,\ell m}$	channels (App. D)	time integrals	Sec. 11.2	—
$C_\ell^{(22)}$ (mono)	—	μ integral, evaluated	Sec. 12.4	—
(E2), (E3), $T^{(\mathrm{ii})}$	formulas	loop integrals	analytic collapse only	numerical test
$\kappa = \Theta_c^{(3)}$, $C_\ell^{(13)}$ (mono)	—	μ integral, evaluated	R-9–R-11, SQ-3, H-13, L-1, L-2	analytic residues; UV
zero-mode invariance	—	—	PW-3, R-11; quadrupole drops identically	analytic proof (general $\mathbf{q}$)

The (13) term and the $O(\epsilon^4)$ coefficient. The contracted third-order kernel is assembled and validated (Secs. 11.6–11.9); the monochromatic $C_\ell^{(13)}$ is finite (numerical cancellation of the collinear residues, Sec. 12.5) and evaluated at $k_0 = 1$, $\eta_* = 2$, $\eta_o = 8$ (not at $\eta_o = 4$). In units of σ^4 (Table 7): $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = -5.398,\ -9.064,\ -1.463 \times 10^{-3}$ and $\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = 8.860,\ -7.198,\ 3.862 \times 10^{-2}$ for $\ell = 2, 4, 10$, against $C_\ell^{(11)}/\sigma^2 = 0.9546,\ 0.1751,\ 9.68 \times 10^{-6}$. These are evaluations by quadrature (relative accuracy of the (13) part $\leq 8 \times 10^{-5}$); open: an analytic proof of the residue cancellation and the UV behaviour for power-law spectra (Sec. 14).

Delicate points.

- *Dilation and squeezed limits.* Constant and gradient modes of ζ_1 must be unobservable; the exact dilation family is the $k = 0$ gauge test (Sec. 13.3); the squeezed limit of K_2 is a dilation of K_1, $K_2(\mathbf{k}_1 \to 0, \mathbf{k}_2) = -\mathbf{k} \cdot \partial_{\mathbf{k}} K_1 + O(k_1)$ (Sec. 11.4), and the soft region of the loop is a total derivative that vanishes only for $P \propto k^{-3}$.
- *Zero modes.* The $k \to 0$ limit of the second-order kernels does not exist (48); the tadpole convention averages over angles first. In the assembled kernel only the angle-averaged tadpole can enter, and the remaining freedom (a residual time reparametrisation) drops out of κ to 10^{-13} (Sec. 11.7); an analytic proof for general momenta is open.
- $\mu \to \pm 1$. In (22), $K \to 0$ is integrable for $\ell \geq 2$ (Sec. 12.3). In (13) at $q = k$, C_3 has a simple pole in μ (46); six channels of κ have $1/(1-\mu)$ poles whose residues cancel numerically, so the integrand has a finite limit (Sec. 12.5); an analytic residue computation is open.
- *UV behaviour of the perfect fluid.* After emission the fluid is continued as a perfect fluid (no free streaming); the acoustic oscillations persist undamped, $\Delta_\ell \to$ oscillatory as $k \to \infty$ (Sec. 7), and the (22) kernel reaches $K = 2k_0$, so $\frac{1}{4}C_\ell^{(22)}/C_\ell^{(11)}$ grows steeply with ℓ for a monochromatic spectrum (Sec. 12.4): the expansion in $\epsilon^2\sigma^2$ is ℓ dependent.

- *IR for tilted spectra.* The IR region of the one-loop terms cancels between $\frac{1}{4}C^{(22)}$ and $\frac{1}{3}\operatorname{Re}C^{(13)}$ only up to a total derivative $\propto (3+\mathbf{k}\cdot\partial_{\mathbf{k}})^2 P$; for $n_s \neq 1$ the loop is IR sensitive (Sec. 11.9). The monochromatic spectrum has no IR region.

4 The exact observable

4.1 Conformal frame

Write $g_{\mu\nu} = a^2\hat{g}_{\mu\nu}$ with $a = \eta$. Null geodesics of g and $\hat{g}$ coincide as curves; if $p^\mu = \mathrm{d}x^\mu/\mathrm{d}\varsigma$ is affine for g then $\hat{p}^\mu = a^2 p^\mu$ is affine for $\hat{g}$. With the conformal fluid variables (3),

$$E \equiv -u_\mu p^\mu = \hat{E}/a, \qquad \hat{E} \equiv -\hat{g}_{\mu\nu}\hat{u}^\mu\hat{p}^\nu. \tag{13}$$

The radiation stress tensor is traceless, hence conformally covariant: $\hat{T}^{\mu\nu} \equiv a^6 T^{\mu\nu} = (\hat{\rho}+\hat{P})\hat{u}^\mu\hat{u}^\nu + \hat{P}\hat{g}^{\mu\nu}$ with $\hat{\rho} = \alpha\hat{T}^4$, $\hat{P} = \hat{\rho}/3$, and $\hat{\nabla}_\mu\hat{T}^{\mu\nu} = 0$ *exactly*, where $\hat{\nabla}$ is the Levi-Civita connection of $\hat{g}$. Everything in this section therefore holds verbatim for the hatted system, whose background is Minkowski space with a constant fluid; we drop the hats in the lemmas (they hold for any metric) and restore them when we parametrise.

4.2 Two lemmas

Let u^μ be a unit timelike field and decompose its gradient in the standard way,

$$\nabla_\nu u_\mu = \sigma_{\mu\nu} + \varpi_{\mu\nu} + \frac{1}{3}\theta\ \perp_{\mu\nu} - u_\nu a_\mu, \qquad \perp_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \quad a_\mu = u^\nu\nabla_\nu u_\mu, \quad \theta = \nabla_\mu u^\mu, \tag{14}$$

with $\sigma_{\mu\nu}$ the symmetric trace-free part and $\varpi_{\mu\nu}$ (vorticity) the antisymmetric part of $\perp_\mu{}^\alpha\ \perp_\nu{}^\beta\nabla_\beta u_\alpha$ [2]. For a null vector p^μ write $p^\mu = E(u^\mu + e^\mu)$ with $E = -u\cdot p$, $u\cdot e = 0$, $e\cdot e = 1$, and define $\mathrm{d}l \equiv E\,\mathrm{d}\varsigma$ along the ray (the proper length element of the ray as measured in the local rest frame of u).

Lemma 4.1 (kinematic redshift identity)**.** *Along an affinely parametrised null geodesic, $p^\nu\nabla_\nu p^\mu = 0$,*

$$\frac{\mathrm{d}\ln E}{\mathrm{d}l} = -\Big(\frac{\theta}{3} + \sigma_{\mu\nu}e^\mu e^\nu + a_\mu e^\mu\Big). \tag{15}$$

Proof. $\mathrm{d}E/\mathrm{d}\varsigma = p^\nu\nabla_\nu(-u_\mu p^\mu) = -p^\mu p^\nu\nabla_\nu u_\mu$. Insert (14): the σ term gives $E^2\sigma_{ee}$ ($\sigma \perp u$); the ϖ term vanishes identically (antisymmetric tensor contracted with $p^\mu p^\nu$), so *vorticity drops out with no assumption*; the θ term gives $\frac{1}{3}\theta E^2\ \perp_{ee}= \frac{1}{3}\theta E^2$; the acceleration term gives $-E^2[u\cdot(u+e)][a\cdot(u+e)] = E^2 a\cdot e$. Divide by E^2. □

Lemma 4.2 (radiation-fluid identities)**.** *For the fluid (1), the conservation law $\nabla_\mu T^{\mu\nu} = 0$ alone (no field equations) is equivalent to*

$$\theta = -3\,u^\mu\partial_\mu\ln T, \qquad a_\mu = -\perp_\mu{}^\nu\partial_\nu\ln T. \tag{16}$$

Proof. $u_\nu\nabla_\mu T^{\mu\nu} = 0$ gives $u\cdot\partial\rho + (\rho+P)\theta = 0$ and $\perp^\nu{}_\alpha\nabla_\mu T^{\mu\alpha} = 0$ gives $(\rho+P)a^\nu + \perp^{\nu\mu}\partial_\mu P = 0$. With $\rho = \alpha T^4$, $P = \rho/3$ one has $\rho + P = \frac{4}{3}\alpha T^4 = T\,\mathrm{d}P/\mathrm{d}T$ and $\mathrm{d}\rho/\mathrm{d}T = 3(\rho+P)/T$, which turn the two equations into (16). Both ingredients are needed: for $P = w\rho$, $\rho \propto T^4$ one finds $a = -\frac{4w}{1+w}\perp\partial\ln T$ and $\theta = -\frac{4}{1+w}u\cdot\partial\ln T$, which reduce to (16) iff $w = \frac{1}{3}$ (App. G, X-2). □

4.3 The theorem

Theorem 4.3. *Let the photon be emitted at the event e where its past-directed null geodesic last meets the hypersurface $T = T_*$, with an isotropic Planck spectrum in the fluid rest frame (assumption (O1)), and let it be received at the event o by an observer comoving with the fluid. Then the observed temperature in the direction $\hat{\mathbf{n}}$ is*

$$\boxed{T_{\rm obs}(\hat{\mathbf{n}}) = T_* \, \frac{E_o}{E_e} = T_o \, \exp[-S(\hat{\mathbf{n}})], \qquad S(\hat{\mathbf{n}}) \equiv \int_e^o \sigma_{\mu\nu} e^\mu e^\nu \, \mathrm{d}l} \tag{17}$$

where T_o is the fluid temperature at o and the integral runs along the ray.

Proof. Since $e^\mu \perp_\mu {}^\nu = e^\nu$, Lemma 4.2 gives $\theta/3 + a \cdot e = -(u+e)\cdot\partial \ln T = -E^{-1} p \cdot \partial \ln T = -\mathrm{d}\ln T/\mathrm{d}l$. Lemma 4.1 then reads $\mathrm{d}(\ln E - \ln T)/\mathrm{d}l = -\sigma_{ee}$; p is future directed, so ς increases from e to o and $\ln(E_o/E_e) = \ln(T_o/T_e) - S$. With $T_e = T_*$ this is $T_* E_o/E_e = T_o e^{-S}$. Finally, E_o/E_e is independent of the photon energy (the geodesic equation is homogeneous in p), so by Liouville's theorem a Planck spectrum at T_* in the fluid frame at e is received as a Planck spectrum at $T_* E_o/E_e$ in the fluid frame at o; this is what $T_{\rm obs}(\hat{\mathbf{n}})$ means. □

Remark 4.4 (conformal invariance). Theorem 4.3 applies to (g, u, T) and to $(\hat{g}, \hat{u}, \hat{T})$ alike. Since $E = \hat{E}/a$ and $T = \hat{T}/a$ with the same a_e, a_o, the two statements agree iff $S = \hat{S}$: the shear integral is conformally invariant and may be computed entirely in the hatted frame, where the background shear vanishes identically.

Corollary 4.5. *With $\Theta(\hat{\mathbf{n}}) \equiv T_{\rm obs}(\hat{\mathbf{n}})/\langle T_{\rm obs}\rangle_{\rm sky} - 1$ and $\langle\cdot\rangle_{\rm sky} = \dfrac{1}{4\pi}\int \mathrm{d}\Omega$ in the observer's rest frame,*

$$1 + \Theta(\hat{\mathbf{n}}) = \frac{e^{-S(\hat{\mathbf{n}})}}{\langle e^{-S}\rangle_{\rm sky}} \qquad \textit{exactly.} \tag{18}$$

Consequences.

(i) All of the usual emission-surface (intrinsic temperature and $T_{\rm phys} = T_*$ shift), Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing and time-delay physics is contained in S: the only inputs are the fluid shear field $\sigma_{\mu\nu}$, the ray $x^\mu(\lambda)$, and the two endpoints. In particular the perturbation of the emission point along the ray does *not* require a separate "intrinsic δT" term; $T_e = T_*$ has been used exactly.

(ii) T_o is a scalar at the observer event, independent of $\hat{\mathbf{n}}$; it drops out of Θ identically. No expansion of T_o or of $\langle T_{\rm obs}\rangle_{\rm sky}$ in terms of a background temperature is ever needed.

(iii) The observer's frame enters only through the identification of $\hat{\mathbf{n}}$ with a photon direction at o. Any two admissible frames differ by a realisation-dependent rotation, which leaves C_ℓ invariant. For the Fermi–Walker frame and the canonical triad the rotation is rigid and starts at $O(\epsilon^2)$, so that $C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ are even separately the same in both (Sec. 4.5).

(iv) The emission surface enters only through the endpoint e of the integral: the location of the last crossing of $T = T_*$ along the past ray. Perturbatively there is a single crossing and $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$.

Remark 4.6 (validity). The proof uses the differential form of $\nabla \cdot T = 0$ and C^1 fields along the ray; it is not defined across a shock (irrelevant perturbatively). Isotropic Planck emission in the fluid frame at $T = T_*$ is self-consistent (the photons *are* the fluid there); after emission the "fluid" (u, T) entering S is the model's perfect-fluid continuation, as the problem statement prescribes. The identities also hold for $\rho = 3P + c$, $P = CT^4 - c/4$ (a Λ-like constant), not needed here.

4.4 Observer, clock, emission event and direction map

Observer worldline (assumption (O2)): the fluid element labelled by its position at its initial event,

$$\mathbf{x}_o(\eta) = \mathbf{x}^{\mathrm{lab}} + \int_{\eta_{\mathrm{init}}}^{\eta} \mathbf{v}(\eta', \mathbf{x}_o(\eta'))\, \mathrm{d}\eta', \qquad 1 + \delta_T(\eta_{\mathrm{init}}, \mathbf{x}^{\mathrm{lab}}) = \eta_{\mathrm{init}}/\eta_i, \tag{19}$$

where the second relation locates the initial event on $T_{\mathrm{phys}} = T_i$ ($\bar{T} \propto 1/\eta$). *Clock*: proper time elapsed since the initial event,

$$\Delta\tau = \int_{\eta_{\mathrm{init}}}^{\eta_o} \frac{a(\eta)}{\hat{u}^0(\eta, \mathbf{x}_o(\eta))}\, \mathrm{d}\eta = \frac{1}{2}(\bar{\eta}_o^2 - \eta_i^2) \quad \text{held fixed} \qquad \Longrightarrow \qquad \eta_o = \bar{\eta}_o + \delta\eta_o[\zeta], \tag{20}$$

with $\delta\eta_o$ direction independent (a functional of the fields on the worldline; explicit through second order with all η_i dependence in (91)–(92)). Labelling the element instead by its position as $\eta \to 0$ selects the element displaced by $\int_0^{\eta_{\mathrm{init}}} \mathbf{v} = O(\epsilon\, \eta_i^2)$; the two definitions coincide as $\eta_i \to 0$ (Sec. 8.2). *Emission event*: the ray meets $T_{\mathrm{phys}} = T_*$, i.e.

$$1 + \delta_T(\eta_e, \mathbf{x}(\eta_e)) = \eta_e/\eta_*. \tag{21}$$

Canonical triad: $\{e_a\}_{a=1,2,3}$ is the Gram–Schmidt orthonormalisation (order $1, 2, 3$) of the coordinate basis ∂_i with respect to $\hat{g}$, orthogonal to $\hat{u}$, at the observer event. *Direction map*: the photon received from the sky direction $\hat{\mathbf{n}}$ (components n^a in the triad) has

$$\hat{p}_o = \hat{E}_o(\hat{u}_o - n^a e_a), \qquad k^i(\eta_o) \equiv \frac{\mathrm{d}x^i}{\mathrm{d}\eta}\Big|_o = \frac{\hat{p}_o^i}{\hat{p}_o^0}, \tag{22}$$

so that in the background $k^i = -n^i$ and the ray is (9). With the ray parametrised by η, $k^\mu = \mathrm{d}x^\mu/\mathrm{d}\eta = (1, k^i)$, and using $\sigma \perp u$, $\hat{E}/\hat{p}^0 = -\hat{u}_\mu k^\mu$,

$$S(\hat{\mathbf{n}}) = \int_{\eta_e(\hat{\mathbf{n}})}^{\eta_o} \mathrm{d}\eta\, \frac{\hat{\sigma}_{\mu\nu} k^\mu k^\nu}{-\hat{u}_\mu k^\mu}\Big|_{x^i(\eta)}. \tag{23}$$

The perturbative evaluation of (23) (ray, endpoints, triad, clock, all to the required order) is the subject of Secs. 8–9.

4.5 Fermi–Walker frame versus canonical triad

Construction. Let $\{u, E_a\}$ be an orthonormal tetrad along the observer worldline, with E_a prescribed at the initial event (the "specified initial orientation") and Fermi–Walker transported,

$$u^\nu \nabla_\nu E_a^{\ \mu} = (u^\mu a_\nu - a^\mu u_\nu) E_a^{\ \nu}, \qquad a^\mu = u^\nu \nabla_\nu u^\mu, \tag{24}$$

which preserves orthonormality and $E_a \perp u$ and is the non-rotating frame of an accelerated observer. At the observer event o, $\{E_a(o)\}$ and the canonical triad $\{e_a(o)\}$ are two orthonormal bases of the same three-space $u(o)^\perp$, so

$$E_a(o) = \mathsf{R}_{ab}\, e_b(o), \qquad \mathsf{R} \in SO(3), \qquad \mathsf{R} = \exp(\epsilon \mathsf{J}), \quad \mathsf{J} = \mathsf{J}_1 + \frac{\epsilon}{2}\mathsf{J}_2 + \dots \text{ antisymmetric}, \tag{25}$$

with J a realisation-dependent random matrix (it depends on the fields along the worldline through (24) and on the initial orientation). The integration of (24) only determines R, and the following shows that R drops out of C_ℓ, and here even of $C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ separately.

Invariance of C_ℓ under any frame rotation. A direction with components n^a in the FW frame has components $(\mathsf{R}^{\mathrm{T}} n)^a$ in the canonical triad, so $\Theta^{\mathrm{FW}}(\hat{\mathbf{n}}) = \Theta^{\mathrm{can}}(\mathsf{R}^{-1}\hat{\mathbf{n}})$ and, realisation by realisation,

$$a_{\ell m}^{\mathrm{FW}} = \sum_{m'} D^{\ell}_{mm'}(\mathsf{R})\, a_{\ell m'}^{\mathrm{can}} \qquad \Longrightarrow \qquad \sum_m |a_{\ell m}^{\mathrm{FW}}|^2 = \sum_m |a_{\ell m}^{\mathrm{can}}|^2, \tag{26}$$

because the Wigner matrix $D^\ell(\mathsf{R})$ is unitary. This holds exactly, hence order by order in ϵ: the ϵ^4 coefficient of $\sum_m |a_{\ell m}|^2$, $\sum_m [\frac{1}{4}|a_{\ell m}^{(2)}|^2 + \frac{1}{3}\,\mathrm{Re}\, a_{\ell m}^{(1)} a_{\ell m}^{(3)*}]$, is invariant, and C_ℓ is the same in both frames. Consequences: (a) the "specified initial orientation and FW transport" may be replaced by the canonical triad without loss for C_ℓ; (b) $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$ must be computed in one and the same frame (we use the canonical triad throughout); (c) $\sum_m |a_{1m}|^2$ is invariant too, so removing the dipole after normalising is frame independent. Statistical isotropy, and hence $\langle a_{\ell m}\rangle = 0$ for $\ell \geq 1$, holds in both frames.

The FW frame does not rotate at first order. For first-order scalar perturbations $\mathsf{J}_1 = 0$. The angular velocity of the canonical triad relative to a Fermi–Walker frame is the sum of the fluid vorticity, the antisymmetric part of the spatial gradient of the triad (here $\propto \partial_{[i}(\omega_{j]} + v_{j]})$, the Φ terms being symmetric) and the Thomas precession $\propto \mathbf{a} \times \mathbf{v}$. At first order $\mathbf{v}_1 = \nabla V_1$ and $\omega_1 = 0$, so the first two vanish, and the Thomas term is $O(\epsilon^2)$; the vorticity and $\mathrm{curl}\, \boldsymbol{\omega}$ start at $O(\epsilon^2)$ (Sec. 6.2). Hence $\mathsf{R} = 1 + \frac{\epsilon^2}{2}\mathsf{J}_2 + O(\epsilon^3)$, with J_2 realisation dependent but the same for all $\hat{\mathbf{n}}$ (a *rigid* rotation).

Lemma 4.7 (triad independence)**.** *(a) $\Theta^{(1)}$ and $\Theta^{(2)}$ do not depend on the choice of orthonormal triad at the observer among those that agree at first order. (b) Let two such triads differ by a rigid rotation $\mathsf{R} = 1 + \frac{\epsilon^2}{2}\Omega + O(\epsilon^3)$, $\Omega_{ij} = \epsilon_{ijk}\varpi_k$, with $\boldsymbol{\varpi}$ independent of $\hat{\mathbf{n}}$ ($\boldsymbol{\varpi}$ may depend on the fields at the observer). Then $\Theta^{(3)} \to \Theta^{(3)} + 3(\Omega\hat{\mathbf{n}})\cdot\nabla_{\hat{\mathbf{n}}}\Theta^{(1)} = \Theta^{(3)} + 3i\,(\boldsymbol{\varpi}\cdot\hat{\mathbf{L}})\Theta^{(1)}$, $\hat{\mathbf{L}} = -i\hat{\mathbf{n}}\times\nabla_{\hat{\mathbf{n}}}$ (up to the sign convention of Ω), and $C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ are separately unchanged.*

Proof. The triad enters only through the map from the observed direction to the initial photon direction, $\hat{\mathbf{n}} \to \mathsf{R}\hat{\mathbf{n}}$. A change of the direction at $O(\epsilon^2)$ changes $\Theta = \epsilon\Theta^{(1)} + \ldots$ first at $O(\epsilon^3)$: $\Theta(\mathsf{R}\hat{\mathbf{n}}) - \Theta(\hat{\mathbf{n}}) = \frac{\epsilon^3}{2}(\Omega\hat{\mathbf{n}})\cdot\nabla_{\hat{\mathbf{n}}}\Theta^{(1)} + O(\epsilon^4)$, i.e. $\delta\Theta^{(3)} = 3(\Omega\hat{\mathbf{n}})\cdot\nabla_{\hat{\mathbf{n}}}\Theta^{(1)}$, with $(\Omega\hat{\mathbf{n}})\cdot\nabla_{\hat{\mathbf{n}}} = \boldsymbol{\varpi}\cdot(\hat{\mathbf{n}}\times\nabla_{\hat{\mathbf{n}}})$. For each realisation $\sum_m a_{\ell m}^{(1)*}[(\boldsymbol{\varpi}\cdot\hat{\mathbf{L}})a^{(1)}]_{\ell m}$ is real ($\hat{\mathbf{L}}$ is Hermitian and acts within each ℓ, $\boldsymbol{\varpi}$ real), so the change of $\sum_m a_{\ell m}^{(1)*} a_{\ell m}^{(3)}$ is purely imaginary. □

Applied to $\mathsf{R} = 1 + \frac{\epsilon^2}{2}\mathsf{J}_2$: *for this observer $C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ are separately the same in the FW frame and in the canonical triad*, and the integration of (24) is not needed for either.

Remark 4.8 (a generic rotation at first order). If two frames differed by a rotation with $\mathsf{J}_1 \neq 0$ (e.g. an observer whose frame rotates relative to the triad at $O(\epsilon)$), then $a^{(2)} \to a^{(2)} + 2\mathcal{D}_1^\ell a^{(1)}$ with $D^\ell(\mathsf{R}) = 1 + \epsilon\mathcal{D}_1^\ell + \ldots$, compensated by a change of $a^{(3)}$: only the sum $\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\, C_\ell^{(13)}$ would be invariant. An explicit $\ell = 2$ example with exact spin-2 generators gives $\sum_m |a_{2m}^{(2)}|^2 = 321/25$ in one frame and $329107/1225 - 8798\sqrt{6}/1225$ in the other, while $\sum_m [\frac{1}{4}|a^{(2)}|^2 + \frac{1}{3}\,\mathrm{Re}\, a^{(1)} a^{(3)*}] = 107/60$ in both (ledger X-6). This case does not arise for the FW observer of (O2).

Global versus per-direction triads. The canonical triad, Gram–Schmidt orthonormalisation of the *global* coordinate basis, is one triad for the whole sky, and any other global choice differs from it by a rigid rotation. A triad constructed per direction (Gram–Schmidt in the frame adapted to each $\hat{\mathbf{n}}$, e.g. the $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ split of the term lists) differs from it by an $\hat{\mathbf{n}}$-dependent rotation, which is not a rotation of the sky map: it changes κ (at the 0.3% level in the monochromatic case, R-12) and must not be used at third order.

4.6 Linear order: recovery of the Sachs–Wolfe formula

At $O(\epsilon)$ the endpoints and the ray may be taken at their background values, since S itself starts at $O(\epsilon)$. With $\hat{g} = \mathrm{diag}(-(1+2\epsilon\Psi), (1-2\epsilon\Phi)\delta_{ij})$, $\hat{u}^\mu = (1-\epsilon\Psi, \epsilon v^i)$, $\hat{T} = \bar{\hat{T}}(1+\epsilon\delta_T)$, the covariant definitions (14) give (App. G, X-5)

$$\hat{\theta} = \epsilon(\partial\cdot v - 3\Phi'), \qquad \hat{a}_i = \epsilon(v_i' + \partial_i\Psi), \qquad \hat{\sigma}_{ij} = \epsilon[\partial_{(i}v_{j)} - \frac{1}{3}\delta_{ij}\partial\cdot v], \qquad \hat{\sigma}_{0\mu} = 0, \tag{27}$$

and Lemma 4.2 becomes the linearised continuity and Euler equations

$$\delta_T' = \Phi' - \frac{1}{3}\partial\cdot v, \qquad v_i' = -\partial_i(\Psi + \delta_T). \tag{28}$$

With $e = -\hat{\mathbf{n}}$ and $D = \partial_\eta - \hat{\mathbf{n}}\cdot\nabla$ along the background ray one has $n^i n^j \partial_i v_j = \hat{\mathbf{n}}\cdot v' - D(\hat{\mathbf{n}}\cdot v)$, $\hat{\mathbf{n}}\cdot\nabla f = f' - Df$; inserting (28),

$$\hat{\sigma}_{1,ee} = n^i n^j \partial_i v_j - \frac{1}{3}\partial\cdot v = D(\delta_T + \Psi - \hat{\mathbf{n}}\cdot v) - (\Phi' + \Psi') \qquad \text{(on shell)}. \tag{29}$$

Integrating along the ray from $(\eta_*, \bar{\mathbf{x}}(\eta_*))$ to $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$,

$$\boxed{L_1 \equiv -S_1 = [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_e - [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_o + \int_{\eta_*}^{\bar{\eta}_o} (\Phi' + \Psi')\,\mathrm{d}\eta} \tag{30}$$

which is the standard Sachs–Wolfe [1] + Doppler + integrated Sachs–Wolfe formula with the emission-surface shift (the δ_T at e arises because $\bar{T} \propto 1/\eta$ and the emission surface is $T_{\mathrm{phys}} = T_*$, not $\eta = \eta_*$) and with the observer terms: a monopole $-[\delta_T + \Psi]_o$ and the observer dipole $+\hat{\mathbf{n}}\cdot v_o$. Both are removed from $\Theta^{(1)}$ by the normalisation and the dipole subtraction, but they multiply other terms at higher order and are kept (Sec. 4.7). Equation (29) has been verified symbolically and the same identity holds through third order at the integrand level (App. G, R-1, X-9).

4.7 Normalisation: exact algebra on the logarithm

Define $\Lambda \equiv -S$ and $M \equiv e^\Lambda$, expanded as in (6):

$$\Lambda = \epsilon L_1 + \frac{\epsilon^2}{2}L_2 + \frac{\epsilon^3}{6}L_3, \qquad M = 1 + \epsilon m_1 + \frac{\epsilon^2}{2}m_2 + \frac{\epsilon^3}{6}m_3, \qquad m_1 = L_1, \quad m_2 = L_2 + L_1^2, \quad m_3 = L_3 + 3L_1L_2 + L_1^3. \tag{31}$$

By Corollary 4.5, $1 + \Theta = M/\bar{M}$ with $\bar{M} \equiv \langle M\rangle_{\mathrm{sky}}$. Writing $\langle m\rangle_{\mathrm{sky}}$ for the sky monopole of a map ($\langle m\rangle_{\mathrm{sky}} = m_{00}/\sqrt{4\pi}$, a random variable correlated with the map) and expanding $1/\bar{M}$, one finds for $\ell \geq 1$

$$\boxed{\begin{aligned} &\Theta^{(1)}_{\ell m} = m_{1,\ell m}, \qquad \Theta^{(2)}_{\ell m} = m_{2,\ell m} - 2\langle m_1\rangle_{\mathrm{sky}} m_{1,\ell m}, \\ &\Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1\rangle_{\mathrm{sky}} m_{2,\ell m} - 3\langle m_2\rangle_{\mathrm{sky}} m_{1,\ell m} + 6\langle m_1\rangle^2_{\mathrm{sky}} m_{1,\ell m} \end{aligned}} \tag{32}$$

(the monopole $\ell = 0$ of Θ vanishes identically). The measured dipole is removed afterwards, i.e. (12) is used for $\ell \geq 2$ only; by Sec. 4.5 this is frame independent. All observer terms (the monopole $-[\delta_T + \Psi]_o$ and the dipole $\hat{\mathbf{n}} \cdot v_o$ of L_1, the clock shift $\delta\eta_o$, the observer position) are retained inside L_1, L_2, L_3 and are removed only through (32); products such as $\langle m_1 \rangle_{\rm sky} m_1$ and L_1^2 keep their imprint at $\ell \geq 2$.

5 Primordial variable and initial data

5.1 The non-linear primordial variable

Definition 5.1. In Poisson gauge, with the non-linear δ_T and Φ of (8), (7),

$$\zeta_{\rm nl}(\mathbf{x}) \equiv \lim_{\eta \to 0} \Big[\ln\left(1 + \delta_T(\eta, \mathbf{x})\right) + \frac{1}{2} \ln\left(1 - 2\Phi(\eta, \mathbf{x})\right) \Big] = \lim_{\eta \to 0} \Big[\frac{1}{4} \ln(1 + \delta) + \frac{1}{2} \ln(1 - 2\Phi) \Big]. \tag{33}$$

Why this is the uniform-density curvature perturbation. Since $\rho = \bar{\rho}(\eta)(1 + \delta)$ with $\bar{\rho} \propto \eta^{-4}$, the uniform-density hypersurface $\rho = \bar{\rho}(\tilde{\eta})$ is the surface of Poisson time $\eta_P(\mathbf{x}) = \tilde{\eta}\,[1 + \delta_T(\eta_P, \mathbf{x})]$. Its induced metric is $\gamma_{ij} = g_{ij} + g_{00}\partial_i\eta_P\partial_j\eta_P + 2g_{0(i}\partial_{j)}\eta_P$ with $g = \eta_P^2 \hat{g}$, i.e.

$$\gamma_{ij} = \tilde{\eta}^2 (1 + \delta_T)^2 \Big[(1 - 2\Phi)\delta_{ij} + h_{ij} + O(\tilde{\eta}^2) \Big], \tag{34}$$

because $\partial_i \eta_P = \tilde{\eta}\,\partial_i \delta_T = O(\tilde{\eta})$. Maldacena's parametrisation of the uniform-density spatial metric, $\gamma_{ij} = a^2 e^{2\zeta} (e^{\gamma})_{ij}$ with $\gamma_{ii} = 0 = \partial_i \gamma_{ij}$, contains *two* independent variables, ζ and the tensor γ_{ij}; the limit of h_{ij} as $\tilde{\eta} \to 0$ is γ_{ij}, and it is not fixed by $\zeta_{\rm nl}$ or by regularity (see below). The initial-data definition (Definition 5.2) sets $\gamma_{ij} = 0$, i.e. $h_{ij} \to 0$ as $\tilde{\eta} \to 0$; with it $\gamma_{ij} \to a(\tilde{\eta})^2 e^{2\zeta_{\rm nl}} \delta_{ij}$, so $\zeta_{\rm nl}$ is exactly the ζ of Salopek–Bond and Maldacena [3, 4]. (On these slices the fluid is also comoving in the limit, since the regular velocity potential vanishes as $\eta \to 0$, Sec. 6.1.) The spatial coordinates are those of Poisson gauge; the residual freedom of the conformally flat form, a constant rescaling $\mathbf{x} \to B\mathbf{x}$, is fixed by $\zeta^{(2)}(0) = 0$ (Sec. 11.8).

Definition 5.2 (initial data: Gaussian $\zeta_{\rm nl}$, no primordial vectors or tensors)**.** The initial data are those of the regular solution (Sec. 5.2) with

$$\zeta_{\rm nl} = \epsilon\,\zeta_1 \quad \text{exactly}, \qquad \zeta_1 \text{ a Gaussian random field with spectrum } P(k), \tag{35}$$

i.e. $\zeta^{(2)}(\eta \to 0) = 0$ and $\zeta^{(3)}(\eta \to 0) = 0$ identically in $\mathbf{x}$ in the expansion (6), and, at every order n,

$$h_{n,ij}(\eta \to 0, \mathbf{x}) = 0, \qquad \omega_{n,i}(\eta \to 0, \mathbf{x}) \text{ and } v^T_{n,i}(\eta \to 0, \mathbf{x}) \text{ finite (no primordial vector mode)}. \tag{36}$$

The first condition is the statement that the exponentiated uniform-density curvature perturbation is Gaussian [4]; in the language of [10] it is $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$ ($f_{\rm NL} = 0$ for the variable appearing in $e^{2\zeta}$). It fixes the scalar sector only. The second condition, $\gamma_{ij} = 0$ at every order, is an independent part of the definition.

5.2 Regularity and initial data at finite η_i

Regular means: at every order the solution of the Einstein–fluid system is analytic in η at $\eta = 0$. What regularity does and does not remove (operators of (56)):

- scalars: the decaying mode $u_2 \propto (\cos x + x \sin x)/x^3 \sim \eta^{-3}$ ($x = k\eta/\sqrt{3}$) is removed at every order; the regular homogeneous mode $U(x) = 3u_1(x)$ remains and its amplitude is fixed by $\zeta^{(n)}(0) = 0$;
- vectors: the homogeneous solution of $\mathcal{L}_\omega \omega = 0$ is $\omega \propto \eta^{-2}$ (with v^T then fixed by the transverse $0i$ constraint), so there is no regular homogeneous vector mode and regularity alone removes it; (36) states the same as part of the definition;
- tensors: the homogeneous solutions of $\mathcal{L}_h h = 0$ are $j_0(k\eta) = \sin(k\eta)/(k\eta)$ and $y_0(k\eta)$. The mode $j_0(k\eta)$ is analytic at $\eta = 0$ and is *not* removed by regularity; it is a free primordial gravitational wave at order n, and it is excluded by (36), which sets its amplitude to zero. The particular solution $\int_0^\eta G_h K_h$ vanishes at $\eta = 0$ ($\propto \frac{8}{9}\eta^2$, App. B) and is the entire h_n.

With these choices the regular solution is uniquely fixed by Definition 5.2.

Initial data at the finite time η_i ($\bar{T}(\eta_i) = T_i$) are *defined* as the restriction of the regular solution to $\eta = \eta_i$. They satisfy the Hamiltonian and momentum constraints at every order by construction, since the regular solution solves all ten Einstein equations for $\eta > 0$ and the constraints propagate (contracted Bianchi identity, verified to $O(\epsilon^3)$, App. G, EF-5). Every Green's-function integral below is written as $\int_0^\eta$. This is *not* an $\eta_i \to 0$ limit: for $\eta \geq \eta_i$,

$$\int_0^\eta G(\eta, \eta')\, S(\eta')\, \mathrm{d}\eta' = \int_{\eta_i}^\eta G(\eta, \eta')\, S(\eta')\, \mathrm{d}\eta' + X^{\mathrm{hom}}_{[\eta_i]}(\eta), \tag{37}$$

where $X^{\mathrm{hom}}_{[\eta_i]}$ is the homogeneous solution whose value and derivative at η_i equal those of the left-hand side. Since the data at η_i are *defined* as the restriction of the regular solution, (37) is an identity at finite η_i, and the initial data at η_i are exactly $X_n(\eta_i)$, $X_n'(\eta_i)$ of the $\int_0$ form. Finite η_i enters the observable only through the observer's initial event, which is kept explicitly (Sec. 8.2). This is how the instruction "do not indiscriminately set higher-order perturbations to zero" is implemented: $X_2(\eta_i)$, $X_3(\eta_i)$ are non-zero, non-local functionals of ζ_1 determined by (35) and the constraints.

5.3 The conditions at second and third order

Expanding (33) with (6) (App. G, EF-8),

$$\zeta^{(1)} = \frac{1}{4}\delta_1 - \Phi_1 = \delta_{T,1} - \Phi_1, \tag{38}$$

$$\zeta^{(2)} = \frac{1}{4}\delta_2 - \frac{1}{4}\delta_1^2 - \Phi_2 - 2\Phi_1^2, \tag{39}$$

$$\zeta^{(3)} = \frac{1}{4}\delta_3 - \frac{3}{4}\delta_1\delta_2 + \frac{1}{2}\delta_1^3 - \Phi_3 - 6\Phi_1\Phi_2 - 8\Phi_1^3. \tag{40}$$

At $\eta \to 0$ the first-order regular solution has (Sec. 6.1) $\Phi_1 = \Psi_1 \to -\frac{2}{3}\zeta_1$, $\delta_1 \to \frac{4}{3}\zeta_1$, $v_1 \to 0$.

Second order. In the kernel notation of Sec. 6.2 ($X_{2,\mathbf{k}} = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} K_X(\mathbf{q}, \mathbf{k}-\mathbf{q}; \eta)\zeta_{\mathbf{q}}\zeta_{\mathbf{k}-\mathbf{q}}$, $\mathbf{p} \equiv \mathbf{k} - \mathbf{q}$, local products of first-order fields being constant kernels at $\eta \to 0$), the 00 constraint gives $\delta_2(0) = \frac{1}{3}\lim(\eta^2 S_{00}) - 2\Psi_2(0)$ with $\lim_{\eta\to 0} \eta^2 K_{S_{00}} = \frac{32}{3}$, the traceless ij equation gives

$\Psi_2(0) = \Phi_2(0) - K_Q(0)$ with

$$K_Q(\mathbf{q},\mathbf{p};0) = -\frac{13k^4 - 6k^2(p^2+q^2) + 9(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{4k^2 - 6kq\mu + 9q^2\mu^2 - 3q^2}{k^2}, \qquad \mu \equiv \hat{\mathbf{k}}\cdot\hat{\mathbf{q}}, \tag{41}$$

and $\zeta^{(2)}(0) = 0$ then yields

$$\boxed{K_{\Phi_2}(\mathbf{q},\mathbf{p};0) = \frac{1}{3}K_Q(\mathbf{q},\mathbf{p};0) - \frac{8}{27} = -\frac{7k^4 - 2k^2(p^2+q^2) + 3(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{2k^2 - 2kq\mu + 3q^2\mu^2 - q^2}{k^2}} \tag{42}$$

(the reader may verify: $\frac{1}{4}[\frac{32}{9} + 2K_Q(0) - 2\Phi_2] - \frac{1}{4}\cdot\frac{16}{9} - \Phi_2 - 2\cdot\frac{4}{9} = 0$). In position space, with $\Phi_1 = -\frac{2}{3}\zeta_1$,

$$\Phi_2(0) = -2\Phi_1^2 - \nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) + 3\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{43}$$

$$\Psi_2(0) - \Phi_2(0) = 4\Phi_1^2 + 3\nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) - 9\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{44}$$

and $\delta_2(0)$, $v_2(0)$ follow from the constraints (App. A): $\delta_2(0) = \dfrac{32}{9} - 2\Psi_2(0)$ in kernel form, $v_{2,i} = \partial_i V_2 + v_{2,i}^T$ with $V_2 = K^0_{\Phi_2}\,\eta + O(\eta^3)$ and $\omega_2, v_2^T = O(\eta)$, so $v_2(0) = 0$ (small-η series in App. B). Equations (43)–(44) agree *exactly* with the radiation-era superhorizon initial conditions of Pitrou–Uzan–Bernardeau [10] (their (3.12b,c) with $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$, after exchanging their names for the two potentials). Bartolo–Matarrese–Riotto [9] (their (7.30)) agree in the local and gradient terms but differ in the velocity-quadrupole term, which they evaluate with the late-time velocity rather than its $\eta \to 0$ limit (App. G, EF-11).

Third order. Only the q-contracted third-order fields are needed for $C_\ell^{(13)}$ (Sec. 6.3). With $\Phi_{3c}(k,\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, K_{\Phi_{3c}}(k,q,\mu;\eta)$, the regular solution has $\Phi_{3c}(k,0) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, C_3(k;q,\mu)$ and $\zeta^{(3)}(0) = 0$ gives

$$\boxed{C_3(k;q,\mu) = -\frac{16(2k^5 + 12k^4q\mu - 42k^3q^2\mu^2 - 10k^3q^2 + 36k^2q^3\mu^3 + 32k^2q^3\mu + 3kq^4\mu^4 - 42kq^4\mu^2 + kq^4 + 8q^5\mu)}{9\,k\,(k^2 - 2kq\mu + q^2)^2}} \tag{45}$$

($p^2 = k^2 - 2kq\mu + q^2$; invariant form and the intermediate $Q_{3c}(0)$, $\lim \eta^2 K_{3c,S_{00}}$ in App. A). The p^{-4} denominator is the Poisson inversion of the traceless equation. At $q = k$ the numerator vanishes only to first order in p^2, so $C_3 \sim p^{-2}$:

$$C_3(k;k,\mu) = -\frac{4(3\mu^3 + 39\mu^2 - 45\mu + 7)}{9(\mu - 1)} = \frac{16}{9(1-\mu)} - \frac{56}{3} + O(1-\mu), \tag{46}$$

a simple pole in μ at $\mu = 1$ ($p \to 0$). For a smooth $P(q)$ the d^3q integral converges (d^3p/p^2 is integrable), but at $q = k$ exactly, as for the monochromatic spectrum, $\int \mathrm{d}\mu\, C_3$ diverges logarithmically unless the pole cancels in the full contracted map $\Theta_c^{(3)}$. (At $q = k$, $\mu \to 1$ means $\hat{\mathbf{p}} \perp \hat{\mathbf{k}}$, so the $(3\mu_p^2 - 1)$ angular structure of the p^{-2} terms does not average away.) In the complete contracted map this pole cancels: the p^{-2} part of the long second-order mode of the pair $(\mathbf{k}, -\mathbf{q})$ is a residual $K = 0$ time reparametrisation of Poisson gauge, and the six channel residues it produces cancel identically (Proposition 12.2, proved in Sec. 12.6).

5.4 The exact dilation family and the zero-mode (tadpole) ambiguity

The flat FRW solution written in rescaled coordinates $\tilde{\eta} = \lambda\eta$, $\tilde{\mathbf{x}} = \beta\mathbf{x}$ (λ, β constants) is, in the $(\eta, \mathbf{x})$ chart, the exact Poisson-gauge configuration

$$1 + 2\Psi = \lambda^4, \qquad 1 - 2\Phi = \lambda^2\beta^2, \qquad 1 + \delta = \lambda^{-4}, \qquad v^i = 0, \qquad \zeta_{\rm nl} = \ln\beta, \tag{47}$$

which solves $E_{\mu\nu} = 0$ through $O(\epsilon^3)$ for arbitrary λ, β (App. G, EF-6) and non-perturbatively (EF-21). We write $\lambda = 1 + \epsilon a_1 + \frac{\epsilon^2}{2}a_2 + \frac{\epsilon^3}{6}a_3$ in the coefficient convention (6) and $\beta = e^{\epsilon c}$. It is the $k = 0$ member of the solution space: a constant $\zeta_1 = c$ corresponds to $\beta = e^{\epsilon c}$ (so that $\zeta_{\rm nl} = \epsilon c$ exactly) and, by (38), $a_1 = -c/3$, giving $\Psi_1 = \Phi_1 = -\frac{2}{3}c$, $\delta_1 = \frac{4}{3}c$ in agreement with the $k \to 0$ limit of the first-order transfer functions. At second order the family gives $\Psi_2 = 2a_2 + \frac{2}{3}c^2$, $\Phi_2 = -a_2 - \frac{7}{9}c^2$, $\delta_2 = -4a_2 + \frac{20}{9}c^2$. At second order, however, the $k = 0$ equations fix only $\delta_2 = -2\Psi_2 + \frac{32}{9}c^2$ and leave Φ_2, Ψ_2 free: this is the residual gauge freedom of Poisson gauge at zero momentum (the free coefficient a_2 of the time rescaling λ, cf. Weinberg's adiabatic modes [5, 6]). Correspondingly, the $k \to 0$ limit of the finite-k kernels (41)–(42) at fixed $\mathbf{q}$ *does not exist*:

$$K_{\Phi_2}(\mathbf{q}, \mathbf{k} - \mathbf{q}; 0) = -\frac{8}{9} + \frac{8q\mu}{9k} - \frac{4q^2(3\mu^2 - 1)}{9k^2}, \tag{48}$$

and similarly for K_Q; the divergent pieces form a quadrupolar/dipolar residual time-reparametrisation mode at $k = 0$ (not a constant shear), and they vanish after averaging over the direction of $\mathbf{q}$ (Sec. 11.8). The angular average taken *before* the limit,

$$\langle\Phi_2(0)\rangle_{\rm ang} = -\frac{8}{9}c^2, \qquad \langle\Psi_2(0)\rangle_{\rm ang} = +\frac{8}{9}c^2, \qquad \delta_2 = \frac{16}{9}c^2, \tag{49}$$

lies on the family (47) with $\lambda = e^{-\epsilon c/3}$ ($a_2 = c^2/9$), $\beta = e^{\epsilon c}$, i.e. $1 + \delta_T = e^{\zeta/3}$, $1 - 2\Phi = e^{4\zeta/3}$, $1 + 2\Psi = e^{-4\zeta/3}$ (the member with $\Psi_2 = \Phi_2$ instead has $a_2 = -\frac{13}{27}c^2$, $\Phi_2 = \Psi_2 = -\frac{8}{27}c^2$, $\delta_2 = \frac{112}{27}c^2$).

This matters because the third-order map contains *mean-field* (tadpole) terms $\langle X_2\rangle = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q) K_X(\mathbf{q}, -\mathbf{q}; \eta)$, the $\mathbf{k} = 0$ components of the second-order fields (Sec. 6.3). They are evaluated as the $\hat{\mathbf{q}}$-averaged limits (closed forms $\mathrm{tad}_X(q, \eta)$ in App. A; e.g. $\mathrm{tad}_{\Phi_2} \to -\frac{8}{9}$, $\mathrm{tad}_{Q_2} \to -\frac{16}{9}$, $\mathrm{tad}_{\delta_2} \to \frac{16}{9}$ as $\eta \to 0$). This is not a convention: the $\hat{\mathbf{q}}$-average at fixed k removes every negative power of k, only the averaged pair kernel enters the contracted map, the remaining isotropic freedom is the residual time reparametrisation $F(\eta)$, and the map is invariant under every zero mode. The complete argument is given once, in Sec. 11.8; the numerical tests are in Sec. 13.3. At the level of the field equations the consistency is already verified by the squeezed limit of (45),

$$\lim_{q\to 0} C_3(k; q, \mu) = -\frac{32}{9}, \tag{50}$$

which is precisely the $c^2\zeta_s$ coefficient $6 \times \frac{1}{2}(\frac{4}{3}c)^2 \times (-\frac{2}{3})$ of the short mode modulated by the long-mode member $\lambda = e^{-\epsilon c/3}$, $\beta = e^{\epsilon c}$ ($1 - 2\Phi = \lambda^2\beta^2(1 - 2\Phi_s)$, App. G, EF-14).

object	defined in	meaning
$x = k\eta/\sqrt{3}$	(52)	dimensionless sound-horizon variable
$T_\Phi, T_\delta, T_{\delta_T}, T_V$	(53)	first-order transfer functions ($X_1 = \zeta_1 T_X$)
$\mathcal{L}_\Phi, \mathcal{L}_h, \mathcal{L}_\omega$	(56)	universal linear operators (same at every order)
$S^{(n)}_{\mu\nu}$; $S_{\rm TL}, S^L_0, S^T_{0i}, S^V_{ij}, S^{TT}_{ij}, S_\Phi$	(54), (55)	order-n sources and their SVT projections
$K_X(\mathbf{q},\mathbf{p};\eta)$, $\mathbf{p} = \mathbf{k} - \mathbf{q}$	(58)	second-order kernel of X per $\zeta_\mathbf{q}\zeta_\mathbf{p}$
$K_Q, K_{S_{00}}, K_{S_{0z}}, K_{S_\Phi}, K_\omega, K_{v^T}, K_h$	(59), App. A	the explicit second-order kernels
$K^0_{\Phi_2}$	(42) (Sec. 5.3)	initial amplitude of Φ_2 fixed by $\zeta^{(2)}(0) = 0$
u_1, u_2, U, G_Φ, G_h	(60)	homogeneous solutions and Green's functions
$\mathcal{K}_{\Phi_2}, \mathcal{K}_h$	(61), App. B	time integrals for Φ_2, h_2 (closed forms)
$X_{3c}(k,\eta)$, $K_{3c,X}(k,q,\mu;\eta)$	(62)	q-contracted third-order fields and kernels
$\mathrm{tad}_X(q,\eta)$	(64)	mean-field (tadpole) functions $\langle X_2\rangle$
$C_3(k;q,\mu)$	(45)	third-order superhorizon amplitude of Φ_{3c}

Table 1: Named objects of the perturbative solution.

6 Perturbative solution of the Einstein–fluid equations

This section summarises the solution of $E_{\mu\nu} = G_{\mu\nu}[\eta^2\hat{g}] - 8\pi G\, T_{\mu\nu} = 0$ through $O(\epsilon^3)$ in the parametrisation of Sec. 2, obtained by computer algebra with truncated power series in ϵ (App. F; checks EF-$*$ of App. G). The Christoffel symbols, Ricci and Einstein tensors are computed directly from $g = \eta^2\hat{g}$ as series in ϵ; the background part of every component vanishes identically and the contracted Bianchi identity $g^{\mu\alpha}\nabla_\alpha E_{\mu\nu} + g_{\nu\beta}\nabla_\mu T^{\mu\beta} = 0$ holds identically through $O(\epsilon^3)$. Table 1 collects the named objects used in the rest of the paper; each is defined once in the text and the long expressions are in App. A.

6.1 First order

In the frame $\mathbf{k} = k\hat{\mathbf{z}}$ the first-order equations are

$$
\begin{aligned}
E^{(1)}_{ij}\ (i\neq j,\ \text{traceless}): &\quad -k^2(\Phi_1 - \Psi_1) = 0 \ \Rightarrow\ \Psi_1 = \Phi_1,\\
E^{(1)}_{0i}: &\quad 2ik_i(\Phi_1' + \Psi_1/\eta) + 4v_{1,i}/\eta^2 = 0 \ \ [4/\eta^2 = 8\pi G a^2(\bar\rho + \bar P)] \ \Rightarrow\ v_1^T = 0, \quad v_{1,i} = \partial_i V_1,\ V_1 =\\
E^{(1)}_{00}: &\quad \delta_1 = -\frac{2}{3}\eta^2 k^2\Phi_1 - 2\eta\Phi_1' - 2\Psi_1,\\
\text{trace}: &\quad \Phi_1'' + \frac{4}{\eta}\Phi_1' + \frac{k^2}{3}\Phi_1 = 0.
\end{aligned}
\tag{51}
$$

The velocity is irrotational and the conservation equations hold identically on this solution (App. G, EF-3). With

$$x \equiv k\eta/\sqrt{3} \tag{52}$$

the regular solution, normalised by $\zeta_1 = \lim_{\eta\to 0}(\delta_{T,1} - \Phi_1)$, is $X_1 = \zeta_1 T_X(x)$ with

$$
\begin{aligned}
T_\Phi(x) &= -\frac{2(\sin x - x\cos x)}{x^3} \ \xrightarrow{x\to 0}\ -\frac{2}{3}, \qquad \Psi_1 = \Phi_1,\\
T_\delta(x) &= 4T_{\delta_T}(x) = -4\cos x + \frac{8\sin x}{x} + \frac{8\cos x}{x^2} - \frac{8\sin x}{x^3} \ \to\ \frac{4}{3},\\
T_V(x) &= \eta\,\frac{x^2\sin x + 2x\cos x - 2\sin x}{x^3} \ \to\ \frac{\eta}{3} \qquad (v_{1,i} = \partial_i V_1,\ V_1 = \zeta_1 T_V).
\end{aligned}
\tag{53}
$$

The second solution $(\cos x + x\sin x)/x^3 \sim \eta^{-3}$ is discarded by regularity. Note $\delta_{T,1} \to \zeta_1/3$, so $\zeta_1 = \delta_{T,1} - \Phi_1 = \frac{1}{3}\zeta_1 + \frac{2}{3}\zeta_1$ as it should. We write $T \equiv T_\Phi$, $\dot{T} \equiv \partial_\eta T_\Phi$, and $T_q \equiv T_\Phi(q\eta/\sqrt{3})$ etc. in the kernels.

6.2 Second order: operator structure

At order $n \geq 2$ every component of $E^{(n)}_{\mu\nu} = n!\,[\epsilon^n]E_{\mu\nu}$ splits as

$$E^{(n)}_{\mu\nu} = \mathrm{Lin}_{\mu\nu}[X_n] + S^{(n)}_{\mu\nu}, \tag{54}$$

where Lin is the *same* linear operator as in (51) and $S^{(n)}_{\mu\nu}$ is built from lower-order fields ($S^{(2)}$ quadratic in X_1; $S^{(3)}$ contains X_1X_2 and X_1^3 terms; the position-space expressions have 6–20 resp. 71–127 terms per component and are available in machine-readable form, App. F). Decomposing $S^{(n)}_{\mu\nu}$ into scalar, vector and tensor parts (frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the x–z plane),

$$S_{\mathrm{TL}} \equiv S_{zz} - \frac{1}{2}(S_{xx}+S_{yy}), \quad S^L_0 \equiv S_{0z}, \quad S^T_{0i} \equiv (S_{0x}, S_{0y}), \quad S^V \equiv S_{xz}, \quad S^{TT} \equiv S_{xx} - S_{yy}, \quad S_{\mathrm{tr}} \equiv S_{ii}, \tag{55}$$

the ten equations become (all verified by computer algebra, EF-4):

$$\begin{aligned}
\text{traceless scalar:}\quad & Q_n \equiv \Phi_n - \Psi_n = S_{\mathrm{TL}}/k^2 \quad (= -\frac{3}{2}\nabla^{-2}(\partial_i\partial_j\nabla^{-2} - \frac{1}{3}\delta_{ij})S_{ij} \text{ in position space}),\\
00:\quad & \delta_n = \frac{\eta^2}{3}S_{00} - \frac{2}{3}\eta^2k^2\Phi_n - 2\eta\Phi_n' - 2\Psi_n,\\
0i \text{ longitudinal:}\quad & v^L_{n,i} = \partial_i V_n, \qquad V_n = -\frac{\eta^2}{4}\Big[2(\Phi_n' + \Psi_n/\eta) + \frac{S^L_0}{ik}\Big],\\
0i \text{ transverse:}\quad & \frac{1}{2}\nabla^2\omega_{n,i} - \frac{4}{\eta^2}(\omega_{n,i} + v^T_{n,i}) = S^T_{0i} \quad [8\pi G a^2(\rho + P) = 4/\eta^2],\\
ij \text{ vector:}\quad & \mathcal{L}_\omega\,\omega_n = \frac{2}{ik}S^V, \qquad \mathcal{L}_\omega \equiv \partial_\eta + \frac{2}{\eta},\\
ij \text{ TT:}\quad & \mathcal{L}_h\,h_{n,ij} = -2\Pi_{ij,kl}(\hat{\mathbf{k}})S_{kl}, \qquad \mathcal{L}_h \equiv \partial_\eta^2 + \frac{2}{\eta}\partial_\eta + k^2,\\
\text{trace:}\quad & \mathcal{L}_\Phi\,\Phi_n = S_\Phi \equiv -\frac{S_{\mathrm{tr}} - S_{00}}{6} + \frac{Q_n'}{\eta} - \frac{k^2}{3}Q_n, \qquad \mathcal{L}_\Phi \equiv \partial_\eta^2 + \frac{4}{\eta}\partial_\eta + \frac{k^2}{3},
\end{aligned} \tag{56}$$

where $\Pi_{ij,kl} = P_{ik}P_{jl} - \frac{1}{2}P_{ij}P_{kl}$, $P_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$, and in the trace equation δ_n and Ψ_n have been eliminated with the 00 and traceless equations. Thus at every order the scalar sector reduces to *one* forced oscillator $\mathcal{L}_\Phi\Phi_n = S_\Phi$, after which Ψ_n, δ_n, V_n are algebraic; the vector sector to a first-order equation; the tensor sector to a forced wave equation.

Vector sector (relativistic Kelvin theorem). Vorticity-freedom of v^i is *not* implied at second order. What the equations imply (verified: the algebraic ω_2 so obtained satisfies the ij-vector equation identically) is the circulation theorem for the isentropic radiation fluid, whose enthalpy per particle is $\propto T$:

$$\partial_{[i}[(1+\delta_T)\,\hat{u}_{j]}] = 0, \qquad \hat{u}_j \equiv \hat{g}_{j\mu}\hat{u}^\mu = \hat{u}^0[\omega_j + (1-2\Phi)v_j + h_{jk}v^k] \tag{57}$$

(lower index, so that ω_j enters). Consequently ω_2 is *algebraic* in the first-order fields and v^T_2 follows from the transverse $0i$ constraint; no vector Green's function is needed (the vector ODE has $G_\omega = (\eta'/\eta)^2$ and gives the same result).

Kernels. Second-order fields are written as

$$X_{2,\mathbf{k}}(\eta)=\int\frac{\mathrm{d}^3q}{(2\pi)^3}\,K_X(\mathbf{q},\mathbf{p};\eta)\,\zeta_{\mathbf{q}}\zeta_{\mathbf{p}},\qquad \mathbf{p}\equiv\mathbf{k}-\mathbf{q},\qquad \mu\equiv\hat{\mathbf{k}}\cdot\hat{\mathbf{q}}=\frac{k^2+q^2-p^2}{2kq},\tag{58}$$

and all scalar kernels are given in the *invariant form* $K_X(k,q,p;\eta)$ (functions of the three magnitudes), equivalently in the (k,q,μ) form. With $T_q=T_\Phi(q\eta/\sqrt{3})$, $\dot T_q=\partial_\eta T_q$, the second-order kernels obtained from (56) with $S^{(2)}$ are (full forms in App. A):

$$\begin{aligned}
K_Q&=T_qT_p\Big[-4+\frac{6\mu q}{k}+\frac{3q^2}{k^2}-\frac{9\mu^2q^2}{k^2}\Big]+\Big[\eta(T_p\dot T_q+T_q\dot T_p)+\eta^2\dot T_q\dot T_p\Big]\Big[\frac{2\mu q}{k}+\frac{q^2}{k^2}-\frac{3\mu^2q^2}{k^2}\Big],\\
K_\omega&=\frac{2i}{k^2}\Big[\eta\,(q^2-p^2)\,T_qT_p+\eta^2(q^2T_q\dot T_p-p^2T_p\dot T_q)\Big]\quad(\omega_{2,i}=\int P_{ij}(\hat{\mathbf{k}})q_j\,K_\omega\,\zeta\zeta;\ \text{antisymmetric}\\
K_h&=4\Big[3T_qT_p+\eta(\dot T_qT_p+T_q\dot T_p)+\eta^2\dot T_q\dot T_p\Big]=4\Big[2T_qT_p+(\eta\dot T+T)_q(\eta\dot T+T)_p\Big],\\
K_{S_{00}},\ K_{S_{0z}},\ K_{S_\Phi},\ K_{v^T}&:\ \text{App. A};\qquad v^T_{2,i}=\int P_{ij}q_jK_{v^T}\zeta\zeta,\qquad K_{v^T}=-\Big(\frac{\eta^2k^2}{8}+1\Big)K_\omega-\frac{\eta^2}{4}K_{S^T_{0i}}.
\end{aligned}\tag{59}$$

The tensor source K_h is identical to the radiation-era kernel of Kohri–Terada [12] (their eq. (8)) and Baumann et al. [13] (their (11)–(12)) with $w=\frac{1}{3}$, $\Phi=\Psi$ (App. G, EF-9). K_{S_Φ} is finite as $\eta\to0$ (no η^{-2} terms), so the particular integral of the trace equation vanishes at $\eta=0$; $\lim_{\eta\to0}\eta^2K_{S_{00}}=\frac{32}{3}$ and $K_Q(\eta\to0)$ is (41). The vector and tensor sources vanish as $\mathbf{p}\to-\mathbf{q}$ ($k\to0$) because of the projectors, which is what makes the $K\to0$ corner of the (22) loop integrable (Sec. 10.8).

Green's functions and the regular solution. With $x=k\eta/\sqrt{3}$, $x'=k\eta'/\sqrt{3}$,

$$\begin{aligned}
u_1(x)&=\frac{\sin x-x\cos x}{x^3},\qquad u_2(x)=\frac{\cos x+x\sin x}{x^3},\qquad U(x)\equiv3u_1(x)=-\frac{3}{2}T_\Phi(x)\ \to1,\qquad W\equiv u_1u_2'-\\
G_\Phi(k;\eta,\eta')&=\frac{u_1(x')u_2(x)-u_1(x)u_2(x')}{W(\eta')}=\frac{\sqrt{3}}{k}\,\frac{x'}{x^3}\Big[(1+xx')\sin(x-x')-(x-x')\cos(x-x')\Big]\ \xrightarrow{k\to0}\ \frac{\eta'}{3}\Big(1-\frac{\eta'^3}{\eta^3}\\
G_h(k;\eta,\eta')&=\frac{\eta'}{\eta}\,\frac{\sin k(\eta-\eta')}{k},
\end{aligned}\tag{60}$$

satisfying $\mathcal{L}G=0$, $G(\eta,\eta)=0$, $\partial_\eta G|_{\eta'=\eta}=1$. The regular second-order scalar and tensor fields are then

$$\begin{aligned}
\Phi_{2,\mathbf{k}}(\eta)&=\int\frac{\mathrm{d}^3q}{(2\pi)^3}\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_{\Phi_2}(k,q,p;\eta),\qquad \mathcal{K}_{\Phi_2}\equiv\int_0^\eta\mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{S_\Phi}(k,q,p;\eta')+K^0_{\Phi_2}(k,q,p)\,U(k\eta/\sqrt{3}),\\
h_{2,ij,\mathbf{k}}(\eta)&=\Pi_{ij,kl}(\hat{\mathbf{k}})\int\frac{\mathrm{d}^3q}{(2\pi)^3}q_kq_l\,\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_h(k,q,p;\eta),\qquad \mathcal{K}_h\equiv\int_0^\eta\mathrm{d}\eta'\,G_h(k;\eta,\eta')K_h(k,q,p;\eta'),
\end{aligned}\tag{61}$$

with $K^0_{\Phi_2}$ the initial amplitude (42) fixed by Gaussianity, and $\Psi_2=\Phi_2-Q_2$, δ_2, V_2 algebraic from (56). The lower limit 0 is equivalent to η_i together with the regular data at η_i, by (37). *Both time integrals are evaluated in closed form* (App. B): $\mathcal{K}_{\Phi_2}$ and $\mathcal{K}_h$ are elementary functions plus $\mathrm{Si}(w\eta)$ and the entire function $\mathrm{Cin}(z)=\int_0^z(1-\cos t)\,\mathrm{d}t/t$ at the frequencies $w=(\pm k\pm q\pm p)/\sqrt{3}$ (for Φ_2) and $k\pm(q\pm p)/\sqrt{3}$ (for h_2); $\mathcal{K}_h$ equals $4/k^2$ times the radiation-era kernel of Kohri–Terada [12] (their (22), with a modulus in the logarithm). Hence no unevaluated integral remains at second order.

Checks at second order. Bianchi propagation of the constraints (EF-5); exact dilation family (47) (EF-6); tensor source = Kohri–Terada (EF-9); superhorizon Φ_2, Ψ_2 = Pitrou–Uzan–Bernardeau (EF-10); vorticity structure (57) (EF-7); an independently coded re-derivation of the equations and kernels (EF-15–EF-21); the closed forms inserted into the full second-order Einstein and conservation equations (CF-3). Summary in Sec. 13.2.

6.3 Third order: contracted kernels

The (13) spectrum needs only the correlation of a third-order field with ζ_1, i.e. the *q-contracted* third-order field,

$$\langle X_{3,\mathbf{k}}\,\zeta_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\,P(k)\,X_{3c}(k,\eta), \qquad X_{3c}(k,\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\,K_{3c,X}(k,q,\mu;\eta), \quad (62)$$

where the kernel sums the three Wick pairings (it is *not* symmetrised under $\mathbf{q}\to-\mathbf{q}$, so only $p=|\mathbf{k}-\mathbf{q}|$ appears). The contracted source $K_{3c,S}$ is built from $S^{(3)}$ by the rules

$$A_1B_1C_1 \;\to\; A_qB_{-q}C_k + A_qC_{-q}B_k + B_qC_{-q}A_k, \qquad A_1B_2 \;\to\; 2A_q\,K_B(-\mathbf{q},\mathbf{k}) + A_k\,\langle B_2\rangle(\eta), \quad (63)$$

in which $Q_2, \delta_2, V_2, v_2^T, \omega_2$ enter through their explicit kernels and Φ_2, h_2 through the closed-form time integrals $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ with $(a,b,c)=(|\mathbf{q}_1|,|\mathbf{q}_2|,|\mathbf{q}_1+\mathbf{q}_2|)$ kept as symbols. The mean-field terms

$$\langle X_2\rangle(\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\,\mathrm{tad}_X(q,\eta), \qquad \mathrm{tad}_X(q,\eta)\equiv\langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}, \quad (64)$$

vanish for vectors, tensors and velocities by isotropy; for the scalars $Q_2, S_{00}, S_\Phi, \Phi_2, \delta_2$ the angle-averaged $\mathbf{k}\to 0$ limits are given in closed form in App. A (e.g. $\mathrm{tad}_{\Phi_2}(q,\eta) = -\frac{8}{9} + \int_0^\eta \frac{\eta'}{3}(1-\frac{\eta'^3}{\eta^3})\mathrm{tad}_{S_\Phi}(q,\eta')\mathrm{d}\eta' = -\frac{8}{9} - \frac{28}{405}q^2\eta^2 + \ldots$), the $\hat{\mathbf{q}}$-average being the unique $k\to 0$ limit (Sec. 11.8).

The contracted third-order equations have the same operators (56):

$$Q_{3c} = K_{3c,Q}, \qquad \delta_{3c} = \frac{\eta^2}{3}K_{3c,S_{00}} - \frac{2}{3}\eta^2k^2\Phi_{3c} - 2\eta\Phi'_{3c} - 2\Psi_{3c}, \qquad V_{3c} = -\frac{\eta^2}{4}\Big[2(\Phi'_{3c}+\Psi_{3c}/\eta) + \frac{K_{3c,S_{0z}}}{ik}\Big],$$
$$\mathcal{L}_\Phi\Phi_{3c} = K_{3c,S_\Phi}, \qquad \Phi_{3c}(k,\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\Big[\int_0^\eta \mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{3c,S_\Phi}(k,q,\mu;\eta') + C_3(k;q,\mu)\,U(k\eta/\sqrt{3})\Big], \quad (65)$$

with C_3 the amplitude (45) fixed by $\zeta^{(3)}(0)=0$ and K_{3c,S_Φ} finite at $\eta\to 0$. The contracted vector and tensor sources $(S_{0x}, S_{xz}, S_{xx}-S_{yy})$ are produced as well but integrate to zero over the $\mathbf{q}$ angles by isotropy (S_{0y}, S_{yz}, S_{xy} vanish identically). The kernels $K_{3c,X}$ have 157–830 terms and are not reproduced; they are available in machine-readable form (App. F). The η' integral in (65) is [unevaluated integral: $\int_0^\eta G_\Phi K_{3c,S_\Phi}\,\mathrm{d}\eta'$, whose integrand contains the closed-form $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ at the momenta (q,k,p); it is a declared single time integral with complete integrand and limits]. The un-contracted kernel $K_3(\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3)$ is not needed for $C_\ell^{(13)}$ and has not been produced.

Summary. First order: closed forms (53). Second order: all fields in closed form, including the time integrals $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ (App. B). Third order: contracted sources and amplitude C_3 in closed form; Φ_{3c} is one declared time integral. The tadpoles are the $\hat{\mathbf{q}}$-averaged $k\to 0$ limits, which exist; the only residual freedom is a time reparametrisation, under which the map is invariant (Sec. 11.8).

7 First order: the pipeline worked through, $\Delta_\ell(k)$ and $C_\ell^{(11)}$

The pipeline at first order. Every order is computed by the same five steps; at first order each is one line, and Secs. 8–9 repeat them at second and third order.

(1) *Shear on the ray.* $\hat\sigma_{1,ij} = \partial_{(i}v_{j)} - \frac{1}{3}\delta_{ij}\partial\cdot v$; the integrand of (23) on the background ray is $I_1 = n^i n^j \partial_i v_{1,j} - \frac{1}{3}\partial\cdot v_1$ (27).

(2) *Line-of-sight functional.* Endpoints and ray at their background values: $L_1 = -\int_{\eta_*}^{\bar\eta_o} I_1\,\mathrm{d}\eta$, which on shell is the Sachs–Wolfe form (30) including the observer terms.

(3) *Exponential and normalisation.* $m_1 = L_1$ (31); $\Theta^{(1)}_{\ell m} = m_{1,\ell m}$ for $\ell \geq 1$ (32).

(4) *Fields.* The regular first-order solution (53) with ζ_1 as the only random variable.

(5) *Harmonic transform and Wick contraction.* Rule (A-s) of Sec. 10.3 gives $\Delta_\ell(k)$, and $C_\ell^{(11)} = \frac{2}{\pi}\int k^2 P \Delta_\ell^2$.

The rest of this section carries out steps (4)–(5) explicitly and derives the closed form of the ISW integral.

Conventions. Observer at $\mathbf{x}_o = 0$, observation direction $\hat{\mathbf{n}}$, background ray $\mathbf{x}(\eta) = \hat{\mathbf{n}}\,\chi(\eta)$, $\chi = \eta_o - \eta$, emission at η_*, $\chi_* = \eta_o - \eta_*$ (at first order $\eta_o = \bar\eta_o$). We write $x = k\eta/\sqrt{3}$, $s = k\eta$, $s_* = k\eta_*$, $s_o = k\eta_o$, $w = s_o - s_* = k\chi_*$. The first-order map, from S_1 (Sec. 4.6), is

$$\Theta_1(\hat{\mathbf{n}}) = [\delta_{T1} + \Psi_1 - \hat{\mathbf{n}}\cdot\mathbf{v}_1]_e - [\delta_{T1} + \Psi_1 - \hat{\mathbf{n}}\cdot\mathbf{v}_1]_o + \int_{\eta_*}^{\eta_o} (\Phi_1' + \Psi_1')\,\mathrm{d}\eta, \tag{66}$$

with the regular first-order solution ($\Phi_1 = \Psi_1 = \zeta\,T_\Phi$, $\delta_{T1} = \zeta\,T_{\delta T}$, $\mathbf{v}_1 = \nabla V_1$, $V_1 = \zeta T_V$)

$$T_\Phi = -\frac{2j_1(x)}{x}, \quad T_{\delta T} = -\cos x + 2j_0(x) + T_\Phi, \quad kT_V = \sqrt{3}\,(\sin x - 2j_1(x)), \quad 2\partial_\eta T_\Phi = \frac{4k}{\sqrt{3}}\frac{j_2(x)}{x}. \tag{67}$$

Define the emission (Sachs–Wolfe) source $A_{\rm SW}(x) := T_{\delta T} + T_\Phi = -\cos x + 2j_0(x) - 4j_1(x)/x$.

Rayleigh rules. From (10) and $\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}\,e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}} = (ik)^{-1}\partial_\chi e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}}$: $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^p e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}} \to 4\pi i^\ell\, j_\ell^{(p)}(k\chi)\, Y^*_{\ell m}(\hat{\mathbf{k}})$. Observer terms carry no phase ($\chi = 0$): since $j_\ell(0) = \delta_{\ell 0}$ and $j_\ell'(0) = \delta_{\ell 1}/3$, the observer monopole $[\delta_{T1}+\Psi_1]_o$ contributes only to $\ell = 0$ and the observer Doppler term $\hat{\mathbf{n}}\cdot\mathbf{v}_1(\eta_o, 0) = ik\,(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})\,T_V(x_o)\zeta_{\mathbf{k}}$ only to $\ell = 1$.

Proposition 7.1 (First-order transfer function)**.** $a^{(1)}_{\ell m} = 4\pi i^\ell \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\zeta_{\mathbf{k}}\,Y^*_{\ell m}(\hat{\mathbf{k}})\,\Delta_\ell(k)$ *with real*

$$\Delta_\ell(k) = A_{\rm SW}(x_*)\,j_\ell(w) - kT_V(x_*)\,j_\ell'(w) - \delta_{\ell 0}\,A_{\rm SW}(x_o) + \frac{1}{3}\delta_{\ell 1}\,kT_V(x_o) + I_\ell(k), \qquad I_\ell(k) = \frac{4}{\sqrt{3}}\int_{s_*}^{s_o} \frac{j_2(s/\sqrt{3})}{s/\sqrt{3}}\,j_\ell(s_o - s \tag{68}$$

For $\ell \geq 1$, $\Theta^{(1)}_{\ell m} = m_{1,\ell m}$ (32) *and*

$$C_\ell^{(11)} = \frac{2}{\pi}\int_0^\infty k^2\,\mathrm{d}k\,P(k)\,\Delta_\ell(k)^2, \tag{69}$$

from $\langle\zeta_{\mathbf{k}}\zeta^*_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}-\mathbf{k}')P$, $i^\ell(-i)^\ell = 1$ *and* $\sum_m |Y_{\ell m}|^2 = (2\ell+1)/4\pi$*:* $C_\ell = \frac{(4\pi)^2}{2\ell+1}\frac{2\ell+1}{4\pi}\frac{4\pi}{(2\pi)^3}\int k^2 P \Delta_\ell^2$.

Closed form of the ISW integral. Write $J(s) = j_2(s/\sqrt{3})/(s/\sqrt{3})$ (entire), $Y(s) = y_2(s/\sqrt{3})/(s/\sqrt{3}) = -27/s^4 - 3/(2s^2) + O(1)$, $a_\pm = 1 \pm 1/\sqrt{3}$ and $\mathrm{Cin}(z) = \int_0^z (1-\cos t)\,\mathrm{d}t/t = \gamma + \ln z - \mathrm{Ci}\, z$.

Theorem 7.2 (ISW integral, all ℓ)**.**

$$I_\ell = E_\ell(s_*, s_o) + A_\ell(s_o)[\,\mathrm{Ci}\, a_+ s_o - \mathrm{Ci}\, a_+ s_* - \mathrm{Ci}\, a_- s_o + \mathrm{Ci}\, a_- s_*] + B_\ell(s_o)[\,\mathrm{Si}\, a_+ s_o - \mathrm{Si}\, a_+ s_* - \mathrm{Si}\, a_- s_o + \mathrm{Si}\, a_- s_*] + C_\ell(s_o)[\mathrm{Cin}(a_+ w) - \mathrm{Cin}(a_- w)] + D_\ell(s_o)[\,\mathrm{Si}\, a_+ w + \mathrm{Si}\, a_- w], \quad (70)$$

$$A_\ell = \frac{2}{\sqrt{3}} \operatorname*{Res}_{s=0} Y(s)\, y_\ell(s_o - s) = \sqrt{3}\,[3y_\ell'''(s_o) + y_\ell'(s_o)], \quad B_\ell = -\frac{2}{\sqrt{3}} \operatorname*{Res}_{s=0} Y(s)\, j_\ell(s_o - s) = -\sqrt{3}\,[3j_\ell'''(s_o) + j_\ell'(s_o)], \quad (71)$$

$$C_\ell = -\frac{2}{\sqrt{3}} \operatorname*{Res}_{u=0} y_\ell(u)\, Y(s_o - u), \qquad D_\ell = -\frac{2}{\sqrt{3}} \operatorname*{Res}_{u=0} y_\ell(u)\, J(s_o - u), \quad (72)$$

and E_ℓ elementary: Laurent polynomials in s_*, s_o, w times $\sin, \cos$ *of $a_\pm s_*$, $a_\pm s_o$, w, $s_*/\sqrt{3}$, $s_o/\sqrt{3}$.*

Derivation. With $g(x) = h_2^{(1)}(x)/x$ and $h = h_\ell^{(1)}$, the integrand is $\frac{1}{\sqrt{3}}(g + \bar{g})(s/\sqrt{3})\,(h + \bar{h})(s_o - s)$, a sum of four terms $e^{ibs}R(s)$ with $b \in \{\pm a_\pm\}$ (never zero: no resonance) and R rational with poles only at $s = 0$ (order 4, from $y_2(x)/x$) and $s = s_o$ (order $\ell + 1$, from y_ℓ). Partial fractions and the recursion $\int e^{ibs} s^{-m}\mathrm{d}s = -\frac{e^{ibs}}{(m-1)s^{m-1}} + \frac{ib}{m-1}\int e^{ibs} s^{1-m}\mathrm{d}s$ reduce everything to elementary terms plus $\int e^{ibs}\mathrm{d}s/s = \mathrm{Ci}|b|s + i\,\mathrm{sgn}(b)\,\mathrm{Si}|b|s$, whose total coefficient is $\mathrm{Res}[e^{ibs}R]$, i.e. the residue of the product of the *full* Hankel functions. Because j_ℓ and $j_2(x)/x$ are entire, the four products pair up ($A^- = -A^+$, $B^- = -B^+$, $C^- = -C^+$, $D^- = D^+$) and only the y-parts survive, giving the residues above. At the observer end $u = s_o - s \to 0$ the coefficient of $\ln u$ is $\propto \mathrm{Re}\,\mathrm{Res}_{u=0}[j_\ell(u)g] = 0$, so the endpoint is regular and the Cin combination appears with no finite-part ambiguity. □

Checks ($\ell = 0, \ldots, 6$; ledger FO-3): every Ci/Si coefficient of the brute-force antiderivative equals the residue formula to 10^{-169} (40-digit evaluation), E_ℓ contains no $\mathrm{Ci}, \mathrm{Si}, \ln, \gamma$; the closed form agrees with quadrature to relative 5×10^{-18} (five parameter sets). Explicit coefficients for $\ell = 2$ ($z = s_o/\sqrt{3}$) are

$$A_2 = \frac{2\sqrt{3}}{s_o^6}(s_o^5 \sin s_o + 7s_o^4 \cos s_o - 36 s_o^3 \sin s_o - 126 s_o^2 \cos s_o + 270 s_o \sin s_o + 270 \cos s_o),$$

$$B_2 = -\frac{2\sqrt{3}}{s_o^6}(s_o^5 \cos s_o - 7s_o^4 \sin s_o - 36 s_o^3 \cos s_o + 126 s_o^2 \sin s_o + 270 s_o \cos s_o - 270 \sin s_o),$$

$$C_2 = \frac{4\sqrt{3}}{3z^6}(z^3 \sin z + 6z^2 \cos z - 15 z \sin z - 15 \cos z), \qquad D_2 = \frac{4\sqrt{3}}{3z^6}(z^3 \cos z - 6z^2 \sin z - 15 z \cos z + 15 \sin z);$$

the coefficients for $\ell \le 6$ and the elementary parts E_ℓ (56–464 terms for $\ell = 0..6$) are machine generated (App. F).

Monochromatic spectrum. For $P = 2\pi^2\sigma^2\delta(k - k_0)/k_0^2$ ($\langle\zeta^2\rangle = \sigma^2$),

$$C_\ell^{(11)} = 4\pi\sigma^2\Big[A_{\mathrm{SW}}(x_{0*})\, j_\ell(k_0\chi_*) - k_0 T_V(x_{0*})\, j_\ell'(k_0\chi_*) - \delta_{\ell 0} A_{\mathrm{SW}}(x_{0o}) + \frac{1}{3}\delta_{\ell 1} k_0 T_V(x_{0o}) + I_\ell(k_0)\Big]^2, \quad (73)$$

$x_{0*} = k_0\eta_*/\sqrt{3}$, $x_{0o} = k_0\eta_o/\sqrt{3}$, with $I_\ell(k_0)$ from Theorem 7.2. A real-space check (two-point function of the map built directly from the kernel vs $\sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell$, including $\ell = 0, 1$ observer terms) agrees to 6×10^{-15}. Example ($\sigma^2 = 1$, $\eta_* = 0.5$, $\eta_o = 2$, $k_0 = 4$): $\Delta_{0..4} = 0.44006$, -0.33896, 0.27562, 0.22303, 0.118 $C_{2,3,4} = 0.95460$, 0.62508, 0.17507.

Limit $k_0\eta_* \to 0$ (Sachs–Wolfe). Superhorizon the constraints give $\Phi_1' = 0$, $\delta_1 = -2\Psi_1$, i.e. $\delta_{T1} = -\Psi_1/2$; with $\zeta = \delta_{T1} - \Phi_1$ (38) and $\Phi_1 = \Psi_1$: $\Phi_1 = -\frac{2}{3}\zeta$ and

$$[\delta_{T1} + \Psi_1]_e \to \frac{1}{2}\Psi_1 = -\frac{1}{3}\zeta \qquad \text{(RD; cf. } -\frac{1}{5}\zeta \text{ in matter domination)}, \tag{74}$$

while the emission Doppler term vanishes ($kT_V \sim x$). Hence at fixed s_o, for $\ell \geq 2$, $\Delta_\ell \to -\frac{1}{3}j_\ell(s_o) + I^0_\ell(s_o)$, $I^0_\ell = I_\ell|_{s_*=0}$ (closed form: Theorem 7.2 with $s_* \to 0$: the poles of E_ℓ in s_* cancel against the small-argument expansions of the s_*-dependent terms, $\mathrm{Ci}\, a_+ s_* - \mathrm{Ci}\, a_- s_* \to \ln(a_+/a_-)$; checked against quadrature). For a scale-invariant $P = 2\pi^2 A_s/k^3$ this is an exact plateau, $\ell(\ell{+}1)C_\ell/2\pi A_s = 2\ell(\ell{+}1)\int_0^\infty \frac{\mathrm{d}s_o}{s_o}\Delta_\ell(s_o)^2$, independent of η_o. The SW term alone gives $1/9$ ($\int \mathrm{d}y\, j_\ell^2/y = 1/[2\ell(\ell{+}1)]$, the RD analogue of $1/25$). In RD, however, Φ decays after horizon entry and every observed mode has entered by η_o ($s_o \gtrsim \ell$), so the ISW term is not small: $\int 2T_\Phi'\,\mathrm{d}\eta = +\frac{4}{3}$. Numerically (quadrature of the closed form):

ℓ	2	3	4	6	10	20	40	100	∞
SW only	.1111	.1111	.1111	.1111	.1111	.1111	.1111	.1111	1/9
SW+ISW	.3772	.4474	.5003	.5757	.6649	.7629	.8290	.8770	.91377

The large-ℓ value follows from the flat-sky limit: with the line-of-sight source $\kappa(s) = -\frac{1}{3}\delta(s) + 2\,\mathrm{d}T_\Phi/\mathrm{d}s$, $\ell(\ell+1)C_\ell/2\pi A_s \to \int_0^1 \mathrm{d}\mu\,|\tilde\kappa(\mu)|^2$, and using $j_1(x)/x = \frac{1}{4}\int_{-1}^1 (1-t^2)e^{ixt}\mathrm{d}t$,

$$\tilde\kappa(\mu) = \int \kappa\, e^{i\mu s}\mathrm{d}s = 1 - 2b^2 - b(1-b^2)\ln\left|\frac{1+b}{1-b}\right| + i\pi b(1-b^2)\,\theta(1-b), \qquad b = \sqrt{3}\,\mu, \tag{75}$$

($\tilde\kappa(0) = -\frac{1}{3} + \frac{4}{3} = 1$), giving 0.913769 (ISW alone 0.865111); verified against a numerical transform.

Limit $k_0 \to \infty$. $A_{\rm SW}(x) = -\cos x + O(1/x)$, $kT_V = \sqrt{3}\sin x + O(1/x)$ and $I_\ell = O(1/(k\eta_*))$ (no resonance, $a_\pm \neq 0$), so

$$\Delta_\ell \to -\cos x_{0*}\, j_\ell(k_0\chi_*) - \sqrt{3}\sin x_{0*}\, j_\ell'(k_0\chi_*) + \delta_{\ell 0}\cos x_{0o} + \frac{1}{\sqrt{3}}\delta_{\ell 1}\sin x_{0o}, \tag{76}$$

(acoustic oscillation with Doppler/SW amplitude ratio $1/c_s = \sqrt{3}$). For fixed $\ell \geq 2$, $C_\ell^{(11)} \to \frac{4\pi\sigma^2}{(k_0\chi_*)^2}[\cos x_{0*}\sin(k_0\chi_* - \frac{\ell\pi}{2}) + \sqrt{3}\sin x_{0*}\cos(k_0\chi_* - \frac{\ell\pi}{2})]^2 \to 0$; the observer terms keep $C_0 \to 4\pi\sigma^2\cos^2 x_{0o}$, $C_1 \to \frac{4\pi}{3}\sigma^2\sin^2 x_{0o}$. Checked at $k = 300, 3000$.

Small k (gauge modes). Expanding in k (ISW integrated term by term): $\Delta_0 = -\frac{k^4\chi_*^3(\eta_o+3\eta_*)}{810} + O(k^6)$, $\Delta_1 = \frac{2k^3\chi_*^2(\eta_o+2\eta_*)}{405} + O(k^5)$: the k^0, k^2 terms of Δ_0 and the k term of Δ_1 cancel between emission, observer and ISW, as required by the invisibility of constant and gradient ζ modes at first order.

Sky monopole. $\langle m_1\rangle_{\rm sky} = \frac{1}{4\pi}\int \mathrm{d}\Omega\, m_1 = a_{00}^{(1)}/\sqrt{4\pi}$, so

$$\langle m_1\rangle_{\rm sky} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\,\Delta_0(k), \qquad \Delta_0(k) = A_{\rm SW}(x_*)\,j_0(w) + kT_V(x_*)\,j_1(w) - A_{\rm SW}(x_o) + I_0(k), \quad (77)$$

observer monopole included; $\Delta_0 = O(k^4)$. Then $\langle\langle m_1\rangle_{\rm sky}^2\rangle = \frac{1}{2\pi^2}\int k^2 P\,\Delta_0^2\,\mathrm{d}k = C_0^{(11)}/4\pi$, monochromatic $\sigma^2\Delta_0(k_0)^2$ (example above: 0.193652). For $\ell \geq 1$,

$$\langle\langle m_1\rangle_{\rm sky}\, a_{\ell m}^{(1)*}\rangle = 4\pi(-i)^\ell \int \frac{k^2\mathrm{d}k}{(2\pi)^3} P\,\Delta_0\,\Delta_\ell \int \mathrm{d}\Omega_k\, Y_{\ell m}(\hat{\mathbf{k}}) = 0, \quad (78)$$

since $\int Y_{\ell m}\mathrm{d}\Omega = \sqrt{4\pi}\,\delta_{\ell 0}\delta_{m0}$ (statistical isotropy; any P). This is a statement about the Gaussian first-order fields only; at higher order $\langle m_1\rangle_{\rm sky}$ multiplies $m_{\ell m}$ in $\Theta^{(2)}_{\ell m}$ (32) and its correlations with products of fields do not vanish. Numerical check: the sky average of the kernel equals $\Delta_0(k_0)$ for every $\hat{\mathbf{k}}$ to 5×10^{-15}, and $|\langle\langle m_1\rangle_{\rm sky} a^*_{\ell m}\rangle| < 10^{-14}$ for $\ell = 1..3$.

8 The second-order map $\Theta^{(2)}$

This section turns the exact observable (23) into explicit line-of-sight functionals L_1, L_2 (this section) and L_3 (Sec. 9) of the Poisson-gauge fields of Sec. 6, with the normalisation of Sec. 4.7. These are steps (1)–(3) of the pipeline of Sec. 7 at second order; the complete term lists are machine generated and reproduced in full in App. C. The whole chain has been tested end to end against a non-perturbative solver in a different gauge (Sec. 13.4).

Write the integrand of (23) as a function of the spacetime point and of the tangent $k^\mu = (1, k^i)$,

$$\mathcal{I}(x;k) \equiv \frac{\hat\sigma_{\mu\nu}k^\mu k^\nu}{-\hat u_\mu k^\mu}, \qquad S(\hat{\mathbf{n}}) = \int_{\eta_e(\hat{\mathbf{n}})}^{\eta_o} \mathrm{d}\eta\, \mathcal{I}(\eta, \mathbf{x}(\eta); k(\eta)), \qquad \Lambda \equiv -S. \quad (79)$$

Since $\hat\sigma$ vanishes on the background, $\mathcal{I} = O(\epsilon)$, and at first order on the background ray ($k = (1, -\hat{\mathbf{n}})$)

$$I_1 \equiv \mathcal{I}_1|_{k^i=-n^i} = n^i n^j \partial_i v_1^j - \frac{1}{3}\partial_i v_1^i = \hat\sigma_{1,ee} \quad (80)$$

(cf. (29)). The *expansion scheme* has three ingredients, each expanded in ϵ as in (6): (i) the perturbed ray $x^i(\eta) = \bar x^i(\eta) + \Delta x^i(\eta)$ with perturbed tangent $k^i(\eta)$; (ii) the perturbed endpoints, $\eta_o = \bar\eta_o + \delta\eta_o$ (clock) with the observer at $\bar{\mathbf{x}}_o + \delta\mathbf{x}_o$, and $\eta_e = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$ (emission surface); (iii) Taylor expansion of every field about the background ray (9), so that the final functionals involve only fields at $(\eta, \bar{\mathbf{x}}(\eta))$, $\eta_* \leq \eta \leq \bar\eta_o$, at the background observer event $(\bar\eta_o, \bar{\mathbf{x}}_o)$ and at the background emission point $(\eta_*, \bar{\mathbf{x}}(\eta_*))$.

8.1 Ray, triad and direction map

Geodesic equation. With the ray parametrised by η, the null geodesic equation of $\hat g$ is, exactly,

$$\frac{\mathrm{d}k^i}{\mathrm{d}\eta} = G^i \equiv -\hat\Gamma^i_{\mu\nu}k^\mu k^\nu + \hat\Gamma^0_{\mu\nu}k^\mu k^\nu\, k^i, \qquad \frac{\mathrm{d}x^i}{\mathrm{d}\eta} = k^i. \quad (81)$$

The background $\hat g$ is Minkowski, so the background ray is straight, $k^i = -n^i$. We write $k^i = -n^i + \epsilon k_1^i + \frac{\epsilon^2}{2}k_2^i + \ldots$, $\Delta x^i = \epsilon\Delta x_1^i + \frac{\epsilon^2}{2}\Delta x_2^i + \ldots$, and use the decomposition along $\hat{\mathbf{n}}$: $\Delta x_\parallel \equiv \hat{\mathbf{n}}\cdot\Delta\mathbf{x}$

(radial: time delay) and $\Delta\mathbf{x}_\perp \equiv \Delta\mathbf{x} - \hat{\mathbf{n}}\,\Delta x_\parallel$ (transverse: lensing), likewise for k. At first order ($|\hat{\mathbf{n}}|= 1$)

$$G_1^i = -\partial_i(\Phi_1+\Psi_1) + 2n^i\,\hat{\mathbf{n}}\cdot\nabla(\Phi_1+\Psi_1) - n^i(\Phi_1'+\Psi_1'), \qquad \hat{\mathbf{n}}\cdot\vec{G}_1 = -D(\Phi_1+\Psi_1), \tag{82}$$

whose transverse part $-\nabla_\perp(\Phi_1+\Psi_1)$ is the lensing deflection. $G_2^i/2$ consists of the second-order Christoffel symbols on the background ray plus $\Gamma_1 \times k_1$ cross terms (82–85 terms per component, App. F); the positional shift $\Delta x_1^j \partial_j G_1^i$ is added explicitly in (88).

Canonical triad. The Gram–Schmidt triad of Sec. 4.4 (order $1,2,3$) is needed through second order, since $\mathcal{I} = O(\epsilon)$ (in the coefficient convention $X = \epsilon X_1 + \frac{\epsilon^2}{2}X_2$):

$$\begin{aligned} e_a{}^0 &= \epsilon v_1^a + \epsilon^2[\frac{1}{2}(v_2^a+\omega_{2,a}) - (\Phi_1+2\Psi_1)v_1^a], \\ e_a{}^i &= \delta_a^i\{1+\epsilon\Phi_1+\epsilon^2[\frac{3}{2}\Phi_1^2+\frac{1}{2}\Phi_2-\frac{1}{4}h_{2,aa}+\frac{1}{2}(v_1^a)^2]\} + \epsilon^2\begin{cases} v_1^a v_1^i, & i>a, \\ -\frac{1}{2}h_{2,ai}, & i<a, \end{cases} \end{aligned} \tag{83}$$

(no sum over a; the asymmetry between $i>a$ and $i<a$ is that of the Gram–Schmidt order). Verified: $\hat{g}(e_a,e_b)=\delta_{ab}$, $\hat{g}(e_a,\hat{u})=0$ through third order (App. G, R-2).

Direction map. From (22), $k^i(\eta_o) = (\hat{u}^i - n^a e_a{}^i)/(\hat{u}^0 - n^a e_a{}^0)$ with all fields at the actual observer event. Expanding, with fields at that event,

$$k^i_{1,\mathrm{init}} = v_1^i - n^i(\hat{\mathbf{n}}\cdot\mathbf{v}_1) - n^i(\Phi_1+\Psi_1), \tag{84}$$

i.e. aberration $\mathbf{v}_{1\perp}$ in the transverse direction and $|\vec{k}|= 1+\Phi_1+\Psi_1$ radially (the null condition $\hat{\mathbf{n}}\cdot\vec{k}_1 = -(\Phi_1+\Psi_1)$ holds at every η, by (82)). At second order, displayed for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ ($\perp = x,y$; general $\hat{\mathbf{n}}$: 23–26 terms per component),

$$\begin{aligned} \frac{1}{2}k^\perp_{2,\mathrm{init}} &= \frac{1}{2}v_2^\perp + v_1^\perp v_1^z + \frac{1}{2}h_{2,\perp z}, \\ \frac{1}{2}k^z_{2,\mathrm{init}} &= -\frac{1}{2}\Psi_2 - \frac{1}{2}\Phi_2 - \frac{3}{2}\Phi_1^2 - \Phi_1\Psi_1 + \frac{1}{2}\Psi_1^2 + \frac{1}{4}h_{2,zz} - \frac{1}{2}\omega_{2,z} + \frac{1}{2}|\mathbf{v}_{1\perp}|^2. \end{aligned} \tag{85}$$

The map $\hat{\mathbf{n}} \mapsto k^\mu(\eta_o)$ is null through third order (App. G, R-3).

Ray functionals. Integrating (81) backwards from the observer and transporting the observer event to $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$ gives, with all fields on the background ray,

$$k_1^i(\eta) = k^i_{1,\mathrm{init}}|_{\bar{o}} - \int_\eta^{\bar{\eta}_o} G_1^i\,\mathrm{d}\eta', \tag{86}$$

$$\Delta x_1^i(\eta) = \delta x_{o1}^i + n^i\,\delta\eta_{o1} - \int_\eta^{\bar{\eta}_o} k_1^i\,\mathrm{d}\eta', \tag{87}$$

$$\frac{1}{2}k_2^i(\eta) = \frac{1}{2}k^i_{2,\mathrm{init}}|_{\bar{o}} + \delta\eta_{o1}\,\partial_\eta k^i_{1,\mathrm{init}}|_{\bar{o}} + \delta x^j_{o1}\,\partial_j k^i_{1,\mathrm{init}}|_{\bar{o}} - \delta\eta_{o1}G_1^i(\bar{\eta}_o) - \int_\eta^{\bar{\eta}_o}[\frac{1}{2}G_2^i + \Delta x_1^j\partial_j G_1^i]\mathrm{d}\eta', \tag{88}$$

$$\frac{1}{2}\Delta x_2^i(\eta) = \frac{1}{2}\delta x^i_{o2} + \frac{1}{2}n^i\,\delta\eta_{o2} - \delta\eta_{o1}\,k_1^i(\bar{\eta}_o) - \int_\eta^{\bar{\eta}_o}\frac{1}{2}k_2^i\,\mathrm{d}\eta', \tag{89}$$

where $|_{\bar o}$ means fields at $(\bar\eta_o, \bar{\mathbf{x}}_o)$. The derivative terms in (88) transport the initial direction from the actual to the background observer event, and G_2^i contains $k_1(\eta)$. The $n^i\delta\eta_o$ terms in Δx express that the photon reaching the later observer event passes $\bar\eta_o$ displaced by $-k\,\delta\eta_o$.

Remark 8.1 (sign of the tangent integrals). Since $\mathrm{d}k/\mathrm{d}\eta = G$ and the integrals run from η to the observer, $k(\eta) = k(\bar\eta_o) - \int_\eta^{\bar\eta_o} G$, hence the minus signs in (86) and (88). With them the null condition $\hat{\mathbf{n}}\cdot\mathbf{k}_1 = -(\Phi_1+\Psi_1)$ holds at every η and the shear and direct forms of Sec. 8.6 agree; the opposite sign violates both at $O(1)$ (checks R-3, PW-4).

8.2 Observer event: clock and displacement at finite η_i

The observer is the fluid element (19), labelled by its position $\bar{\mathbf{x}}_o \equiv \mathbf{x}^{\rm lab}$ at its initial event $\eta_{\rm init} = \eta_i(1+\delta_T)$. Expanding (19) and (20) with $1/\hat u^0 = 1 + \epsilon\Psi_1 + \epsilon^2[\frac{1}{2}\Psi_2 - \frac{1}{2}\Psi_1^2 - \frac{1}{2}|\mathbf{v}_1|^2] + O(\epsilon^3)$, fields on the worldline expanded about the label, $F(\eta,\mathbf{x}_o(\eta)) = F(\eta,\bar{\mathbf{x}}_o) + \epsilon\,\delta\mathbf{x}_{o1}(\eta)\cdot\nabla F + \ldots$, and the coefficient convention (6) for $\eta_{\rm init}$, η_o and $\delta\mathbf{x}_o$, one finds, keeping every η_i dependence,

$$\eta_{\rm init} = \eta_i + \epsilon\,\eta_i\delta_{T,1} + \frac{\epsilon^2}{2}\,\eta_i[\delta_{T,2} + 2\eta_i\delta_{T,1}\delta'_{T,1}] + O(\epsilon^3) \qquad \text{(fields at } (\eta_i,\bar{\mathbf{x}}_o)\text{)},$$
$$\delta x^i_{o1}(\eta) = \int_{\eta_i}^{\eta} v_1^i(\eta',\bar{\mathbf{x}}_o)\,\mathrm{d}\eta', \qquad \frac{1}{2}\delta x^i_{o2} = \int_{\eta_i}^{\bar\eta_o}[\frac{1}{2}v_2^i + \delta x^j_{o1}(\eta)\partial_j v_1^i]\mathrm{d}\eta + \delta\eta_{o1}v_1^i|_{\bar o} - \eta_i\,\delta_{T,1}v_1^i|_{(\eta_i,\bar{\mathbf{x}}_o)}, \tag{90}$$

with $\delta x^i_{o1} \equiv \delta x^i_{o1}(\bar\eta_o)$, and

$$\delta\eta_{o1} = -\frac{1}{\bar\eta_o}\int_{\eta_i}^{\bar\eta_o}\eta\,\Psi_1(\eta,\bar{\mathbf{x}}_o)\,\mathrm{d}\eta \;+\; \frac{\eta_i^2}{\bar\eta_o}\,\delta_{T,1}(\eta_i,\bar{\mathbf{x}}_o), \tag{91}$$
$$\delta\eta_{o2} = -\frac{2}{\bar\eta_o}\Big\{\frac{1}{2}\delta\eta_{o1}^2 + \bar\eta_o\,\delta\eta_{o1}\Psi_1|_{\bar o} + \int_{\eta_i}^{\bar\eta_o}\eta[\frac{1}{2}\Psi_2 - \frac{1}{2}\Psi_1^2 - \frac{1}{2}|\mathbf{v}_1|^2 + \delta x^j_{o1}(\eta)\partial_j\Psi_1]_{(\eta,\bar{\mathbf{x}}_o)}\mathrm{d}\eta\Big\}$$
$$+\frac{\eta_i^2}{\bar\eta_o}\Big[\delta_{T,2} + \delta_{T,1}^2 + 2\Psi_1\delta_{T,1} + 2\eta_i\,\delta_{T,1}\delta'_{T,1}\Big]_{(\eta_i,\bar{\mathbf{x}}_o)}, \tag{92}$$

both integrals running along the observer worldline (sympy derivation and checks: ledger R-8). The explicit $O(\eta_i^2)$ terms come from the initial event (the term in square brackets, and $\eta_i\delta_{T,1}$ in $\eta_{\rm init}$) and from the lower limits. Since the regular fields and their time derivatives are bounded at $\eta = 0$ (per Fourier mode, uniformly in k), every η_i-dependent piece is $O(\eta_i^2)$ uniformly, and the limit $\eta_i \to 0$ is obtained by setting the lower limits to 0 and dropping the boundary terms. In this limit the label at the initial event and the label as $\eta \to 0$ coincide; at finite η_i they select fluid elements a distance $\int_0^{\eta_{\rm init}}\mathbf{v} = O(\epsilon\,\eta_i^2)$ apart, and because the two selections weight the fluid elements differently (by the Jacobian of the map between the labels, which is correlated with the fields), the resulting C_ℓ differ at $O(\epsilon^4\eta_i^2)$. *All perturbative evaluations in this paper (the kernels of Sec. 11 and the monochromatic results) use the $\eta_i \to 0$ limit*; the end-to-end comparison of Sec. 13.4 keeps the clock terms at $\eta_i = 0.01$ and the label as $\eta \to 0$, matching the non-perturbative solver, whose comoving label is the $\eta \to 0$ position. For the dilation family (Sec. 5.4) these give $\eta_o = \bar\eta_o/\lambda$ exactly through second order for any a_2 *and any* η_i (R-8). The clock and the displacement are direction independent random variables; they affect the map only through (87)–(89), the transport terms of (88) and the endpoint terms below.

8.3 Emission event

Solving (21), $1+\delta_T(\eta_e,\mathbf{x}(\eta_e))=\eta_e/\eta_*$, with $\mathbf{x}(\eta_e)=\bar{\mathbf{x}}(\eta_*)-\hat{\mathbf{n}}\,\delta\eta_e+\Delta\mathbf{x}(\eta_*)+\dots$,

$$\delta\eta_{e1}=\eta_*\,\delta_{T,1}|_*, \qquad \delta\eta_{e2}=\eta_*\Big[\delta_{T,2}+2\,\delta\eta_{e1}\,\delta'_{T,1}+2(\Delta\mathbf{x}_1(\eta_*)-\hat{\mathbf{n}}\,\delta\eta_{e1})\cdot\nabla\delta_{T,1}\Big]_*, \tag{93}$$

with $|_*$ the background emission point $(\eta_*,\bar{\mathbf{x}}(\eta_*))$. $\delta\eta_{e1}$ is the first-order emission-surface shift ("temperature at fixed time" bookkeeping of standard derivations); $\delta\eta_{e2}$ contains the second-order temperature perturbation and the lensed and time-delayed position of the first-order one.

8.4 Expansion about the background ray

Let $F(\eta)\equiv\mathcal{I}(\eta,\mathbf{x}(\eta);k(\eta))$. The three steps are: (1) *position*: $F=\mathcal{I}+\Delta x^j\partial_j\mathcal{I}+\frac{1}{2}\Delta x^j\Delta x^l\partial_j\partial_l\mathcal{I}$, all at $(\eta,\bar{\mathbf{x}}(\eta))$, with $k=k(\eta)$ inserted in $\mathcal{I}$; (2) *endpoints*:

$$\int_{\eta_e}^{\eta_o}F\,\mathrm{d}\eta=\int_{\eta_*}^{\bar\eta_o}F\,\mathrm{d}\eta+\sum_{j\ge1}\Big[\frac{\delta\eta_o^j}{j!}D^{j-1}F\Big]_{\bar\eta_o}-\sum_{j\ge1}\Big[\frac{\delta\eta_e^j}{j!}D^{j-1}F\Big]_{\eta_*}, \tag{94}$$

where D is the total derivative along the ray in the background representation ($D=\partial_\eta-\hat{\mathbf{n}}\cdot\nabla$ on fields, $\mathrm{d}k_1/\mathrm{d}\eta\to G_1$, $\mathrm{d}\Delta x_1/\mathrm{d}\eta\to k_1$, …); (3) at the observer endpoint the functionals take their initial values ($k_1\to k_{1,\mathrm{init}}|_{\bar o}$, $\Delta\mathbf{x}_1\to\delta\mathbf{x}_{o1}+\hat{\mathbf{n}}\,\delta\eta_{o1}$, …), while at the emission endpoint they appear as $k_1(\eta_*)$, $\Delta x_1(\eta_*)$. The outcome is stored as lists of terms, each labelled *ray* (integrated over $[\eta_*,\bar\eta_o]$ on the background ray), *obs* (at $(\bar\eta_o,\bar{\mathbf{x}}_o)$) or *emi* (at $(\eta_*,\bar{\mathbf{x}}(\eta_*))$). No equation of motion is used in building the lists; $|\hat{\mathbf{n}}|=1$ is imposed only in the $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ "split" versions.

First order. $L_1=-\int_{\eta_*}^{\bar\eta_o}I_1\,\mathrm{d}\eta$ (3 terms for $\hat{\mathbf{n}}=\hat{\mathbf{z}}$, 15 for general $\hat{\mathbf{n}}$), which on shell equals the Sachs–Wolfe form (30) including the observer monopole and dipole (App. G, R-1).

8.5 L_2: named pieces

Collecting the ϵ^2 terms,

$$\frac{1}{2}L_2=-\int_{\eta_*}^{\bar\eta_o}\mathrm{d}\eta\Big[\frac{1}{2}\mathcal{I}_2|_{k=(1,-\hat{\mathbf{n}})}+k_1^i\frac{\partial\mathcal{I}_1}{\partial k^i}\Big|_{k=(1,-\hat{\mathbf{n}})}+\Delta x_1^j\partial_jI_1\Big]-\delta\eta_{o1}\,I_1|_{\bar o}+\delta\eta_{e1}\,I_1|_*, \tag{95}$$

where $\frac{1}{2}\mathcal{I}_2$ is the ϵ^2 part of $\mathcal{I}$ at fixed point and fixed background tangent and $k_1^i\partial\mathcal{I}_1/\partial k^i=-2k_1^i\hat\sigma_{1,ij}n^j$. Table 2 lists the pieces. Two simplifications occur in the $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ form: the radial direction perturbation $k_1^z=-(\Phi_1+\Psi_1)$, the $(1-2\Phi)$ of $\hat\sigma_{ij}$ and the factor $1/(-\hat u\cdot k)=1-\Psi-\hat{\mathbf{n}}\cdot\mathbf{v}+\dots$ cancel exactly (as they must for σ_{ee} with e a unit vector), so no undifferentiated Φ_1,Ψ_1 survive; and ω_2,Φ_2,Ψ_2 drop out of the second-order shear (ω cancels between $\partial_{(i}\hat u_{j)}$ and $\hat\Gamma^0_{ij}\hat u_0$). These fields *do* appear in L_3 and in the direct form below.

Explicitly, for $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ (all fields first order unless indexed, $\perp=x,y$, $\partial_\perp\cdot\mathbf{v}_\perp=\partial_xv^x+\partial_yv^y$, $I_1=\partial_zv_1^z-\frac{1}{3}\partial_iv_1^i$, and $I_1[\mathbf{v}_2]$ the same with $\mathbf{v}_2$):

$$\begin{aligned}\frac{1}{2}L_2=\int_{\eta_*}^{\bar\eta_o}\mathrm{d}\eta\Big\{&\underbrace{-\frac{1}{4}h'_{2,zz}-\frac{1}{2}I_1[\mathbf{v}_2]}_{\text{(a)}}\ \underbrace{+\mathbf{v}_\perp\cdot(\frac{1}{3}\mathbf{v}'_\perp-\partial_z\mathbf{v}_\perp-\nabla_\perp v^z)+v^z(\frac{1}{3}\partial_\perp\cdot\mathbf{v}_\perp-\frac{2}{3}v^{z\prime}-\frac{2}{3}\partial_zv^z)}_{\text{(b)}}\\&\underbrace{+\vec{k}_{1\perp}\cdot(\partial_z\mathbf{v}_\perp+\nabla_\perp v^z)}_{\text{(c)}}\ \underbrace{-\Delta\mathbf{x}_{1\perp}\cdot\nabla_\perp I_1}_{\text{(d)}}\ \underbrace{-\Delta x_1^z\,\partial_zI_1}_{\text{(e)}}\Big\}\ \underbrace{-\delta\eta_{o1}I_1|_{\bar o}}_{\text{(f)}}\ \underbrace{+\delta\eta_{e1}I_1|_*}_{\text{(g)}},\end{aligned} \tag{96}$$

piece	where	physical meaning	$\hat{\mathbf{n}}=\hat{\mathbf{z}}$	general $\hat{\mathbf{n}}$
(a) 2nd-order shear	ray	intrinsic second-order shear of the fluid: v_2 and the tensor shear $\frac{1}{2}h_2'$ (ISW-like tensor term)	5	22
(b) quadratic local	ray	products of first-order fields at the same point: non-linearity of $\hat{\sigma}_{\mu\nu}k^\mu k^\nu/(-\hat{u}\cdot k)$	10	111
(c) direction (k_1)	ray	aberration at the observer ($\mathbf{v}_{1\perp}$ in $k_{1,\mathrm{init}}$) and accumulated deflection, changing the projection σ_{kk}	4	21
(d) lensing ($\Delta\mathbf{x}_{1\perp}$)	ray	first-order shear sampled at the transversely displaced point	6	$45^\dagger$
(e) time delay ($\Delta x_{1\parallel}$)	ray	first-order shear sampled at the radially displaced point (Shapiro delay, clock shift, radial observer displacement)	3	
(f) clock ($\delta\eta_{o1}$)	obs	observation at a perturbed time: shift of the upper limit	3	15
(g) emission ($\delta\eta_{e1}$)	emi	emission on $T_{\mathrm{phys}}=T_*$ rather than $\eta=\eta_*$: shift of the lower limit	3	15
total			34	229

Table 2: Pieces of $\frac{1}{2}L_2$ (shear form) and their numbers of terms. †For symbolic $\hat{\mathbf{n}}$, (d) and (e) are not separated and the 45 terms count both. The observer displacement $\delta\mathbf{x}_{o1}$ enters through $\Delta\mathbf{x}_1$ in (d), (e) (and through k_2 only at third order). Full expressions: App. C.2.

with $h_{2,zz}=-h_{2,xx}-h_{2,yy}$, k_1, Δx_1 from (86)–(87), $\delta\eta_{o1}$ from (91), $\delta\eta_{e1}=\eta_*\delta_{T,1}|_*$. For general $\hat{\mathbf{n}}$, (a) reads $-\frac{1}{2}I_1[\mathbf{v}_2]-\frac{1}{4}n^in^jh'_{2,ij}$ and (d)+(e) $=-\Delta\mathbf{x}_1\cdot\nabla I_1$. No equation of motion has been used; $\mathbf{v}'$ may be eliminated with the Euler equation (28).

8.6 Direct (Sachs–Wolfe) form and equivalence

The alternative organisation integrates the photon energy directly. Since $\hat{E}=\hat{p}^0(-\hat{u}_\mu k^\mu)$ and $\mathrm{d}\ln\hat{p}^0/\mathrm{d}\eta=-\hat{\Gamma}^0_{\mu\nu}k^\mu k^\nu$,

$$\Lambda=\ln\frac{T_{\mathrm{obs}}}{T_o}=-\int_{\eta_e}^{\eta_o}\hat{\Gamma}^0_{\mu\nu}k^\mu k^\nu\,\mathrm{d}\eta+\Big[\ln(-\hat{u}_\mu k^\mu)-\ln(1+\delta_T)\Big]_e^o, \tag{97}$$

expanded with the same ray, endpoints and Taylor rules. The boundary function is $\Psi+\hat{\mathbf{n}}\cdot\mathbf{v}-\delta_T+O(\epsilon^2)$, and at first order (97) gives (30) after one integration by parts. At second order, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$,

with $\mathcal{G}_1 \equiv \Phi_1' - \Psi_1' + 2\partial_z\Psi_1 = -(\hat{\Gamma}^0_{kk})_1$ and $B_1 \equiv \Psi_1 - \delta_{T,1} + v_1^z$:

$$\begin{aligned}\frac{1}{2}L_2^{\rm dir} = \int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta \Big\{ &\frac{1}{2}(\Phi_2' - \Psi_2') + \partial_z\Psi_2 - \frac{1}{4}h_{2,zz}' + \frac{1}{2}\partial_z\omega_{2,z} \\ &+ 2\Phi_1\Phi_1' + 2\Phi_1\partial_z\Psi_1 + 2\Psi_1\Psi_1' - 2\Psi_1\partial_z\Psi_1 \\ &- 2\vec{k}_{1\perp}\cdot\nabla_\perp\Psi_1 + \Delta\mathbf{x}_1\cdot\nabla\mathcal{G}_1\Big\} \\ + \Big[&\frac{1}{2}(\Psi_2 - \delta_{T,2} + \omega_{2,z} + v_2^z) - \Phi_1 v^z - \Psi_1^2 - \Psi_1 v^z + \frac{1}{2}\delta_{T,1}^2 - \frac{1}{2}|\mathbf{v}_\perp|^2 \\ &+ \delta\eta_{o1}(\mathcal{G}_1 + B_1') + \delta\mathbf{x}_{o1}\cdot\nabla B_1\Big]_{\bar{o}} \\ + \Big[&-\frac{1}{2}(\Psi_2 - \delta_{T,2} + \omega_{2,z} + v_2^z) + \Phi_1 v^z + \Psi_1^2 + \Psi_1 v^z - \frac{1}{2}\delta_{T,1}^2 - \frac{1}{2}|\mathbf{v}_\perp|^2 \\ &- \delta\eta_{e1}(\mathcal{G}_1 + DB_1) - \Delta\mathbf{x}_1(\eta_*)\cdot\nabla B_1 + \vec{k}_{1\perp}(\eta_*)\cdot\mathbf{v}_\perp\Big]_*, \end{aligned} \tag{98}$$

76 terms (179 for general $\hat{\mathbf{n}}$; App. C.3). (At the observer the displacement $\delta\mathbf{x}_{o1}$ is separate and B_1 is differentiated at fixed point, B_1'; at the emitter the point moves along the ray with η_e, so the total derivative DB_1 appears.) Its pieces have the familiar names: second-order ISW/Rees–Sciama, vector and tensor integrand $\frac{1}{2}(\Phi_2' + \Psi_2' + \ldots)$ after integration by parts, quadratic ISW corrections, transverse deflection, lensing and Shapiro delay of the ISW integrand, second-order Sachs–Wolfe and Doppler at both ends with the clock, displacement and emission-surface shifts.

Equivalence. The two forms are equal, not merely equivalent on average: (i) at the integrand level along an arbitrary perturbed ray, $-\hat{\Gamma}^0_{kk} + \mathrm{d}_\eta[\ln(-\hat{u}\cdot k) - \ln\hat{T}] + \hat{\sigma}_{kk}/(-\hat{u}\cdot k) + \hat{\theta}\hat{g}_{kk}/(3(-\hat{u}\cdot k)) \equiv 0$ modulo $\hat{\nabla}_\mu\hat{T}^{\mu\nu} = 0$ at orders 1, 2, 3 (App. G, X-9); (ii) at the assembled level, $\Lambda^{\rm dir} - \Lambda^{\rm shear}$ vanishes piece by piece (ray, obs, emi) at orders 1, 2 *and* 3 modulo the conservation equations and their derivatives (R-4); (iii) numerically, for the plane wave of Sec. 13.4, the two forms agree to 4×10^{-10} (L_2) and 9×10^{-7} (L_3), the level of the quadrature (PW-4). With $\mathbf{v} = 0$, emission at fixed coordinate time and the observer at the origin, (98) reduces term by term to the second-order Sachs–Wolfe result of Boubekeur et al. [18], the only remainder being $\int[\Delta\mathbf{x}_1\cdot\nabla(\Phi_1' + \Psi_1') + 2\Phi_1\Phi_1' - 2\Psi_1\Psi_1']$, which vanishes under their matter-era assumption $\Phi_1' = \Psi_1' = 0$ (R-7).

8.7 From L_2 to $\Theta^{(2)}$

By (31) and (32),

$$m_2 = L_2 + L_1^2, \qquad \Theta^{(2)}_{\ell m} = m_{2,\ell m} - 2\langle m_1\rangle_{\rm sky}\, m_{1,\ell m} \quad (\ell \geq 1), \tag{99}$$

with L_2 twice the expression (96) (or (98)) and the second-order fields of Sec. 6.2 inserted. In the language of Sec. 10: L_1^2 and the emission-surface piece (g) are products of type-A maps on the sky (type B); the local pieces (a), (b) are line-of-sight integrals of fields at one spacetime point (type C, with the second-order kernels of (59) in (a)); the direction, lensing and time-delay pieces (c)–(e) are nested unequal-time products (type D); the clock piece (f) is a direction-independent random variable times a quadratic polynomial in $\hat{\mathbf{n}}$ built from fields at the observer, and contributes to $\ell \leq 2$ only. In Fourier space this gives $\Theta^{(2)}(\hat{\mathbf{n}}) = \int \frac{\mathrm{d}^3k_1\mathrm{d}^3k_2}{(2\pi)^6}\zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2}K_2(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}})$, with K_2 the sum of the channel kernels written out in Sec. 11.2 and App. D; their harmonic transforms and $C_\ell^{(22)}$ follow in Secs. 11.3–11.5.

9 The third-order map $\Theta^{(3)}$

The third-order map is built with exactly the machinery of Sec. 8: the integrand $\mathcal{I}$ of (79) expanded to $O(\epsilon^3)$ with $k^i = -n^i + \epsilon k_1^i + \frac{\epsilon^2}{2} k_2^i$ (the k_3 and Δx_3 terms do not enter, since $\mathcal{I} = O(\epsilon)$), the position expansion to second order in $\Delta\mathbf{x} = \epsilon\Delta\mathbf{x}_1 + \frac{\epsilon^2}{2}\Delta\mathbf{x}_2$, the endpoint formula (94) with $\delta\eta_o = \epsilon\delta\eta_{o1} + \frac{\epsilon^2}{2}\delta\eta_{o2}$ and $\delta\eta_e = \epsilon\delta\eta_{e1} + \frac{\epsilon^2}{2}\delta\eta_{e2}$, and the functionals (86)–(89), (91)–(93). The new ingredients relative to L_2 are the second-order functionals k_2, $\Delta\mathbf{x}_2$, $\delta\eta_{o2}$, $\delta\eta_{e2}$, the triad and direction map at second order (83), (85), the second-order fields (including ω_2, Φ_2, Ψ_2, which now appear in products), and the third-order fields in the local shear.

9.1 L_3: named groups

The expansion gives $\frac{1}{6}L_3$ as 359 terms in 32 groups for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$. Table 3 organises the groups by the three endpoints and by physical content; *every term of every group is displayed in App. C.4*, so that $\Theta^{(3)}$ is fully explicit: L_3 from App. C.4, L_1, L_2 from App. C, and (101) below. The structure is that of L_2 promoted by one order: every second-order piece (a)–(g) of Table 2 reappears (1) with one first-order field replaced by a second-order field, (2) multiplied by one more first-order factor (a field, a functional k_1, $\Delta\mathbf{x}_1$, or a shift $\delta\eta_{o1}, \delta\eta_{e1}$), or (3) with a first-order functional replaced by its second-order counterpart (k_2, $\Delta\mathbf{x}_2$, $\delta\eta_{o2}$, $\delta\eta_{e2}$). The one genuinely new piece is the third-order shear,

$$\frac{1}{6}L_3 \supset -\frac{1}{6}\int_{\eta_*}^{\bar{\eta}_o} \Big(I_1[\mathbf{v}_3] + \frac{1}{2}n^i n^j h'_{3,ij}\Big)\mathrm{d}\eta \qquad (5 \text{ terms for } \hat{\mathbf{n}} = \hat{\mathbf{z}}), \tag{100}$$

and, as at second order, ω_3, Φ_3, Ψ_3 do not appear in the shear form (they do in the direct form). The second-order functionals carry the second-order physics of the ray: k_2 contains the second-order aberration (85), the transport of the initial direction to the background event, and the second-order deflection $\int[\frac{1}{2}G_2 + \Delta\mathbf{x}_1\cdot\nabla G_1]$ (lens–lens coupling and the deflection by second-order potentials, ω_2, h_2); $\Delta\mathbf{x}_2$ is the second-order displacement (post-Born lensing and time delay); $\delta\eta_{e2}$ the emission surface at second order including the lensed position of $\delta_{T,1}$ (93); $\delta\eta_{o2}$ the second-order clock (92).

9.2 From L_3 to $\Theta^{(3)}$

By (31) and (32),

$$m_3 = L_3 + 3L_1L_2 + L_1^3, \qquad \Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1\rangle_{\rm sky} m_{2,\ell m} - 3\langle m_2\rangle_{\rm sky} m_{1,\ell m} + 6\langle m_1\rangle^2_{\rm sky} m_{1,\ell m} \quad (\ell \geq 1), \tag{101}$$

obtained from the exponential and ratio series (App. G, R-5). Only the correlation of $\Theta^{(3)}$ with $\Theta^{(1)}$ is needed, $C_\ell^{(13)}$ of (105), decomposed as in (E0) of Sec. 10.6.

What survives the self-contraction. In $\langle m_1\, m_3^*\rangle$ one ζ of m_3 is paired with m_1 and the remaining two are contracted with each other. Qualitatively the groups of Table 3 and the products $3L_1L_2$, L_1^3 fall into four classes (their explicit evaluation is the subject of Sec. 11):

(i) *Variance × map.* Products of three first-order objects (cubic local terms; k_1^2, $(\Delta\mathbf{x}_1)^2$, $k_1\Delta\mathbf{x}_1$, $\delta\eta_{o1}^2$, $\delta\eta_{e1}^2$, …; first-order parts of L_1L_2 and L_1^3): contracting two of the three factors leaves a one-point variance or an equal-sky-point correlation times a first-order map, structure (i)

group (fields)	meaning	terms
ray (integrated over $[\eta_*, \bar{\eta}_o]$)		
local (3)	third-order shear (100)	5
local (2+1), local (1+1+1)	non-linearity of $\mathcal{I}$: second×first, cubic	57, 55
$k_{1\perp}$ (1+1), (2); $k_{1\perp}^2$ (1)	direction perturbation acting on quadratic / second-order shear; quadratic in the deflection	22, 6, 8
$k_{2\perp}$ (1), k_2^z (1)	second-order direction (aberration, transport, post-Born deflection) acting on I_1	4, 3
lensing $\Delta\mathbf{x}_{1\perp}$ (1+1), (2); $(\Delta\mathbf{x}_{1\perp})^2$ (1)	quadratic / second-order shear at the lensed point; second-order Taylor term (convergence-type)	50, 10, 9
lensing $\Delta\mathbf{x}_{2\perp}$ (1)	second-order (post-Born) transverse displacement	6
time delay Δx_1^z (1+1), (2); $(\Delta x_1^z)^2$ (1)	as lensing, radial	26, 5, 3
time delay Δx_2^z (1)	second-order radial displacement	3
mixed $\Delta\mathbf{x}_{1\perp}\Delta x_1^z$, $k_{1\perp}\Delta\mathbf{x}_{1\perp}$, $k_{1\perp}\Delta x_1^z$ (1)	cross terms of displacement and direction	6, 8, 4
obs (at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$)		
$\delta\eta_{o2}$ (1)	second-order clock	3
$\delta\eta_{o1}$ (1+1), (2)	clock acting on quadratic / second-order shear at the observer	6, 5
$\delta\eta_{o1}\delta x_{o1}$ (1)	clock × observer displacement	9
$\delta\eta_{o1}^2$ (1)	second-order Taylor term of the upper limit (two groups)	3+6
emi (at $(\eta_*, \bar{\mathbf{x}}(\eta_*))$)		
$\delta\eta_{e2}$ (1)	second-order emission surface	3
$\delta\eta_{e1}$ (1+1), (2)	emission shift acting on quadratic / second-order shear	10, 5
$\delta\eta_{e1}k_{1\perp}$, $\delta\eta_{e1}\Delta\mathbf{x}_{1\perp}$, $\delta\eta_{e1}\Delta x_1^z$ (1)	emission shift × direction / lensing / time delay at η_*	4, 6, 3
$\delta\eta_{e1}^2$ (1)	second-order Taylor term of the lower limit	6
total	32 groups	359

Table 3: The groups of $\frac{1}{6}L_3$ (shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$); all terms are listed in App. C.4. "(1+1)" etc. gives the orders of the explicit fields; functionals and shifts are counted separately. The direct form has 480 terms (69 groups) for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$.

of Sec. 10.6. The $(\Delta\mathbf{x}_{1\perp})^2\partial\partial I_1$ group, for instance, produces the convergence-type term $-\ell(\ell+1)R\,C_\ell$ of the lensing control case (Sec. 10.7), now with the radiation-era deflection correlated with the source.

(ii) *Tadpole (mean field).* A second-order field X_2 times a first-order factor, with the first-order factor paired with m_1: the contraction rule $A_1B_2 \to A_k\langle B_2\rangle$ of (63) gives the mean field $\langle X_2\rangle(\eta)$ (a function of time only) times a first-order map. Vector, tensor and velocity tadpoles vanish by isotropy, so only $\langle\Phi_2\rangle$, $\langle\Psi_2\rangle$, $\langle\delta_{T,2}\rangle$ survive, and the groups that contain them are: the local (2+1) group (Φ_2, Ψ_2 undifferentiated); the second-order clock $\delta\eta_{o2}$ (Ψ_2 on the worldline, (92)); the second-order emission shift $\delta\eta_{e2}$ ($\delta_{T,2}$, (93)); and the radial direction functionals k_2^z, Δx_2^z (through $k^z_{2,\mathrm{init}} \supset -\frac{1}{2}(\Psi_2+\Phi_2)$, (85), and $\partial_\eta\langle\Phi_2+\Psi_2\rangle$ in G_2^z). Hence

$$\Theta^{(3)}_{c,\mathrm{tad}}(\mathbf{k};\hat{\mathbf{n}}) = \sum_{X\in\{\Phi_2,\Psi_2,\delta_{T,2}\}} \mathcal{T}_X[\langle X_2\rangle](\mathbf{k};\hat{\mathbf{n}}), \tag{102}$$

where $\mathcal{T}_X$ is the linear map obtained by replacing $X_2(\eta,\mathbf{x}) \to \langle X_2\rangle(\eta)$ in the listed groups of App. C.4 (and in $3L_1L_2$), keeping the first-order factor as a Fourier mode $\mathbf{k}$. Individual

tadpoles shift under the residual time reparametrisation of Sec. 11.8; only their sum with the clock, emission-surface and normalisation terms is invariant.

(iii) *Cross-contracted.* The same products with the first-order factor contracted with one leg of X_2 (rule $A_1B_2 \to 2A_qK_B(-\mathbf{q},\mathbf{k})$), and the cross-pairings of three first-order factors at different points: a loop convolution, structure (ii) of Sec. 10.6.

(iv) *Third-order fields.* The local shear (100) (and $\Phi_3, \Psi_3, \delta_{T,3}$ in the direct form), which enter only through the contracted fields X_{3c} of (62), including the amplitude C_3 (45): structure (iii).

The normalisation terms $-3\langle m_1\rangle_{\rm sky}m_2 - 3\langle m_2\rangle_{\rm sky}m_1 + 6\langle m_1\rangle^2_{\rm sky}m_1$ of (101) supply the remaining pieces of (E0).

A worked contraction: the emission-shift tadpole. To show what a (13) contribution looks like, take the group "emission shift $\delta\eta_{e2}$" (3 terms, App. C.4): by the endpoint formula (94), $\frac{1}{6}L_3 \supset \frac{1}{2}\delta\eta_{e2}\, I_1|_*$, and by (93) $\delta\eta_{e2} \supset \eta_*\delta_{T,2}|_*$, so $L_3 \supset 3\eta_*\,\delta_{T,2}\, I_1|_*$. For one Fourier mode, $v_{1,i} = ik_iT_V\zeta_{\mathbf{k}}$ gives $I_1 = n^in^j\partial_iv_{1,j} - \frac{1}{3}\partial\cdot v_1 = -\frac{2}{3}k^2T_V(k,\eta)P_2(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})\, e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}\zeta_{\mathbf{k}}$. The contraction $A_1B_2 \to A_k\langle B_2\rangle$ then contributes to the contracted map (120)

$$\Theta_c^{(3)}(\mathbf{k};\hat{\mathbf{n}}) \supset -2\eta_*\,\langle\delta_{T,2}\rangle(\eta_*)\,k^2T_V(k,\eta_*)\,P_2(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})\,e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi_*}, \qquad \langle\delta_{T,2}\rangle(\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\Big[\frac{1}{4}\mathrm{tad}_{\delta_2}(q,\eta) - \frac{3}{16}T_\delta(q\eta/\sqrt{3})^2 \tag{103}$$

using $\delta_{T,2} = \frac{1}{4}\delta_2 - \frac{3}{16}\delta_1^2$ and tad_{δ_2} of App. A, i.e. $h_2(k,\eta) \supset -2\eta_*\langle\delta_{T,2}\rangle k^2T_V\,\delta(\eta-\eta_*)$ in (121), and by (A-L) $\Delta_\ell^{(3c)} \supset -2\eta_*\langle\delta_{T,2}\rangle(\eta_*)k^2T_V(k,\eta_*)\sum_{J=\ell,\ell\pm2}(2J+1)\left(\begin{smallmatrix}\ell & 2 & J\\ 0 & 0 & 0\end{smallmatrix}\right)^2 i^{J-\ell}j_J(k\chi_*)$. Physically this is the first-order emission-time Doppler/shear term with the emission surface displaced by the mean temperature perturbation; it is matched by the $\langle\Psi_2\rangle$ in $\delta\eta_{o2}$ and by $-3\langle m_2\rangle_{\rm sky}m_1$, as the zero-mode argument requires. The cross-contracted part of the same group, $2A_qK_{\delta_{T,2}}(-\mathbf{q},\mathbf{k})$, is a structure-(ii) loop term. The *obs* groups are polynomials in $\hat{\mathbf{n}}$ of bounded degree (all fields at the observer event, direction-independent shifts), so their direct contribution to $C_\ell^{(13)}$ is confined to a few low multipoles; through the products $3L_1L_2$, L_1^3 and the normalisation the observer quantities nevertheless affect all ℓ.

10 Loop structure and harmonic machinery

This section turns direction-space kernels into $C_\ell^{(rs)}$. It acts on the maps of Secs. 7–9 through the normalisation algebra of Sec. 4.7; all rules are in the single convention (9)–(10) (phase $e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}$, factor i^ℓ).

10.1 Loop structure

Write each order as a direction-space kernel acting on the Gaussian field,

$$\Theta^{(r)}(\hat{\mathbf{n}}) = \int\prod_{i=1}^{r}\frac{\mathrm{d}^3k_i}{(2\pi)^3}\,\zeta_{\mathbf{k}_1}\cdots\zeta_{\mathbf{k}_r}\,K_r(\mathbf{k}_1,\ldots,\mathbf{k}_r;\hat{\mathbf{n}}), \qquad K_{r,\ell m}(\mathbf{k}_1,\ldots) \equiv \int\mathrm{d}\Omega_{\hat{\mathbf{n}}}Y^*_{\ell m}(\hat{\mathbf{n}})K_r, \tag{104}$$

with K_r symmetric in its momenta (symmetrise if necessary). Wick's theorem for $\ell \geq 1$ then gives

$$C_\ell^{(11)} = \frac{1}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k)\,|K_{1,\ell m}(\mathbf{k})|^2,$$
$$C_\ell^{(22)} = \frac{2}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k_1\,\mathrm{d}^3k_2}{(2\pi)^6} P(k_1)P(k_2)\,|K_{2,\ell m}(\mathbf{k}_1,\mathbf{k}_2)|^2,$$
$$C_\ell^{(13)} = \frac{3}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k)\,K_{1,\ell m}(\mathbf{k})\Big[\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\,K_{3,\ell m}(\mathbf{k},\mathbf{q},-\mathbf{q})\Big]^*. \tag{105}$$

The factor 2 in $C^{(22)}$ counts the two connected pairings of a symmetric K_2; the third pairing $\langle\zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2}\rangle$ inside $\Theta^{(2)}$ has $\mathbf{k}_1+\mathbf{k}_2=0$ and feeds only $\ell=0$. The factor 3 in $C^{(13)}$ counts the three choices of which ζ inside $\Theta^{(3)}$ pairs with $\Theta^{(1)}$; the remaining pair is contracted at $\mathbf{q}_j=-\mathbf{q}_k$, so the "tadpole" configuration $K_3(\mathbf{k},\mathbf{q},-\mathbf{q})$ *is* the entire (13) term and must be kept (this is where the mean-field terms of Sec. 6.3 enter). Both prefactors have been re-derived independently and confirmed on the lensing control case below (App. G, H-6). Equation (105) uses $\langle a^{(1)}a^{(3)*}\rangle$; only $\mathrm{Re}\,C_\ell^{(13)}$ enters (12). With the contracted fields of (62), $3\int P(q)K_3(\mathbf{k},\mathbf{q},-\mathbf{q})$ is the q-contracted map $\Theta_c^{(3)}(\mathbf{k};\hat{\mathbf{n}})$ built from X_{3c} and the contraction rules (63).

10.2 Direction convention

A field on the background ray (9) (observer at $\bar{\mathbf{x}}_o=0$) reads $f(\eta,\hat{\mathbf{n}}\chi)=\int\frac{\mathrm{d}^3k}{(2\pi)^3}f_{\mathbf{k}}\,e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}$, and all rules below follow from the Rayleigh expansion (10). Because $Y_{\ell m}(-\hat{\mathbf{n}})=(-1)^\ell Y_{\ell m}(\hat{\mathbf{n}})$, expressing the same maps through the propagation direction $-\hat{\mathbf{n}}$ (phase $e^{-i\mathbf{k}\cdot(-\hat{\mathbf{n}})\chi}$, factors $(-i)^\ell$) multiplies every $a_{\ell m}$ by $(-1)^\ell$ at every order; the transfer functions Δ_ℓ (the factor i^ℓ versus $(-i)^\ell$ absorbs the sign) and all $C_\ell^{(rs)}$ are unchanged. One convention must be used for $\Theta^{(1)},\Theta^{(2)},\Theta^{(3)}$ together; we use $\hat{\mathbf{n}}$ throughout. Gaunt integrals are

$$\mathcal{G}^{\ell\,\ell_1\ell_2}_{m\,m_1m_2} \equiv \int \mathrm{d}\Omega\, Y^*_{\ell m}Y_{\ell_1m_1}Y_{\ell_2m_2} = (-1)^m\sqrt{\frac{(2\ell+1)(2\ell_1+1)(2\ell_2+1)}{4\pi}}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ -m & m_1 & m_2\end{pmatrix}. \tag{106}$$

10.3 Type A: linear line-of-sight maps

A map of the form

$$\Theta_A(\hat{\mathbf{n}}) = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta\sum_{s\geq0} g_s(k,\eta)\,(i\,\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^s\,e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi(\eta)} \tag{107}$$

(endpoint terms are $\delta(\eta-\eta_1)$ pieces of g_s; the monomials are written as $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^s$ because $i\hat{\mathbf{k}}=\nabla/k$, so a real map has real g_s) has

$$\text{(A-}s\text{)}\qquad a_{\ell m} = 4\pi\,i^\ell\int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}Y^*_{\ell m}(\hat{\mathbf{k}})\,\Delta_\ell(k),\qquad \Delta_\ell(k)=\int\mathrm{d}\eta\sum_s g_s(k,\eta)\,j_\ell^{(s)}(k\chi(\eta)), \tag{108}$$

i.e. $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^s e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}\to j_\ell^{(s)}(x)$, $x=k\chi$, since $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}}=k^{-1}\partial_\chi e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}}$. Helicity pieces (vectors ω_i,v_i^T contracted with $\hat{\mathbf{n}}$, tensors $h_{ij}n^in^j$): in the frame $\hat{\mathbf{k}}=\hat{\mathbf{z}}$ with $e^{(\pm1)}=(\hat{\mathbf{x}}\pm i\hat{\mathbf{y}})/\sqrt{2}$, $e_{ij}^{(\pm2)}=$

$e_i^{(\pm1)}e_j^{(\pm1)}$, $Q_{\pm1}=\hat{\mathbf{n}}\cdot e^{(\pm1)}$, $Q_{\pm2}=e_{ij}^{(\pm2)}n^in^j$,

$$\text{(A-}\lambda\text{)}\qquad Q_\lambda e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}=\sum_\ell i^\ell\sqrt{4\pi(2\ell+1)}\,\hat\Delta_\ell^{(\lambda)}(x)\,Y_{\ell\lambda}(\hat{\mathbf{n}}),$$
$$\hat\Delta_\ell^{(\pm1)}=\pm i\sqrt{\frac{\ell(\ell+1)}{2}}\frac{j_\ell(x)}{x},\quad \hat\Delta_\ell^{(\pm2)}=-\frac{1}{2}\sqrt{\frac{(\ell+2)!}{(\ell-2)!}}\frac{j_\ell(x)}{x^2}, \tag{109}$$

and in a general frame $a_{\ell m}^{(\lambda)}=4\pi\,i^\ell\int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}^{(\lambda)}(-1)^\lambda[{}_{-\lambda}Y_{\ell m}(\hat{\mathbf{k}})]^*\Delta_\ell^{(\lambda)}(k)$, $\Delta_\ell^{(\lambda)}=\int\mathrm{d}\eta\,g^{(\lambda)}\hat\Delta_\ell^{(\lambda)}(k\chi)$. (Relative to the propagation-direction convention only $\hat\Delta^{(\pm1)}$ changes sign, because $Q_{\pm1}$ is odd in $\hat{\mathbf{n}}$.) Scalar, vector and tensor pieces of the same $\mathbf{k}$ do not cross-correlate. For two type-A maps

$$C_\ell^{AA'}=\frac{2}{\pi}\int k^2\mathrm{d}k\,P(k)\,\Delta_\ell^A(k)\Delta_\ell^{A'}(k)^*,\qquad \langle A\rangle_{\rm sky}=\int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\Delta_0^A(k),$$
$$\langle\langle A\rangle_{\rm sky}\langle A'\rangle_{\rm sky}\rangle=\frac{1}{2\pi^2}\int k^2\mathrm{d}k\,P\,\Delta_0^A\Delta_0^{A'}=\frac{C_0^{AA'}}{4\pi}, \tag{110}$$

and $\langle\langle A\rangle_{\rm sky}\,a_{\ell m}^{A'}\rangle=0$ for $\ell\geq1$ by isotropy. The first-order map $m_1=L_1$ of (30) is of type A (with $s\leq1$ and endpoint terms), so $C_\ell^{(11)}$ is (110) with $A=A'=m_1$ (closed form: Sec. 7, Proposition 7.1).

Same-point correlators in closed form. The rules below need the zero-lag correlator of two type-A maps, $\xi^{XY}(0)\equiv\langle X(\hat{\mathbf{n}})Y(\hat{\mathbf{n}})\rangle=\sum_\ell\frac{2\ell+1}{4\pi}C_\ell^{XY}$. The ℓ sum need not be performed: writing $x(k,\nu)\equiv\int\mathrm{d}\eta\sum_s g_s^X(k,\eta)(i\nu)^s e^{ik\nu\chi(\eta)}$ for the plane-wave profile of X ($\nu=\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}$), the Rayleigh expansion gives $x(k,\nu)=\sum_\ell(2\ell+1)i^\ell\Delta_\ell^X(k)P_\ell(\nu)$, and Legendre orthogonality (the addition theorem, $\sum_\ell(2\ell+1)j_\ell(z)^2=1$, in its general form) yields

$$\xi^{XY}(0)=\int\frac{\mathrm{d}^3k}{(2\pi)^3}P(k)\,\frac{1}{2}\int_{-1}^1\mathrm{d}\nu\;x(k,\nu)\,y(k,\nu)^*=\frac{1}{2\pi^2}\int k^2\mathrm{d}k\,P(k)\sum_\ell(2\ell+1)\Delta_\ell^X\Delta_\ell^{Y*}, \tag{111}$$

a single ν integral of explicitly known profiles (for endpoint terms elementary, for line-of-sight terms a double time integral of $e^{ik\nu(\chi_1-\chi_2)}$, whose ν integral is $2j_0(k(\chi_1-\chi_2))$ times polynomials). The angular-gradient version $\xi^{\nabla X\cdot\nabla Y}(0)=\sum_\ell\frac{\ell(\ell+1)(2\ell+1)}{4\pi}C_\ell^{XY}$ is the same with $x\,y^*$ replaced by $(1-\nu^2)\,\partial_\nu x\,\partial_\nu y^*$ (since $\nabla_{\hat{\mathbf{n}}}f(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})=f'(\nu)(\hat{\mathbf{k}}-\nu\hat{\mathbf{n}})$).

10.4 Type B: products of maps at the same point

For $B=AA'$ and $B'=CD$ with A,A',C,D type-A maps, $b_{\ell m}=\sum\mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2}w(\ell,\ell_1,\ell_2)a_{\ell_1m_1}a'_{\ell_2m_2}$, where $w=1$ for a plain product and

$$w_\nabla(\ell,\ell_1,\ell_2)=\frac{1}{2}[\ell_1(\ell_1+1)+\ell_2(\ell_2+1)-\ell(\ell+1)] \tag{112}$$

for $\nabla_{\hat{\mathbf{n}}}A\cdot\nabla_{\hat{\mathbf{n}}}A'$ (w_∇ is symmetric in $\ell_1\leftrightarrow\ell_2$). Wick's theorem with $\langle a_{\ell m}^X a_{\ell'm'}^Y\rangle=(-1)^mC_\ell^{XY}\delta_{\ell\ell'}\delta_{m,-m'}$ gives, for $\ell\geq1$,

$$\text{(B1)}\qquad C_\ell^{BB'}=\sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell&\ell_1&\ell_2\\0&0&0\end{pmatrix}^2 w\,w'\Big[C_{\ell_1}^{AC}C_{\ell_2}^{A'D}+C_{\ell_1}^{AD}C_{\ell_2}^{A'C}\Big], \tag{113}$$

with $\ell+\ell_1+\ell_2$ even; *both* Wick pairings appear and reduce to a single term with a factor 2 only when $A=A'$ (the third pairing $\langle AA'\rangle\langle CD\rangle$ feeds $\ell=0$ only). Monopole×map terms, with $\langle X\rangle_{\rm sky}\cdot 1$ treated as an $\ell=0$ map, give for $\ell\geq 1$

$$\begin{aligned}
&\text{(i)} \quad \frac{1}{2\ell+1}\sum_m \langle|\langle X\rangle_{\rm sky}\, y_{\ell m}|^2\rangle = \langle\langle X\rangle_{\rm sky}^2\rangle\, C_\ell^{YY},\\
&\text{(ii)} \quad \frac{1}{2\ell+1}\sum_m \langle (AA')_{\ell m}(\langle X\rangle_{\rm sky}\, y_{\ell m})^*\rangle = \langle\langle A\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C_\ell^{A'Y} + \langle\langle A'\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C_\ell^{AY},\\
&\text{(iii)} \quad \frac{1}{2\ell+1}\sum_m \langle(\langle X\rangle_{\rm sky} y_{\ell m})(\langle Z\rangle_{\rm sky} w_{\ell m})^*\rangle = \langle\langle X\rangle_{\rm sky}\langle Z\rangle_{\rm sky}\rangle C_\ell^{YW}.
\end{aligned} \tag{114}$$

Hence, for the type-B part of m_2 and with (32),

$$\frac{1}{4}C_\ell^{(22)} = \frac{1}{4}\Big[C_\ell^{m_2m_2} - 4\ \text{Re (ii)}|_{AA'\to m_2,\ X=Y=m_1} + 4\langle\langle m_1\rangle_{\rm sky}^2\rangle C_\ell^{(11)}\Big]. \tag{115}$$

Type D (nested, unequal-time products such as $\int \mathrm{d}\eta_1 \int_{\eta_1}\mathrm{d}\eta_2\, W\,(\delta_{ij}-n_in_j)\partial_i X(\eta_1)\partial_j Y(\eta_2)$, the lensing-type terms) reduce to (B1) applied to "shell maps" $\tilde X(\hat{\mathbf{n}}) = X(\eta_1,\hat{\mathbf{n}}\chi_1)$ with unequal-time spectra $C_\ell^{xy}(\eta,\eta') = \frac{2}{\pi}\int k^2\mathrm{d}k\, P\, x(k,\eta)y(k,\eta')^* j_\ell(k\chi)j_\ell(k\chi')$, using the exact identity $(\delta - \hat{\mathbf{n}}\hat{\mathbf{n}})_{ij}\partial_i X\partial_j Y = (\chi_1\chi_2)^{-1}\nabla_{\hat{\mathbf{n}}}\tilde X\cdot\nabla_{\hat{\mathbf{n}}}\tilde Y$ (weight w_∇) and $\hat{\mathbf{n}}\cdot\partial X = \partial_\chi\tilde X$ for radial contractions ($j_\ell\to kj_\ell'$, weight 1).

10.5 Type C: convolution kernels

A genuinely second-order map

$$\Theta_C(\hat{\mathbf{n}}) = \int\frac{\mathrm{d}^3k_1\mathrm{d}^3k_2}{(2\pi)^6}\zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2}\int\mathrm{d}\eta\, K(k_1,k_2,K,\eta)\,\mathcal{Q}(\hat{\mathbf{n}};\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\hat{\mathbf{K}})\,e^{i\mathbf{K}\cdot\hat{\mathbf{n}}\chi}, \qquad \mathbf{K}=\mathbf{k}_1+\mathbf{k}_2, \tag{116}$$

with $\mathcal{Q}=\sum q_{abc}(k_1,k_2,\mu)(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}_1)^a(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}_2)^b(\hat{\mathbf{n}}\cdot\hat{\mathbf{K}})^c$ a polynomial of degree d (transverse and TT projectors contracted with $\hat{\mathbf{n}}$ are of this form), has

$$K_{2,\ell m}(\mathbf{k}_1,\mathbf{k}_2) = \int\mathrm{d}\eta\, K\sum_{L=|\ell-d|}^{\ell+d}\sum_M 4\pi\, i^L\, j_L(K\chi)\, Y_{LM}^*(\hat{\mathbf{K}})\,\langle\ell m|\mathcal{Q}|LM\rangle, \qquad \langle\ell m|\mathcal{Q}|LM\rangle = \int\mathrm{d}\Omega\, Y_{\ell m}^*\,\mathcal{Q}\,Y_{LM}, \tag{117}$$

a finite $3j$ sum generated by repeated use of $(\hat{\mathbf{n}}\cdot\hat{\mathbf{v}})Y_{\ell m} = \frac{4\pi}{3}\sum_{m''}Y_{1m''}^*(\hat{\mathbf{v}})\sum_{\ell'=\ell\pm1,m'}\mathcal{G}^{\ell'\ell 1}_{m'mm''}Y_{\ell'm'}$. In the frame $\hat{\mathbf{K}}=\hat{\mathbf{z}}$, $\mathbf{k}_{1,2}$ in the x–z plane ($\cos\alpha_1 = (k_1+k_2\mu)/K$, $\cos\alpha_2 = (k_2+k_1\mu)/K$, $\mu=\hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$), $K_{2,\ell m}\to\tilde K_{\ell m}(k_1,k_2,\mu) = \int\mathrm{d}\eta\, K\sum_L i^L\sqrt{4\pi(2L+1)}j_L(K\chi)\langle\ell m|\mathcal{Q}|L0\rangle$ with $|m|\leq d$, and isotropy reduces the angular integrals of (105) to

$$\begin{aligned}
&\text{(C1)} \quad C_\ell^{(22)} = \frac{1}{4\pi^4(2\ell+1)}\int k_1^2\mathrm{d}k_1\, k_2^2\mathrm{d}k_2\, P(k_1)P(k_2)\int_{-1}^1\mathrm{d}\mu\sum_m|\tilde K_{\ell m}(k_1,k_2,\mu)|^2,\\
&\text{(C2)} \quad \frac{1}{2\ell+1}\sum_m\langle C_{\ell m}B_{\ell m}^*\rangle = \frac{1}{4\pi^4(2\ell+1)}\int k_1^2\mathrm{d}k_1k_2^2\mathrm{d}k_2\, PP\int\mathrm{d}\mu\sum_m\tilde K_{\ell m}^C\tilde K_{\ell m}^{B*},
\end{aligned} \tag{118}$$

where for a type-B map $\tilde K_{\ell m}^B = \frac{1}{2}\sum\mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2}[\tilde A_{\ell_1m_1}(k_1)\tilde A'_{\ell_2m_2}(k_2) + (A\leftrightarrow A')]$, $\tilde A_{\ell_1m_1}(k) = 4\pi\, i^{\ell_1}Y_{\ell_1m_1}^*(\hat{\mathbf{k}})\Delta_{\ell_1}^A(k)$ at the frame angles. A non-symmetric kernel must be symmetrised as $\frac{1}{2}(K+K^{\rm T})$ *before* the m-sum. Two-time kernels use the double Rayleigh expansion instead of (117); (C1) is unchanged.

10.6 The (13) term

From (32),

$$\text{(E0)}\qquad C_\ell^{(13)} = T_{3,\ell} - 3T_{\langle 1\rangle_{\rm sky}2,\ell} - 3T_{\langle 2\rangle_{\rm sky}1,\ell} + 6\langle\langle m_1\rangle_{\rm sky}^2\rangle C_\ell^{(11)}, \qquad T_{3,\ell} \equiv \frac{1}{2\ell+1}\sum_m \langle m_{1,\ell m} m^*_{3,\ell m}\rangle, \tag{119}$$

and $T_{\langle 1\rangle_{\rm sky}2}$, $T_{\langle 2\rangle_{\rm sky}1}$ the corresponding cross terms with $\langle m_1\rangle_{\rm sky} m_2$, $\langle m_2\rangle_{\rm sky} m_1$.

The contracted map and $\Delta_\ell^{(3c)}$ (defined once). The q-contracted third-order map is

$$\Theta_c^{(3)}(\mathbf{k};\hat{\mathbf{n}}) \equiv 3\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, K_3(\mathbf{k},\mathbf{q},-\mathbf{q};\hat{\mathbf{n}}), \qquad C_\ell^{(13)} = \frac{1}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k)\, K_{1,\ell m}(\mathbf{k})\, \Theta^{(3)*}_{c,\ell m}(\mathbf{k}), \tag{120}$$

i.e. the pairing factor 3 of (105) is included in $\Theta_c^{(3)}$ once and for all. After the $\hat{\mathbf{q}}$ integration it depends on $\hat{\mathbf{n}}$ only through $\nu \equiv \hat{\mathbf{n}}\cdot\hat{\mathbf{k}}$; its part with all fields at one point of the ray (structure (iii) below) is written as

$$\Theta_c^{(3)}(\mathbf{k};\hat{\mathbf{n}}) \supset \int \mathrm{d}\eta \sum_L h_L(k,\eta)\, P_L(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})\, e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}, \qquad h_L(k,\eta) = \int \frac{q^2\mathrm{d}q}{2\pi^2} P(q)\, h_L(k,q,\eta), \tag{121}$$

with $h_L(k,q,\eta)$ the Legendre coefficients in ν of the $\hat{\mathbf{q}}$-averaged kernel, $h_L = \frac{2L+1}{2}\int_{-1}^1 \mathrm{d}\nu\, P_L(\nu)\, F(\nu)$ if the $\hat{\mathbf{q}}$-average of the integrand is $F(\nu)e^{ik\nu\chi}$. Every such Legendre series is handled by

$$\text{(A-}L\text{)}\qquad h(k,\eta) P_L(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}) e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} \;\Rightarrow\; \Delta_\ell = \int \mathrm{d}\eta\, h \sum_{J=|\ell-L|}^{\ell+L} (2J+1) \begin{pmatrix} \ell & L & J \\ 0 & 0 & 0 \end{pmatrix}^2 i^{J-\ell} j_J(k\chi), \tag{122}$$

and the transfer function of the contracted map is

$$\Delta_\ell^{(3c)}(k) \equiv \int \mathrm{d}\eta \sum_L h_L(k,\eta) \sum_J (2J+1) \begin{pmatrix} \ell & L & J \\ 0 & 0 & 0 \end{pmatrix}^2 i^{J-\ell} j_J(k\chi), \qquad \Theta^{(3)}_{c,\ell m} = 4\pi\, i^\ell Y^*_{\ell m}(\hat{\mathbf{k}})\, \Delta_\ell^{(3c)}(k). \tag{123}$$

The $\mathbf{q}$-angular reduction of products of fields at two points of the ray is

$$\text{(E1)}\qquad \int \frac{\mathrm{d}\Omega_q}{4\pi} g(\hat{\mathbf{q}}\cdot\hat{\mathbf{k}})(\hat{\mathbf{n}}\cdot\hat{\mathbf{q}})^b e^{i\mathbf{q}\cdot\hat{\mathbf{n}}\Delta\chi} = \sum_{L'} i^{L'}(2L'+1) j_{L'}(q\Delta\chi) \sum_{L''} \beta_{L'bL''}\, g_{L''}\, P_{L''}(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}), \tag{124}$$

with $g_{L''} = \frac{1}{2}\int \mathrm{d}\mu\, P_{L''}(\mu) g(\mu)$ and $x^b P_{L'} = \sum_{L''} \beta_{L'bL''} P_{L''}$; for $\Delta\chi = 0$ only $L' = 0$ survives. The three structures in m_3 are then:

(iii) genuine third-order kernels (local in the ray): $T_{3,\ell}^{(\rm iii)} = \frac{2}{\pi}\int k^2\mathrm{d}k\, P\, \Delta_\ell(k)\Delta_\ell^{(3c)}(k)^*$;

(i) products of three type-A maps ABC: $T_{3,\ell}^{(\rm i)} = C_\ell^{m_1 A}\xi^{BC}(0) + C_\ell^{m_1 B}\xi^{AC}(0) + C_\ell^{m_1 C}\xi^{AB}(0)$, with $\xi^{XY}(0)$ in the closed form (111) (and $\xi^{\nabla B\cdot\nabla C}(0)$ likewise; $\nabla B\nabla C : \nabla\nabla A \to -\ell(\ell+1)R_{BC}C_\ell^{m_1 A}$, $R_{BC} = \frac{1}{2}\xi^{\nabla B\cdot\nabla C}(0)$);

(ii) type-A $\times$ type-C products $A\cdot C$: $T_{3,\ell}^{(\rm ii)} = C_\ell^{m_1 A}\langle\langle C\rangle\rangle + 2\cdot\frac{2}{\pi}\int k^2\mathrm{d}k\, P\, \Delta_\ell^{m_1}\Delta_\ell^{AC*}$, $\langle\langle C\rangle\rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k) K^C(\mathbf{k},-\mathbf{k};\hat{\mathbf{n}})$ and Δ_ℓ^{AC} the transfer function of $\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q) K^A(\mathbf{q};\hat{\mathbf{n}}) K^C(\mathbf{k},-\mathbf{q};\hat{\mathbf{n}})$ obtained from (E1) with $\Delta\chi = \chi_A - \chi_C$ and (A-L).

The normalisation pieces, with $\bar{K}_2(\mathbf{q}_1,\mathbf{q}_2) \equiv K_{2,00}(\mathbf{q}_1,\mathbf{q}_2)/\sqrt{4\pi}$ the sky-average kernel of m_2, are

$$\text{(E2)}\quad T_{\langle 2\rangle_{\rm sky}1,\ell} = \langle\!\langle m_2\rangle\!\rangle C_\ell^{(11)} + \frac{1}{\pi^3}\int k^2\mathrm{d}k\, k'^2\mathrm{d}k'\, PP\,\Delta_\ell(k)\Delta_\ell(k')^* \int \mathrm{d}\mu\, P_\ell(\mu)\bar{K}_2(k\hat{\mathbf{k}}, -k'\hat{\mathbf{k}}')^*, \quad \langle\!\langle m_2\rangle\!\rangle = \frac{1}{2\pi^2}\int k^2$$

$$\text{(E3)}\quad T_{\langle 1\rangle_{\rm sky}2,\ell} = \frac{2}{2\pi^2}\int q^2\mathrm{d}q\, P(q)\Delta_0(q)\,\frac{2}{\pi}\int k^2\mathrm{d}k\, P(k)\Delta_\ell(k)\bar{\Delta}_\ell^{m_2}(k;q)^*, \qquad \bar{\Delta}_\ell(k;q) = \frac{\frac{1}{2}\int \mathrm{d}\mu\, K_{2,\ell 0}(k\hat{\mathbf{z}},\mathbf{q})}{i^\ell\sqrt{4\pi(2\ell+1)}}, \tag{125}$$

and the same objects give the cross term of (115). Formulas (E2), (E3) and $T^{(\mathrm{ii})}$ are derived by Wick counting in the same way as the verified ones; (E3) collapses to (114)(ii) for a type-B m_2 (analytic check). They have not been tested numerically (status table, Sec. 3). The explicit h_L, K^C, $\bar{K}_2$ of our map are assembled in Sec. 11 from the term lists of Sec. 9 and App. C.4.

10.7 Control case: lensed C_ℓ

For the remapping $\tilde{\Theta} = \Theta + \nabla\phi\cdot\nabla\Theta + \frac{1}{2}\nabla^i\phi\nabla^j\phi\nabla_i\nabla_j\Theta$ with $\Theta = \epsilon\Theta_1$, $\phi = \epsilon\phi_1$ one has $\Theta^{(2)} = 2\nabla\phi\cdot\nabla\Theta$, $\Theta^{(3)} = 3\nabla\phi\nabla\phi{:}\nabla\nabla\Theta$, and the bookkeeping (12) with (B1) (gradient weight (112)) and the (i)-type pairing gives exactly

$$\frac{1}{4}C_\ell^{(22)} = \sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}^2 w_\nabla(\ell,\ell_1,\ell_2)^2\Big[C_{\ell_1}^{\phi\phi}C_{\ell_2}^{\Theta\Theta} + C_{\ell_1}^{\phi\Theta}C_{\ell_2}^{\Theta\phi}\Big], \qquad \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = -\ell(\ell{+}1)R\,C_\ell^{\Theta\epsilon} \tag{126}$$

$R = \frac{1}{2}\langle|\nabla\phi|^2\rangle$. Because w_∇ is symmetric in $\ell_1 \leftrightarrow \ell_2$, the correlated cross term $C^{\phi\Theta}C^{\Theta\phi}$ carries the same weight as the uncorrelated one. For uncorrelated ϕ,Θ these are Hu's (62)–(63) [16, 17], with $F_{\ell\ell_1\ell_2} = w_\nabla\sqrt{(2\ell+1)(2\ell_1+1)(2\ell_2+1)/4\pi}\,\left(\begin{smallmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{smallmatrix}\right)$; the cross term is the extension to correlated ϕ,Θ, relevant here because the deflection correlates with SW/ISW. This was verified algebraically and by Monte Carlo on the sphere (App. G, H-5), validating the prefactors of (105), the two-pairing structure of (B1) and the monopole×map rules.

10.8 Monochromatic reduction: definitions

The monochromatic primordial spectrum is defined with the variance $\langle\zeta_1^2\rangle = \sigma^2$ held fixed as k_0 varies:

$$P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\,\delta(k{-}k_0) \quad\Longleftrightarrow\quad \langle\zeta_1^2(\mathbf{x})\rangle = \sigma^2, \qquad \int k^2\mathrm{d}k\, P\,F(k) \to 2\pi^2\sigma^2F(k_0), \quad \int\frac{\mathrm{d}^3k}{(2\pi)^3}P\,F(\mathbf{k}) \to \sigma^2\langle F(k_0\hat{\mathbf{k}})\rangle \tag{127}$$

It is applied only after the full observable has been assembled for general $P(k)$. The expansion parameter is then $\epsilon^2\sigma^2$. Under (127) the general formulas collapse to (all transfer functions at k_0):

$$C_\ell^{AA'} = 4\pi\sigma^2\Delta_\ell^A\Delta_\ell^{A'*}, \qquad \langle\langle A\rangle_{\rm sky}\langle A'\rangle_{\rm sky}\rangle = \sigma^2\Delta_0^A\Delta_0^{A'}, \qquad \xi^{XY}(0) = \sigma^2\,\frac{1}{2}\int_{-1}^{1}\mathrm{d}\nu\, x(k_0,\nu)y(k_0,\nu)^*;$$

$$\text{(B1)}:\ C_\ell^{BB'} = (4\pi\sigma^2)^2\sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}^2 ww'[\Delta_{\ell_1}^A\Delta_{\ell_1}^C\Delta_{\ell_2}^{A'}\Delta_{\ell_2}^D + \Delta_{\ell_1}^A\Delta_{\ell_1}^D\Delta_{\ell_2}^{A'}\Delta_{\ell_2}^C];$$

$$\text{(C1)}:\ C_\ell^{(22)} = \frac{\sigma^4}{2\ell+1}\int_{-1}^{1}\mathrm{d}\mu\sum_{|m|\le d}|\tilde K_{\ell m}(k_0,k_0,\mu)|^2, \quad K = k_0\sqrt{2(1+\mu)},\ \cos\alpha_1 = \cos\alpha_2 = \sqrt{(1+\mu)/2};$$

$$\text{(13)(iii)}:\ T_{3,\ell}^{\rm (iii)} = 4\pi\sigma^2\,\Delta_\ell(k_0)\,\Delta_\ell^{(3c)}(k_0)^*, \qquad \Delta_\ell^{(3c)}(k_0) = \sigma^2\int\mathrm{d}\eta\sum_L h_L(k_0,k_0,\eta)\sum_J(2J+1)\begin{pmatrix}\ell & L & J\\ 0 & 0 & 0\end{pmatrix}^2 i^{J-\ell}j_J$$

$$\langle\langle m_1\rangle_{\rm sky}^2\rangle = \sigma^2\Delta_0^2, \quad \langle\!\langle m_2\rangle\!\rangle = \sigma^2\bar K_2(k_0\hat{\mathbf{k}},-k_0\hat{\mathbf{k}}), \quad \text{(E2) cross} = 4\pi\sigma^4|\Delta_\ell|^2\int\mathrm{d}\mu\,P_\ell(\mu)\bar K_2(k_0\hat{\mathbf{k}},-k_0\hat{\mathbf{k}}')^*, \quad \text{(E3)} = 8\pi \tag{128}$$

where in (13)(iii) the pairing factor 3 is inside $\Delta_\ell^{(3c)}$ by (120) and $|\mathbf{k}\pm\mathbf{q}| = k_0\sqrt{2(1\pm\mu)}$.

Convergence of the ℓ_1,ℓ_2 sums (analytic bound). For the monochromatic spectrum every transfer function is a finite combination of $j_\ell^{(s)}(k_0\chi)$ with $0\le\chi\le\chi_{\max} = \chi_*$ (endpoint and line-of-sight terms; the observer terms only touch $\ell\le 1$). From $j_\ell(x) = \frac{x^\ell}{2^{\ell+1}\ell!}\int_{-1}^{1}e^{ixt}(1-t^2)^\ell\mathrm{d}t$ one has, for real $x\ge 0$,

$$|j_\ell(x)| \le \frac{x^\ell}{(2\ell+1)!!} \le \Big(\frac{e\,x}{2\ell+1}\Big)^\ell, \qquad |j_\ell'(x)| \le \Big(\frac{\ell}{x}+1\Big)\frac{x^\ell}{(2\ell+1)!!}, \tag{129}$$

using $\ln(2\ell+1)!! \ge \int_0^\ell\ln(2t+1)\mathrm{d}t \ge \ell\ln[(2\ell+1)/e]$. Hence, with $X\equiv k_0\chi_*$ and $\mathcal{N}_A$ the total absolute weight of the map ($\sum_s\int|g_s|\mathrm{d}\eta$ with the factor $(\ell/X+1)$ for $s=1$), $|\Delta_\ell^A(k_0)| \le \mathcal{N}_A(\ell/X+1)(eX/(2\ell+1))^\ell$, which for $\ell\ge\ell_c\equiv\lceil eX\rceil$ is $\le\mathcal{N}_A(\ell/X+1)\,2^{-\ell}$: the decay is super-exponential. In (B1) the sum over ℓ_2 at fixed ℓ,ℓ_1 is bounded with $\sum_{\ell_2}(2\ell_2+1)\left(\begin{smallmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{smallmatrix}\right)^2 = 1$, and $|w| \le (\ell_1+\ell+1)^2$ for $|\ell_1-\ell_2|\le\ell$, while $|\Delta_\ell|\le\mathcal{N}$ for all ℓ ($|j_\ell|, |j_\ell'|\le 1$). Denoting by $C_\ell^{BB'}[L]$ the sum truncated at $\ell_1,\ell_2\le L$, a geometric-series estimate of the omitted terms gives, for $L\ge\max(\ell_c,10)+\ell$,

$$|C_\ell^{BB'} - C_\ell^{BB'}[L]| \le 8\pi\sigma^4\,\mathcal{N}_A\mathcal{N}_{A'}\mathcal{N}_C\mathcal{N}_D\,(L+\ell+1)^6\,(L/X+1)^2\,4^{\ell-L}, \tag{130}$$

so the sums are finite to any prescribed accuracy with an explicit, a-priori cut-off; no numerical convergence criterion is used. The same bound applies to the type-D shell maps, and the type-C sums (117) are finite ($|L-\ell|\le d$).

Delicate points (treated in Secs. 11 and 12). (1) $\mu\to-1$ in (C1), where $K\to 0$, $\mathrm{d}\mu = K\mathrm{d}K/k_0^2$ and $j_L(K\chi)\to\delta_{L0}$: a kernel $\sim K^{-p}$ converges only for $p<1$, so every $1/K^2$ (Poisson inversion) must come with a vanishing numerator (treated in Sec. 12.3: bounded for $\ell\ge 2$). (2) $\mu\to\pm1$ in the (13) loop, where $|\mathbf{k}\mp\mathbf{q}|\to 0$ inside $K_3(\mathbf{k},\mathbf{q},-\mathbf{q})$ and C_3; at $q=k$, C_3 has a simple pole (46), and the monochromatic (13) integral exists because the pole cancels in $\Theta_c^{(3)}$ (Proposition 12.2). (3) The $K=0$ tadpoles $\langle\!\langle m_2\rangle\!\rangle$, $\langle\!\langle C\rangle\!\rangle$ require kernels exactly at $\mathbf{K}=0$, where the $k\to 0$ limit does not exist without angular averaging (48); they enter only through (32), where the sky-mean normalisation cancels the direction-independent mean of e^Λ, and are kept symbolic until combined; the $\hat{\mathbf{q}}$-averaged limit exists and is the correct one (Sec. 11.8). (4) $\langle\zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2}\rangle$ inside $\Theta^{(2)}$ feeds only $\ell=0$, consistent with the monopole normalisation.

11 Assembly of the power spectrum

This section turns the map of Secs. 4, 8, 9 into explicit kernels K_1, K_2 (and, when available, K_{3c}) and assembles $C_\ell^{(11)}$, $C_\ell^{(22)}$, $C_\ell^{(13)}$ with the rules of Sec. 10. Every statement below is either derived in the text or quoted with the check that established it (App. G; code in App. F).

Conventions of this section. Observer label at the origin, ray $\bar{\mathbf{x}}(\eta) = \hat{\mathbf{n}}\chi$, $\chi = \eta_o - \eta$ (Sec. 2; $\hat{\mathbf{n}}$ = observation direction), so a field on the ray carries $e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi}$. Rayleigh: $e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = 4\pi\sum_{\ell m} i^\ell j_\ell(k\chi)Y_{\ell m}(\hat{\mathbf{n}})Y^*_{\ell m}(\hat{\mathbf{k}})$, and $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^s e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} \to 4\pi i^\ell j_\ell^{(s)}(k\chi)Y^*_{\ell m}(\hat{\mathbf{k}})$, the convention of Sec. 10. We write $\mu_a \equiv \hat{\mathbf{n}}\cdot\hat{\mathbf{k}}_a$, $E_a \equiv e^{ik_a\mu_a\chi}$, and $\dot{T} \equiv \partial_\eta T$.

11.1 First-order kernel, sky monopole and observer objects

From (30) with $\Phi_1 = \Psi_1 = \zeta T_\Phi$, $\delta_{T1} = \zeta T_{\delta_T}$, $\mathbf{v}_1 = i\mathbf{k}\,T_V\zeta$ (53),

$$K_1(\mathbf{k};\hat{\mathbf{n}}) = [T_{\delta_T}+T_\Phi-ik\mu T_V]_{\eta_*}E(\eta_*)-[T_{\delta_T}+T_\Phi-ik\mu T_V]_{\eta_o}+\int_{\eta_*}^{\eta_o}\mathrm{d}\eta\,2\dot{T}_\Phi\,E(\eta), \qquad \mu=\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}, \tag{131}$$

including the observer monopole and dipole. Its multipoles are

$$K_{1,\ell m}(\mathbf{k}) = 4\pi\, i^\ell\,\Delta_\ell(k)\,Y^*_{\ell m}(\hat{\mathbf{k}}), \tag{132}$$

with the real transfer function $\Delta_\ell(k)$ of Proposition 7.1 (closed form of the ISW part: Theorem 7.2); the observer terms are $-\delta_{\ell0}(T_{\delta_T}+T_\Phi)(k,\eta_o)+\frac{1}{3}\delta_{\ell1}kT_V(k,\eta_o)$. ($K_1$ rebuilt from Δ_ℓ, $\ell\le 40$, equals the numerical line-of-sight kernel to 7×10^{-15}; the sign of the observer dipole is fixed by the comparison with the non-perturbative planar solver of Sec. 13.4, $a_{10}[\Theta_1] = 2.0812147132$ in both.) The sky monopole entering (32) is the $\ell=0$ projection,

$$\langle m_1\rangle_{\mathrm{sky}} = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\,\zeta_{\mathbf{k}}\,\Delta_0(k), \qquad \langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle = \frac{1}{2\pi^2}\int k^2\mathrm{d}k\,P(k)\,\Delta_0(k)^2 = \frac{C_0^{(11)}}{4\pi}, \tag{133}$$

with $\Delta_0 = O(k^4)$ as $k\to0$ (Sec. 7).

Observer clock. The observer measures at fixed proper time $\Delta\tau = \int_{\eta_{\mathrm{init}}}^{\eta_o} a\,\mathrm{d}\eta/\hat{u}^0$ from the initial-data surface (Sec. 4.4, (20); second order in (92)). At first order $1/\hat{u}^0 = 1+\Psi_1$ on the worldline, so $\Delta\tau = \int\eta(1+\epsilon\Psi_1)\mathrm{d}\eta$; holding it fixed gives $\bar{\eta}_o\,\delta\eta_{o1} + \int_0^{\bar{\eta}_o}\eta\Psi_1(\eta,\mathbf{0})\mathrm{d}\eta = 0$ up to an initial-surface term $O(\eta_i^2)\to0$. Per mode ($\delta\eta_{o1} = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\,\delta\eta_o(k)$), using $\int xT_\Phi(x)\mathrm{d}x = 2\sin x/x$,

$$\delta\eta_o(k) = -\frac{1}{\eta_o}\int_0^{\eta_o}\eta\,T_\Phi(k,\eta)\,\mathrm{d}\eta = \frac{6}{k^2\eta_o}\Big(1-\frac{\sin x_o}{x_o}\Big) \xrightarrow{k\to0} \frac{\eta_o}{3}, \qquad x_o\equiv\frac{k\eta_o}{\sqrt{3}}. \tag{134}$$

The clock shift is the time integral of the *lapse* Ψ_1, not of δ_{T1} (the two differ for $k\eta_o\sim1$). The limit $\eta_o/3$ is the constant-mode value $\eta_o = \bar{\eta}_o/\lambda$, $\lambda = 1-c/3+O(c^2)$, of the exact dilation family (Sec. 5.4, (47)).

Observer displacement. The tagged worldline (Sec. 4.4; label = position as $\eta\to0$) moves by $\delta\mathbf{x}_{o1} = \int_0^{\eta_o}\mathbf{v}_1(\eta,\mathbf{0})\mathrm{d}\eta = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\,i\mathbf{k}\,X_o(k)$ with

$$X_o(k) = \int_0^{\eta_o}T_V(k,\eta)\,\mathrm{d}\eta = \frac{3}{k^2}\Big(\frac{2\sin x_o}{x_o}-\cos x_o-1\Big) \xrightarrow{k\to0} \frac{\eta_o^2}{6}. \tag{135}$$

Emission-surface shift. From $1+\delta_T(\eta_e,\mathbf{x}(\eta_e))=\eta_e/\eta_*$ (21),

$$\delta\eta_{e1}=\eta_*\,\delta_{T1}(\eta_*,\bar{\mathbf{x}}_*)\quad\Rightarrow\quad\delta\eta_e(\mathbf{k})=\eta_*\,T_{\delta_T}(k,\eta_*)\,E(\eta_*).\tag{136}$$

At first order $\delta\eta_{o1}$ and $\delta\mathbf{x}_{o1}$ do not enter $\Theta^{(1)}$ (S has no background part); they first appear in K_2.

11.2 The second-order kernel K_2

Factor bookkeeping. With $\Lambda=-S=\epsilon L_1+\dfrac{\epsilon^2}{2}L_2+\dots$ (31), the term list of Sec. 8 is $L_2/2$; call its kernel $K_{L_2/2}(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}})$ (not symmetric). Then $m_2=L_2+L_1^2$, and by (32) $\Theta^{(2)}_{\ell m}=m_{2,\ell m}-2\langle m_1\rangle_{\rm sky}m_{1,\ell m}$ for $\ell\geq1$. With $\Theta=\epsilon\Theta^{(1)}+\dfrac{\epsilon^2}{2}\Theta^{(2)}$ and $\Theta^{(2)}=\iint\zeta\zeta K_2$ (104), the symmetric kernel is

$$\boxed{K_2(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}})=2\,\mathrm{Sym}[K_{L_2/2}]+K_1(\mathbf{k}_1)K_1(\mathbf{k}_2)-\Delta_0(k_1)K_1(\mathbf{k}_2)-\Delta_0(k_2)K_1(\mathbf{k}_1)}\qquad(\ell\geq1),\tag{137}$$

$\mathrm{Sym}[f]=\dfrac{1}{2}[f(\mathbf{k}_1,\mathbf{k}_2)+f(\mathbf{k}_2,\mathbf{k}_1)]$; the factor 2 converts $L_2/2$ into L_2, the product K_1K_1 is the L_1^2 of m_2, and $-2\langle m_1\rangle_{\rm sky}m_1$ symmetrised gives the last two terms (with (133)). The $\ell=0$ part of (137) is not the monopole of $\Theta^{(2)}$ (which vanishes) and is never used. We use the *direct* (redshift) form (98) of L_2, which equals the shear form modulo the conservation equations; the two give the same K_2 to 6×10^{-15} at generic momenta (Sec. 11.5).

Fourier rules and named objects. A first-order field X_1 on the ray contributes, in slot $a\in\{1,2\}$, $\zeta_{\mathbf{k}_a}T_X(k_a,\eta)E_a$, with $\partial_j\to ik_{a,j}$, $\hat{\mathbf{n}}\cdot\mathbf{v}_1\to ik_a\mu_aT_V$. Geometry: $\mu\equiv\hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$, $\mathbf{K}\equiv\mathbf{k}_1+\mathbf{k}_2$, $K_n\equiv\hat{\mathbf{n}}\cdot\mathbf{K}=k_1\mu_1+k_2\mu_2$, $P_\perp\equiv\mathbf{k}_{1\perp}\cdot\mathbf{k}_{2\perp}=k_1k_2(\mu-\mu_1\mu_2)$ ($\mathbf{k}_\perp=\mathbf{k}-\hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbf{k})$), $E_K\equiv E_1E_2=e^{iK_n\chi}$. Two first-order combinations recur: the direct-form line-of-sight integrand and boundary function,

$$\begin{aligned}J_a&\equiv2ik_a\mu_aT_\Phi(k_a,\eta)E_a\quad(=2\partial_n\Psi_1=-\Gamma^0_{kk}|_1),\\ U_a&\equiv[T_\Phi+ik_a\mu_aT_V-T_{\delta_T}](k_a,\eta)\,E_a\quad(=\Psi_1+\hat{\mathbf{n}}\cdot\mathbf{v}_1-\delta_{T1}),\end{aligned}\tag{138}$$

the kernels of $\mathcal{G}_1$ and B_1 of (98), so that $L_1=\int_{\eta_*}^{\eta_o}J\,\mathrm{d}\eta+[U]_e^o$, equal to (30) after one integration by parts. The perturbed photon direction and position along the ray, $\mathbf{k}_1(\eta)=\mathbf{k}_1^{\rm init}-\int_\eta^{\eta_o}\mathbf{G}_1\mathrm{d}\eta'$ and $\Delta\mathbf{x}_1(\eta)=\delta\mathbf{x}_{o1}+\hat{\mathbf{n}}\,\delta\eta_{o1}-\int_\eta^{\eta_o}\mathbf{k}_1\mathrm{d}\eta'$ ((86), (87); for the sign of the $\mathbf{G}_1$ integral see Remark 8.1), are, per mode a,

$$\begin{aligned}\mathbf{k}_{1\perp}&=i\mathbf{k}_{a\perp}\,\mathcal{F}_a(\eta),&\mathcal{F}_a(\eta)&\equiv T_V(k_a,\eta_o)+2\int_\eta^{\eta_o}\mathrm{d}\eta'\,T_\Phi(k_a,\eta')E_a(\eta'),\\ \Delta\mathbf{x}_{1\perp}&=i\mathbf{k}_{a\perp}\,\mathcal{D}_{\perp,a}(\eta),&\mathcal{D}_{\perp,a}(\eta)&\equiv X_o(k_a)-\int_\eta^{\eta_o}\mathrm{d}\eta'\,\mathcal{F}_a(\eta'),\\ \hat{\mathbf{n}}\cdot\Delta\mathbf{x}_1&=\mathcal{D}_{r,a}(\eta),&\mathcal{D}_{r,a}(\eta)&\equiv ik_a\mu_aX_o(k_a)+\delta\eta_o(k_a)+2\int_\eta^{\eta_o}\mathrm{d}\eta'\,T_\Phi(k_a,\eta')E_a(\eta'),\end{aligned}\tag{139}$$

($\mathcal{F}$: aberration at the observer plus accumulated deflection; $\mathcal{D}_\perp$: lensing displacement; $\mathcal{D}_r$: radial displacement = Shapiro delay + clock + observer displacement), while $\hat{\mathbf{n}}\cdot\mathbf{k}_1=-(\Phi_1+\Psi_1)$ is the null condition. A second-order field contributes its pair kernel times E_K (Sec. 6, (58) with $\mathbf{q}\to\mathbf{k}_1$,

$\mathbf{p} \to \mathbf{k}_2$, $\mathbf{k} \to \mathbf{K}$):

$$\Phi_2 \equiv \mathcal{K}_{\Phi_2}(K, k_1, k_2; \eta), \quad \Psi_2 \equiv \mathcal{K}_{\Phi_2} - K_Q,$$
$$\delta_{T2} \equiv \frac{1}{4} K_{\delta_2} - \frac{3}{16} T_\delta(k_1) T_\delta(k_2), \quad V_2 \ (\mathbf{v}_2^L = \nabla V_2) \text{ from (56)},$$
$$\hat{\mathbf{n}} \cdot P(\hat{\mathbf{K}}) \mathbf{k}_a = k_a \mu_a - \frac{K_n \, \mathbf{K} \cdot \mathbf{k}_a}{K^2},$$
$$\hat{\mathbf{n}} \cdot \mathbf{v}_2^T = \frac{1}{2}[\hat{\mathbf{n}} \cdot P(\hat{\mathbf{K}}) \mathbf{k}_1 \, K_{v^T}^{(12)} + \hat{\mathbf{n}} \cdot P(\hat{\mathbf{K}}) \mathbf{k}_2 \, K_{v^T}^{(21)}], \quad \hat{\mathbf{n}} \cdot \boldsymbol{\omega}_2 = \frac{1}{2}[\hat{\mathbf{n}} \cdot P \mathbf{k}_1 \, K_\omega^{(12)} + \hat{\mathbf{n}} \cdot P \mathbf{k}_2 \, K_\omega^{(21)}],$$
$$h_{2,nn} = \Big[(\hat{\mathbf{n}} \cdot P \mathbf{k}_1)^2 - \frac{1}{2} \Big(1 - \frac{K_n^2}{K^2} \Big) \Big(k_1^2 - \frac{(\mathbf{K} \cdot \mathbf{k}_1)^2}{K^2} \Big) \Big] \mathcal{H}, \qquad \mathcal{H} \equiv \mathcal{K}_h(K, k_1, k_2; \eta), \tag{140}$$

with $K_X^{(ab)} \equiv K_X(\mathbf{k}_a, \mathbf{k}_b)$, $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ the time integrals (61) (closed forms, App. B), K_Q, K_ω, K_{v^T} from (59), K_{δ_2} the 00-constraint kernel (App. A), and δ_{T2} from $1 + \delta_T = (1 + \delta)^{1/4}$.

11.3 Channels of K_2

$K_2 = \sum_c K^c$ is organised by the harmonic type of Sec. 10 (B: products of maps at one point; C: single-time convolution kernels with phase E_K; D: nested two-time or endpoint×functional products). "Ray" channels are $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$ of the expression; "obs" channels are evaluated at η_o, $\chi = 0$ (so $E_a = 1$); "emi" channels at η_*, χ_*. $\sum_{(ab)}$ denotes the sum over $(a, b) = (1, 2), (2, 1)$. Each compact form below was checked symbolically against the machine-generated kernel of App. D (residual 0), and each machine kernel against the numerical engine ($\leq 1.8 \times 10^{-15}$).

Type B (products and normalisation).

$$K^{\mathrm{B,prod}} = K_1(\mathbf{k}_1) K_1(\mathbf{k}_2) - \Delta_0(k_1) K_1(\mathbf{k}_2) - \Delta_0(k_2) K_1(\mathbf{k}_1),$$
$$K_{1\times1}^{\mathrm{B,obs}} = \Big[P_\perp T_{V1} T_{V2} - 2 T_{\Phi1} T_{\Phi2} + T_{\delta_T 1} T_{\delta_T 2} - 2i (T_{\Phi1} T_{V2} k_2 \mu_2 + T_{\Phi2} T_{V1} k_1 \mu_1) \Big]_{\eta_o},$$
$$K_{1\times1}^{\mathrm{B,emi}} = \Big[E_1 E_2 \Big(P_\perp T_{V1} T_{V2} + 2 T_{\Phi1} T_{\Phi2} - T_{\delta_T 1} T_{\delta_T 2} + 2i (T_{\Phi1} T_{V2} k_2 \mu_2 + T_{\Phi2} T_{V1} k_1 \mu_1) \Big) \Big]_{\eta_*}, \tag{141}$$

with $T_{Xa} \equiv T_X(k_a, \cdot)$: the quadratic part of the boundary function $\ln(-\hat{u} \cdot k) - \ln(1 + \delta_T)$ at the two endpoints. The two brackets are not opposite: the $P_\perp T_{V1} T_{V2}$ ($v_{1\perp}^2$) term has the same sign at both ends.

Type C (single-time second-order fields and ray-local products).

$$K_{\Phi_2\Psi_2}^{\mathrm{C,ray}} = E_K [\dot{\Phi}_2 - \dot{\Psi}_2 + 2i K_n \Psi_2], \qquad K_{\Psi_2}^{\mathrm{C,obs/emi}} = \pm E_K \Psi_2,$$
$$K_{\delta_{T2}}^{\mathrm{C,obs/emi}} = \mp E_K \, \delta_{T2}, \qquad K_{V_2}^{\mathrm{C,obs/emi}} = \pm E_K \, i K_n V_2,$$
$$K_{v_2^T + \omega_2}^{\mathrm{C,obs/emi}} = \pm E_K \, \hat{\mathbf{n}} \cdot (\mathbf{v}_2^T + \boldsymbol{\omega}_2) \quad (\text{helicity } \pm 1), \qquad K_{\omega_2}^{\mathrm{C,ray}} = E_K \, i K_n \, \hat{\mathbf{n}} \cdot \boldsymbol{\omega}_2,$$
$$K_{h_2}^{\mathrm{C,ray}} = -\frac{1}{2} E_K \, \dot{h}_{2,nn} \quad (\text{helicity } \pm 2), \qquad K_{1\times1}^{\mathrm{C,ray}} = 4 E_1 E_2 (T_{\Phi1} \dot{T}_{\Phi2} + T_{\Phi2} \dot{T}_{\Phi1}), \tag{142}$$

upper sign at the observer, lower at emission. The ray terms are $-\Gamma_{kk}^0$ evaluated on the second-order fields ($\frac{1}{2}(\Phi_2' - \Psi_2') + \partial_n \Psi_2 + \frac{1}{2} \partial_n \omega_{2,n} - \frac{1}{4} h_{2,nn}'$ in $L_2/2$; the $\partial_n \Psi_2$ term is the second-order SW+ISW after integration by parts) and the local product $2\Phi_1 \Phi_1' + 2\Psi_1 \Psi_1' + 2(\Phi_1 - \Psi_1) \partial_n \Psi_1$. Grouping for the numerics (Table 5): $\Phi_2\Psi_2$ = ray $\Phi_2\Psi_2$ + endpoint Ψ_2; vector = endpoint $v_2^T + \omega_2$ + ray ω_2.

Type D (lensing, time delay, aberration+deflection, emission-surface shift, clock, observer displacement).

$$
\begin{aligned}
K^{\mathrm{D}}_{\mathrm{lens}} &= \int \mathrm{d}\eta \sum_{(ab)} -P_\perp \mathcal{D}_{\perp,a} J_b + \Big[\sum_{(ab)} P_\perp \mathcal{D}_{\perp,a} U_b\Big]_{\eta_*} && (\Delta\mathbf{x}_{1\perp}\cdot\nabla_\perp J,\ -\Delta\mathbf{x}_{1\perp}\cdot\nabla_\perp U|_e),\\
K^{\mathrm{D}}_{\mathrm{delay}} &= \int \mathrm{d}\eta \sum_{(ab)} ik_b\mu_b\, \mathcal{D}_{r,a} J_b - \Big[\sum_{(ab)} ik_b\mu_b\, \mathcal{D}_{r,a} U_b\Big]_{\eta_*} && (\Delta x_{1r}\,\partial_n J,\ -\Delta x_{1r}\,\partial_n U|_e),\\
K^{\mathrm{D}}_{\mathrm{defl}} &= \int \mathrm{d}\eta \sum_{(ab)} 2P_\perp \mathcal{F}_a T_\Phi(k_b) E_b - \Big[\sum_{(ab)} P_\perp \mathcal{F}_a T_V(k_b) E_b\Big]_{\eta_*} && (-2\mathbf{k}_{1\perp}\cdot\nabla_\perp\Psi_1,\ \mathbf{k}_{1\perp}\cdot\mathbf{v}_{1\perp}|_e),\\
K^{\mathrm{D}}_{\mathrm{emi}} &= -\Big[\sum_{(ab)} \delta\eta_e(\mathbf{k}_a)\,(J_b + \mathbb{D}U_b)\Big]_{\eta_*}, \qquad \mathbb{D}\equiv\partial_\eta - ik_b\mu_b && (\text{lower-endpoint shift}),\\
K^{\mathrm{D}}_{\mathrm{clock}} &= \Big[\sum_{(ab)} \delta\eta_o(k_a)\,(J_b + \dot{U}_b)\Big]_{\eta_o} && (\text{upper-endpoint shift at fixed } \mathbf{x}),\\
K^{\mathrm{D}}_{\mathrm{disp}} &= -\Big[\sum_{(ab)} X_o(k_a)\,(\mathbf{k}_a\cdot\mathbf{k}_b)\,U_b\Big]_{\eta_o} && (\delta\mathbf{x}_{o1}\cdot\nabla U|_o),
\end{aligned}
\tag{143}
$$

with $\mathbb{D} = \partial_\eta - \hat{\mathbf{n}}\cdot\nabla$ the derivative along the background ray; here, and in $\dot{U}_b$, ∂_η is the field derivative at fixed $\mathbf{x}$, i.e. it acts on the transfer functions only, not on E_b (J_b, U_b evaluated at the time of the term). Explicitly, the ray lensing kernel is $-2iP_\perp\sum_{(ab)}\mathcal{D}_{\perp,a}T_\Phi(k_b)k_b\mu_b E_b$ and the ray time delay $-2\sum_{(ab)}\mathcal{D}_{r,a}T_\Phi(k_b)k_b^2\mu_b^2 E_b$. The radial shift $\hat{\mathbf{n}}\,\delta\eta_{o1}$ and the observer displacement also enter through $\mathcal{D}_r$, $\mathcal{D}_\perp$ (139); the channels "clock" and "disp" contain only the explicit endpoint pieces.

Remark 11.1 (observer channels of the direct form). At the observer $\chi = 0$, so an obs channel is a polynomial in $\hat{\mathbf{n}}$ of the degree of its μ_a, K_n, $P_\perp$ content: $K^{\mathrm{C,obs}}$, $K^{\mathrm{D}}_{\mathrm{clock}}$, $K^{\mathrm{D}}_{\mathrm{disp}}$ have degree ≤ 1 and contribute to $\ell \le 1$ only; $K^{\mathrm{B,obs}}_{1\times1}$ has degree 2 and contributes to $\ell \le 2$. For $\ell \ge 2$ the observer motion and clock act only through $\mathcal{F}, \mathcal{D}_\perp, \mathcal{D}_r$ and the type-B products. (Consistent with the vanishing clock and displacement diagonals in Table 5.) The split into channels depends on the form of L_2 (in the shear form the explicit clock piece has degree 2, Sec. 8.7); only the sum K_2 is form independent.

The complete machine-generated set (fifteen channels, plus B) is App. D.

11.4 Squeezed limits: dilation and special conformal transformation

Let $\mathcal{D} \equiv \mathbf{k}_2\cdot\partial_{\mathbf{k}_2}$ (at fixed $\hat{\mathbf{k}}_2$, $\hat{\mathbf{n}}$) and, for a vector $\mathbf{p}$, $\mathcal{S}_{\mathbf{p}} \equiv (\mathbf{k}_2\cdot\partial_{\mathbf{k}_2})(\mathbf{p}\cdot\partial_{\mathbf{k}_2}) - \frac{1}{2}(\mathbf{p}\cdot\mathbf{k}_2)\nabla^2_{\mathbf{k}_2}$.

Theorem 11.2 (squeezed limit of the second-order map)**.** *For $\eta_i \to 0$ and every $\hat{\mathbf{n}}$ (all multipoles, observer terms and normalisation included),*

$$
\begin{aligned}
K_2(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}}) &= -\mathcal{D}K_1(\mathbf{k}_2;\hat{\mathbf{n}}) - \mathcal{S}_{\mathbf{k}_1}K_1(\mathbf{k}_2;\hat{\mathbf{n}}) + O(k_1^2),\\
\lim_{\mathbf{k}_1\to0} K_2(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}}) &= -\mathbf{k}_2\cdot\partial_{\mathbf{k}_2}K_1(\mathbf{k}_2;\hat{\mathbf{n}}).
\end{aligned}
\tag{144}
$$

(For finite η_i add $+\frac{\eta_i^2}{6}(\mathbf{k}_1\cdot\mathbf{k}_2)K_1(\mathbf{k}_2;\hat{\mathbf{n}})$, a translation of the observer label.) Since in RD K_1 depends on k only through $k\eta_o$, $k\eta_*$ and μ, $\mathcal{D}K_1 = (\eta_o\partial_{\eta_o} + \eta_*\partial_{\eta_*})K_1$.

Status. Derived from the adiabatic-mode transformation of the full observable, independently of K_2: the long mode $\zeta_L = c + \mathbf{q}\cdot\mathbf{x}$ is the Poisson-gauge image of FRW under $\tilde\eta = \eta - \frac{\eta}{3}\zeta_L$, $\tilde{\mathbf{x}} = (1+\zeta_L)\mathbf{x} - \frac{x^2}{2}\mathbf{q} - \frac{\eta^2}{6}\mathbf{q}$; every ingredient of $T_{\rm obs} = T_o e^{-S}$ (shear, null geodesic, $T_{\rm phys} = T_*$ surface, tagged worldline, proper-time clock, sky normalisation) is covariant, and at $\eta \to 0$ the map is conformal with $\tilde\zeta_S(\tilde{\mathbf{x}}) = \zeta_S(\mathbf{x}(\tilde{\mathbf{x}}))$ exactly. Verified on the full K_2 (137) (SQ-2): $\|K_2(s\hat{\mathbf{e}}, \mathbf{k}_2) + \mathcal{D}K_1\|$ (rms over $\ell \geq 2$) $= 3.18 \times 10^{-1,-2,-3}$ for $s = 10^{-2,-3,-4}$ ($|\mathbf{k}_2| = 1.07$; the dilation itself has rms 11.25), exactly linear in s; the k_1-odd part $[K_2(s\hat{\mathbf{e}}) - K_2(-s\hat{\mathbf{e}})]/2s$ equals $-\mathcal{S}_{\hat{\mathbf{e}}}K_1$ to 1.5–6×10^{-5} ($|\mathbf{k}_2| = 1.07$) and 1.6–2.2×10^{-4} ($|\mathbf{k}_2| = 2.44$) per multipole, the accuracy of the finite-difference Hessian. Individual channels are $O(10)$ at $s \to 0$ (e.g. ray V_2-containing terms 11.0, time delay 13.9) and cancel to the dilation.

Why a constant ζ is a dilation and not zero. A constant $\zeta_{\rm nl} = c$ is pure gauge: the early-time spatial metric $a^2 e^{2\zeta}\mathrm{d}\mathbf{x}^2$ is FRW in the coordinates $\tilde{\mathbf{x}} = e^c\mathbf{x}$, and the observer (at the origin), its proper-time clock and the emission temperature T_* are invariant. But $\zeta_{\rm nl} = c + \zeta_S$ is then the configuration $\tilde\zeta_S(\tilde{\mathbf{x}}) = \zeta_S(e^{-c}\tilde{\mathbf{x}})$ in FRW coordinates, i.e. the *same* observable evaluated on the short field *relabelled* to e^c times longer comoving wavelengths: $\Theta[c + \zeta_S] = \Theta[\zeta_S(e^{-c}\cdot)]$, whose $O(c)$ term is $-c\,\mathbf{k}\cdot\partial_{\mathbf{k}}K_1$. Physically, the long mode stretches every short wavelength relative to the only other scales in the problem, the emission and observation times fixed by $T_{\rm phys} = T_*$ and by the proper-time clock; since the RD map depends on $k\eta_*$, $k\eta_o$ only, this is the same as $-c(\eta_o\partial_{\eta_o} + \eta_*\partial_{\eta_*})$. For a *fixed* Gaussian realisation of ζ_S the relabelled field is a different realisation, so $K_2(0, \mathbf{k}_2)$ cannot vanish; "unobservable" means only that the long mode is equivalent to a relabelling (Maldacena's consistency relation for the map), whose effect on the statistics vanishes iff k^3P is scale invariant. The separate CPV-type pieces (amplitude, emission-time shift $\eta_e = \eta_*(1 + c/3)$, clock $\eta_o(1 + c/3)$, coordinate length of the ray, field cross terms, monopole normalisation) do not survive individually. At $O(k_1)$ the observer velocity, drift, emission-surface tilt, gradient lensing/time delay and the squeezed vector mode cancel likewise, leaving the SCT relabelling $\mathcal{S}_{\mathbf{k}_1}$; physical effects start at $O(k_1^2)$.

Consequence for the IR of the loop (second-order half). Let q_* separate soft from hard momenta and $\sigma_L^2 \equiv \int_{k<q_*} \frac{\mathrm{d}^3k}{(2\pi)^3}P(k)$. In (105) the regions $k_1 < q_*$ and $k_2 < q_*$ contribute equally; with (144) ($O(k_1)$ terms vanish by parity after the angular integral) and (132),

$$\frac{1}{4}C_\ell^{(22)}|_{\rm soft} = \sigma_L^2\,\frac{1}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k}{(2\pi)^3}P(k)\,|\mathcal{D}K_{1,\ell m}(\mathbf{k})|^2 = \sigma_L^2\,\frac{2}{\pi}\int k^2\mathrm{d}k\,P(k)\,[k\partial_k\Delta_\ell(k)]^2 + O(q_*^2\sigma_L^2), \tag{145}$$

the second-order dilation response. It is positive, so for $P \propto k^{-3}$ ($\sigma_L^2 \propto \int \mathrm{d}q/q$) $C_\ell^{(22)}$ alone is logarithmically IR divergent and nothing inside $C^{(22)}$ can cancel it. The cancellation requires the (13) soft region with $K_3(\mathbf{q}, -\mathbf{q}, \mathbf{k}) \to \mathcal{D}^2K_1(\mathbf{k})$, derived from the same adiabatic-mode transformation and verified on the full contracted kernel (Theorem 11.3), after which the sum is $\sigma_L^2 \cdot \frac{1}{2} \cdot \frac{2}{\pi}\int k^2\mathrm{d}k\,\Delta_\ell^2(3+\mathcal{D})^2P$: a total derivative in $\ln k$ that vanishes for scale-invariant k^3P, and leaves an IR sensitivity $\propto \sigma_L^2$ times the running of k^3P otherwise (Sec. 11.9, (159)). For the monochromatic spectrum of Sec. 12 there is no soft region; the corner $K = |\mathbf{k}_1 + \mathbf{k}_2| \to 0$ treated there is a different limit.

11.5 $C_\ell^{(22)}$

With K_2 symmetric, the two connected pairings give ((105), (C1) of (118), frame $\hat{\mathbf{K}}=\hat{\mathbf{z}}$, $\mathbf{k}_{1,2}$ in the x–z plane)

$$\begin{aligned} C_\ell^{(22)} &= \frac{2}{2\ell+1}\sum_m \int \frac{\mathrm{d}^3k_1\mathrm{d}^3k_2}{(2\pi)^6} P(k_1)P(k_2)|K_{2,\ell m}|^2 \\ &= \frac{1}{4\pi^4(2\ell+1)} \int k_1^2\mathrm{d}k_1\, k_2^2\mathrm{d}k_2\, P(k_1)P(k_2) \int_{-1}^{1}\mathrm{d}\mu \sum_m |\tilde{K}_{\ell m}(k_1,k_2,\mu)|^2, \end{aligned} \tag{146}$$

and it enters C_ℓ as $\frac{1}{4}C_\ell^{(22)}$ (12). With $\tilde{K}_{\ell m}=\sum_c \tilde{K}^c_{\ell m}$, $C_\ell^{(22)}=\sum_{c,c'} G_{cc'}(\ell)$, $G_{cc'}$ being (146) with $|\tilde{K}|^2 \to \tilde{K}^c_{\ell m}\tilde{K}^{c'*}_{\ell m}$ (cross terms included). The radial functions of each type are:

B×B: $\Delta_\ell(k)$ only:

$$C_\ell^{\mathrm{BB}} = C_\ell^{m_1^2} - 4\langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle C_\ell^{(11)}, \qquad C_\ell^{m_1^2} = 2\sum_{\ell_1\ell_2} \frac{(2\ell_1+1)(2\ell_2+1)}{4\pi} \begin{pmatrix} \ell & \ell_1 & \ell_2 \\ 0 & 0 & 0 \end{pmatrix}^2 C_{\ell_1}^{(11)} C_{\ell_2}^{(11)} \tag{147}$$

((113), $\ell_{1,2}\geq 0$ including the observer monopole and dipole), and $-4=-8$ (cross with m_1^2, (114)(ii)) $+4$ (i). The endpoint 1×1 channels are products of shell maps with radial functions $T_X(k,\eta_e)\, j_\ell^{(s)}(k\chi_e)$, $s\leq 1$ (gradient weight w_∇ for $P_\perp$), $\eta_e\in\{\eta_*,\eta_o\}$.

C: $\tilde{K}^C_{\ell m} = \int \mathrm{d}\eta\, K^c(k_1,k_2,K;\eta) \sum_{|L-\ell|\leq d} i^L \sqrt{4\pi(2L+1)}\, j_L(K\chi)\, \langle \ell m|\mathcal{Q}|L0\rangle$ (117), $|m|\leq d$, with K^c the pair kernels of (140) ($\Phi_2, \dot{\Phi}_2, \Psi_2, \dot{\Psi}_2, \delta_{T2}, V_2, K_{v^T}, K_\omega, \mathcal{H}, \dot{\mathcal{H}}$, or $T_\Phi \dot{T}_\Phi$ for the ray-local product) and $\mathcal{Q}$ the polynomial in $\mu_1, \mu_2, K_n/K$ of (142) (degree $d\leq 2$; helicity pieces equivalently by (A-λ) (109)).

D: the functionals (139) are expanded in Legendre series of μ_a, $\mathcal{F}_a=\sum_\ell (2\ell+1) i^\ell \mathcal{F}_\ell(k_a;\eta) P_\ell(\mu_a)$ etc., with radial functions

$$\begin{aligned} \mathcal{F}_\ell(k;\eta) &= T_V(k,\eta_o)\delta_{\ell 0} + 2\int_\eta^{\eta_o} \mathrm{d}\eta'\, T_\Phi(k,\eta') j_\ell(k\chi'), \\ \mathcal{D}_{\perp\ell}(k;\eta) &= X_o(k)\delta_{\ell 0} - \int_\eta^{\eta_o} \mathrm{d}\eta'\, \mathcal{F}_\ell(k;\eta'), \\ \mathcal{D}_{r\ell}(k;\eta) &= \delta\eta_o(k)\delta_{\ell 0} + \frac{1}{3}kX_o(k)\delta_{\ell 1} + 2\int_\eta^{\eta_o} \mathrm{d}\eta'\, T_\Phi(k,\eta') j_\ell(k\chi'), \end{aligned} \tag{148}$$

paired with the shell functions $T_\Phi(k_b,\eta) j_\ell^{(s)}(k_b\chi)$ (from J_b) or $[T_\Phi - T_{\delta_T}](k_b,\eta_*) j_\ell(k_b\chi_*)$, $kT_V(k_b,\eta_*) j_\ell'(k_b\chi_*)$ (from U_b); the projection is the bipolar (double Rayleigh) sum of Sec. 10.5 with the $P_\perp$ contractions carrying the weight w_∇ (type-D identity of Sec. 10.4), and (146) is unchanged.

Implementation. The full K_2 is projected on $Y_{\ell m}$ by Gauss–Legendre sky quadrature in the $\hat{\mathbf{K}}=\hat{\mathbf{z}}$ frame at each μ node; this realises the C and D reductions exactly and reproduces the B channels' finite $3j$ sum (diagonal G_{BB} to 4×10^{-12} for $\ell\leq 12$, 5×10^{-9} for $\ell\leq 20$ at $\eta_o=8$). No closed form of the C/D μ-integrands beyond (117), (148) has been produced.

Verified properties of K_2 (App. G): (i) shear form vs direct form of L_2 at generic non-collinear $\mathbf{k}_1 = (0.7, -0.2, 0.4)$, $\mathbf{k}_2 = (-0.3, 0.9, 0.5)$: relative difference 6×10^{-15} pointwise in $\hat{\mathbf{n}}$ (the shear form contains only v_2, h_2 among second-order fields, the direct form $\Phi_2, \Psi_2, \delta_{T2}, V_2, v_2^T, \omega_2, h_2$: a test of all second-order $K \neq 0$ kernels, including vector and tensor sectors); (ii) plane wave $\zeta = \cos(k(x + x_o))$ vs the non-perturbative planar solver, four configurations ($\eta_* = 2$, $\eta_o = 4, 8$, $x_o = 0, \pi/2$), $a_{\ell 0}[\Theta_1]$, $a_{\ell 0}[\Theta_2]$, $\ell \leq 12$: max deviation 1.2×10^{-8}, 1.0×10^{-9}, 6.5×10^{-9}, 7.0×10^{-10}, all within the solver's quoted errors (e.g. $a_{20}[\Theta_2] = -4.2224942187$ vs -4.2224942193); (iii) symmetry $K_2(\mathbf{k}_1, \mathbf{k}_2) = K_2(\mathbf{k}_2, \mathbf{k}_1)$ to 9×10^{-15}; (iv) reality $K_2(-\mathbf{k}_1, -\mathbf{k}_2; \hat{\mathbf{n}}) = K_2(\mathbf{k}_1, \mathbf{k}_2; \hat{\mathbf{n}})^*$ exactly; (v) squeezed limits, Theorem 11.2.

11.6 Contracted third-order kernel

Definition and factors. With $\Theta^{(3)}(\hat{\mathbf{n}}) = \iiint \zeta\zeta\zeta\, K_3$ (104) (K_3 symmetric), the only part of K_3 that enters $C_\ell^{(13)}$ is its contraction with one power spectrum. We define the *contracted third-order kernel* as the mean response of $\Theta^{(3)}$ to one Fourier mode,

$$\kappa(\mathbf{k}; \hat{\mathbf{n}}) \equiv \left\langle \frac{\delta\Theta^{(3)}(\hat{\mathbf{n}})}{\delta\zeta_{\mathbf{k}}} \right\rangle = 3 \int \frac{\mathrm{d}^3 q}{(2\pi)^3} P(q)\, K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}}) \equiv 3 K_{3c}(\mathbf{k}; \hat{\mathbf{n}}), \tag{149}$$

($\delta\zeta_{\mathbf{k}'}/\delta\zeta_{\mathbf{k}} = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}')$); the factor 3 counts the three ways of choosing the leg that carries $\mathbf{k}$, so κ is the contracted map $\Theta_c^{(3)}$ of (120), $K_{3c} = \kappa/3$ is the single-pairing contraction, and the transfer function $\Delta_\ell^{(3c)}$ of (123) (which includes the factor 3) is the Legendre coefficient of κ:

$$\kappa(\mathbf{k}; \hat{\mathbf{n}}) = \sum_\ell (2\ell + 1)\, i^\ell\, \Delta_\ell^\kappa(k)\, P_\ell(\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}), \qquad \Delta_\ell^\kappa \equiv \Delta_\ell^{(3c)}, \qquad C_\ell^{(13)} = \frac{2}{\pi} \int k^2 \mathrm{d}k\, P(k)\, \Delta_\ell(k)\, \Delta_\ell^\kappa(k)^*. \tag{150}$$

In practice κ is computed by polarisation: for $\zeta = a\, e^{i\mathbf{k}\cdot\mathbf{x}} + b\, e^{i\mathbf{q}\cdot\mathbf{x}} + c\, e^{-i\mathbf{q}\cdot\mathbf{x}}$ with $a^2 = b^2 = c^2 = 0$ the coefficient of abc in $\Theta^{(3)}$ is $\mathrm{coef}_{abc}(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}}) = 3!\, K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}})$, hence

$$\kappa(\mathbf{k}; \hat{\mathbf{n}}) = \frac{1}{2} \int \frac{\mathrm{d}^3 q}{(2\pi)^3} P(q)\, \mathrm{coef}_{abc}(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}}). \tag{151}$$

$\Theta^{(3)}$ is (101), $m_3 = L_3 + 3L_1 L_2 + L_1^3$, with L_3 in the shear form (App. C.4; the groups are listed with the contraction rule in App. E). After the $\hat{\mathbf{q}}$ integration κ depends on $\hat{\mathbf{n}}$ only through $\nu = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}}$.

Second-order observer objects. The third-order map needs the clock and the worldline at second order. In the limit $\eta_i \to 0$ of (90), (92) (label at the initial event; all fields on the worldline $\mathbf{x} = \mathbf{0}$),

$$\begin{aligned} \frac{1}{2}\delta\eta_{o2} &= -\frac{1}{\eta_o}\Big[\frac{1}{2}\delta\eta_{o1}^2 + \eta_o\, \delta\eta_{o1} \Psi_1|_o + \int_0^{\eta_o} \eta \left(\frac{1}{2}\Psi_2 - \frac{1}{2}\Psi_1^2 - \frac{1}{2}|\mathbf{v}_1|^2 + \mathbf{X}_1 \cdot \nabla\Psi_1\right) \mathrm{d}\eta\Big], \\ \frac{1}{2}\delta\mathbf{x}_{o2} &= \int_0^{\eta_o} \Big[\frac{1}{2}\mathbf{v}_2 + (\mathbf{X}_1 \cdot \nabla)\mathbf{v}_1\Big] \mathrm{d}\eta + \delta\eta_{o1}\, \mathbf{v}_1|_o, \qquad \mathbf{X}_1(\eta) \equiv \int_0^\eta \mathbf{v}_1\, \mathrm{d}\eta', \end{aligned} \tag{152}$$

with $\delta\eta_{o1}$, $\delta\mathbf{x}_{o1} = \mathbf{X}_1(\eta_o)$ of (134), (135). (Check: for the dilation family, $\lambda = 1 + \epsilon a_1 + \frac{\epsilon^2}{2} a_2$, $a_1 = -c/3$, the first line gives $\frac{1}{2}\delta\eta_{o2} = \eta_o(c^2/9 - a_2/2)$, the second-order coefficient of $\bar{\eta}_o/\lambda$.) The fields in (152) include the tadpoles: in the configuration $(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ the worldline integral of Ψ_2 contains the pair $(\mathbf{q}, -\mathbf{q})$ multiplying $K_1(\mathbf{k})$, and the pairs $(\mathbf{k}, \pm\mathbf{q})$ multiplying first-order objects of $\mp\mathbf{q}$. The second-order ray functionals $\mathbf{k}_2$, $\Delta\mathbf{x}_2$ are (88), (89) with the sign of Remark 8.1, and the emission shift $\delta\eta_{e2}$ is (93).

Channel classes. Let $\langle YZ\rangle \equiv \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, Y[\mathbf{q}]\, Z[-\mathbf{q}]$ denote the contraction of two first-order objects (fields or ray functionals, possibly at different points of the ray), $X[\mathbf{k}]$ the kernel of a first-order object ($\zeta_{\mathbf{k}}\, T_X(k,\eta)E$ with the Fourier rules of Sec. 11.2), $X_2^{\mathrm{s}}(\mathbf{q}_1,\mathbf{q}_2) \equiv K_X(\mathbf{q}_1,\mathbf{q}_2) + K_X(\mathbf{q}_2,\mathbf{q}_1)$ the pair kernel of a second-order object and $\langle X_2\rangle = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q) K_X(\mathbf{q},-\mathbf{q})$ its tadpole (64). Applying (151) monomial by monomial gives three classes.

(i) *Cubic first-order products* ABC (one self-contraction): the leg $\mathbf{k}$ sits on one factor and the other two are contracted,

$$\kappa^{(\mathrm{i})}[ABC] = A[\mathbf{k}]\,\langle BC\rangle + B[\mathbf{k}]\,\langle AC\rangle + C[\mathbf{k}]\,\langle AB\rangle, \tag{153}$$

a variance (equal-point for local products, an unequal-point correlation along the ray for products with ray functionals) times a first-order map. Example: the cube L_1^3 of m_3 gives $3K_1(\mathbf{k};\hat{\mathbf{n}})\langle L_1L_1\rangle(\hat{\mathbf{n}})$, and $\langle L_1L_1\rangle = \langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle + \sum_{\ell\geq 1}\frac{2\ell+1}{4\pi}C_\ell^{(11)}$ is direction independent, $= \xi^{(11)}(0)$ (111).

(ii) *First* $\times$ *second order* A_1B_2 (including first-order fields times second-order ray functionals, clock, displacement and emission shift):

$$\kappa^{(\mathrm{ii})}[A_1B_2] = \underbrace{A[\mathbf{k}]\,\langle B_2\rangle}_{\text{tadpole pair}} + \underbrace{\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, A[\mathbf{q}]\, B_2^{\mathrm{s}}(\mathbf{k},-\mathbf{q})}_{\text{cross-contracted}}. \tag{154}$$

The tadpole term multiplies the free leg only, so only the $\hat{\mathbf{q}}$-averaged tadpole enters κ (Sec. 11.8). The cross term contains the second-order kernels at the momenta $(\mathbf{k},-\mathbf{q})$, $|\mathbf{k}-\mathbf{q}|= p$, with all their Poisson inversions p^{-2}, p^{-4}.

(iii) *Third-order fields*, $X_3 \to K_{3c,X}(\mathbf{q}) + K_{3c,X}(-\mathbf{q})$, i.e. the contracted solution X_{3c} of (62), (65): $\Phi_{3c} = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)[\int_0^\eta G_\Phi K_{3c,S_\Phi}\mathrm{d}\eta' + C_3\, U(k\eta/\sqrt{3})]$, $\Psi_{3c} = \Phi_{3c} - Q_{3c}$, $V_{3c} = -\frac{\eta^2}{4}[2(\Phi'_{3c} + \Psi_{3c}/\eta) + K_{3c,S_{0z}}/(ik)]$. In the shear form only the velocity enters (the contracted vector and tensor parts vanish by isotropy), $\mathbf{v}_{3c} = i\mathbf{k}V_{3c}$, through the same projection as at first order:

$$\begin{aligned}
\kappa^{(\mathrm{iii})}(\mathbf{k};\hat{\mathbf{n}}) &= \int_{\eta_*}^{\eta_o}\mathrm{d}\eta\,\frac{2}{3}k^2\,P_2(\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})\,V_{3c}(k,\eta)\,e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi},\\
\Delta_\ell^{(\mathrm{iii})}(k) &= \int_{\eta_*}^{\eta_o}\mathrm{d}\eta\,\frac{2}{3}k^2 V_{3c}(k,\eta)\sum_{J=|\ell-2|}^{\ell+2}(2J+1)\begin{pmatrix}\ell & 2 & J\\ 0 & 0 & 0\end{pmatrix}^2 i^{J-\ell}j_J(k\chi),
\end{aligned} \tag{155}$$

by (A-L) (122) with $L=2$. (In the direct form, $\Phi_{3c}, \Psi_{3c}, \delta_{T3c}, V_{3c}$ appear instead; the two forms agree, below.)

The normalisation terms of (101) contribute, for $\ell \geq 1$ (with $K^{m_2} \equiv 2\,\mathrm{Sym}[K_{L_2/2}] + K_1K_1$ the kernel of m_2 and $\bar{K}_2$ its sky average, Sec. 10.6),

$$\begin{aligned}
\kappa_{\mathrm{norm}} = &-6\int_{\mathbf{q}} P\,\Delta_0(q)\,K^{m_2}(\mathbf{k},-\mathbf{q};\hat{\mathbf{n}}) - 6\int_{\mathbf{q}} P\,\bar{K}_2(\mathbf{k},-\mathbf{q})\,K_1(\mathbf{q};\hat{\mathbf{n}})\\
&- 3\langle\langle m_2\rangle\rangle K_1(\mathbf{k};\hat{\mathbf{n}}) + 6\langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle K_1(\mathbf{k};\hat{\mathbf{n}}),
\end{aligned} \tag{156}$$

$\int_{\mathbf{q}} \equiv \int \frac{\mathrm{d}^3q}{(2\pi)^3}$; inserted in (150) they give the four terms of (E0) (119) with (E2), (E3) (125). The complete list of groups (L_1: 3 terms; $\frac{1}{2}L_2$: 34; $\frac{1}{6}L_3$: 359 in 32 groups) with their classes is App. E;

the engine evaluates every group in a per-direction frame $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{n}})$, builds the ray functionals by spectral cumulative integration on a Chebyshev grid and the third-order fields from Green's-function quadrature.

The spatial triad. The direction map (22) contains, from second order on, the spatial triad of the observer (83) through $k_{2,\mathrm{init}}$ (85). By Lemma 4.7 any global triad that agrees at first order gives the same $C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$, whereas a per-direction triad must not be used (Sec. 4.5). We use the canonical global triad, evaluated in global components with general n^a for each $\hat{\mathbf{n}}$ and then rotated into the per-direction frame of the engine. (In the shear form the observer contains no k_2 term, so the canonical triad enters only through the ray functional $\mathbf{k}_2$; the direct-form observer terms written in the $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ split implicitly use the per-direction triad and must be re-expressed before they are compared, R-12.)

11.7 $C_\ell^{(13)}$

Combining (150) with the classes,

$$C_\ell^{(13)} = \frac{2}{\pi} \int k^2 \mathrm{d}k\, P(k)\, \Delta_\ell(k) \Big[\Delta_\ell^{(\mathrm{i})} + \Delta_\ell^{(\mathrm{ii})} + \Delta_\ell^{(\mathrm{iii})} + \Delta_\ell^{\mathrm{norm}}\Big]^*(k), \tag{157}$$

where each $\Delta_\ell^{(\cdot)}$ is the Legendre coefficient of the corresponding part of κ; equivalently, in the language of Sec. 10.6, $C_\ell^{(13)} = T_{3,\ell} - 3T_{\langle 1\rangle_{\mathrm{sky}}2,\ell} - 3T_{\langle 2\rangle_{\mathrm{sky}}1,\ell} + 6\langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle C_\ell^{(11)}$ (E0), with T_3 the m_3 part ($T^{(\mathrm{i})}$, $T^{(\mathrm{ii})}$, $T^{(\mathrm{iii})}$) and the normalisation pieces from (156). The radial functions are named objects: $\Delta_\ell^{(\mathrm{iii})}$ in (155) (one declared time integral inside V_{3c}); $\Delta_\ell^{(\mathrm{i})}$ from the variances $\xi^{XY}(0)$ (111) and the unequal-point correlations of the ray functionals (reduction (E1) with $\Delta\chi \neq 0$); $\Delta_\ell^{(\mathrm{ii})}$ from the tadpoles $\langle X_2\rangle(\eta)$ (closed forms, App. A) and from the second-order pair kernels at $(\mathbf{k}, -\mathbf{q})$, reduced by (E1) and (A-L). None of the $\mathbf{q}$ integrals has been done in closed form; for the monochromatic spectrum they collapse to one angular integral (Sec. 12.5).

Validation of κ (App. G, R-9–R-11, EF-22, H-13).

1. *Shear form = direct form at third order.* The two term lists of $\frac{1}{6}L_3$, evaluated with the contracted third-order fields (65), the second-order functionals, clock, displacement and emission shift, give the same $\hat{\mathbf{q}}$-azimuth-averaged $\mathrm{coef}_{abc}(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}})$ to 2×10^{-11} absolute (3×10^{-14} relative) at $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{q}} = 0.31$ and -0.62. Since the direct form contains $\Phi_{3c}, \Psi_{3c}, \delta_{T3c}, V_{3c}$ and the shear form only V_{3c}, this tests the contracted third-order solution (third-order conservation), both term lists and all third-order ray objects at once. At lower order: K_1 from the L_1 lists equals the type-A expansion with Δ_ℓ to 2×10^{-15}; $\frac{1}{2}L_2$ shear = direct for three generic momenta, all pairs including the $K = 0$ pair, to 10^{-13}; the second-order null condition holds for the computed $\mathbf{k}_2$ to 10^{-13}.
2. *Residual-gauge invariance.* At $K = 0$ Poisson gauge leaves a time reparametrisation $\eta \to \eta + \frac{\epsilon^2}{2}\tau(\eta)$ free; it shifts the isotropic tadpoles, $\delta\langle\Psi_2\rangle = \tau' + \tau/\eta$, $\delta\langle\Phi_2\rangle = -\tau/\eta$, $\delta\langle\delta_2\rangle = -4\tau/\eta$ (ζ_{nl} unchanged), and the third-order fields by $\delta\Phi_3 = 3\tau T'_\Phi + 6(\tau/\eta)T_\Phi$, $\delta\Psi_3 = 3\tau T'_\Phi + 6(\tau' + \tau/\eta)T_\Phi$, $\delta V_3 = 3\tau T'_V + 3\tau' T_V$. (a) The contracted sources with shifted tadpoles, solved with the same Green's function and the shifted amplitude $\delta C_3 = \delta\Phi_3(0)$, reproduce these shifts to

10^{-13}: the tadpole bookkeeping of the third-order solution is gauge covariant. (b) The fully assembled κ (all tadpole routes: local 2+1 products, radial $\mathbf{k}_2$ and $\Delta\mathbf{x}_2$, clock $\delta\eta_{o2}$, emission shift $\delta\eta_{e2}$, class (iii), $3L_1L_2$ and $-3\langle m_2\rangle_{\rm sky}m_1$) is unchanged to 6×10^{-14} for $\tau=0.1\eta$ and 3×10^{-13} for $\tau=0.1\eta^3$ (scale of κ: 4×10^2), while individual channels change by $O(1$–$4)$ (time delay with $\Delta\mathbf{x}_2$: 3.6; clock: 2.3; class (iii): 1.35; local 2+1 and radial $\mathbf{k}_2$: 1.18 each; emission shift: 0.15).

3. *Reality.* $\Im C_\ell^{(13)}\sim10^{-14}$ in the monochromatic evaluation (relative to $\mathrm{Re}\,C_\ell^{(13)}$, $\ell\le12$).

11.8 Zero modes and the tadpoles

This is the single authoritative discussion of the $k=0$ (zero-mode) freedom of Poisson gauge and of the tadpole prescription; Secs. 5.4, 6.3, 13.3 and 14 refer to it (ledger ZM-1–ZM-5).

(1) The residual freedom at $k=0$. Every candidate homogeneous coordinate change was applied exactly to FRW plus generic first-order fields, keeping $\partial_i\omega_i=0$ and h_{ij} transverse-traceless (ZM-1).

- *Time reparametrisation* $\eta\to\eta+F(\eta)$, $F=O(\eta)$: it induces $\Psi=F'+F/\eta$, $\Phi=-F/\eta$, $\delta=-4F/\eta$ and leaves $\zeta_{\rm nl}$, $h(0)$ and the observer label unchanged. It is a genuine residual freedom; the residual time rescaling a_2 of the dilation family (47) is its member $F\propto\eta$. ($\eta\to\eta+c_0$ gives $\Psi\propto c_0/\eta$ and is excluded by regularity.)
- *Dilation, constant shear, constant rotation, constant translation*: each changes $\zeta^{(2)}$ (e.g. by $\mathbf{x}\cdot\nabla\zeta_1$), $h(0)$ or the label, and is therefore fixed by $\zeta^{(2)}(0)=0$, $h_n(0)=0$ (Definition 5.2) and the labelling (O2).
- *Time-dependent translations* $\mathbf{c}(\eta)$ *and rotations* $\mathbf{w}(\eta)$ with $\mathbf{c}(0)=\mathbf{w}(0)=0$: pure gauge, absorbed by the observer's worldline and triad.

Hence the only residual $k=0$ freedom is the family $F(\eta)$. For the vector and tensor sectors (ZM-5): the tensor mode $j_0(k\eta)$ is regular (indicial roots $0,-1$), so regularity does not exclude it and $h_n(0)=0$ must be, and is, part of the initial-data definition (this also removes the constant-shear zero mode); for $k\neq0$ the only homogeneous vector mode is $\omega=-8C_v/(k^2\eta^2)$, singular and carried by primordial vorticity, so regularity is equivalent to $\omega_n=v_n^T=0$ initially; at $k=0$ the only vector mode is $\boldsymbol{\omega}=-\mathbf{v}=\mathbf{c}'(\eta)$, pure gauge.

(2) The $k\to0$ limit of the second-order kernels. The divergent term $-4q^2(3\mu^2-1)/(9k^2)$ of (48) is *not* a constant shear (ZM-2). At $O(k^{-2})$, $K_Q=-(3\mu^2-1)q^2K_h(q,q)/(4k^2)$ (the tensor source), $K^{(-2)}_{S_\Phi}=(K^{(-2)}_Q)'/\eta$, and at $\eta=0$ the k^{-2} parts obey $Q=3\Phi$. The k^{-2} and k^{-1} parts of $(\Phi_2,\Psi_2,\delta_2,V_2)$ satisfy exactly $\Phi=G/\eta$, $\Psi=-(G'+G/\eta)$, $\delta=4\Phi$, $V=\eta\Phi$ for one function G, i.e. they form a quadrupolar/dipolar *time-reparametrisation* mode ($F=-G$ with a quadratic or linear dependence on position), whose induced homogeneous shear vanishes at $\eta=0$. The average over the direction of $\mathbf{q}$ at fixed k removes every negative power of k (the k^{-1} part of ω_2 averages to zero as well), so the tadpole limit exists and equals the angle-averaged values $\mathrm{tad}_{Q_2}\to-\dfrac{16}{9}$, $\mathrm{tad}_{\Phi_2}\to-\dfrac{8}{9}$, $\eta^2\mathrm{tad}_{S_{00}}\to\dfrac{32}{3}$ of App. A. In the assembled kernel the question does not even arise for the anisotropic part: in (154) the tadpole multiplies first-order objects of the free leg $\mathbf{k}$ only, so only the $\hat{\mathbf{q}}$-averaged $K=0$ pair kernel enters κ. The angular average is therefore not a convention.

(3) Invariance of the map. To second order the map is invariant under every zero mode (ZM-3): with all ingredients transformed (observer event, emission surface, triad, ray functionals), $\Delta L_1 = 0$ and ΔL_2 is a rigid rotation of the sky plus an exact total derivative whose boundary terms cancel. This was verified for time rescaling and a general cubic $F(\eta)$ (no rotation), for the dilation and $\mathbf{c}(\eta)$ (no change), for a constant shear (rotation $(-S_{23}, S_{13}, -S_{12})/2$, the Gram–Schmidt prediction), for $\mathbf{w}(\eta)$ (rotation $\mathbf{w}(\eta_o)$), and for the quadrupolar and dipolar modes of (2); deliberately wrong controls fail. A rigid rotation drops out of the $C_\ell^{(rs)}$ (Sec. 4.5).

(4) The physical tadpole. The isotropic tadpole has three functions $(\langle\Phi_2\rangle, \langle\Psi_2\rangle, \langle\delta_2\rangle)(\eta)$ subject to two $K = 0$ constraints; $\zeta^{(2)}(0) = 0$ with the 00 constraint $\delta_2 = -2\Psi_2 + \frac{32}{9}c^2$ fixes the dilation direction, and the remaining freedom is exactly the family $F(\eta)$, whose linear member $F = a\eta$ shifts $(\Psi_2, \Phi_2, \delta_2)$ by $(2a, -a, -4a)$. The two isotropic tadpole members of Sec. 5.4 (the angle-averaged one (49) and the one with $\Psi_2 = \Phi_2$) differ by $|a| = 16/27$ along this family: the choice between them is pure gauge (ZM-4). At third order the reparametrised solution again solves the field equations with $\zeta^{(3)}(0) = 0$ and $h(0) = 0$, and the regular solution is unique, so the third-order fields shift by the corresponding gauge transformation (173). The sourced part of the tadpole is not a gauge mode: its source is the $\hat{\mathbf{q}}$-averaged $K = 0$ stress of the short modes, the physical back-reaction on the homogeneous expansion; it is kept and is not expected to cancel. In the shear form L_2 contains no tadpole field (the shear of a homogeneous perturbation vanishes), so the tadpoles act only through the third-order routes listed in item 2 of Sec. 11.7.

(5) Status at third order. Invariance of $\Theta^{(3)}$ under $F(\eta)$ follows from the same multilinear identity as at second order together with the covariance of the third-order solution (EF-22); it was not run separately as a symbolic third-order test. It is supported numerically by the invariance of the fully assembled κ under $\tau = 0.1\eta$ and $0.1\eta^3$ (R-11) and by the plane-wave zero-mode test PW-3, and analytically, for the long-mode poles, by Sec. 12.6.

11.9 Infrared and ultraviolet structure of the loop

Write $\kappa(\mathbf{k}; \hat{\mathbf{n}}) = \int \frac{q^2 \mathrm{d}q}{2\pi^2} P(q)\, \tilde{\kappa}(\mathbf{k}, q; \hat{\mathbf{n}})$, $\tilde{\kappa} \equiv 3\langle K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}})\rangle_{\hat{\mathbf{q}}}$, and $\Delta_\ell^{\tilde{\kappa}}(k, q)$ for its Legendre coefficients.

Theorem 11.3 (squeezed limit at third order)**.** *For $\eta_i \to 0$, with $\mathcal{D} = \mathbf{k}\cdot\partial_{\mathbf{k}}$,*

$$K_3(\mathbf{q}, -\mathbf{q}, \mathbf{k}; \hat{\mathbf{n}}) = \mathcal{D}^2 K_1(\mathbf{k}; \hat{\mathbf{n}}) + O(q^2), \qquad \textit{hence} \qquad \Delta_\ell^{\tilde{\kappa}}(k, q) = 3\,\mathcal{D}^2 \Delta_\ell(k) + O(q^2). \tag{158}$$

Derivation. As for Theorem 11.2, two long modes $\mathbf{k}_a, \mathbf{k}_b$ compose to the adiabatic transformation with $c = \zeta_L$ to second order; the composite of the dilation $e^{-c\mathcal{D}}$ and the special conformal transformation gives, to all orders in c, $K_{m+1}(0, \ldots, 0, \mathbf{k}) = (-\mathcal{D})^m K_1(\mathbf{k})$, and with two long legs $K_3(\mathbf{k}_a, \mathbf{k}_b, \mathbf{k}) = \mathcal{D}^2 K_1 + (\mathcal{S}_{\mathbf{k}_a} + \mathcal{S}_{\mathbf{k}_b})\mathcal{D}K_1 + O(k_a^2, k_b^2, k_a k_b)$ (plus an η_i^2 label term). For $\mathbf{k}_b = -\mathbf{k}_a = -\mathbf{q}$ the $O(q)$ terms cancel. Physically: a constant $\zeta = c$ is the same universe in the coordinates $\mathbf{x} \to e^c\mathbf{x}$, so $\Theta[c + \zeta_S] = \Theta[\zeta_S(e^{-c}\,\cdot)]$, whose $O(c^2)$ term is $\frac{1}{2}c^2\mathcal{D}^2$ of the first-order map. *Numerical verification on the full assembled κ* (including normalisation; $k = 1$, $\eta_* = 2$, $\eta_o = 8$; SQ-3):

	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$
$q = 0.100$	25.523	−15.785	−10.342	−0.218
$q = 0.050$	25.685	−16.210	−10.255	0.197
$q = 0.025$	25.732	−16.322	−10.240	0.301
$3\mathcal{D}^2\Delta_\ell$	25.748	−16.360	−10.235	0.335

The deviations decrease by a factor 4.0 per halving of q ($\ell = 2$: 0.575, 0.151, 0.038; $\ell = 4$: 0.553, 0.138, 0.034), i.e. as q^2.

Combined infrared behaviour. Let q_* separate soft from hard momenta, $\sigma_L^2 = \int_{q<q_*} \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)$. By (158) the soft region of $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}$ is $\sigma_L^2 \frac{2}{\pi} \int k^2 \mathrm{d}k\, P\, \Delta_\ell\, \mathcal{D}^2 \Delta_\ell$; adding the soft region (145) of $\frac{1}{4}C_\ell^{(22)}$ and using $(\mathcal{D}\Delta)^2 + \Delta\mathcal{D}^2\Delta = \frac{1}{2}\mathcal{D}^2(\Delta^2)$ and two integrations by parts in $\ln k$,

$$\Big[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}\Big]_{\text{soft}} = \sigma_L^2\,\frac{2}{\pi}\int \mathrm{d}\ln k\;\frac{1}{2}\,\Delta_\ell(k)^2\,\frac{\mathrm{d}^2}{\mathrm{d}(\ln k)^2}(k^3P(k)) + O(q_*^2\sigma_L^2), \tag{159}$$

provided the boundary terms $\Delta_\ell^2\, k^3P$, $\Delta_\ell\Delta_\ell' k^4 P$ vanish at $k \to 0$ and $k \to \infty$ (true for $\ell \geq 2$ and the power laws below: $\Delta_\ell = O(k^\ell)$ at small k and $O(k^{-1})$ at large k). Each half separately is $\propto \sigma_L^2$ with a coefficient of definite sign; the sum is a total second derivative of k^3P. For a power law $P \propto k^{n_s-4}$, $\mathrm{d}^2_{\ln k}(k^3P) = (n_s-1)^2 k^3 P$ and

$$\Big[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}\Big]_{\text{soft}} = \frac{1}{2}(n_s-1)^2\,\sigma_L^2\,C_\ell^{(11)} + O(q_*^2\sigma_L^2), \qquad \sigma_L^2 \propto \int_0^{q_*} \mathrm{d}q\, q^{n_s-2}. \tag{160}$$

Hence: for $n_s = 1$ the coefficient of the (logarithmically divergent) σ_L^2 vanishes identically and the loop is IR finite; for $n_s > 1$, σ_L^2 converges and the loop is IR finite; for $n_s < 1$, σ_L^2 diverges as $q_{\min}^{n_s-1}$ and the one-loop spectrum has a genuine IR sensitivity $\frac{1}{2}(n_s-1)^2\sigma_L^2 C_\ell^{(11)}$ (the response to a long-wavelength dilation of a spectrum that is not scale invariant). The $O(q^2)$ remainder converges at $q \to 0$ for $n_s > -1$. The monochromatic spectrum has no soft region. The formula (159) is derived; it has not been tested numerically for a tilted spectrum.

Ultraviolet behaviour (numerical only). At fixed $k = 1$ ($\eta_* = 2$, $\eta_o = 8$) the $\hat{\mathbf{q}}$-averaged integrand $\Delta_\ell^{\tilde{\kappa}}(k,q)$ is

ℓ	$q \to 0$	$q = 1$	$q = 2$	$q = 3$	$q = 4$
2	−16.36	−4.68	2.04	5.96	11.00
3	−10.24	−25.17	−5.49	−7.06	−0.50
4	0.34	−18.33	−6.92	−15.64	−0.08

($q \to 0$: the squeezed value $3\mathcal{D}^2\Delta_\ell$; the $\hat{\mathbf{q}}$ quadrature needs $\gtrsim 2q\chi_* + 8$ azimuthal nodes). It stays $O(10)$ and oscillatory in q up to $q = 4k$, although individual channels (lensing, time delay, clock, normalisation) are each $O(10)$ and cancel. Power counting supports $O(q^0)$: inside the horizon the potentials decay, $T_\Phi \sim (q\eta)^{-2}$, but δ_T and v stay $O(1)$ (no damping in a perfect fluid); the same-point variance of the local terms is removed by the monopole normalisation at leading order. *If* $\tilde{\kappa} = O(q^0)$ as $q \to \infty$, the q integral $\int \mathrm{d}q\, q^2 P(q)\tilde{\kappa}$ converges in the UV for $n_s < 1$ and is logarithmically divergent for $n_s = 1$, like $\langle\zeta^2\rangle$ itself; a physical UV cut-off (diffusion damping, absent in the perfect fluid) is then needed. No analytic large-q expansion of K_{3c} has been made; the UV statement is unresolved (Sec. 14).

12 Monochromatic spectrum

12.1 Definition and first order

The primordial spectrum is a shell at k_0 with the variance held fixed (Sec. 10.8),

$$P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\,\delta(k-k_0), \qquad \langle\zeta_1^2\rangle = \sigma^2, \tag{161}$$

so the expansion parameter is $\epsilon^2\sigma^2$ and every momentum integral collapses by (127). In RD the only other scales are η_* and η_o, so the results depend on $k_0\eta_*$, $k_0\eta_o$ only. At first order (Sec. 7)

$$C_\ell^{(11)} = 4\pi\sigma^2\,\Delta_\ell(k_0)^2, \tag{162}$$

with Δ_ℓ in closed form (Proposition 7.1, Theorem 7.2); the sky monopole has $\langle\langle m_1\rangle_{\rm sky}^2\rangle = \sigma^2\Delta_0(k_0)^2$.

12.2 Collapse of $C_\ell^{(22)}$ to a single integral

Inserting (161) in (146) fixes $k_1 = k_2 = k_0$; the only remaining variable is the angle $\mu = \hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$, or equivalently the magnitude of the total momentum $K = |\mathbf{k}_1+\mathbf{k}_2| = k_0\sqrt{2(1+\mu)} \in [0, 2k_0]$, $\mathrm{d}\mu = K\,\mathrm{d}K/k_0^2$:

$$C_\ell^{(22)} = \frac{\sigma^4}{2\ell+1}\int_{-1}^{1}\mathrm{d}\mu\sum_m |\tilde K_{\ell m}(k_0,k_0,\mu)|^2 = \frac{\sigma^4}{(2\ell+1)\,k_0^2}\int_0^{2k_0} K\,\mathrm{d}K\sum_{c,c'}\sum_m \tilde K^{c}_{\ell m}\,\tilde K^{c'*}_{\ell m}\Big|_K, \tag{163}$$

where $\tilde K^c_{\ell m}$ is the multipole of channel c of (141)–(143) in the frame $\hat{\mathbf{K}} = \hat{\mathbf{z}}$, $\mathbf{k}_{1,2} = k_0(\pm\sin\alpha, 0, \cos\alpha)$, $\cos\alpha = K/2k_0$ (the prefactor follows from (C1) of (118) with $\int k^2\mathrm{d}k\,P \to 2\pi^2\sigma^2$). The integrand is analytic in K (not in μ) at $\mu = -1$, which is why K is the integration variable. The type-B×B part is closed (all at k_0):

$$C_\ell^{\rm BB} = 2(4\pi\sigma^2)^2\sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}^2\Delta^2_{\ell_1}\Delta^2_{\ell_2} - 16\pi\sigma^4\Delta_0^2\Delta_\ell^2. \tag{164}$$

12.3 The corner $K \to 0$

At $\mu \to -1$ the two momenta become antiparallel and the second-order fields are evaluated at vanishing total momentum, where the Poisson-gauge inversions ∇^{-2}, ∇^{-4} act. This is the delicate point (1) of Sec. 10.8.

Proposition 12.1 (behaviour at $K \to 0$)**.** *Let $|\mathbf{k}_1|, |\mathbf{k}_2|$ be fixed and $K \to 0$, with $c = \hat{\mathbf{K}}\cdot\hat{\mathbf{k}}_1$. The pair kernels of Sec. 6 behave as*

$$\begin{aligned}
&K_Q,\ K_{S_\Phi},\ K^0_{\Phi_2} \sim K^{-2}\ (\textit{coefficient} \propto (3c^2-1)k_1^2:\ \partial_i\partial_j\nabla^{-4}S_{ij}), \qquad \textit{hence}\quad \Phi_2, \Psi_2 \sim K^{-2};\\
&K_\omega,\ K_{v^T} \sim K^{-1}\ \textit{with exactly opposite coefficients}, \qquad K_\omega + K_{v^T} \sim K^{+1};\\
&K_{S_{0z}} \sim K\ \Rightarrow\ V_2 \sim K^{-2},\ \mathbf{v}_2 = \nabla V_2 \sim K^{-1}; \qquad K_{S_{00}},\ K_h \sim K^0.
\end{aligned} \tag{165}$$

A term $f(K)\,\mathcal{P}_d(\hat{\mathbf{n}})\,E_K$ of K_2 with $f \sim K^{-p}$ and $\mathcal{P}_d$ a polynomial of degree d in $\hat{\mathbf{n}}$ contributes to multipole ℓ only through $j_L(K\chi) \sim (K\chi)^L$ with $|L-\ell| \le d$, i.e. as $K^{\ell-d-p}$. Every term of the type-C channels (142) satisfies

$$d + p \le 2, \tag{166}$$

namely Ψ_2, $\dot\Phi_2 - \dot\Psi_2$, δ_{T2} (which contains $-\frac{1}{2}\Psi_2$ through the 00 constraint): $d = 0, p \le 2$; $\partial_n\Psi_2$: $d = 1, p = 1$; $\hat{\mathbf{n}}\cdot\boldsymbol{\omega}_2$, $\hat{\mathbf{n}}\cdot\mathbf{v}_2^T$, $\partial_n V_2$: $d = 1, p = 1$; $h_{2,nn}$, $\partial_n\omega_{2,n}$: $p = 0$; and the B and D channels contain no inverse powers of K. Hence $K_{2,\ell m} = O(1)$ as $K \to 0$ for $\ell \ge 2$ (a finite, $\hat{\mathbf{K}}$-dependent limit), while the $\ell = 1$ and $\ell = 0$ projections of individual channels are $O(K^{-1})$ and $O(K^{-2})$. With $\mathrm{d}\mu = K\mathrm{d}K/k_0^2$ the integrand of (163) is bounded at $K = 0$ for every $\ell \ge 2$: $C_\ell^{(22)}$ converges without any cancellation between channels. The dipole would be logarithmically divergent channel by channel.

	$\eta_o = 8$			$\eta_o = 4$	
ℓ	$C_\ell^{(11)}/\sigma^2$	$C_\ell^{(22)}/\sigma^4$	$\frac{1}{4}C_\ell^{(22)}/\sigma^4$	$C_\ell^{(11)}/\sigma^2$	$\frac{1}{4}C_\ell^{(22)}/\sigma^4$
2	9.546006061×10^{-1}	5.703017520×10^{1}	1.425754380×10^{1}	8.613350843×10^{-3}	3.993324543×10^{0}
3	6.250779591×10^{-1}	1.151253897×10^{1}	2.878134742×10^{0}	8.052093896×10^{-3}	6.124256484×10^{-1}
4	1.750712875×10^{-1}	7.463675108×10^{0}	1.865918777×10^{0}	1.116078495×10^{-3}	5.287309571×10^{-2}
5	1.933453283×10^{-2}	4.520572234×10^{0}	1.130143058×10^{0}	7.175773518×10^{-5}	7.780704277×10^{-3}
6	1.770349639×10^{-4}	2.088249491×10^{0}	5.220623728×10^{-1}	2.780479034×10^{-6}	1.672196841×10^{-3}
7	3.174580556×10^{-4}	1.496197895×10^{0}	3.740494738×10^{-1}	7.318262641×10^{-8}	2.619102919×10^{-4}
8	2.304634771×10^{-4}	1.047895782×10^{0}	2.619739455×10^{-1}	1.403749744×10^{-9}	2.798933064×10^{-5}
9	6.045085961×10^{-5}	4.967381754×10^{-1}	1.241845438×10^{-1}	$2.057008620\times10^{-11}$	2.143254526×10^{-6}
10	9.679528515×10^{-6}	1.603390998×10^{-1}	4.008477495×10^{-2}	$2.382758584\times10^{-13}$	1.234364265×10^{-7}
11	1.105691361×10^{-6}	3.789202401×10^{-2}	9.473006002×10^{-3}	$2.239446102\times10^{-15}$	5.557386369×10^{-9}
12	9.699082749×10^{-8}	8.010458273×10^{-3}	2.002614568×10^{-3}	$1.743328105\times10^{-17}$	$2.015547066\times10^{-10}$
13	6.823169755×10^{-9}	2.205827422×10^{-3}	5.514568556×10^{-4}	$1.143109189\times10^{-19}$	$6.028691151\times10^{-12}$
14	$3.962943779\times10^{-10}$	7.797776236×10^{-4}	1.949444059×10^{-4}	$6.402183756\times10^{-22}$	$1.515307705\times10^{-13}$
15	$1.940877947\times10^{-11}$	2.540234250×10^{-4}	6.350585624×10^{-5}	$3.099033167\times10^{-24}$	$3.249712209\times10^{-15\dagger}$
16	$8.146618828\times10^{-13}$	6.723015045×10^{-5}	1.680753761×10^{-5}	$1.309717407\times10^{-26}$	$6.022021672\times10^{-17\dagger}$
17	$2.968887381\times10^{-14}$	1.441356324×10^{-5}	3.603390811×10^{-6}	$4.875242191\times10^{-29}$	$9.746048238\times10^{-19\dagger}$
18	$9.494896918\times10^{-16}$	2.557045655×10^{-6}	6.392614138×10^{-7}	$1.610736317\times10^{-31}$	$1.390051124\times10^{-20\dagger}$
19	$2.688913155\times10^{-17}$	3.834413557×10^{-7}	9.586033893×10^{-8}	$4.755727176\times10^{-34}$	$1.758712758\times10^{-22\dagger}$
20	$6.795223637\times10^{-19}$	4.946249047×10^{-8}	1.236562262×10^{-8}	$1.262423388\times10^{-36}$	$2.002188565\times10^{-24\dagger}$

Table 4: Monochromatic spectrum, $k_0 = 1$, $\eta_* = 2$: evaluation of the exact expressions by quadrature. The $O(\epsilon^4)$ contribution to C_ℓ is $\epsilon^4\sigma^4[\frac{1}{4}C_\ell^{(22)}/\sigma^4 + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}/\sigma^4]$; the (13) term and the sum are in Table 7 ($\eta_o = 8$). †: below the double-precision cancellation floor, not reliable (see text).

Derivation and status. The leading coefficients (165) are obtained by symbolic expansion of the second-order kernels with $|\mathbf{k}_2|^2 = k_1^2 - 2Kk_1c + K^2$; the power counting is elementary. Numerically the total integrand $\sum_m |\tilde{K}_{\ell m}|^2$ at the smallest node $K = 0.0018$ is 1.335 ($\ell = 2$), 63.1 ($\ell = 4$) for $\eta_o = 8$ and 12.01, 0.995 for $\eta_o = 4$, constant to three digits down from $K = 0.04$; odd ℓ vanish as K^2 (at $K = 0$ the configuration is invariant under $\mathbf{k}_1 \leftrightarrow -\mathbf{k}_2$); and even $\ell = 0$ is finite in the total (213.5 at $\eta_o = 8$), i.e. the K^{-2} pieces of individual channels cancel in the monopole; this cancellation is not needed for $\ell \geq 2$. This corner is unrelated to the IR region of Sec. 11.4: both legs stay hard ($k_1 = k_2 = k_0$).

12.4 Evaluation at two parameter points

Table 4 gives the *evaluation of the exact expressions* (162), (163) *by quadrature* at $k_0 = 1$, $\eta_* = 2$ and $\eta_o = 8$ ($k_0\chi_* = 6$) or $\eta_o = 4$ ($k_0\chi_* = 2$). The time integrals ($\mathcal{K}_{\Phi_2}, \mathcal{K}_h$ (61), which have closed forms but are evaluated numerically for speed, and the functionals (139)) are done on a Chebyshev grid with Gauss–Legendre Green's-function integrals, the K integral by n_K-point Gauss–Legendre on $[0, 2k_0]$, and the multipoles by an $n_\theta \times 2n_\theta$ Gauss–Legendre sky grid. These are numbers, not closed forms.

Reliability. The quadrature is converged to $\leq 1.5 \times 10^{-11}$ relative for $\ell \leq 20$ at $\eta_o = 8$ and to $\leq 4 \times 10^{-11}$ for $\ell \leq 12$ at $\eta_o = 4$; entries marked † lie below the double-precision cancellation floor (settings and the type-B control: ledger L-0). The kernel has passed the checks of Sec. 11.5, and the assembly is confirmed by the Monte Carlo of Sec. 12.8.

ℓ	B prod	B 1×1	$\Phi_2\Psi_2$	V_2	δ_{T2}	h_2	ray 1×1	defl	emi shift	lens	delay	cross
2	0.025	0.052	0.156	0.007	0.005	0.008	0.011	0.505	0.000	0.992	0.196	−0.957
3	0.089	0.009	0.365	0.016	0.004	0.006	0.042	1.776	0.002	3.302	0.346	−4.958
4	0.103	0.014	0.297	0.014	0.005	0.003	0.051	1.851	0.002	3.590	0.216	−5.145
6	0.100	0.032	0.231	0.018	0.009	0.001	0.110	3.002	0.005	4.825	0.231	−7.567
8	0.016	0.045	0.117	0.010	0.009	0.000	0.080	3.204	0.003	3.134	0.316	−5.934
10	0.003	0.029	0.190	0.023	0.056	0.000	0.082	6.014	0.014	3.381	0.893	−9.686
15	0.001	0.035	0.107	0.071	0.097	0.000	0.020	4.834	0.392	0.600	0.379	−5.538
20	0.002	0.023	0.011	0.020	0.016	0.000	0.001	0.775	0.305	0.013	0.012	−0.178

Table 5: Diagonal channel-group contributions $G_{gg}(\ell)/C_\ell^{(22)}$ for $k_0 = 1$, $\eta_* = 2$, $\eta_o = 8$ (rounded to 10^{-3}). Groups: B prod = $K^{\mathrm{B,prod}}$; B 1×1 = endpoint products; type C columns as in (142); defl, emi shift, lens, delay = $K^{\mathrm{D}}_{\mathrm{defl}}$, $K^{\mathrm{D}}_{\mathrm{emi}}$, $K^{\mathrm{D}}_{\mathrm{lens}}$, $K^{\mathrm{D}}_{\mathrm{delay}}$ (ray + emission parts). Not shown: vector ($< 5 \times 10^{-4}$), clock and displacement (0). "cross" $= 1 - \sum_g G_{gg}/C_\ell^{(22)}$.

Structure. Table 5 lists diagonal channel-group contributions $G_{gg}(\ell)/C_\ell^{(22)}$ at $\eta_o = 8$ (cross terms $g \neq g'$ in the last column). There are strong cancellations: the lensing (transverse displacement) and aberration+deflection (transverse direction) remappings are individually several times $C_\ell^{(22)}$ and largely cancel against each other through the cross terms, as expected for the remapping of a map that is itself generated along the ray. The vector $(\omega_2 + v_2^T)$ diagonal is $< 5 \times 10^{-4}$ and the observer clock and displacement diagonals vanish identically for $\ell \geq 2$ (Remark 11.1). Because the first-order spectrum is band limited ($k_0\chi_* = 6$, so $C_\ell^{(11)}$ falls steeply above $\ell \approx 6$) while K reaches $2k_0$ in (22), the ratio $\frac{1}{4}C_\ell^{(22)}/C_\ell^{(11)}$ grows rapidly with ℓ (15 at $\ell = 2$, $\sim 10^3$ at $\ell = 6$–8 for $\eta_o = 8$); the perturbative ordering in $\epsilon^2\sigma^2$ is therefore ℓ dependent in this toy spectrum.

12.5 The (13) term

Collapse. With (161) every momentum in κ sits on the shell, $|\mathbf{k}| = |\mathbf{q}| = k_0$, and by (128) the $\mathbf{q}$ integral of (151) becomes an angular average, $\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q) F \to \sigma^2 \langle F \rangle_{\hat{\mathbf{q}}}$:

$$\kappa(k_0\hat{\mathbf{k}}; \hat{\mathbf{n}}) = \sigma^2 \frac{1}{2} \int_{-1}^{1} \mathrm{d}\mu \, \frac{1}{2} \langle \mathrm{coef}_{abc}(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}}) \rangle_\varphi, \qquad C_\ell^{(13)} = 4\pi\sigma^2 \, \Delta_\ell(k_0) \, \Delta_\ell^\kappa(k_0)^*, \tag{167}$$

where $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{q}}$, φ is the azimuth of $\hat{\mathbf{q}}$ about $\hat{\mathbf{k}}$, and Δ_ℓ^κ ($\propto \sigma^2$) is the Legendre coefficient of κ (150); in units of σ^4, $C_\ell^{(13)} = 4\pi\Delta_\ell\Delta_\ell^{\kappa *}$ with Δ_ℓ^κ per unit σ^2. We write $\Delta_\ell^\kappa = \frac{1}{2}\int \mathrm{d}\mu \, \Delta_\ell^\kappa(\mu)$, with $\Delta_\ell^\kappa(\mu)$ the Legendre coefficient (in $\nu = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}}$) of $\frac{1}{2}\langle \mathrm{coef}_{abc} \rangle_\varphi$ at fixed μ. The integrand is even in μ (the configuration $(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ is symmetric under $\mathbf{q} \to -\mathbf{q}$), and the sky monopole enters through $\langle \langle m_1 \rangle_{\mathrm{sky}}^2 \rangle = \sigma^2\Delta_0^2$ and $\langle\langle m_2 \rangle\rangle = -7.46928\,\sigma^2$ ($\eta_o = 8$).

The collinear limit $\mu \to \pm 1$: statement. At $\mu \to 1$ the pair $(\mathbf{k}, -\mathbf{q})$ has total momentum $p = |\mathbf{k} - \mathbf{q}| = k_0\sqrt{2(1-\mu)} \to 0$ (at $\mu \to -1$ the pair $(\mathbf{k}, \mathbf{q})$; the integrand is even in μ, and the two limits agree to 1.5×10^{-14}). Every Poisson-inverted second-order object of that pair carries $p^{-2} = [2k_0^2(1-\mu)]^{-1}$, e.g. the amplitude $C_3 \to 16/[9(1-\mu)]$ (46), so individual channels of $\Delta_\ell^\kappa(\mu)$ have simple poles and their μ-integrals diverge logarithmically. Define the channel residues $R_{c,\ell} = \lim_{\mu\to 1}(1-\mu)\Delta_{\ell,c}^\kappa(\mu)$. Exactly six channels have non-zero residues: the third-order velocity $\mathbf{v}_{3c} = \nabla V_{3c}$ (class (iii), including C_3U), the radial time delay with $\Delta\mathbf{x}_2$, the second-order clock $\delta\eta_{o2}$, the second-order emission shift $\delta\eta_{e2}$, the local 2+1 products and the radial $\mathbf{k}_2$.

channel	$\ell=0$	1	2	3	4	5	6	7	8
class (iii) $\mathbf{v}_{3c}$	−0.344109	0.007989	0.011686	−0.058296	−0.075801	−0.058979	−0.034176	−0.015961	−0.006274
time delay, radial $\Delta\mathbf{x}_2$	0.354859	−0.050939	−0.280538	0.059482	0.081315	0.053080	0.020982	0.003707	−0.001550
clock $\delta\eta_{o2}$	0	0	0.239315	0	0	0	0	0	0
emission $\delta\eta_{e2}$	−0.010750	0.042950	0.029538	−0.001186	−0.005514	0.005899	0.013194	0.012254	0.007824
local 2+1	0.009983	−0.040507	0.024701	0.019679	0.004434	−0.002613	−0.003785	−0.002757	−0.001508
radial $\mathbf{k}_2$	−0.009983	0.040507	−0.024701	−0.019679	−0.004434	0.002613	0.003785	0.002757	0.001508
sum (extracted)	-6×10^{-11}	2×10^{-11}	-2×10^{-11}	3×10^{-10}	2×10^{-10}	6×10^{-10}	-7×10^{-11}	-1×10^{-10}	-8×10^{-10}

Table 6: Residues $R_{c,\ell}$ of the six singular channels of the monochromatic (13) integrand at $\mu\to1$ ($k_0=1$, $\eta_*=2$, $\eta_o=8$, per unit σ^2), Richardson extraction with the finite part subtracted. The exact sum is zero (Sec. 12.6); the last row is the sum of the extracted values.

Proposition 12.2 (collinear residues)**.** *The six residues cancel identically, for every* $\hat{\mathbf{n}}$ *and hence for every* ℓ*; the integrand* $\Delta_\ell^\kappa(\mu)$ *is analytic in* $1-\mu$ *at* $\mu=1$*, with no* $(1-\mu)^{-1}$ *and no* $(1-\mu)^{-1/2}$ *term, and the monochromatic* $C_\ell^{(13)}$ *is finite without regularisation.*

The proof (Sec. 12.6) shows that the p^{-2} part of the long second-order mode is a residual $K=0$ time reparametrisation of Poisson gauge, to which the observable is blind, and computes the residues explicitly. Table 6 gives them at $k_0=1$, $\eta_*=2$, $\eta_o=8$ (per unit σ^2), extracted numerically by Richardson extrapolation: the per-channel integrand is evaluated at nine points $1-\mu=0.04,\dots,6.25\times10^{-4}$ and fitted by $R/(1-\mu)$ + a polynomial of degree 6 in $1-\mu$, so that the finite part is subtracted. The extracted residues agree with the analytic formulas (175) to $\le 9\times10^{-11}$ in every channel; all other channels (lensing, aberration, transverse $\mathbf{k}_2$ and $\Delta\mathbf{x}_2$, products, normalisation) have $|R|\le 5\times10^{-10}$; the sum is $\le 8\times10^{-10}$, i.e. 2.3×10^{-9} of the largest residue, for $\ell=0,\dots,8$ (ledger L-1). The total integrand approaches its limit linearly in $1-\mu$ (a fitted $(1-\mu)^{-1/2}$ coefficient is $\le 2\times10^{-5}$, consistent with zero at $\sim2\sigma$; pure polynomial fits leave residuals 10^{-11}–3×10^{-10}), with

$$F_\ell=\lim_{\mu\to1}\Delta_\ell^\kappa(\mu)=-52.7128,\ 28.7446,\ -6.2552,\ -18.9101,\ -3.2837,\ -13.2462,\ 0.3738,\ 1.1248,\ 2.9794 \qquad (\ell=0, \tag{168}$$

The $\sqrt{3}$ resonances ($|\mathbf{k}\mp\mathbf{q}|=2k_0/\sqrt{3}$, $\mu=\pm\frac{1}{3}$) are harmless at finite η: the Green's-function integrals are entire functions of the frequencies (cf. App. B, Cin and Si are entire), so the integrand is analytic there.

Evaluation. Δ_ℓ^κ is evaluated by Gauss–Legendre quadrature in μ (no regularisation), uniform quadrature in φ and Gauss–Legendre quadrature in ν for the Legendre projection, at $k_0=1$, $\eta_*=2$, $\eta_o=8$. The φ integral needs $N_\varphi\ge16$ (the phases $e^{i\mathbf{q}_\perp\cdot\hat{\mathbf{n}}_\perp\chi}$ with $k_0\chi_*=6$; at $\mu=0.5$, $\Delta_2^\kappa(\mu)=-0.970,\ -2.82625,\ -2.82632$ for $N_\varphi=8,16,32$). Two settings, $(N_\mu,N_\varphi,N_\nu)=(16,16,32)$ and $(12,20,32)$, agree in $\frac{1}{3}\operatorname{Re}C_\ell^{(13)}$ to $\le 8\times10^{-5}$ relative for $\ell\le10$, 3×10^{-4} at $\ell=12$ and 2×10^{-3} at $\ell=14$; for $\ell\ge16$ the values ($<10^{-9}$) are not reliable and are irrelevant next to $C_\ell^{(22)}$. $C_\ell^{(11)}=4\pi\Delta_\ell^2$ from the same code agrees with (162) to 3×10^{-14}, and $\Im C_\ell^{(13)}\sim10^{-14}$. The transfer functions of κ are (per unit σ^2)

$$\begin{aligned}\Delta_\ell^\kappa&=-96.795,\ 53.949,\ -4.6756,\ -25.171,\ -18.333,\ -8.3515 \qquad (\ell=0,\dots,5),\\ \Delta_\ell^\kappa&=-1.0090,\ 1.6018,\ 1.5573,\ 0.89536,\ 0.39797 \qquad (\ell=6,\dots,10).\end{aligned} \tag{169}$$

There are strong cancellations between channels: for $\ell=2$ the largest contributions to $\Delta_2^\kappa=-4.68$ are class (iii) (−22.47), aberration+deflection × second-order fields (+12.54), radial $\mathbf{k}_2$ (−11.40),

clock (+11.14), time delay × second order (+7.00), lensing × second order (−6.76), transverse $\mathbf{k}_2$ (+5.52), $-3\langle m_2\rangle_{\rm sky} m_1$ (+3.96) and $3L_1L_2$ (−4.00). The (13) term was not evaluated at $\eta_o = 4$.

12.6 The collinear residues are a residual gauge mode

This subsection proves Proposition 12.2 (ledger L-5, L-6). Momenta are in units of $k = |\mathbf{k}| = |\mathbf{q}|$, $p = |\mathbf{k} - \mathbf{q}|$, $p^2 = 2k^2(1-\mu)$; $\mu \to -1$ follows by $\mathbf{q} \to -\mathbf{q}$.

Step 1: the pole of a long second-order mode. Consider the pair kernels of Sec. 6 for two first-order modes of equal magnitude s and total momentum $P \to 0$, and write $T = T_\Phi(s\eta/\sqrt{3})$, $T' = \partial_\eta T$. Symbolic expansion gives: only K_Q, K_{S_Φ} and $K^0_{\Phi_2}$ have a pole; it is a single P^{-2} term (no P^{-4}); $K_{S_{00}}$, $K_{S_{0z}}$, K_h have no pole, and K_ω, K_{v^T} vanish at equal magnitudes. Explicitly

$$K_Q = \frac{A_Q(\eta)}{P^2} + O(1), \qquad A_Q = s^2[2T^2 + (T+\eta T')^2], \qquad \mathrm{pole}(K_{S_\Phi}) = \frac{A'_Q}{\eta}, \qquad \mathrm{pole}(K^0_{\Phi_2}) = \frac{4s^2}{9} = \frac{A_Q(0)}{3}, \tag{170}$$

the second relation exact once T'' is eliminated with the first-order equation. Define $\tau(\eta)$ as the regular solution of

$$(\eta^2\tau)' = -\eta^2 A_Q, \qquad \tau = -\sqrt{3}\, s\, \frac{x^3(2x - \sin 2x) - 8x^2\cos^2 x + 8x\sin 2x + 4\cos 2x - 4}{x^5} = -\frac{4}{9}s^2\eta + O(\eta^3), \qquad x = \tag{171}$$

With the $k \to 0$ Green's function $G_\Phi \to \frac{\eta'}{3}(1 - \eta'^3/\eta^3)$ (60) and one integration by parts, the pole parts of the second-order fields are exactly

$$(\Phi_2,\ \Psi_2,\ \delta_2,\ \delta_{T2})\Big|_{P^{-2}} = \Big(-\frac{\tau}{\eta},\ \tau' + \frac{\tau}{\eta},\ -\frac{4\tau}{\eta},\ -\frac{\tau}{\eta}\Big)\frac{1}{P^2}, \qquad V_2 = -\frac{\tau}{P^2} + O(P^{-1}), \tag{172}$$

and V_2 enters only through $\mathbf{v}_2 = \nabla V_2 = O(P^{-1})$. These are the field shifts of the residual $K = 0$ time reparametrisation $\eta \to \eta + \frac{\epsilon^2}{2}\tau/P^2$ of Poisson gauge (Sec. 11.8, with $F = \tau/P^2$). A constant spatial dilation could also appear at $K = 0$; it is excluded because the pole of $K^0_{\Phi_2}$ equals $A_Q(0)/3$, which is the statement $\zeta^{(2)}(0) = 0$. (Numerically: $P^2\times$ the pair profiles of the engine tend to (172) with error $O(P^2)$, 4×10^{-8} at $P = 10^{-4}$.)

Step 2: the third-order poles are the same gauge mode. Set $q = k$, $\mu = 1 - p^2/2k^2$, $K_{\Phi_2} = a/p^2 + b + cp^2$, $K_h = h_0 + h_1 p^2$ in the contracted sources $K_{3c,X}$. All of them have monomial denominators p^2, hence only a simple p^{-2} pole, and the pole depends only on $a = -\tau/\eta$ and on $T(k)$: the regular parts $b, c, h_{0,1}$ and the tadpoles drop out. Using $\tau' = -A_Q - 2\tau/\eta$, each pole equals exactly $2/p^2$ times the source generated by the gauge shift of the first-order fields of the leg $\mathbf{k}$,

$$\delta\Phi_3 = 3\tau T' + \frac{6\tau}{\eta}T, \quad \delta\Psi_3 = 3\tau T' + 6\Big(\tau' + \frac{\tau}{\eta}\Big)T, \quad \delta V_3 = 3\tau T'_V + 3\tau' T_V, \quad \delta\delta_3 = 3\tau T'_\delta - \frac{12\tau}{\eta}T_\delta \tag{173}$$

($T = T_\Phi(k\eta/\sqrt{3})$ etc.; the factor 2 is that of $2A(q)B_2$ in (63)). The initial amplitude matches: $2\lim_{\eta\to 0}[3\tau T' + 6\tau T/\eta] = 32k^2/9$, i.e. $C_3 \to (32k^2/9)/p^2 = 16/[9(1-\mu)]$, which is (46). Since the regular solution is unique, the poles of $\Phi_{3c}, \Psi_{3c}, \delta_{3c}, V_{3c}$ are exactly $2/p^2$ times (173).

Step 3: only the poles matter, and they act as a gauge transformation. Each term of κ contains a single second-order factor. Scalars are Laurent series in p^2; gradients and $\mathbf{v}_2$ give p_j/p^2, and after the azimuthal average about $\hat{\mathbf{k}}$ these are either odd (and vanish) or come with $\mathbf{p}\cdot\hat{\mathbf{k}} = p^2/2k$, which makes them $O(1)$. Hence the residue is the response of the map of the leg $\mathbf{k}$ to the homogeneous pole fields alone, i.e. the response to the time reparametrisation with parameter

$$\tilde{\tau}(\eta) \equiv \frac{\tau(\eta)}{2k^2} \qquad (\text{since } p^{-2} = [2k^2(1-\mu)]^{-1}). \tag{174}$$

The same structure (the integrand depends only on p^2, $\mathbf{p}\cdot\hat{\mathbf{k}} = p^2/2k$ and even powers of $p_\perp$) excludes a $(1-\mu)^{-1/2}$ term.

Step 4: explicit residues. Let $I_1(\eta,\nu) = -k^2(\nu^2 - \frac{1}{3})T_V(\eta)\, e^{ik\nu\chi}$ be the first-order shear integrand on the ray ($\nu = \hat{\mathbf{n}}\cdot\hat{\mathbf{k}}$; $\Delta_\ell = \frac{1}{2i^\ell}\int \mathrm{d}\nu\, P_\ell \cdot$ for the Legendre coefficients). The responses of the six channels are

$$\begin{aligned} R_{\rm clock} &= 3\tilde{\tau} I_1|_{\eta_o}, \qquad R_{\rm emi} = -3\tilde{\tau} I_1|_{\eta_*}, \qquad R_{\rm TD} = 3\int_{\eta_*}^{\eta_o} \tilde{\tau}\,(ik\nu)\, I_1\, \mathrm{d}\eta, \\ R_{v_3} &= -3\int_{\eta_*}^{\eta_o} [\tilde{\tau}\,\partial_\eta I_1 + \tilde{\tau}' I_1]\mathrm{d}\eta, \qquad R_{2+1} = 6\int_{\eta_*}^{\eta_o} \tilde{\tau}' I_1\, \mathrm{d}\eta = -R_{\mathbf{k}_2,r}, \end{aligned} \tag{175}$$

where ∂_η in R_{v_3} acts on T_V only. They come, respectively, from $\delta\eta_{o2} \to -\tilde{\tau}(\eta_o)$, $\delta\eta_{e2} \to -\tilde{\tau}(\eta_*)$, $\Delta x_{2,\parallel} \to -\tilde{\tau}(\eta)$, δV_3 of (173), and $\hat{\mathbf{n}}\cdot\mathbf{k}_2 \to -(\Phi_2+\Psi_2) = -\tilde{\tau}'$. The clock residue is pure $\ell = 2$, $\Delta_2 = \frac{2}{5}k^2\tilde{\tau}(\eta_o)T_V(\eta_o) = 0.2393147529$ at the parameter point of Table 6. Since $\mathrm{d}e^{ik\nu\chi}/\mathrm{d}\eta = -ik\nu\, e^{ik\nu\chi}$,

$$R_{v_3} + R_{\rm TD} = -3\int_{\eta_*}^{\eta_o} \frac{\mathrm{d}}{\mathrm{d}\eta}[\tilde{\tau} I_1(\eta,\nu)]\mathrm{d}\eta = -3[\tilde{\tau} I_1]_{\eta_o} + 3[\tilde{\tau} I_1]_{\eta_*} = -R_{\rm clock} - R_{\rm emi}, \tag{176}$$

and $R_{2+1} + R_{\mathbf{k}_2,r} = 0$: the six residues cancel identically in ν, hence for every ℓ. □

Physical reason and status. Under $\eta \to \eta + \tilde{\tau}(\eta)$ at fixed $\mathbf{x}$ the observer's proper-time clock, the $T_{\rm phys} = T_*$ surface, the label and the canonical triad are invariant, while the ray, the clock reading and the emission time shift by $\tilde{\tau}$ and $\mathbf{v}$ acquires a Lie-derivative shift; Θ is a scalar, so the total response vanishes, and the six channels are exactly those that respond to the residual gauge mode (item 2 of Sec. 11.7). Status: Steps 1 and 2 are symbolic identities (computer algebra); the structural statements of Step 3 are analytic and confirmed numerically; the formulas (175) were written by hand from the group structures of the term lists and agree with the engine's channels to $\le 1.2\times10^{-14}$ ($\ell = 0,\dots,8$), and their cancellation is an exact identity. The gauge response of κ to $\tilde{\tau}$ through the tadpole and third-order routes is $O(0.36)$ per channel and 1.2×10^{-14} in total. The argument presupposes that the term lists of L_3 represent $\Theta^{(3)}$ correctly, which is established independently (R-9, PW-2).

12.7 The combination $\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}$

Table 7 gives the $O(\epsilon^4)$ coefficient of C_ℓ at $k_0 = 1$, $\eta_* = 2$, $\eta_o = 8$, next to $C_\ell^{(11)}$. It is an *evaluation* of the exact expressions (163), (167) by quadrature, not a closed-form result. Both terms are computed in the same frame (the canonical triad, Sec. 4.5); only their sum is independent of the observer frame.

ℓ	$C_\ell^{(11)}/\sigma^2$	$\frac{1}{4}C_\ell^{(22)}/\sigma^4$	$\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}/\sigma^4$	$[\frac{1}{4}C_\ell^{(22)}+\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}]/\sigma^4$	ratio
2	9.546006×10^{-1}	1.425754×10^{1}	-5.39801×10^{0}	$+8.85953\times10^{0}$	$+9.28$
3	6.250780×10^{-1}	2.878135×10^{0}	-2.35152×10^{1}	-2.06370×10^{1}	-33.0
4	1.750713×10^{-1}	1.865919×10^{0}	-9.06394×10^{0}	-7.19802×10^{0}	-41.1
5	1.933453×10^{-2}	1.130143×10^{0}	-1.37218×10^{0}	-2.42041×10^{-1}	-12.5
6	1.770350×10^{-4}	5.220624×10^{-1}	-1.58643×10^{-2}	$+5.06198\times10^{-1}$	2.9×10^{3}
7	3.174581×10^{-4}	3.740495×10^{-1}	-3.37230×10^{-2}	$+3.40326\times10^{-1}$	1.1×10^{3}
8	2.304635×10^{-4}	2.619739×10^{-1}	-2.79364×10^{-2}	$+2.34038\times10^{-1}$	1.0×10^{3}
9	6.045086×10^{-5}	1.241845×10^{-1}	-8.22586×10^{-3}	$+1.15959\times10^{-1}$	1.9×10^{3}
10	9.679529×10^{-6}	4.008477×10^{-2}	-1.46307×10^{-3}	$+3.86217\times10^{-2}$	4.0×10^{3}
11	1.105691×10^{-6}	9.473006×10^{-3}	-1.8408×10^{-4}	$+9.28892\times10^{-3}$	8.4×10^{3}
12	9.699083×10^{-8}	2.002615×10^{-3}	-1.7666×10^{-5}	$+1.98495\times10^{-3}$	2.0×10^{4}

Table 7: The $O(\epsilon^4)$ coefficient of the monochromatic C_ℓ, $k_0 = 1$, $\eta_* = 2$, $\eta_o = 8$: evaluation by quadrature. "ratio" $= [\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}]/C_\ell^{(11)}$ in units of σ^2, so that $C_\ell = \epsilon^2\sigma^2 C_\ell^{(11)}/\sigma^2\,[1 + \epsilon^2\sigma^2 \times \text{ratio}]$. Relative accuracy of the (13) column: 8×10^{-5} ($\ell \le 10$), 3×10^{-4} ($\ell = 12$); of the (22) column: Sec. 12.4.

Remarks. (a) For $\ell = 13, 14$, $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = -1.351\times10^{-6}$, -8.49×10^{-8} (accuracy 10^{-3}), i.e. 2.5×10^{-3} and 4×10^{-4} of $\frac{1}{4}C_\ell^{(22)}$; for $\ell \ge 15$, $|\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}| < 5\times10^{-9}$ and the combination equals $\frac{1}{4}C_\ell^{(22)}$ of Table 4 to 10^{-4} relative. (b) Signs: the ϵ^4 term is a correction to a positive quantity, not a power spectrum, and it can be negative: $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} < 0$ for all evaluated $\ell \ge 2$, and the combination is negative for $\ell = 3, 4, 5$, where the (13) term dominates. At $\ell = 2$, $\frac{1}{3}\,\mathrm{Re}\,C^{(13)}$ cancels 38% of $\frac{1}{4}C^{(22)}$. Positivity of C_ℓ through this order requires $\epsilon^2\sigma^2 < 1/|\text{ratio}|$, e.g. $\epsilon^2\sigma^2 < 0.030$ at $\ell = 3$ and < 0.024 at $\ell = 4$. (c) For $\ell \ge 6$ the first-order spectrum is tiny (the shell at k_0 with $k_0\chi_* = 6$ is band limited) while the loop reaches total momenta up to $2k_0$; the ϵ^4 terms then exceed $C_\ell^{(11)}$ by three to four orders of magnitude per unit $\epsilon^2\sigma^2$, so the perturbative ordering is strongly ℓ dependent for this toy spectrum (Sec. 12.4). (d) Reliability: the kernel κ has passed the checks of Sec. 11.7 (form equivalence at third order, residual-gauge invariance, reality) and the squeezed test of Sec. 11.9; the finiteness of the monochromatic number rests on Proposition 12.2, proved in Sec. 12.6; no end-to-end test of $C_\ell^{(13)}$ itself (e.g. Monte Carlo) exists. (e) $\eta_o = 4$ was not evaluated for (13).

12.8 Monte-Carlo cross-check of $C_\ell^{(22)}$

The quadrature of Sec. 12.4 was cross-checked by a Monte Carlo that shares only the kernel evaluator K_2 with it (ledger L-4). The field is a finite sum of $N = 20$ plane waves on the shell, $\zeta = \sum_{j=1}^{N}(\sigma/\sqrt{N})[a_j\cos(\mathbf{k}_j\cdot\mathbf{x}) + b_j\sin(\mathbf{k}_j\cdot\mathbf{x})]$, $|\mathbf{k}_j| = k_0$, isotropic random directions, a_j, b_j standard normal; $\Theta^{(2)}$ is summed over all ordered pairs of modes with the full K_2 (normalisation term included, checked realisation by realisation), the $a_{\ell m}$ are obtained by sky quadrature, and $C_\ell^{(22)}$ is estimated from 45 direction sets $\times$ 4000 amplitude draws (1.8×10^5 realisations), omitting the $i = j$ pairs and rescaling by $N/(N-1)$, which makes the estimator unbiased. No $\hat{\mathbf{K}} = \hat{\mathbf{z}}$ frame, no μ or K quadrature, no Wick formula and no m-sum identity are used.

ℓ	2	3	4	6	8
$\frac{1}{4}C_\ell^{(22)}/\sigma^4$, Monte Carlo	14.174(100)	2.8564(208)	1.8724(99)	0.5201(32)	0.2603(26)
$\frac{1}{4}C_\ell^{(22)}/\sigma^4$, quadrature (Table 4)	14.2575	2.87813	1.86592	0.522062	0.261974

For $\ell = 10$: $3.9864(25) \times 10^{-2}$ vs 4.00848×10^{-2}. All $\ell = 1, \ldots, 11$ agree within 1.1σ at 0.3–0.7% precision ($\ell = 12$: $+2.0\sigma$, 1.4%); the errors are correlated across ℓ (common direction sets). $C_\ell^{(11)}$ is reproduced to 0.1–0.2%. Sensitivity: a wrong Wick factor would be a 50% effect, and omitting the normalisation term $-2\langle m_1\rangle_{\rm sky} m_1$ would shift $C_3^{(22)}$ by 4.2%, a 6σ discrepancy. The test therefore confirms the harmonic projection, the Wick factor, the factor $\frac{1}{4}$, the normalisation and the monochromatic reduction; since it uses the same K_2 evaluator, it does not independently test K_2 itself (which is tested in Sec. 11.5).

13 Checks

This section summarises the verifications of the chain exact observable → field equations → line-of-sight functionals → observer bookkeeping → harmonic machinery. Each item cites check identifiers of the ledger (App. G), where the numbers are recorded; the code is listed in App. F. The classification analytic / numerical / pending is collected in the status table of Sec. 3. Checks of the assembled $C_\ell^{(22)}$ and of the monochromatic evaluation are described with those results (Secs. 11.5, 12.4).

13.1 The exact formula

- *Algebraic identities* (analytic). Lemma 4.1 in a generic Lorentzian metric with no symmetry, for generic u and null p; Lemma 4.2 for general ρ, P, with the converse that the identities hold iff $w = \frac{1}{3}$ for $P = w\rho$, $\rho \propto T^4$; the off-shell identity behind Theorem 4.3 vanishes exactly on shell (X-1–X-3).
- *Exact solutions.* FRW and Milne: $T_{\rm obs} = T_o$ symbolically; Bianchi I with a radiation test fluid (Kasner $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and non-Kasner exponents), where the anisotropy is $O(1)$: $|T_{\rm obs}^{\rm exact} - T_o e^{-S}| \lesssim 2 \times 10^{-31}$, i.e. at quadrature precision (X-4). Linear Poisson gauge: the Sachs–Wolfe + Doppler + ISW formula with observer terms is recovered (X-5).
- *Non-perturbative.* In an exact planar solver the photon energies are integrated from the geodesic equation and the fluid temperatures are read off independently at emission and observation; $\max_{\hat{\mathbf{n}}} |T_{\rm obs}/T_o - e^{-S}| = 7.5 \times 10^{-15}$ at $\epsilon = 0.01$ and 1.4×10^{-14} at $\epsilon = 0.1$ ($\Theta_{\rm rms} = 0.08$), and $\leq 7.5 \times 10^{-14}$ in every production run (X-10).
- *Symbolic, through third order* (analytic). The photon energy computed from $\hat{\Gamma}^0_{kk}$ (direct route) and e^{-S} agree (a) at the integrand level along an arbitrary perturbed ray at orders 1, 2, 3 modulo $\hat{\nabla}_\mu \hat{T}^{\mu\nu} = 0$ only (X-7–X-9), and (b) at the level of the assembled, background-ray-expanded functionals, piece by piece (ray, obs, emi), orders 1, 2, 3 (R-4). The two assembled forms agree numerically for the plane wave below to 4×10^{-10} (L_2) and 9×10^{-7} (L_3) (PW-4).

13.2 Einstein equations

- *Internal* (analytic). Background parts vanish; the contracted Bianchi identity holds identically through $O(\epsilon^3)$, so the constraints propagate; the first-order system reproduces the

standard equations and the conservation laws hold on shell; the exact dilation family solves the equations through $O(\epsilon^3)$ (EF-1–EF-6).

- *Independent re-derivation* (analytic). A separately coded computation gives $E_{\mu\nu}$ term by term identical for all ten components at orders 1, 2, 3 (EF-15; same method, independently coded, hence complemented by the literature controls). It confirms the linear solution, the operator structure at orders 2 and 3, all second-order kernels (agreement to 10^{-164} at random points), the algebraic ω_2 from the relativistic Kelvin theorem and its uniqueness, the superhorizon Φ_2, the third-order contraction rules (brute-force Wick), the contracted third-order sources, and C_3 re-derived at three random points to 30 digits (EF-16–EF-20).

- *Literature controls.* Kohri–Terada [12]: $K_h = 4f_{\rm RD}$, relative difference 5×10^{-29} (EF-9, EF-17); Pitrou–Uzan–Bernardeau [10]: scalar sources (A.1)–(A.8) except the trace source (A.6), whose printed form is visibly garbled (our trace source is fixed by two independent computations and by the exact dilation family), and the superhorizon initial data (3.12b,c) exactly (EF-10, EF-18); Lu et al. [15]: the radiation-era vector kernel is identical to K_ω (EF-19).

- *Closed forms* (high precision). The closed-form $\mathcal{K}_h$ equals Kohri–Terada (22) [with the modulus in the logarithm] to 10^{-15} (25 digits with rational input) and their late-time form (25) exactly; the closed-form Φ_2 agrees with 60-digit quadrature to 3.5×10^{-40}; all second-order fields assembled from the closed forms satisfy the full second-order Einstein and conservation equations with relative residual 5.5×10^{-50} (CF-1–CF-3); the printed form of App. B was re-checked against quadrature (CF-4).

- *Plane-wave fields.* Orders-1–3 plane-wave fields generated from the same field equations satisfy the conservation equations to $10^{-27}/5 \times 10^{-21}$ (orders 2/3) and reproduce the non-perturbative maps (Sec. 13.4), which tests the equations together with the initial-data prescription $\zeta_{\rm nl}^{(2)}(0) = \zeta_{\rm nl}^{(3)}(0) = 0$ (PW-1).

13.3 Gauge invariance at $k = 0$: the dilation family

The observable is defined by physical labels only: $T_{\rm phys}$ surfaces (emission and initial event), proper time along a fluid worldline, the photon direction relative to an orthonormal frame, and the integral of the scalar $\sigma_{\mu\nu}e^\mu e^\nu$ along a geometric curve. Under a change of coordinates each of these is invariant, except that the canonical triad (built from ∂_i) rotates; by Sec. 4.5 that rotation drops out of C_ℓ. The perturbative chain, however, is written in Poisson gauge, and gauge invariance is a non-trivial test of it. The exact dilation family (47) is a pure coordinate transformation of FRW and is the $k = 0$ test:

- *The family* solves the field equations through $O(\epsilon^3)$ (EF-6) and non-perturbatively (EF-21); it carries $\zeta_{\rm nl} = \ln\beta$ exactly and, with $\beta = e^{\epsilon c}$, is the $k \to 0$ limit of the regular solution with a free residual time rescaling a_2 at second order.

- *Cancellation in the observable* (analytic). For $\lambda = 1 + \epsilon a_1 + \dfrac{\epsilon^2}{2}a_2$, $a_1 = -c/3$, a_2 free, $\beta = e^{\epsilon c}$: $\hat\sigma_{\mu\nu} \equiv 0$ through third order; the assembled L_1, L_2, L_3 vanish in the shear *and* the direct form, with $\delta\eta_{e1} = c\eta_*/3$, $\delta\eta_{e2} = (-a_2 + \dfrac{2}{9}c^2)\eta_*$ inserted (the exact $\eta_*(1/\lambda - 1)$ series); the clock gives $\eta_o = \bar\eta_o/\lambda$ through second order for any a_2 and any η_i, hence $T_o = \bar T(\bar\eta_o)$ and $\Theta = 0$ (R-6, R-8). In the e^{-S} formulation the anisotropy part is automatic (S involves only derivatives of $\hat u$), so the non-trivial content is the direct form and the monopole at fixed proper time.

- *Squeezed limit of the initial data.* $C_3(q \to 0) = -\dfrac{32}{9}$, the value predicted by the dilation family (EF-14, EF-20).
- *Zero modes* (analytic and numerical; Sec. 11.8). The only residual $k = 0$ freedom is the time reparametrisation $F(\eta)$; the map is invariant to second order under every zero mode, up to a rigid rotation and a total derivative (ZM-1–ZM-3); the two isotropic tadpole members differ by a pure-gauge shift (ZM-4). Numerically: regenerating the plane-wave fields with a different zero-mode assignment ($\Psi_{2,0} = \Phi_{2,0} + c_g\eta$, a residual time reparametrisation) changes $\Phi_{2,0}, \Psi_{2,0}, \delta_{2,0}$ and all third-order fields at $O(1)$ ($\Psi_{2,0}(8)$: $-1.21 \to +4.79$) and $\delta\eta_{o2}$ ($4.411 \to -11.589$), but every $a_{\ell 0}$ of $\Theta^{(1,2,3)}$, the sky monopoles and T_o are unchanged to $\leq 2 \times 10^{-14}$ (PW-3); the amplitude C_3, summed over the two orderings of the contracted momenta, shifts by 8 per unit shift of $\langle\Phi_2\rangle$ (EF-20): individual tadpoles are gauge dependent, the observable is not. For general momenta the contracted third-order solution is covariant to 10^{-13} (EF-22) and the fully assembled κ is unchanged to 6×10^{-14} ($\tau = 0.1\eta$) and 3×10^{-13} ($\tau = 0.1\eta^3$) (R-11). The third-order invariance under $F(\eta)$ is inferred from the multilinear identity and these tests; it was not run as a separate symbolic third-order check.
- *Collinear residues* (analytic; Sec. 12.6). The p^{-2} part of a long second-order mode is the residual time reparametrisation, and the six collinear residues of the (13) integrand cancel identically (L-5, L-6; numerically L-1).
- *First-order gradient mode.* $\Delta_0 = O(k^4)$, $\Delta_1 = O(k^3)$ as $k \to 0$ (FO-3): constant and gradient modes are invisible at first order.

13.4 Gauge invariance at finite k: plane wave versus the non-perturbative solver

Set-up. For $\zeta = \epsilon\cos x$ (single plane wave, $k = 1$) the full planar Einstein–radiation system was solved non-perturbatively in *comoving-orthogonal gauge*, $\mathrm{d}s^2 = -\alpha^2\mathrm{d}t^2 + A^2\mathrm{d}x^2 + B^2(\mathrm{d}y^2 + \mathrm{d}z^2)$ with $u = \alpha^{-1}\partial_t$ and the lapse fixed algebraically by Lemma 4.2, $\alpha = e^{-\zeta}(AB^2)^{1/3}$, $T_{\mathrm{phys}} = 1/\alpha$; regular non-linear initial data at $t_i = 0.01$, ray tracing to the hypersurface $T_{\mathrm{phys}} = T_*$, the observer clock at fixed proper time from the $T_{\mathrm{phys}} = T_i$ crossing, and the normalisation of Corollary 4.5; $\Theta^{(1,2,3)}$ were extracted from 16 amplitudes $|\epsilon| \leq 0.1$ by even/odd polynomial fits, with an uncertainty defined as the spread between fit subsets (NP-2). Independently, the analytic chain of this paper was evaluated in *Poisson gauge*: the field equations reduced to the plane wave (fields to third order as exact Green's-function integrals, 40-digit arithmetic), the symbolic-$\hat{\mathbf{n}}$ shear-form lists L_1, L_2, L_3 (App. C) with the ray functionals (86)–(89), the emission shifts (93), the clock and displacement of Sec. 8.2 (with the $O(\eta_i^2)$ clock terms at $\eta_i = 0.01$ and the label as $\eta \to 0$, i.e. the solver's comoving label) and the bookkeeping (32) including $\ell = 0, 1$. No fitting of one to the other is involved. Configurations: $\bar{\eta}_o/\eta_* = 4$ ($\bar{\eta}_o = 8$) and 2 ($\bar{\eta}_o = 4$), $\eta_* = 2$; observer at $x_o = 0$ (maximum of ζ; even ℓ only) and $x_o = \pi/2$ (generic; odd ℓ and dipole). Multipoles $a_{\ell 0} = \sqrt{4\pi(2\ell+1)}\,\dfrac{1}{2}\int \Theta P_\ell\,\mathrm{d}\mu$ about the wave vector.

Why this is a gauge-invariance test. The two computations share no coordinate quantity: the slicing (Poisson versus fluid-orthogonal with $\alpha T_{\mathrm{phys}} = 1$), the spatial coordinates, the time of observation and the field variables all differ. For example, the coordinate time of observation is $\eta_o = 8 + 0.91167\,\epsilon + 4.41138\,\epsilon^2/2$ in Poisson gauge but $t_o = 8 + 2.54456\,\epsilon - 3.2885\,\epsilon^2$ in the solver ($\bar{\eta}_o = 8$, $x_o = 0$). What agrees must therefore be the gauge-invariant content: the sky map defined by physical labels. By planar symmetry both canonical triads are aligned with the symmetry axes,

configuration	order	ℓ	analytic chain (Poisson gauge)	non-perturbative (comoving gauge)	difference
$\bar{\eta}_o = 8$, $x_o = 0$	1	2	-2.1847203553	-2.1847203553	-4.4×10^{-12}
	1	4	$+1.2552456282$	$+1.2552456282$	-3.6×10^{-12}
	1	6	$-4.7973477367 \times 10^{-2}$	$-4.7973477364 \times 10^{-2}$	-3.4×10^{-12}
	2	2	-4.2224942186	-4.2224942193	$+6.5 \times 10^{-10}$
	2	4	$+1.3785439774$	$+1.3785439781$	-7.4×10^{-10}
	2	6	-1.2595382846	-1.2595382817	-2.9×10^{-9}
	3	2	$-1.2889987902 \times 10^{1}$	$-1.2889987968 \times 10^{1}$	$+6.6 \times 10^{-8}$
	3	4	$-4.7070907021 \times 10^{1}$	$-4.7070907098 \times 10^{1}$	$+7.7 \times 10^{-8}$
	3	6	$+6.2905106702$	$+6.2905105363$	$+1.3 \times 10^{-7}$
$\bar{\eta}_o = 4$, $x_o = 0$	1	2	$+2.0752530981 \times 10^{-1}$	$+2.0752530981 \times 10^{-1}$	$+4.3 \times 10^{-12}$
	2	2	-1.3572165380	-1.3572165385	$+5.4 \times 10^{-10}$
	2	4	$-2.2995087658 \times 10^{-1}$	$-2.2995087720 \times 10^{-1}$	$+6.1 \times 10^{-10}$
	3	2	-4.8330628410	-4.8330627544	-8.7×10^{-8}
	3	4	-3.0635108209	-3.0635106964	-1.2×10^{-7}
$\bar{\eta}_o = 8$, $x_o = \pi/2$	1	1	$+2.0812147132$	$+2.0812147132$	$+1.1 \times 10^{-11}$
	1	3	$+2.0917805127$	$+2.0917805128$	-5.1×10^{-11}
	2	2	$+3.0669017921$	$+3.0669017899$	$+2.2 \times 10^{-9}$
	2	4	$+6.5664035508$	$+6.5664035492$	$+1.6 \times 10^{-9}$
	3	1	$-5.1723421459 \times 10^{1}$	$-5.1723420730 \times 10^{1}$	-7.3×10^{-7}
	3	3	$-4.3006733540 \times 10^{1}$	$-4.3006736906 \times 10^{1}$	$+3.4 \times 10^{-6}$
$\bar{\eta}_o = 4$, $x_o = \pi/2$	1	1	$-2.3149107586 \times 10^{-1}$	$-2.3149107587 \times 10^{-1}$	$+3.8 \times 10^{-12}$
	2	2	-2.1076209609	-2.1076209611	$+2.1 \times 10^{-10}$
	3	1	$+1.3059761781 \times 10^{1}$	$+1.3059761844 \times 10^{1}$	-6.4×10^{-8}
	3	3	$-3.6965575489 \times 10^{-1}$	$-3.6965566762 \times 10^{-1}$	-8.7×10^{-8}

Table 8: Plane-wave comparison of the analytic chain (Poisson gauge) with the non-perturbative planar solver (comoving-orthogonal gauge): $a_{\ell 0}$ of $\Theta^{(r)}$, $r = 1, 2, 3$ ($\ell = 1$ is the dipole, kept here because the comparison is made before dipole removal). Full lists ($\ell \leq 12$): App. F.

so the frame rotation of Sec. 4.5 is trivial and the comparison holds multipole by multipole at each order, not only for C_ℓ. The test covers the second- and third-order gauge transformations of all scalar quantities entering the map (the solver has no vector modes and only a homogeneous tensor mode).

Result. Tables 8 and 9: the analytic chain reproduces the non-perturbative maps at all three orders in all four configurations, within the extraction uncertainty: first order to $\sim 10^{-11}$, second order to $\sim 10^{-9}$ in $a_{\ell 0}$, third order to $\sim 10^{-7}$ ($x_o = 0$) and $\sim 10^{-6}$ ($x_o = \pi/2$, where the extraction uncertainty itself is 3×10^{-2}) (PW-2). The second-order map differences for $x_o = 0$ exceed the quoted spread by ~ 4 but sit at the solver's stated noise floor; a test that uses no fitting at all settles this: the residual $R(\epsilon) = \Theta_{\text{non-pert}}(\epsilon) - [\epsilon\Theta^{(1)} + \frac{\epsilon^2}{2}\Theta^{(2)} + \frac{\epsilon^3}{6}\Theta^{(3)}]_{\text{analytic}}$ scales cleanly as ϵ^4 for all 16 amplitudes ($\max|R|/\epsilon^4 = 11.6 \ldots 14.5$ smoothly over $-0.1 \leq \epsilon \leq 0.1$; $|R| = 1.2 \times 10^{-7}$ at $\epsilon = 0.01$), i.e. there is no lower-order remainder (PW-6). The gauge-invariant observer quantities agree as well: T_o and $\langle T_{\text{obs}} \rangle_{\text{sky}}$ through $O(\epsilon^2)$, e.g. for $\bar{\eta}_o = 8$, $x_o = 0$: $T_o = 0.125 - 0.0550223132\,\epsilon + 0.0041441746\,\epsilon^2$ (analytic) vs $0.125 - 0.0550222904\,\epsilon + 0.0041441914\,\epsilon^2$ (fit to the solver), with fit uncertainties 10^{-8}–5×10^{-7}. The analytic evaluation is converged (Chebyshev $N = 48/64/96$ identical to all printed digits).

configuration	$\Theta^{(1)}$	$\Theta^{(2)}$	$\Theta^{(3)}$
$\bar{\eta}_o = 8$, $x_o = 0$	1.5×10^{-10} (3.6×10^{-9})	1.3×10^{-7} (3.2×10^{-8})	4.0×10^{-6} (2.1×10^{-4})
$\bar{\eta}_o = 4$, $x_o = 0$	1.3×10^{-11} (6.8×10^{-12})	4.5×10^{-9} (9.7×10^{-10})	2.7×10^{-7} (4.0×10^{-7})
$\bar{\eta}_o = 8$, $x_o = \pi/2$	2.7×10^{-8} (5.3×10^{-7})	4.0×10^{-7} (9.2×10^{-6})	1.8×10^{-3} (3.1×10^{-2})
$\bar{\eta}_o = 4$, $x_o = \pi/2$	2.1×10^{-11} (6.7×10^{-10})	2.4×10^{-9} (5.4×10^{-9})	3.1×10^{-7} (3.9×10^{-5})
map rms ($\bar{\eta}_o = 8$, $x_o = 0$)	0.71	1.31	14.2

Table 9: Maximum over the sky of $|\Theta^{(r)}_{\text{analytic}} - \Theta^{(r)}_{\text{non-pert}}|$, with the extraction uncertainty of the non-perturbative maps in parentheses.

What it tests. Every link of the chain enters: the second- and third-order field equations and the initial-data prescription, the tadpoles at third order, L_1, L_2, L_3 including lensing, time delay, aberration, the direction map with the canonical triad, the emission surface to second order, the clock and observer displacement to second order, and the normalisation (32). It also fixes two conventions that a different choice would violate at $O(1)$: the sign of the tangent integrals in (86), (88) (the opposite sign violates the null condition by 2.3 and makes the shear and direct forms differ by 2.1 (L_2) and 38 (L_3), PW-4), and the zero-mode weight in the conversion $\delta \to \delta_T$ (PW-7). The label convention matters at finite η_i: with the label taken at the initial event instead of $\eta \to 0$ the observer is a different fluid element (offset 1.7×10^{-5} at $\eta_i = 0.01$), which moves $a_{20}[\Theta^{(2)}]$ by 7×10^{-5}, beyond the solver's uncertainty (PW-5); this is the $O(\eta_i^2)$ effect of Sec. 8.2 and disappears as $\eta_i \to 0$.

13.5 First-order controls

The transfer functions satisfy the first-order equations and constraints exactly; the Rayleigh rules were checked by sphere quadrature (3×10^{-14}, $\ell \le 6$), including the observer monopole ($\ell = 0$ only) and observer Doppler ($\ell = 1$ only); the prefactor of $C_\ell^{(11)}$ and the observer and ISW terms by an independent real-space computation of the correlation function (6×10^{-15}); the ISW closed form against quadrature (2×10^{-18}), its residue formulas for the Ci/Si/Cin coefficients ($< 10^{-169}$) and the vanishing of the $\ln w$ coefficient (regularity at the observer) symbolically; the limits $k\eta_* \to 0$ (Sachs–Wolfe $-\frac{1}{3}$, plateau $A_s/9$), $k \to \infty$ and $k \to 0$; and $\langle \langle m_1 \rangle_{\text{sky}} a_{\ell m} \rangle = 0$ ($\ell \ge 1$), $\langle \langle m_1 \rangle^2_{\text{sky}} \rangle = C_0^{(11)}/4\pi$ (FO-1–FO-4). The first-order map is in addition tested end to end by the plane-wave comparison ($\sim 10^{-11}$).

13.6 Harmonic machinery controls

Rules (A-s), (A-λ), (A-L), the type-C $3j$ sums and the reduction (E1) against direct quadrature (10^{-14}–10^{-15}), re-checked in the observation-direction convention of this paper (H-1, H-7, H-8, H-11); (B1) with both pairings against an exact real-space Wick computation (2.6×10^{-12}) and by Monte Carlo on the sphere (H-3); the type-D Fourier/angular-gradient identity (3×10^{-12}, H-4); the prefactors 2 and 3 of (105) (H-6); the Bessel tail bound (129) (H-12). The decisive control is the lensed C_ℓ (Sec. 10.7): the bookkeeping (12) reproduces Hu's (22) and (13) lensing terms, extended to correlated ϕ, Θ, algebraically (2.5×10^{-16}) and by Monte Carlo of the remapped sky (2–5%, statistical) (H-5). Formulas (E2), (E3) and $T^{(\text{ii})}$ are derived by Wick counting in the same way as the verified ones but have not been tested numerically (H-10).

13.7 What the checks do not cover

- A single plane wave probes only the axisymmetric ($m = 0$ about $\mathbf{k}$) part of the kernels K_r, and only the collinear configurations $\mathbf{k}_i \parallel \mathbf{k}$ of K_2, K_3 (with the $x_o = 0$ configurations restricted to even ℓ); the planar solver contains no vector or tensor modes beyond a homogeneous h, so the ω_2 terms and the non-collinear h_2 terms of the map are untested end to end (the non-collinear second-order kernel is tested by the equality of its shear and direct forms, Sec. 11.2).
- Only $\bar{\eta}_o/\eta_* = 2$ and 4 were compared; the third-order comparison at ratio 4 and $x_o = \pi/2$ is limited by the extraction uncertainty (3×10^{-2} absolute on a map of rms 24).
- The gauge-invariance test at finite k covers scalar modes and collinear momenta; a perturbative check under a general second-order gauge transformation of the assembled L_2 has not been performed.
- (E2), (E3), $T^{(\mathrm{ii})}$ are not used by the monochromatic evaluation and remain untested numerically (H-10).
- The squeezed-limit predictions for K_2 and K_3 derived from the adiabatic-mode transformation (SQ-1) are compared with the assembled kernels in Secs. 11.4 and 11.9 (SQ-2, SQ-3); the IR formula for tilted spectra is not tested.
- The literature agreement for the scalar second-order trace source rests on two independent computations and the dilation family only, since the printed PUB (A.6) is garbled.

14 Unresolved steps and limitations

The following list collects every step of the calculation that is not derived analytically, not tested, or rests on an assumption. Items marked *numerical* have been established by computation only, with the accuracy quoted at the place indicated.

1. *Collinear residues of the* (13) *integrand (resolved; residual caveat).* The cancellation of the six $1/(1-\mu)$ residues at $q = k$ is proved (Sec. 12.6): the pole parts at second and third order are identified with the residual time reparametrisation by computer algebra, and the residues cancel by an exact total-derivative identity. The explicit residue formulas (175) were written by hand from the group structures of the term lists and confirmed against the engine (10^{-14}), not obtained by a symbolic reduction of the lists; the absence of a $(1-\mu)^{-1/2}$ term rests on a structural argument plus numerics.

2. *UV behaviour of the loop for power-law spectra (numerical, partial).* The $\hat{\mathbf{q}}$-averaged integrand of κ is $O(10)$ and oscillatory up to $q = 4k$ (Sec. 11.9); no analytic large-q expansion of K_{3c} (nor of the (22) integrand at large $k_1 \simeq k_2$) has been made. If the integrand is $O(q^0)$, the loop converges in the UV for $n_s < 1$ and is logarithmically divergent for $n_s = 1$, as $\langle\zeta^2\rangle$; the perfect fluid provides no damping scale.

3. *IR for tilted spectra (derived, untested).* The combined soft region (159), $\frac{1}{2}(n_s-1)^2\sigma_L^2 C_\ell^{(11)}$ for a power law (160), follows from the squeezed theorems 11.2 and 11.3, both verified numerically; the combined formula itself (including the vanishing of the boundary terms) has not been tested by a loop computation with a tilted spectrum.

4. *Closed forms versus declared integrals.* Closed: the first-order transfer function and the ISW integral for all ℓ (Theorem 7.2); all second-order fields, including the tensor kernel $\mathcal{K}_h$ and the

special-function part of $\mathcal{K}_{\Phi_2}$ (one coefficient D, App. B); the contracted third-order sources $K_{3c,X}$, the tadpoles and the amplitude C_3. Left as explicit one-dimensional integrals with complete integrands and limits: the third-order Green's integral $\int_0^\eta G_\Phi K_{3c,S_\Phi} \mathrm{d}\eta'$ in Φ_{3c} (65), hence V_{3c} and $\Delta_\ell^{(\mathrm{iii})}$ (155); the ray functionals $\mathcal{F}_\ell$, $\mathcal{D}_{\perp\ell}$, $\mathcal{D}_{r\ell}$ (148) and their second-order analogues; the transfer functions Δ_ℓ^κ of the contracted kernel (a μ-integral of time integrals); the type-C and type-D μ-integrands of $C_\ell^{(22)}$, which are projected on $Y_{\ell m}$ by sky quadrature (no closed form beyond (117), (148)). The loop integrals over k, q, μ are evaluated numerically for the monochromatic spectrum only.

5. *(E2), (E3), $T^{(\mathrm{ii})}$ untested numerically.* These harmonic formulas (Sec. 10.6) are derived by Wick counting and pass an analytic collapse check (H-10). The monochromatic $C_\ell^{(13)}$ does not use them: κ, including its normalisation part (156), is computed directly by polarisation. An evaluation of $C_\ell^{(13)}$ through (E0)–(E3) would be an independent route and has not been made.

6. *Gauge invariance.* Tested: the exact $k = 0$ family (dilation, all orders, analytic; Sec. 13.3); the residual $K = 0$ time reparametrisation $\tau(\eta)$ for the full contracted kernel κ (Sec. 11.7); the finite-k comparison of the planar plane-wave maps with a non-perturbative solver in comoving-orthogonal gauge (Sec. 13.4). Not performed: a perturbative test of the assembled L_2, L_3 under a general second-order gauge transformation at finite momenta. The invariance of $\mathrm{Re}\, C^{(13)}$ under a rigid rotation of the observer triad is proved (Lemma 4.7) but was not checked numerically by rotating the global axes.

7. *Coverage of the end-to-end test.* The single-plane-wave comparison with the non-perturbative solver probes only the axisymmetric ($m = 0$ about $\mathbf{k}$) parts of K_2, K_3 in collinear configurations, scalar modes only, and $\bar{\eta}_o/\eta_* = 2$ and 4 (Sec. 13.7). The contracted kernel κ shares the term lists and field solutions tested there but was not itself compared with the plane-wave chain; the (13) term was evaluated at $\eta_o/\eta_* = 4$ only.

8. *Zero modes and tadpoles (resolved; residual caveat).* The residual $k = 0$ freedom is classified and the map is shown to be invariant to second order under every zero mode; the $\hat{\mathbf{q}}$-averaged tadpole is the unique $k \to 0$ limit (Sec. 11.8). What is not separately established: the invariance of $\Theta^{(3)}$ under $F(\eta)$ is inferred from the multilinear identity and the uniqueness of the regular solution, supported by the numerical τ-tests of the assembled κ (R-11) and PW-3; no separate symbolic third-order run was made.

9. *Exchange of limits and integrations.* Justified: (a) the monochromatic δ-function collapse is applied only to the complete observable, after the kernels K_2 and κ have been assembled (Sec. 10.8); (b) the corner $K \to 0$ of $C_\ell^{(22)}$ is taken inside the μ-integral with the bound of Proposition 12.1 (bounded integrand for $\ell \geq 2$, analytic power counting); (c) the limit $\eta_i \to 0$ is controlled by the exact split (37), with the $O(\eta_i^2)$ corrections displayed (Sec. 8.2); (d) the ℓ-sums are cut off with the analytic Bessel bound (130); (e) $k \to 0$ versus the angular average of the tadpoles: the averaged limit exists and is the only one that enters (item 8); (f) the collinear limit $\mu \to \pm 1$ of the (13) integrand, taken inside the μ-integral: the integrand is analytic in $1-\mu$ there (item 1). Justified numerically only: (g) the squeezed limits $q \to 0$ inside the soft region (Theorems 11.2, 11.3: derived, verified with $O(q^2)$ remainders). Not justified: the interchange of the large-q limit with the time integrals (item 2), and the uniformity in k of the soft expansion used in (159) (item 3). The perturbative series in ϵ itself is used as an asymptotic expansion; its convergence is not addressed.

10. *Physical limitations of the model.* Radiation-only universe ($\Lambda = 0$, no matter, no neutrinos); the fluid is a perfect fluid before and *after* emission (it is continued past T_*, so there is no free streaming, no diffusion damping and no recombination physics); acoustic oscillations are undamped and $\Delta_\ell(k)$ decays only as k^{-1}, which is the origin of the UV issue of item 2; emission is instantaneous on $T_{\rm phys} = T_*$ and isotropic in the fluid frame. The monochromatic spectrum is a toy: band limited at first order but not in the loop, so the ϵ^4 terms dominate $C_\ell^{(11)}$ for $\ell \gtrsim 6$ (Sec. 12.7).

A Explicit kernels (machine generated)

All expressions below are exported directly from the computer-algebra solution of the field equations (App. F). Notation: $T_q \equiv T_\Phi(q\eta/\sqrt{3})$, $\dot T_q \equiv \partial_\eta T_q$, $p = |\mathbf{k}-\mathbf{q}|$, $\mu = \hat{\mathbf{k}}\cdot\hat{\mathbf{q}} = (k^2+q^2-p^2)/(2kq)$; every second-order kernel is per $\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}$ in the sense of (58). Sources are those of $E^{(2)}_{\mu\nu} = 2[\epsilon^2]E_{\mu\nu}$ in the frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the x–z plane. The transfer functions (53) and the dictionary $\dot T_q = \partial_\eta T_\Phi$ with $\partial_\eta = (q/\sqrt{3})\,\mathrm{d}/\mathrm{d}x$ convert everything to closed form in $\sin, \cos$ of $q\eta/\sqrt{3}$, $p\eta/\sqrt{3}$.

A.1 Second-order kernels

K_Q **(traceless scalar,** $Q_2 = \Phi_2 - \Psi_2$**)**

$$
\begin{aligned}
&-\frac{T_pT_q\left(13k^4-6k^2p^2-6k^2q^2+9p^4-18p^2q^2+9q^4\right)}{4k^4}\\
&-\frac{T_p\dot T_q\eta\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}\\
&-\frac{T_q\dot T_p\eta\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}\\
&-\frac{\dot T_p\dot T_q\eta^2\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}
\end{aligned}
\tag{177}
$$

$K_{S_{00}}$ **(source of the 00 constraint)**

$$
\begin{aligned}
&-\frac{2T_pT_q\left(\eta^2k^2+3\eta^2p^2+3\eta^2q^2-12\right)}{\eta^2}-T_p\dot T_q\eta\left(-k^2+p^2+q^2\right)\\
&-T_q\dot T_p\eta\left(-k^2+p^2+q^2\right)+\dot T_p\dot T_q\left(\eta^2k^2-\eta^2p^2-\eta^2q^2+6\right)
\end{aligned}
\tag{178}
$$

$K_{S_{0z}}$ **(longitudinal** $0i$ **source,** S_0^L**)**

$$
\begin{aligned}
&-\frac{2iT_pT_q\left(-\eta^2k^2p^2-\eta^2k^2q^2+\eta^2p^4-2\eta^2p^2q^2+\eta^2q^4-6k^2\right)}{3\eta k}\\
&-\frac{iT_p\dot T_q\left(-2\eta^2k^2p^2+2\eta^2p^4-2\eta^2p^2q^2-27k^2+9p^2-9q^2\right)}{3k}\\
&+\frac{iT_q\dot T_p\left(2\eta^2k^2q^2+2\eta^2p^2q^2-2\eta^2q^4+27k^2+9p^2-9q^2\right)}{3k}+4i\dot T_p\dot T_q\eta k
\end{aligned}
\tag{179}
$$

K_{S_Φ} **(source of $\mathcal{L}_\Phi \Phi_2 = S_\Phi$)**

$$\frac{T_p T_q \left(-7k^6 - 19k^4p^2 - 19k^4q^2 + 7k^2p^4 - 22k^2p^2q^2 + 7k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6\right)}{12k^4} \tag{180}$$
$$+ \frac{T_p \dot{T}_q \eta \left(k^2 + p^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}$$
$$+ \frac{T_q \dot{T}_p \eta \left(k^2 + q^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}$$
$$+ \frac{\dot{T}_p \dot{T}_q \left(-\eta^2 k^6 - 2\eta^2 k^4 p^2 - 2\eta^2 k^4 q^2 + 3\eta^2 k^2 p^4 - 6\eta^2 k^2 p^2 q^2 + 3\eta^2 k^2 q^4 + 12k^4 - 24k^2p^2 - 24k^2q^2 + 36p^4 - 72p^2q^2 -}{12k^4}$$

K_ω **(vector metric, $\omega_{2,i} = \int P_{ij} q_j K_\omega \zeta\zeta$)**

$$-\frac{2iT_pT_q\eta\left(p-q\right)\left(p+q\right)}{k^2} - \frac{2iT_p\dot{T}_q\eta^2p^2}{k^2} + \frac{2iT_q\dot{T}_p\eta^2q^2}{k^2} \tag{181}$$

K_{v^T} **(transverse velocity, $v^T_{2,i} = \int P_{ij} q_j K_{v^T} \zeta\zeta$)**

$$-\frac{iT_pT_q\eta\left(p-q\right)\left(p+q\right)\left(\eta^2k^2-24\right)}{12k^2} - \frac{iT_p\dot{T}_q\eta^2\left(\eta^2k^2p^2+18k^2-24p^2\right)}{12k^2} \tag{182}$$
$$+ \frac{iT_q\dot{T}_p\eta^2\left(\eta^2k^2q^2+18k^2-24q^2\right)}{12k^2}$$

K_h **(tensor source, $h_{2,ij} = \Pi_{ij,kl} q_k q_l \int_0^\eta G_h K_h$)**

$$12T_pT_q + 4T_p\dot{T}_q\eta + 4T_q\dot{T}_p\eta + 4\dot{T}_p\dot{T}_q\eta^2 \tag{183}$$

$K^0_{\Phi_2}$ **(initial amplitude of Φ_2, Eq. (42))**

$$-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4} \tag{184}$$

$K_Q(\eta \to 0)$

$$-\frac{13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4}{9k^4} \tag{185}$$

kernel of the local quadratic part $-\frac{1}{4}\delta_1^2 - 2\Phi_1^2$ of $\zeta^{(2)}$, Eq. (39) ($\to -\frac{4}{3}$ as $\eta \to 0$)

$$-\frac{T_pT_q\left(\eta^4p^2q^2+3\eta^2p^2+3\eta^2q^2+27\right)}{9} - \frac{T_p\dot{T}_q\eta\left(\eta^2p^2+3\right)}{3} - \frac{T_q\dot{T}_p\eta\left(\eta^2q^2+3\right)}{3} - \dot{T}_p\dot{T}_q\eta^2 \tag{186}$$

Algebraic second-order scalars (with S_{00}, S_{0z} the corresponding sources)

$$\delta_2 = \frac{S_{00}\eta^2}{3} - \frac{2\eta^2k^2\Phi_2(\eta)}{3} - 2\eta\frac{d}{d\eta}\Phi_2(\eta) - 2\Psi_2(\eta) \tag{187}$$

$$v_{2,z} = \frac{\eta\left(-S_{0z}\eta - 2i\eta k\frac{d}{d\eta}\Phi_2(\eta) - 2ik\Psi_2(\eta)\right)}{4} \tag{188}$$

A.2 Homogeneous solutions and Green's functions

u_1

$$\frac{3\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{189}$$

u_2

$$\frac{3\sqrt{3}\left(\frac{\sqrt{3}\eta k\sin\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\cos\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{190}$$

$U = 3u_1$

$$\frac{9\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{191}$$

$G_\Phi(k;\eta,\eta')$

$$\frac{\sqrt{3}\eta'\left(\eta'\eta k^2\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)+\sqrt{3}\eta' k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)-\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)+3\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)\right)}{\eta^3k^3} \tag{192}$$

$G_h(k;\eta,\eta')$

$$\frac{\eta'\sin\left(k\left(-\eta'+\eta\right)\right)}{\eta k} \tag{193}$$

$G_\omega(\eta,\eta')$ **(for the ODE form of the vector equation)**

$$\frac{\eta'^2}{\eta^2} \tag{194}$$

A.3 Third-order initial amplitude and intermediate limits

Per $P(q)\,\zeta_{\mathbf{k}}$ in the contracted sense of (62); $p^2 = k^2 - 2kq\mu + q^2$.

$C_3(k;q,\mu)$**, Eq.** (45)

$$-\frac{16\left(2k^5+12k^4\mu q-42k^3\mu^2q^2-10k^3q^2+36k^2\mu^3q^3+32k^2\mu q^3+3k\mu^4q^4-42k\mu^2q^4+kq^4+8\mu q^5\right)}{9k\left(k^2-2k\mu q+q^2\right)^2} \tag{195}$$

C_3 **in invariant form** (k,q,p)

$$-\frac{35k^8+12k^6p^2-84k^6q^2+66k^4p^4-52k^4p^2q^2+66k^4q^4-84k^2p^6+84k^2p^4q^2+20k^2p^2q^4-20k^2q^6+3p^8-12p^6q^2}{9k^4p^4} \tag{196}$$

$Q_{3c}(\eta \to 0)$

$$-\frac{16q\left(44k^4\mu - 142k^3\mu^2q - 18k^3q + 108k^2\mu^3q^2 + 104k^2\mu q^2 + 9k\mu^4q^3 - 126k\mu^2q^3 - 3kq^3 + 24\mu q^4\right)}{9k\left(k^2 - 2k\mu q + q^2\right)^2} \tag{197}$$

$\lim_{\eta\to 0}\eta^2 K_{3c,S_{00}}$

$$-\frac{256kq\left(2k^2\mu - k\mu^2q - 3kq + 2\mu q^2\right)}{3\left(k^2 - 2k\mu q + q^2\right)^2} \tag{198}$$

A.4 Tadpole (mean-field) functions

$\mathrm{tad}_X(q,\eta) = \langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}$, Eq. (64), angle-averaged $\mathbf{k}\to 0$ limits of the second-order kernels (convention of Sec. 5.4); small-η behaviour in brackets.

$\mathrm{tad}_{Q_2}\ [-\frac{16}{9}+\frac{88}{405}q^2\eta^2+\ldots]$

$$\frac{8\left(\eta^6q^6\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)+\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+6\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+18\eta^2q^2\left(16\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)-\right.\right.}{15\eta^6q^6} \tag{199}$$

$\mathrm{tad}_{S_{00}}\ [\frac{32}{3\eta^2}-\frac{272}{45}q^2+\ldots]$

$$\frac{24\left(\frac{\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)}{2}-\frac{\eta^6q^6}{2}-2\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-21\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\eta^4q^4+36\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+126\eta^2q^2\right.}{\eta^8q^6} \tag{200}$$

$\mathrm{tad}_{S_\Phi}\ [-\frac{56}{81}q^2+\ldots]$

$$\frac{4\left(2\sqrt{3}\eta^7q^7\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+12\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-30\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-684\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+1764\sqrt{3}\right.}{45\eta^8q^6} \tag{201}$$

$\mathrm{tad}_{\Phi_2}\ [-\frac{8}{9}-\frac{28}{405}q^2\eta^2+\ldots]$

$$\frac{-116\eta^6q^6+3\eta^3q^3\left(-2\sqrt{3}\eta^2q^2\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-18\eta q\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+180\eta q+33\sqrt{3}\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)\right)+108\eta^2q^2\left(17\cos\left(\frac{2\sqrt{}\right.\right.}{45\eta^6q^6} \tag{202}$$

$\mathrm{tad}_{\delta_2}\ [\frac{16}{9}-\frac{472}{405}q^2\eta^2+\ldots]$

$$\frac{4\left(45\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+19\eta^6q^6-186\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-1944\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+3879\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{}\right.\right.}{45\eta^6q^6} \tag{203}$$

A.5 Not reproduced here

The contracted third-order source kernels $K_{3c,X}$ for $X \in \{Q, S_{00}, S_{0z}, S_\Phi, S_{0x}, S_{xz}, S_{xx} - S_{yy}\}$ (157–830 terms each, containing the closed-form $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ of App. B as symbols), the position-space sources $S^{(2)}_{\mu\nu}$, $S^{(3)}_{\mu\nu}$ and the full $E^{(n)}_{\mu\nu}$ are available in machine-readable form (App. F).

B Closed-form second-order time integrals

The only non-algebraic second-order time integrals are those of (61),

$$\mathcal{K}_{\Phi_2} = P(\eta) + K^0_{\Phi_2}\, U(x_k), \qquad P \equiv \int_0^\eta G_\Phi(k;\eta,\eta')\, K_{S_\Phi}(k,q,p;\eta')\,\mathrm{d}\eta', \qquad \mathcal{K}_h = \int_0^\eta \frac{\eta'}{\eta}\,\frac{\sin k(\eta-\eta')}{k}\, K_h(k,q,p;\eta')\,\mathrm{d}\eta',$$

with $x_a \equiv a\eta/\sqrt{3}$, $T_a \equiv T_\Phi(x_a)$. Both are evaluated in closed form: elementary functions plus $\mathrm{Si}(w\eta)$ and the entire, even function $\mathrm{Cin}(z) = \int_0^z (1-\cos t)\,\mathrm{d}t/t = \gamma_E + \ln|z| - \mathrm{Ci}(|z|)$. Because Cin is entire, the forms need no absolute values, principal values or branch choices; rewriting $\mathrm{Cin} \to \mathrm{Ci}$ reproduces the $\ln|\dots|$ terms of Kohri–Terada [12]. All other second-order fields are algebraic in Φ_2, Φ_2' and first-order transfer functions (56), so *nothing remains as an unevaluated integral at second order* (checks CF-1–CF-4).

Method. The integrands are finite sums $\sum c_{n,w}(k,q,p)\,\eta'^n e^{iw\eta'}$, $-7 \le n \le 0$. Each term is integrated on $[\varepsilon,\eta]$ by the recursion $I_n = [t^{n+1}e^{iwt}]/(n+1) - \frac{iw}{n+1} I_{n+1}$, $I_{-1} = -\mathrm{Cin}(w\eta) + i\,\mathrm{Si}(w\eta) + \ln(\eta/\varepsilon)$. All ε^{-m} and $\ln\varepsilon$ pieces cancel identically (symbolic check) — this is the regularity of the total integrand at $\eta' = 0$ — so the lower limit 0 is attained.

The scalar Φ_2

With u_1, u_2, W of (60), $P = u_2(x_k)\,\mathcal{A}(\eta) - u_1(x_k)\,\mathcal{B}(\eta)$, $\mathcal{A} = \int_0^\eta (u_1/W) K_{S_\Phi}\,\mathrm{d}\eta'$, $\mathcal{B} = \int_0^\eta (u_2/W) K_{S_\Phi}\,\mathrm{d}\eta'$, where $u_1/W = -\eta'\sin x'_k + (k/\sqrt{3})\eta'^2\cos x'_k$, $u_2/W = -\eta'\cos x'_k - (k/\sqrt{3})\eta'^2 \sin x'_k$. The frequencies are

$$w_1 = \frac{k+q+p}{\sqrt{3}}, \quad w_2 = \frac{k+q-p}{\sqrt{3}}, \quad w_3 = \frac{k-q+p}{\sqrt{3}}, \quad w_4 = \frac{q+p-k}{\sqrt{3}} \qquad (\text{all } \ge 0 \text{ by the triangle inequalities}),$$

and the complete special-function part is governed by one coefficient,

$$\mathcal{A} \supset \mathcal{D}\,[\mathrm{Si}(w_1\eta) - \mathrm{Si}(w_2\eta) - \mathrm{Si}(w_3\eta) - \mathrm{Si}(w_4\eta)], \qquad \mathcal{B} \supset \mathcal{D}\,[\mathrm{Cin}(w_2\eta) + \mathrm{Cin}(w_3\eta) - \mathrm{Cin}(w_1\eta) - \mathrm{Cin}(w_4\eta)],$$

$$\mathcal{D}(k,q,p) = \frac{(k^2+p^2+q^2)\,R}{8k^2p^3q^3}, \qquad R = 3k^6 - 3k^4(p^2+q^2) - 3k^2(p^4+q^4) - 2k^2p^2q^2 + 3(p^6+q^6) - 3p^2q^2(p^2+q^2).$$

The remainder of $\mathcal{A}, \mathcal{B}$ is elementary: a sum over the eight products $t(x_k)t(x_q)t(x_p)$, $t \in \{\sin,\cos\}$, of Laurent polynomials in η with powers $\eta^{-7},\dots,\eta^0$, obtained from the displayed source by the recursion above (64 monomials per integral; machine-readable, App. F). Only the $n = 0$ monomials produce $1/w$ factors; they become removable 0/0 at the collinear edges $w_2 = 0$ ($\mu = -1$), $w_3 = 0$ ($\mu = 1$, $q > k$), $w_4 = 0$ ($\mu = 1$, $q < k$). Exact collinear limiting forms are obtained by substituting $p = k+q$, $q-k$, $k-q$ before integrating; zero-frequency terms occur only at $n = -4, -2, 0$, so no $\ln\eta$ appears in collinear Φ_2. There is no $\sqrt{3}$-resonance for Φ_2 (the sound speed enters both G_Φ and the source).

The tensor kernel $\mathcal{K}_h$: complete form

With the frequencies $\varpi_1 = k + \frac{q+p}{\sqrt{3}}$, $\varpi_2 = k + \frac{q-p}{\sqrt{3}}$, $\varpi_3 = k - \frac{q-p}{\sqrt{3}}$, $\varpi_4 = k - \frac{q+p}{\sqrt{3}}$ (only ϖ_4 can vanish), $s_k \equiv k\eta$, $x_q = q\eta/\sqrt{3}$, $x_p = p\eta/\sqrt{3}$:

$$\mathcal{K}_h = \frac{3(p^2+q^2-3k^2)^2}{k\,p^3q^3\,\eta}\Big\{\sin s_k[\mathrm{Cin}(\varpi_1\eta) - \mathrm{Cin}(\varpi_2\eta) - \mathrm{Cin}(\varpi_3\eta) + \mathrm{Cin}(\varpi_4\eta)]$$

$$+\cos s_k[\,\mathrm{Si}(\varpi_1\eta)-\mathrm{Si}(\varpi_2\eta)-\mathrm{Si}(\varpi_3\eta)+\mathrm{Si}(\varpi_4\eta)]\Big\}\;+\;\mathcal{K}_h^{\mathrm{el}},$$

with the elementary part

$$\begin{aligned}
\mathcal{K}_h^{\mathrm{el}} = {} & \cos^2(s_k)\cos(x_p)\cos(x_q)\times\Big\{\Big[\frac{72}{p^2q^2}\Big]\eta^{-2}\Big\}\\
&+\sin^2(s_k)\cos(x_p)\cos(x_q)\times\Big\{\Big[\frac{72}{p^2q^2}\Big]\eta^{-2}\Big\}\\
&+\sin^2(s_k)\sin(x_p)\cos(x_q)\times\Big\{\Big[-\frac{72\sqrt{3}}{p^3q^2}\Big]\eta^{-3}\Big\}\\
&+\sin^2(s_k)\sin(x_p)\sin(x_q)\times\Big\{\Big[\frac{216}{p^3q^3}\Big]\eta^{-4}\\
&\quad+\Big[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\Big]\eta^{-2}\Big\}\\
&+\sin^2(s_k)\sin(x_q)\cos(x_p)\times\Big\{\Big[-\frac{72\sqrt{3}}{p^2q^3}\Big]\eta^{-3}\Big\}\\
&+\sin(x_p)\cos^2(s_k)\cos(x_q)\times\Big\{\Big[-\frac{72\sqrt{3}}{p^3q^2}\Big]\eta^{-3}\Big\}\\
&+\sin(x_p)\sin(x_q)\cos^2(s_k)\times\Big\{\Big[\frac{216}{p^3q^3}\Big]\eta^{-4}\\
&\quad+\Big[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\Big]\eta^{-2}\Big\}\\
&+\sin(x_q)\cos^2(s_k)\cos(x_p)\times\Big\{\Big[-\frac{72\sqrt{3}}{p^2q^3}\Big]\eta^{-3}\Big\}\\
&+\sin(s_k)\times\Big\{\Big[-\frac{12\left(-3k^2+p^2+q^2\right)}{kp^2q^2}\Big]\eta^{-1}\Big\}
\end{aligned}$$

(the printed form, re-assembled and compared with 30-digit quadrature of the defining integral at three points, agrees to 2×10^{-29}, CF-4). It equals $(4/k^2)\times$ Kohri–Terada (22) [with $|\cdot|$ inside its logarithm] to 10^{-15} (25 digits with exact input). At the resonance $q+p=\sqrt{3}k$ ($\varpi_4=0$) nothing happens in the Cin form; the separately integrated resonant form contains an explicit $\ln\eta$. The other resonance $|q-p|=\sqrt{3}k$ lies outside the physical domain.

Algebraic fields and limits

$\Psi_2=\Phi_2-Q_2$, $\delta_2=\dfrac{\eta^2}{3}K_{S_{00}}-\dfrac{2}{3}\eta^2k^2\Phi_2-2\eta\Phi_2'-2\Psi_2$, $\delta_{T,2}=\dfrac{1}{4}\delta_2-\dfrac{3}{16}T_\delta(x_q)T_\delta(x_p)$, $V_2=-\dfrac{\eta^2}{4}[2(\Phi_2'+\Psi_2/\eta)+K_{S_{0z}}/(ik)]$, $\omega_{2,i}=P_{ij}q_jK_\omega$, $v_{2,i}^T=P_{ij}q_jK_{v^T}$, with the kernels of (59) and App. A; Φ_2' is the closed-form derivative ($G_\Phi(\eta,\eta)=0$).

Small η (Frobenius series of the equations with the constraints; checked against the closed forms at 120 digits):

$$\begin{aligned}
\Phi_2 = {} & \left[-\frac{7k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4}{9k^4}\right]\eta^0\\
&+\left[\frac{-21k^4p^2-21k^4q^2+10k^2p^4-28k^2p^2q^2+10k^2q^4+3p^6-3p^4q^2-3p^2q^4+3q^6}{270k^4}\right]\eta^2\\
&+\Big[-\frac{-150k^6p^2-150k^6q^2-65k^4p^4-398k^4p^2q^2-65k^4q^4+80k^2p^6-144k^2p^4q^2-144k^2p^2q^4+80k^2q^6+15p^8+36p^6q^2}{113400k^4}\\
\mathcal{K}_h = {} & \left[\frac{8}{9}\right]\eta^2
\end{aligned}$$

$$+\left[-\frac{2\left(3k^2+p^2+q^2\right)}{135}\right]\eta^4+O(\eta^6)$$

$$\Psi_2=\left[\frac{2\left(3k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{9k^4}\right]\eta^0$$
$$+\left[-\frac{2\left(8k^4p^2+8k^4q^2-5k^2p^4+2k^2p^2q^2-5k^2q^4+3p^6-3p^4q^2-3p^2q^4+3q^6\right)}{135k^4}\right]\eta^2$$
$$+\left[\frac{75k^6p^2+75k^6q^2+55k^4p^4+234k^4p^2q^2+55k^4q^4-75k^2p^6-117k^2p^4q^2-117k^2p^2q^4-75k^2q^6+45p^8+108p^6q^2-3}{56700k^4}\right.$$

$$\delta_2=\left[-\frac{4\left(-5k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{9k^4}\right]\eta^0$$
$$+\left[\frac{2\left(15k^6-41k^4p^2-41k^4q^2-5k^2p^4+2k^2p^2q^2-5k^2q^4+3p^6-3p^4q^2-3p^2q^4+3q^6\right)}{135k^4}\right]\eta^2$$
$$+\left[-\frac{-1095k^6p^2-1095k^6q^2-275k^4p^4-3394k^4p^2q^2-275k^4q^4-25k^2p^6-39k^2p^4q^2-39k^2p^2q^4-25k^2q^6+15p^8+36}{28350k^4}\right.$$

$$V_2=\left[-\frac{7k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4}{9k^4}\right]\eta^1$$
$$+\left[\frac{39k^4p^2+39k^4q^2+6k^2p^4-20k^2p^2q^2+6k^2q^4+3p^6-3p^4q^2-3p^2q^4+3q^6}{270k^4}\right]\eta^3$$
$$+\left[-\frac{375k^6p^2+375k^6q^2+105k^4p^4+134k^4p^2q^2+105k^4q^4+105k^2p^6-169k^2p^4q^2-169k^2p^2q^4+105k^2q^6+15p^8+36p}{113400k^4}\right.$$

Large η. With $\Xi\equiv\ln\frac{(k+q+p)(q+p-k)}{(k+q-p)(k+p-q)}$ and $x_k=k\eta/\sqrt{3}$:

$$\Phi_2\to\frac{1}{\eta^2}\Big\{C_L[\Xi\cos x_k+\pi\sin x_k]+C_0\cos x_k\Big\}+O(\eta^{-3}),\qquad C_L=-\frac{3(k^2+p^2+q^2)R}{8k^4p^3q^3},$$
$$C_0=\frac{3\left[-3k^6-3k^4(p^2+q^2)+3k^2(p^4+q^4)+2k^2p^2q^2+3(p^6+q^6)-3p^2q^2(p^2+q^2)\right]}{2k^4p^2q^2},$$
$$\mathcal{K}_h\to\frac{4}{k^2}\ \text{[Kohri–Terada (25)]}\quad(\sin k\eta\ \log\text{ term}+\Theta(q+p-\sqrt{3}k)\,\pi\cos k\eta\ \text{term}).$$

The next orders (η^{-3} for Φ_2, η^{-2} for $\mathcal{K}_h$) are available in machine-readable form; verified up to $k\eta=10^5$ (remainder scales as the first dropped power). The expansion is non-uniform as $q\to0$ or $p\to0$ and near the collinear edges / $q+p=\sqrt{3}k$.

Evaluators. All second-order fields ($\Phi_2,\Phi_2',\Psi_2,\delta_2,\delta_{T,2},V_2,\omega_2,v_2^T,h_2$) are provided as fast double-precision evaluators (`C4_timeintegrals/second_order_closed.py`): small-η series for $(k+q+p)\eta/\sqrt{3}\le4$, the closed form where it is well conditioned, and Gauss–Legendre quadrature of the Green's-function integral otherwise (the double-precision closed form loses digits through the removable $1/w_j$ collinear poles and the $1/(p^3q^3)$ prefactors). Against the 50-digit closed form at 56 points, including near-collinear and resonant ones, the worst relative error is 3.6×10^{-13} (CF-5).

Verification. Quadrature (50–60 digits) versus the closed forms: P at 11 points including collinear and $\sqrt{3}$ points, $\le3.5\times10^{-40}$; $\mathcal{K}_h$: 25 digits. All closed-form fields inserted into the full second-order Einstein tensor (10 components) and the 4 conservation equations for a plane-wave pair: residual/maximal term $\le5.5\times10^{-50}$ at three points ($k\eta=2.5$, 9.6, 20.1). Small-η series = Frobenius series; Kohri–Terada (22), (25) controls (CF-1–CF-4).

C Complete line-of-sight term lists of L_1, L_2 and L_3

This appendix displays the full perturbative expansion of $\Lambda = -S$ of (23) through third order for the direction $\hat{\mathbf{n}} = \hat{\mathbf{z}}$, as produced by the symbolic expansion of Secs. 8–9 (generator and code: App. F). It is *complete*: together with the field kernels of Sec. 6 and App. A, the ray functionals (86)–(89), the observer shifts (91)–(90) and the emission shifts (93), it defines $\Theta^{(3)}$ via (101). The listed expressions are the coefficients of ϵ^m in Λ, i.e. L_1, $\frac{1}{2}L_2$, $\frac{1}{6}L_3$. Notation: v_n^i, Ψ_n, Φ_n, $\delta_{T,n}$, $\omega_{n,i}$, $h_{n,ij}$ are the order-n Poisson-gauge fields of (7)–(8), with $h_{n,zz} \equiv -h_{n,xx} - h_{n,yy}$ substituted; $k_1^i, k_2^i, \Delta x_1^i, \Delta x_2^i$ are the ray functionals (functions of η on the ray; "(η_*)" marks their value at the emission end); primes are ∂_η. "Ray" groups are integrated over $\eta \in [\eta_*, \bar{\eta}_o]$ with all fields at $(\eta, \bar{\mathbf{x}}(\eta))$; "observer" groups are evaluated at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; "emission" groups at $(\eta_*, \bar{\mathbf{x}}_*)$. For $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ the radial direction functional has been eliminated with the null condition $k_1^z = -(\Phi_1 + \Psi_1)$ and $|\hat{\mathbf{n}}| = 1$ is used. A general direction follows by rotation, since Λ is a scalar functional of the fields and $\hat{\mathbf{n}}$ (Sec. 9); the lists for symbolic $\hat{\mathbf{n}}$ (without $|\hat{\mathbf{n}}| = 1$ imposed) are longer (L_1: 15, $\frac{1}{2}L_2$: 229, $\frac{1}{6}L_3$: 2484 terms) and are available in machine-readable form (App. F). No equation of motion has been used in the lists.

C.1 L_1, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$

(1) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 1 (3 terms)

$$\frac{1}{3}\partial_x v_1^x + \frac{1}{3}\partial_y v_1^y - \frac{2}{3}\partial_z v_1^z$$

i.e. $L_1 = -\int_{\eta_*}^{\bar{\eta}_o} I_1 \, \mathrm{d}\eta$ with I_1 of (80); on shell this equals (30).

C.2 $\frac{1}{2}L_2$, shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$: all terms

(1) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 2 (5 terms)

$$\begin{aligned}&\frac{1}{4}h_{2,xx}{}' + \frac{1}{4}h_{2,yy}{}' + \frac{1}{6}\partial_x v_2^x \\ &\quad + \frac{1}{6}\partial_y v_2^y - \frac{1}{3}\partial_z v_2^z\end{aligned}$$

(2) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$; explicit fields of order 1 (4 terms)

$$\begin{aligned}&k_1^x \partial_z v_1^x + k_1^x \partial_x v_1^z + k_1^y \partial_z v_1^y \\ &\quad + k_1^y \partial_y v_1^z\end{aligned}$$

(3) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 1+1 (10 terms)

$$\begin{aligned}&\frac{1}{3}v_1^x v_1^{x\prime} - v_1^x \partial_z v_1^x - v_1^x \partial_x v_1^z \\ &\quad + \frac{1}{3}v_1^y v_1^{y\prime} - v_1^y \partial_z v_1^y - v_1^y \partial_y v_1^z \\ &\quad + \frac{1}{3}v_1^z \partial_x v_1^x + \frac{1}{3}v_1^z \partial_y v_1^y - \frac{2}{3}v_1^z v_1^{z\prime} \\ &\quad - \frac{2}{3}v_1^z \partial_z v_1^z\end{aligned}$$

(4) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; time delay Δx_1^z; explicit fields of order 1 (3 terms)

$$-\frac{2}{3}\Delta x_1^z \partial_z^2 v_1^z + \frac{1}{3}\Delta x_1^z \partial_x \partial_z v_1^x + \frac{1}{3}\Delta x_1^z \partial_y \partial_z v_1^y$$

(5) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; lensing $\Delta\mathbf{x}_{1\perp}$; explicit fields of order 1 (6 terms)

$$\begin{aligned}&\frac{1}{3}\Delta x_1^x\partial_x^2 v_1^x+\frac{1}{3}\Delta x_1^x\partial_x\partial_y v_1^y-\frac{2}{3}\Delta x_1^x\partial_x\partial_z v_1^z\\&+\frac{1}{3}\Delta x_1^y\partial_y^2 v_1^y+\frac{1}{3}\Delta x_1^y\partial_x\partial_y v_1^x-\frac{2}{3}\Delta x_1^y\partial_y\partial_z v_1^z\end{aligned}$$

(6) observer, at $(\bar{\eta}_o,\bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$; explicit fields of order 1 (3 terms)

$$\frac{1}{3}\delta\eta_{o1}\partial_x v_1^x+\frac{1}{3}\delta\eta_{o1}\partial_y v_1^y-\frac{2}{3}\delta\eta_{o1}\partial_z v_1^z$$

(7) emission, at $(\eta_*,\bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$; explicit fields of order 1 (3 terms)

$$-\frac{1}{3}\delta\eta_{e1}\partial_x v_1^x-\frac{1}{3}\delta\eta_{e1}\partial_y v_1^y+\frac{2}{3}\delta\eta_{e1}\partial_z v_1^z$$

Total: 34 terms.

Table 10: Groups of $\frac{1}{2}L_2$ for symbolic $\hat{\mathbf{n}}$ (term counts).

#	where	content	fields	terms
1	ray	local	2	22
2	ray	direction $\mathbf{k}_1$	1	21
3	ray	local	1+1	111
4	ray	displacement $\Delta\mathbf{x}_1$	1	45
5	obs	clock $\delta\eta_{o1}$	1	15
6	emi	emission shift $\delta\eta_{e1}$	1	15
		total		229

C.3 $\frac{1}{2}L_2$, direct (redshift) form, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$

(1) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; local; explicit fields of order 2 (6 terms)

$$\begin{aligned}&\frac{1}{2}{\Phi_2}'-\frac{1}{2}{\Psi_2}'+\partial_z\Psi_2\\&+\frac{1}{4}{h_{2,xx}}'+\frac{1}{4}{h_{2,yy}}'+\frac{1}{2}\partial_z\omega_{2,z}\end{aligned}$$

(2) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; lensing $\Delta\mathbf{x}_{1\perp}$; explicit fields of order 1 (6 terms)

$$\begin{aligned}&\Delta x_1^x\partial_\eta\partial_x\Phi_1-\Delta x_1^x\partial_\eta\partial_x\Psi_1+2\Delta x_1^x\partial_x\partial_z\Psi_1\\&+\Delta x_1^y\partial_\eta\partial_y\Phi_1-\Delta x_1^y\partial_\eta\partial_y\Psi_1+2\Delta x_1^y\partial_y\partial_z\Psi_1\end{aligned}$$

(3) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; time delay Δx_1^z; explicit fields of order 1 (3 terms)

$$2\Delta x_1^z\partial_z^2\Psi_1+\Delta x_1^z\partial_\eta\partial_z\Phi_1-\Delta x_1^z\partial_\eta\partial_z\Psi_1$$

(4) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$; explicit fields of order 1 (2 terms)

$$-2k_1^x\partial_x\Psi_1-2k_1^y\partial_y\Psi_1$$

(5) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 1+1 (4 terms)

$$\begin{aligned} & 2\Phi_1\Phi_1{}' + 2\Phi_1\partial_z\Psi_1 + 2\Psi_1\Psi_1{}' \\ & - 2\Psi_1\partial_z\Psi_1 \end{aligned}$$

(6) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$; explicit fields of order 1 (3 terms)

$$\delta\eta_{o1}\Phi_1{}' - \delta\eta_{o1}\Psi_1{}' + 2\delta\eta_{o1}\partial_z\Psi_1$$

(7) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$; explicit fields of order 1 (3 terms)

$$- \delta\eta_{e1}\Phi_1{}' + \delta\eta_{e1}\Psi_1{}' - 2\delta\eta_{e1}\partial_z\Psi_1$$

(8) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; local; explicit fields of order 1+1 (6 terms)

$$\begin{aligned} & - \Phi_1 v_1^z - \Psi_1 v_1^z - (\Psi_1)^2 \\ & + \frac{1}{2}(\delta_{T,1})^2 - \frac{1}{2}(v_1^x)^2 - \frac{1}{2}(v_1^y)^2 \end{aligned}$$

(9) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; local; explicit fields of order 2 (4 terms)

$$\begin{aligned} & \frac{1}{2}\Psi_2 - \frac{1}{2}\delta_{T,2} + \frac{1}{2}\omega_{2,z} \\ & + \frac{1}{2}v_2^z \end{aligned}$$

(10) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$; explicit fields of order 1 (3 terms)

$$\delta\eta_{o1}\Psi_1{}' - \delta\eta_{o1}\delta_{T,1}{}' + \delta\eta_{o1}v_1^{z\prime}$$

(11) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; observer displacement $\delta\mathbf{x}_{o1}$; explicit fields of order 1 (9 terms)

$$\begin{aligned} & \delta x_{o1}^x\partial_x\Psi_1 - \delta x_{o1}^x\partial_x\delta_{T,1} + \delta x_{o1}^x\partial_x v_1^z \\ & + \delta x_{o1}^y\partial_y\Psi_1 - \delta x_{o1}^y\partial_y\delta_{T,1} + \delta x_{o1}^y\partial_y v_1^z \\ & + \delta x_{o1}^z\partial_z\Psi_1 - \delta x_{o1}^z\partial_z\delta_{T,1} + \delta x_{o1}^z\partial_z v_1^z \end{aligned}$$

(12) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; local; explicit fields of order 1+1 (6 terms)

$$\begin{aligned} & \Phi_1 v_1^z + \Psi_1 v_1^z + (\Psi_1)^2 \\ & - \frac{1}{2}(\delta_{T,1})^2 - \frac{1}{2}(v_1^x)^2 - \frac{1}{2}(v_1^y)^2 \end{aligned}$$

(13) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; local; explicit fields of order 2 (4 terms)

$$\begin{aligned} & - \frac{1}{2}\Psi_2 + \frac{1}{2}\delta_{T,2} - \frac{1}{2}\omega_{2,z} \\ & - \frac{1}{2}v_2^z \end{aligned}$$

(14) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$; explicit fields of order 1 (6 terms)

$$\begin{aligned} & - \delta\eta_{e1}\Psi_1{}' + \delta\eta_{e1}\partial_z\Psi_1 + \delta\eta_{e1}\delta_{T,1}{}' \\ & - \delta\eta_{e1}\partial_z\delta_{T,1} - \delta\eta_{e1}v_1^{z\prime} + \delta\eta_{e1}\partial_z v_1^z \end{aligned}$$

(15) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; lensing $\Delta\mathbf{x}_{1\perp}$; explicit fields of order 1 (6 terms)

$$\begin{aligned}&- \Delta x_1^x(\eta_*)\partial_x\Psi_1 + \Delta x_1^x(\eta_*)\partial_x\delta_{T,1} - \Delta x_1^x(\eta_*)\partial_x v_1^z\\ &- \Delta x_1^y(\eta_*)\partial_y\Psi_1 + \Delta x_1^y(\eta_*)\partial_y\delta_{T,1} - \Delta x_1^y(\eta_*)\partial_y v_1^z\end{aligned}$$

(16) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; time delay Δx_1^z; explicit fields of order 1 (3 terms)

$$- \Delta x_1^z(\eta_*)\partial_z\Psi_1 + \Delta x_1^z(\eta_*)\partial_z\delta_{T,1} - \Delta x_1^z(\eta_*)\partial_z v_1^z$$

(17) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; direction $\mathbf{k}_{1\perp}$; explicit fields of order 1 (2 terms)

$$k_1^x(\eta_*)v_1^x + k_1^y(\eta_*)v_1^y$$

Total: 76 terms; for symbolic $\hat{\mathbf{n}}$: 179 terms.

C.4 $\frac{1}{6}L_3$, shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$: all 359 terms

(1) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 3 (5 terms)

$$\begin{aligned}&\frac{1}{12}{h_{3,xx}}' + \frac{1}{12}{h_{3,yy}}'\\ &+ \frac{1}{18}\partial_x v_3^x + \frac{1}{18}\partial_y v_3^y\\ &- \frac{1}{9}\partial_z v_3^z\end{aligned}$$

(2) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 2+1 (57 terms)

$$\begin{aligned}&\frac{1}{2}\Phi_1{h_{2,xx}}' + \frac{1}{2}\Phi_1{h_{2,yy}}'\\ &- \frac{1}{3}\Phi_2\partial_x v_1^x - \frac{1}{3}\Phi_2\partial_y v_1^y\\ &+ \frac{2}{3}\Phi_2\partial_z v_1^z - \frac{1}{3}\Psi_2\partial_x v_1^x\\ &- \frac{1}{3}\Psi_2\partial_y v_1^y + \frac{2}{3}\Psi_2\partial_z v_1^z\\ &+ \frac{1}{2}h_{2,xx}{\Phi_1}' - \frac{1}{6}h_{2,xx}\partial_x v_1^x\\ &- \frac{1}{6}h_{2,xx}\partial_y v_1^y + \frac{1}{3}h_{2,xx}\partial_z v_1^z\\ &- \frac{1}{2}h_{2,xz}\partial_z v_1^x + \frac{1}{2}h_{2,yy}{\Phi_1}'\\ &- \frac{1}{6}h_{2,yy}\partial_x v_1^x - \frac{1}{6}h_{2,yy}\partial_y v_1^y\\ &+ \frac{1}{3}h_{2,yy}\partial_z v_1^z - \frac{1}{2}h_{2,yz}\partial_z v_1^y\\ &+ \frac{1}{6}\omega_{2,x}{v_1^x}' + \frac{1}{6}\omega_{2,y}{v_1^y}'\\ &- \frac{1}{6}\omega_{2,z}\partial_x v_1^x - \frac{1}{6}\omega_{2,z}\partial_y v_1^y\\ &- \frac{1}{3}\omega_{2,z}{v_1^z}' + \frac{1}{3}\omega_{2,z}\partial_z v_1^z\\ &+ \frac{1}{4}v_1^x\partial_x h_{2,xx} - \frac{1}{2}v_1^x{h_{2,xz}}'\end{aligned}$$

Table 11: The 32 groups of $\frac{1}{6}L_3$ (shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$), numbered as in the list below.

#	where	content	fields	terms
1	ray	local	3	5
2	ray	local	2+1	57
3	ray	direction $\mathbf{k}_{1\perp}$	2	6
4	ray	direction $\mathbf{k}_{2\perp}$	1	4
5	ray	time delay Δx_1^z	1+1	26
6	ray	local	1+1+1	55
7	ray	direction $\mathbf{k}_{1\perp}$ (squared)	1	8
8	ray	lensing $\Delta\mathbf{x}_{1\perp}$ (squared)	1	9
9	ray	time delay Δx_1^z (squared)	1	3
10	ray	direction k_2^z	1	3
11	ray	time delay Δx_2^z	1	3
12	ray	time delay Δx_1^z	2	5
13	ray	lensing $\Delta\mathbf{x}_{2\perp}$	1	6
14	ray	lensing $\Delta\mathbf{x}_{1\perp}$	2	10
15	ray	direction $\mathbf{k}_{1\perp}$	1+1	22
16	ray	direction $\mathbf{k}_{1\perp}$ × time delay Δx_1^z	1	4
17	ray	direction $\mathbf{k}_{1\perp}$ × lensing $\Delta\mathbf{x}_{1\perp}$	1	8
18	ray	lensing $\Delta\mathbf{x}_{1\perp}$	1+1	50
19	ray	lensing $\Delta\mathbf{x}_{1\perp}$ × time delay Δx_1^z	1	6
20	obs	clock $\delta\eta_{o1}$ (squared)	1	3
21	obs	clock $\delta\eta_{o1}$	2	5
22	obs	clock $\delta\eta_{o2}$	1	3
23	obs	clock $\delta\eta_{o1}$ × observer displacement $\delta\mathbf{x}_{o1}$	1	9
24	obs	clock $\delta\eta_{o1}$	1+1	6
25	obs	clock $\delta\eta_{o1}$ (squared)	1	6
26	emi	emission shift $\delta\eta_{e1}$	2	5
27	emi	emission shift $\delta\eta_{e2}$	1	3
28	emi	emission shift $\delta\eta_{e1}$	1+1	10
29	emi	emission shift $\delta\eta_{e1}$ × direction $\mathbf{k}_{1\perp}$	1	4
30	emi	emission shift $\delta\eta_{e1}$ × lensing $\Delta\mathbf{x}_{1\perp}$	1	6
31	emi	emission shift $\delta\eta_{e1}$ × time delay Δx_1^z	1	3
32	emi	emission shift $\delta\eta_{e1}$ (squared)	1	6
		total		359

Table 12: Groups of $\frac{1}{6}L_3$ for symbolic $\hat{\mathbf{n}}$ (term counts).

#	where	content	fields	terms
1	ray	direction $\mathbf{k}_1$	1	15
2	ray	local	1+1+1	411
3	ray	local	3	22
4	ray	direction $\mathbf{k}_1$	1+1	171
5	ray	direction $\mathbf{k}_1$	2	31
6	ray	direction $\mathbf{k}_2$	1	21
7	ray	local	2+1	404
8	ray	displacement $\Delta\mathbf{x}_1$	1+1	651
9	ray	displacement $\Delta\mathbf{x}_1$	1	90
10	ray	displacement $\Delta\mathbf{x}_2$	1	45
11	ray	displacement $\Delta\mathbf{x}_1$	2	66
12	ray	displacement $\Delta\mathbf{x}_1 \times$ direction $\mathbf{k}_1$	1	63
13	obs	clock $\delta\eta_{o1}$	1	36
14	obs	clock $\delta\eta_{o1}$	2	22
15	obs	clock $\delta\eta_{o2}$	1	15
16	obs	clock $\delta\eta_{o1} \times$ observer displacement $\delta\mathbf{x}_{o1}$	1	45
17	obs	clock $\delta\eta_{o1}$	1+1	60
18	obs	clock $\delta\eta_{o1}$	1	51
19	emi	emission shift $\delta\eta_{e1}$	2	22
20	emi	emission shift $\delta\eta_{e2}$	1	15
21	emi	emission shift $\delta\eta_{e1}$	1+1	111
22	emi	emission shift $\delta\eta_{e1} \times$ direction $\mathbf{k}_1$	1	21
23	emi	displacement $\Delta\mathbf{x}_1 \times$ emission shift $\delta\eta_{e1}$	1	45
24	emi	emission shift $\delta\eta_{e1}$	1	51
		total		2484

$$
\begin{aligned}
&+\frac{1}{4}v_1^x\partial_x h_{2,yy}+\frac{1}{6}v_1^x{\omega_{2,x}}'\\
&+\frac{1}{6}v_1^x v_2^{x\prime}-\frac{1}{2}v_1^x\partial_z v_2^x\\
&-\frac{1}{2}v_1^x\partial_x v_2^z+\frac{1}{6}v_2^x v_1^{x\prime}\\
&-\frac{1}{2}v_2^x\partial_z v_1^x-\frac{1}{2}v_2^x\partial_x v_1^z\\
&+\frac{1}{4}v_1^y\partial_y h_{2,xx}+\frac{1}{4}v_1^y\partial_y h_{2,yy}\\
&-\frac{1}{2}v_1^y{h_{2,yz}}'+\frac{1}{6}v_1^y{\omega_{2,y}}'\\
&+\frac{1}{6}v_1^y v_2^{y\prime}-\frac{1}{2}v_1^y\partial_z v_2^y\\
&-\frac{1}{2}v_1^y\partial_y v_2^z+\frac{1}{6}v_2^y v_1^{y\prime}\\
&-\frac{1}{2}v_2^y\partial_z v_1^y-\frac{1}{2}v_2^y\partial_y v_1^z\\
&+\frac{1}{4}v_1^z{h_{2,xx}}'+\frac{1}{4}v_1^z\partial_z h_{2,xx}\\
&+\frac{1}{4}v_1^z{h_{2,yy}}'+\frac{1}{4}v_1^z\partial_z h_{2,yy}\\
&-\frac{1}{3}v_1^z{\omega_{2,z}}'+\frac{1}{6}v_1^z\partial_x v_2^x\\
&+\frac{1}{6}v_1^z\partial_y v_2^y-\frac{1}{3}v_1^z v_2^{z\prime}\\
&-\frac{1}{3}v_1^z\partial_z v_2^z+\frac{1}{6}v_2^z\partial_x v_1^x\\
&+\frac{1}{6}v_2^z\partial_y v_1^y-\frac{1}{3}v_2^z v_1^{z\prime}\\
&-\frac{1}{3}v_2^z\partial_z v_1^z
\end{aligned}
$$

(3) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$; explicit fields of order 2 (6 terms)

$$
\begin{aligned}
&\frac{1}{2}k_1^x{h_{2,xz}}'+\frac{1}{2}k_1^x\partial_z v_2^x\\
&\quad+\frac{1}{2}k_1^x\partial_x v_2^z+\frac{1}{2}k_1^y{h_{2,yz}}'\\
&\quad+\frac{1}{2}k_1^y\partial_z v_2^y+\frac{1}{2}k_1^y\partial_y v_2^z
\end{aligned}
$$

(4) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; direction $\mathbf{k}_{2\perp}$; explicit fields of order 1 (4 terms)

$$
\begin{aligned}
&\frac{1}{2}k_2^x\partial_z v_1^x+\frac{1}{2}k_2^x\partial_x v_1^z\\
&\quad+\frac{1}{2}k_2^y\partial_z v_1^y+\frac{1}{2}k_2^y\partial_y v_1^z
\end{aligned}
$$

(5) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; time delay Δx_1^z; explicit fields of order 1+1 (26 terms)

$$
\begin{aligned}
&-\Delta x_1^z\left(\partial_z v_1^x\right)^2-\Delta x_1^z\left(\partial_z v_1^y\right)^2\\
&\quad-\frac{2}{3}\Delta x_1^z\left(\partial_z v_1^z\right)^2-\Delta x_1^z v_1^x\partial_z^2 v_1^x
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{3}\Delta x_1^z v_1^x \partial_\eta \partial_z v_1^x - \Delta x_1^z v_1^x \partial_x \partial_z v_1^z \\
&- \Delta x_1^z v_1^y \partial_z^2 v_1^y + \frac{1}{3}\Delta x_1^z v_1^y \partial_\eta \partial_z v_1^y \\
&- \Delta x_1^z v_1^y \partial_y \partial_z v_1^z - \frac{2}{3}\Delta x_1^z v_1^z \partial_z^2 v_1^z \\
&+\frac{1}{3}\Delta x_1^z v_1^z \partial_x \partial_z v_1^x + \frac{1}{3}\Delta x_1^z v_1^z \partial_y \partial_z v_1^y \\
&- \frac{2}{3}\Delta x_1^z v_1^z \partial_\eta \partial_z v_1^z - \frac{2}{3}\Delta x_1^z \partial_z \Phi_1 \partial_x v_1^x \\
&- \frac{2}{3}\Delta x_1^z \partial_z \Phi_1 \partial_y v_1^y + \frac{4}{3}\Delta x_1^z \partial_z \Phi_1 \partial_z v_1^z \\
&- \frac{2}{3}\Delta x_1^z \partial_z \Psi_1 \partial_x v_1^x - \frac{2}{3}\Delta x_1^z \partial_z \Psi_1 \partial_y v_1^y \\
&+\frac{4}{3}\Delta x_1^z \partial_z \Psi_1 \partial_z v_1^z + \frac{1}{3}\Delta x_1^z v_1^{x\prime} \partial_z v_1^x \\
&+\frac{1}{3}\Delta x_1^z \partial_x v_1^x \partial_z v_1^z - \Delta x_1^z \partial_z v_1^x \partial_x v_1^z \\
&+\frac{1}{3}\Delta x_1^z v_1^{y\prime} \partial_z v_1^y + \frac{1}{3}\Delta x_1^z \partial_y v_1^y \partial_z v_1^z \\
&- \Delta x_1^z \partial_z v_1^y \partial_y v_1^z - \frac{2}{3}\Delta x_1^z v_1^{z\prime} \partial_z v_1^z
\end{aligned}
$$

(6) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; local; explicit fields of order 1+1+1 (55 terms)

$$
\begin{aligned}
&- (\Phi_1)^2 \partial_x v_1^x - (\Phi_1)^2 \partial_y v_1^y \\
&\quad + 2(\Phi_1)^2 \partial_z v_1^z + \frac{1}{3}(\Psi_1)^2 \partial_x v_1^x \\
&\quad + \frac{1}{3}(\Psi_1)^2 \partial_y v_1^y - \frac{2}{3}(\Psi_1)^2 \partial_z v_1^z \\
&\quad - \frac{1}{3}(v_1^x)^2 {\Phi_1}' - \frac{1}{3}(v_1^x)^2 {\Psi_1}' \\
&\quad - \frac{1}{3}(v_1^x)^2 \partial_x v_1^x + \frac{1}{3}(v_1^x)^2 \partial_y v_1^y \\
&\quad - (v_1^x)^2 v_1^{z\prime} + \frac{1}{3}(v_1^x)^2 \partial_z v_1^z \\
&\quad - \frac{1}{3}(v_1^y)^2 {\Phi_1}' - \frac{1}{3}(v_1^y)^2 {\Psi_1}' \\
&\quad + \frac{1}{3}(v_1^y)^2 \partial_x v_1^x - \frac{1}{3}(v_1^y)^2 \partial_y v_1^y \\
&\quad - (v_1^y)^2 v_1^{z\prime} + \frac{1}{3}(v_1^y)^2 \partial_z v_1^z \\
&\quad + \frac{2}{3}(v_1^z)^2 {\Phi_1}' + \frac{2}{3}(v_1^z)^2 {\Psi_1}' \\
&\quad + \frac{1}{3}(v_1^z)^2 \partial_x v_1^x + \frac{1}{3}(v_1^z)^2 \partial_y v_1^y \\
&\quad - \frac{2}{3}(v_1^z)^2 v_1^{z\prime} - \frac{4}{3}(v_1^z)^2 \partial_z v_1^z \\
&\quad - \frac{2}{3}\Phi_1 \Psi_1 \partial_x v_1^x - \frac{2}{3}\Phi_1 \Psi_1 \partial_y v_1^y \\
&\quad + \frac{4}{3}\Phi_1 \Psi_1 \partial_z v_1^z - \frac{2}{3}\Phi_1 v_1^x v_1^{x\prime} \\
&\quad + \Phi_1 v_1^x \partial_z v_1^x + \Phi_1 v_1^x \partial_x v_1^z
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{2}{3}\Phi_1 v_1^y v_1^{y\prime} + \Phi_1 v_1^y \partial_z v_1^y \\
&+ \Phi_1 v_1^y \partial_y v_1^z - \frac{1}{3}\Phi_1 v_1^z \partial_x v_1^x \\
&- \frac{1}{3}\Phi_1 v_1^z \partial_y v_1^y + \frac{4}{3}\Phi_1 v_1^z v_1^{z\prime} \\
&+ \frac{2}{3}\Phi_1 v_1^z \partial_z v_1^z - \frac{2}{3}\Psi_1 v_1^x v_1^{x\prime} \\
&+ \Psi_1 v_1^x \partial_z v_1^x + \Psi_1 v_1^x \partial_x v_1^z \\
&- \frac{2}{3}\Psi_1 v_1^y v_1^{y\prime} + \Psi_1 v_1^y \partial_z v_1^y \\
&+ \Psi_1 v_1^y \partial_y v_1^z - \frac{1}{3}\Psi_1 v_1^z \partial_x v_1^x \\
&- \frac{1}{3}\Psi_1 v_1^z \partial_y v_1^y + \frac{4}{3}\Psi_1 v_1^z v_1^{z\prime} \\
&+ \frac{2}{3}\Psi_1 v_1^z \partial_z v_1^z - \frac{2}{3} v_1^x v_1^y \partial_y v_1^x \\
&- \frac{2}{3} v_1^x v_1^y \partial_x v_1^y - \frac{2}{3} v_1^x v_1^z v_1^{x\prime} \\
&- \frac{2}{3} v_1^x v_1^z \partial_z v_1^x - \frac{2}{3} v_1^x v_1^z \partial_x v_1^z \\
&- \frac{2}{3} v_1^y v_1^z v_1^{y\prime} - \frac{2}{3} v_1^y v_1^z \partial_z v_1^y \\
&- \frac{2}{3} v_1^y v_1^z \partial_y v_1^z
\end{aligned}
$$

(7) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$ (squared); explicit fields of order 1 (8 terms)

$$
\begin{aligned}
&- \frac{2}{3}(k_1^x)^2 \partial_x v_1^x + \frac{1}{3}(k_1^x)^2 \partial_y v_1^y \\
&+ \frac{1}{3}(k_1^x)^2 \partial_z v_1^z + \frac{1}{3}(k_1^y)^2 \partial_x v_1^x \\
&- \frac{2}{3}(k_1^y)^2 \partial_y v_1^y + \frac{1}{3}(k_1^y)^2 \partial_z v_1^z \\
&- k_1^x k_1^y \partial_y v_1^x - k_1^x k_1^y \partial_x v_1^y
\end{aligned}
$$

(8) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; lensing $\Delta\mathbf{x}_{1\perp}$ (squared); explicit fields of order 1 (9 terms)

$$
\begin{aligned}
&\frac{1}{6}(\Delta x_1^x)^2 \partial_x^3 v_1^x + \frac{1}{6}(\Delta x_1^x)^2 \partial_x^2 \partial_y v_1^y \\
&- \frac{1}{3}(\Delta x_1^x)^2 \partial_x^2 \partial_z v_1^z + \frac{1}{6}(\Delta x_1^y)^2 \partial_y^3 v_1^y \\
&+ \frac{1}{6}(\Delta x_1^y)^2 \partial_x \partial_y^2 v_1^x - \frac{1}{3}(\Delta x_1^y)^2 \partial_y^2 \partial_z v_1^z \\
&+ \frac{1}{3}\Delta x_1^x \Delta x_1^y \partial_x^2 \partial_y v_1^x + \frac{1}{3}\Delta x_1^x \Delta x_1^y \partial_x \partial_y^2 v_1^y \\
&- \frac{2}{3}\Delta x_1^x \Delta x_1^y \partial_x \partial_y \partial_z v_1^z
\end{aligned}
$$

(9) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; time delay Δx_1^z (squared); explicit fields of order 1 (3 terms)

$$
- \frac{1}{3}(\Delta x_1^z)^2 \partial_z^3 v_1^z + \frac{1}{6}(\Delta x_1^z)^2 \partial_x \partial_z^2 v_1^x
$$

$$+\frac{1}{6}(\Delta x_1^z)^2\partial_y\partial_z^2 v_1^y$$

(10) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; direction k_2^z; explicit fields of order 1 (3 terms)

$$\begin{aligned}&-\frac{1}{3}k_2^z\partial_x v_1^x-\frac{1}{3}k_2^z\partial_y v_1^y\\&+\frac{2}{3}k_2^z\partial_z v_1^z\end{aligned}$$

(11) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; time delay Δx_2^z; explicit fields of order 1 (3 terms)

$$\begin{aligned}&-\frac{1}{3}\Delta x_2^z\partial_z^2 v_1^z+\frac{1}{6}\Delta x_2^z\partial_x\partial_z v_1^x\\&+\frac{1}{6}\Delta x_2^z\partial_y\partial_z v_1^y\end{aligned}$$

(12) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; time delay Δx_1^z; explicit fields of order 2 (5 terms)

$$\begin{aligned}&-\frac{1}{3}\Delta x_1^z\partial_z^2 v_2^z+\frac{1}{4}\Delta x_1^z\partial_\eta\partial_z h_{2,xx}\\&+\frac{1}{4}\Delta x_1^z\partial_\eta\partial_z h_{2,yy}+\frac{1}{6}\Delta x_1^z\partial_x\partial_z v_2^x\\&+\frac{1}{6}\Delta x_1^z\partial_y\partial_z v_2^y\end{aligned}$$

(13) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; lensing $\Delta\mathbf{x}_{2\perp}$; explicit fields of order 1 (6 terms)

$$\begin{aligned}&\frac{1}{6}\Delta x_2^x\partial_x^2 v_1^x+\frac{1}{6}\Delta x_2^x\partial_x\partial_y v_1^y\\&-\frac{1}{3}\Delta x_2^x\partial_x\partial_z v_1^z+\frac{1}{6}\Delta x_2^y\partial_y^2 v_1^y\\&+\frac{1}{6}\Delta x_2^y\partial_x\partial_y v_1^x-\frac{1}{3}\Delta x_2^y\partial_y\partial_z v_1^z\end{aligned}$$

(14) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; lensing $\Delta\mathbf{x}_{1\perp}$; explicit fields of order 2 (10 terms)

$$\begin{aligned}&\frac{1}{6}\Delta x_1^x\partial_x^2 v_2^x+\frac{1}{4}\Delta x_1^x\partial_\eta\partial_x h_{2,xx}\\&+\frac{1}{4}\Delta x_1^x\partial_\eta\partial_x h_{2,yy}+\frac{1}{6}\Delta x_1^x\partial_x\partial_y v_2^y\\&-\frac{1}{3}\Delta x_1^x\partial_x\partial_z v_2^z+\frac{1}{6}\Delta x_1^y\partial_y^2 v_2^y\\&+\frac{1}{4}\Delta x_1^y\partial_\eta\partial_y h_{2,xx}+\frac{1}{4}\Delta x_1^y\partial_\eta\partial_y h_{2,yy}\\&+\frac{1}{6}\Delta x_1^y\partial_x\partial_y v_2^x-\frac{1}{3}\Delta x_1^y\partial_y\partial_z v_2^z\end{aligned}$$

(15) ray, $\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$; explicit fields of order 1+1 (22 terms)

$$\begin{aligned}&-k_1^x\Phi_1\partial_z v_1^x-k_1^x\Phi_1\partial_x v_1^z\\&-k_1^y\Phi_1\partial_z v_1^y-k_1^y\Phi_1\partial_y v_1^z\\&-k_1^x\Psi_1\partial_z v_1^x-k_1^x\Psi_1\partial_x v_1^z\end{aligned}$$

$$
\begin{aligned}
&- k_1^y \Psi_1 \partial_z v_1^y - k_1^y \Psi_1 \partial_y v_1^z \\
&+ \frac{5}{3} k_1^x v_1^x \partial_x v_1^x - \frac{1}{3} k_1^x v_1^x \partial_y v_1^y \\
&+ k_1^x v_1^x v_1^{z\prime} - \frac{4}{3} k_1^x v_1^x \partial_z v_1^z \\
&+ k_1^x v_1^y \partial_y v_1^x + k_1^x v_1^y \partial_x v_1^y \\
&+ k_1^x v_1^z v_1^{x\prime} + k_1^y v_1^x \partial_y v_1^x \\
&+ k_1^y v_1^x \partial_x v_1^y - \frac{1}{3} k_1^y v_1^y \partial_x v_1^x \\
&+ \frac{5}{3} k_1^y v_1^y \partial_y v_1^y + k_1^y v_1^y v_1^{z\prime} \\
&- \frac{4}{3} k_1^y v_1^y \partial_z v_1^z + k_1^y v_1^z v_1^{y\prime}
\end{aligned}
$$

(16) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$ × time delay Δx_1^z; explicit fields of order 1 (4 terms)

$$
\begin{aligned}
&\Delta x_1^z k_1^x \partial_z^2 v_1^x + \Delta x_1^z k_1^x \partial_x \partial_z v_1^z \\
&\quad + \Delta x_1^z k_1^y \partial_z^2 v_1^y + \Delta x_1^z k_1^y \partial_y \partial_z v_1^z
\end{aligned}
$$

(17) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; direction $\mathbf{k}_{1\perp}$ × lensing $\Delta \mathbf{x}_{1\perp}$; explicit fields of order 1 (8 terms)

$$
\begin{aligned}
&\Delta x_1^x k_1^x \partial_x^2 v_1^z + \Delta x_1^x k_1^x \partial_x \partial_z v_1^x \\
&\quad + \Delta x_1^x k_1^y \partial_x \partial_z v_1^y + \Delta x_1^x k_1^y \partial_x \partial_y v_1^z \\
&\quad + \Delta x_1^y k_1^x \partial_y \partial_z v_1^x + \Delta x_1^y k_1^x \partial_x \partial_y v_1^z \\
&\quad + \Delta x_1^y k_1^y \partial_y^2 v_1^z + \Delta x_1^y k_1^y \partial_y \partial_z v_1^y
\end{aligned}
$$

(18) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; lensing $\Delta \mathbf{x}_{1\perp}$; explicit fields of order 1+1 (50 terms)

$$
\begin{aligned}
&- \Delta x_1^x v_1^x \partial_x^2 v_1^z + \frac{1}{3} \Delta x_1^x v_1^x \partial_\eta \partial_x v_1^x \\
&- \Delta x_1^x v_1^x \partial_x \partial_z v_1^x + \frac{1}{3} \Delta x_1^x v_1^y \partial_\eta \partial_x v_1^y \\
&- \Delta x_1^x v_1^y \partial_x \partial_z v_1^y - \Delta x_1^x v_1^y \partial_x \partial_y v_1^z \\
&+ \frac{1}{3} \Delta x_1^x v_1^z \partial_x^2 v_1^x + \frac{1}{3} \Delta x_1^x v_1^z \partial_x \partial_y v_1^y \\
&- \frac{2}{3} \Delta x_1^x v_1^z \partial_\eta \partial_x v_1^z - \frac{2}{3} \Delta x_1^x v_1^z \partial_x \partial_z v_1^z \\
&- \frac{2}{3} \Delta x_1^x \partial_x \Phi_1 \partial_x v_1^x - \frac{2}{3} \Delta x_1^x \partial_x \Phi_1 \partial_y v_1^y \\
&+ \frac{4}{3} \Delta x_1^x \partial_x \Phi_1 \partial_z v_1^z - \frac{2}{3} \Delta x_1^x \partial_x \Psi_1 \partial_x v_1^x \\
&- \frac{2}{3} \Delta x_1^x \partial_x \Psi_1 \partial_y v_1^y + \frac{4}{3} \Delta x_1^x \partial_x \Psi_1 \partial_z v_1^z \\
&+ \frac{1}{3} \Delta x_1^x v_1^{x\prime} \partial_x v_1^x - \Delta x_1^x \partial_x v_1^x \partial_z v_1^x \\
&- \frac{2}{3} \Delta x_1^x \partial_x v_1^x \partial_x v_1^z + \frac{1}{3} \Delta x_1^x v_1^{y\prime} \partial_x v_1^y \\
&- \Delta x_1^x \partial_x v_1^y \partial_z v_1^y - \Delta x_1^x \partial_x v_1^y \partial_y v_1^z \\
&+ \frac{1}{3} \Delta x_1^x \partial_y v_1^y \partial_x v_1^z - \frac{2}{3} \Delta x_1^x v_1^{z\prime} \partial_x v_1^z
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{2}{3}\Delta x_1^x \partial_x v_1^z \partial_z v_1^z + \frac{1}{3}\Delta x_1^y v_1^x \partial_\eta \partial_y v_1^x \\
&- \Delta x_1^y v_1^x \partial_y \partial_z v_1^x - \Delta x_1^y v_1^x \partial_x \partial_y v_1^z \\
&- \Delta x_1^y v_1^y \partial_y^2 v_1^z + \frac{1}{3}\Delta x_1^y v_1^y \partial_\eta \partial_y v_1^y \\
&- \Delta x_1^y v_1^y \partial_y \partial_z v_1^y + \frac{1}{3}\Delta x_1^y v_1^z \partial_y^2 v_1^y \\
&+ \frac{1}{3}\Delta x_1^y v_1^z \partial_x \partial_y v_1^x - \frac{2}{3}\Delta x_1^y v_1^z \partial_\eta \partial_y v_1^z \\
&- \frac{2}{3}\Delta x_1^y v_1^z \partial_y \partial_z v_1^z - \frac{2}{3}\Delta x_1^y \partial_y \Phi_1 \partial_x v_1^x \\
&- \frac{2}{3}\Delta x_1^y \partial_y \Phi_1 \partial_y v_1^y + \frac{4}{3}\Delta x_1^y \partial_y \Phi_1 \partial_z v_1^z \\
&- \frac{2}{3}\Delta x_1^y \partial_y \Psi_1 \partial_x v_1^x - \frac{2}{3}\Delta x_1^y \partial_y \Psi_1 \partial_y v_1^y \\
&+ \frac{4}{3}\Delta x_1^y \partial_y \Psi_1 \partial_z v_1^z + \frac{1}{3}\Delta x_1^y v_1^{x\prime} \partial_y v_1^x \\
&+ \frac{1}{3}\Delta x_1^y \partial_x v_1^x \partial_y v_1^z - \Delta x_1^y \partial_y v_1^x \partial_z v_1^x \\
&- \Delta x_1^y \partial_y v_1^x \partial_x v_1^z + \frac{1}{3}\Delta x_1^y v_1^{y\prime} \partial_y v_1^y \\
&- \Delta x_1^y \partial_y v_1^y \partial_z v_1^y - \frac{2}{3}\Delta x_1^y \partial_y v_1^y \partial_y v_1^z \\
&- \frac{2}{3}\Delta x_1^y v_1^{z\prime} \partial_y v_1^z - \frac{2}{3}\Delta x_1^y \partial_y v_1^z \partial_z v_1^z
\end{aligned}
$$

(19) ray, $\int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta$; lensing $\Delta \mathbf{x}_{1\perp}$ $\times$ time delay Δx_1^z; explicit fields of order 1 (6 terms)

$$
\begin{aligned}
&\frac{1}{3}\Delta x_1^x \Delta x_1^z \partial_x^2 \partial_z v_1^x - \frac{2}{3}\Delta x_1^x \Delta x_1^z \partial_x \partial_z^2 v_1^z \\
&\quad + \frac{1}{3}\Delta x_1^x \Delta x_1^z \partial_x \partial_y \partial_z v_1^y + \frac{1}{3}\Delta x_1^y \Delta x_1^z \partial_y^2 \partial_z v_1^y \\
&\quad - \frac{2}{3}\Delta x_1^y \Delta x_1^z \partial_y \partial_z^2 v_1^z + \frac{1}{3}\Delta x_1^y \Delta x_1^z \partial_x \partial_y \partial_z v_1^x
\end{aligned}
$$

(20) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$ (squared); explicit fields of order 1 (3 terms)

$$
\begin{aligned}
&- \frac{2}{3}\delta\eta_{o1}^2 \partial_z^2 v_1^z + \frac{1}{3}\delta\eta_{o1}^2 \partial_x \partial_z v_1^x \\
&\quad + \frac{1}{3}\delta\eta_{o1}^2 \partial_y \partial_z v_1^y
\end{aligned}
$$

(21) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$; explicit fields of order 2 (5 terms)

$$
\begin{aligned}
&\frac{1}{4}\delta\eta_{o1} h_{2,xx}{}' + \frac{1}{4}\delta\eta_{o1} h_{2,yy}{}' \\
&\quad + \frac{1}{6}\delta\eta_{o1} \partial_x v_2^x + \frac{1}{6}\delta\eta_{o1} \partial_y v_2^y \\
&\quad - \frac{1}{3}\delta\eta_{o1} \partial_z v_2^z
\end{aligned}
$$

(22) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o2}$; explicit fields of order 1 (3 terms)

$$
\frac{1}{6}\delta\eta_{o2} \partial_x v_1^x + \frac{1}{6}\delta\eta_{o2} \partial_y v_1^y
$$

$$- \frac{1}{3}\delta\eta_{o2}\partial_z v_1^z$$

(23) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$ × observer displacement $\delta\mathbf{x}_{o1}$; explicit fields of order 1 (9 terms)

$$\begin{aligned}
&\frac{1}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x^2 v_1^x + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x\partial_y v_1^y \\
&\quad - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x\partial_z v_1^z + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_y^2 v_1^y \\
&\quad + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_x\partial_y v_1^x - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_y\partial_z v_1^z \\
&\quad - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_z^2 v_1^z + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_x\partial_z v_1^x \\
&\quad + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_y\partial_z v_1^y
\end{aligned}$$

(24) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$; explicit fields of order 1+1 (6 terms)

$$\begin{aligned}
&\frac{1}{3}\delta\eta_{o1} v_1^x v_1^{x\prime} + \frac{1}{3}\delta\eta_{o1} v_1^y v_1^{y\prime} \\
&\quad + \frac{1}{3}\delta\eta_{o1} v_1^z \partial_x v_1^x + \frac{1}{3}\delta\eta_{o1} v_1^z \partial_y v_1^y \\
&\quad - \frac{2}{3}\delta\eta_{o1} v_1^z v_1^{z\prime} - \frac{2}{3}\delta\eta_{o1} v_1^z \partial_z v_1^z
\end{aligned}$$

(25) observer, at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$; clock $\delta\eta_{o1}$ (squared); explicit fields of order 1 (6 terms)

$$\begin{aligned}
&\frac{1}{3}\delta\eta_{o1}^2 \partial_z^2 v_1^z + \frac{1}{6}\delta\eta_{o1}^2 \partial_\eta\partial_x v_1^x \\
&\quad - \frac{1}{6}\delta\eta_{o1}^2 \partial_x\partial_z v_1^x + \frac{1}{6}\delta\eta_{o1}^2 \partial_\eta\partial_y v_1^y \\
&\quad - \frac{1}{6}\delta\eta_{o1}^2 \partial_y\partial_z v_1^y - \frac{1}{3}\delta\eta_{o1}^2 \partial_\eta\partial_z v_1^z
\end{aligned}$$

(26) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$; explicit fields of order 2 (5 terms)

$$\begin{aligned}
&- \frac{1}{4}\delta\eta_{e1} h_{2,xx}{}' - \frac{1}{4}\delta\eta_{e1} h_{2,yy}{}' \\
&\quad - \frac{1}{6}\delta\eta_{e1}\partial_x v_2^x - \frac{1}{6}\delta\eta_{e1}\partial_y v_2^y \\
&\quad + \frac{1}{3}\delta\eta_{e1}\partial_z v_2^z
\end{aligned}$$

(27) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e2}$; explicit fields of order 1 (3 terms)

$$\begin{aligned}
&- \frac{1}{6}\delta\eta_{e2}\partial_x v_1^x - \frac{1}{6}\delta\eta_{e2}\partial_y v_1^y \\
&\quad + \frac{1}{3}\delta\eta_{e2}\partial_z v_1^z
\end{aligned}$$

(28) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$; explicit fields of order 1+1 (10 terms)

$$\begin{aligned}
&- \frac{1}{3}\delta\eta_{e1} v_1^x v_1^{x\prime} + \delta\eta_{e1} v_1^x \partial_z v_1^x \\
&\quad + \delta\eta_{e1} v_1^x \partial_x v_1^z - \frac{1}{3}\delta\eta_{e1} v_1^y v_1^{y\prime}
\end{aligned}$$

$$
\begin{aligned}
&+ \delta\eta_{e1} v_1^y \partial_z v_1^y + \delta\eta_{e1} v_1^y \partial_y v_1^z \\
&- \frac{1}{3}\delta\eta_{e1} v_1^z \partial_x v_1^x - \frac{1}{3}\delta\eta_{e1} v_1^z \partial_y v_1^y \\
&+ \frac{2}{3}\delta\eta_{e1} v_1^z v_1^{z\prime} + \frac{2}{3}\delta\eta_{e1} v_1^z \partial_z v_1^z
\end{aligned}
$$

(29) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$ × direction $\mathbf{k}_{1\perp}$; explicit fields of order 1 (4 terms)

$$
\begin{aligned}
&- \delta\eta_{e1} k_1^x(\eta_*) \partial_z v_1^x - \delta\eta_{e1} k_1^x(\eta_*) \partial_x v_1^z \\
&- \delta\eta_{e1} k_1^y(\eta_*) \partial_z v_1^y - \delta\eta_{e1} k_1^y(\eta_*) \partial_y v_1^z
\end{aligned}
$$

(30) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$ × lensing $\Delta\mathbf{x}_{1\perp}$; explicit fields of order 1 (6 terms)

$$
\begin{aligned}
&- \frac{1}{3}\delta\eta_{e1} \Delta x_1^x(\eta_*) \partial_x^2 v_1^x - \frac{1}{3}\delta\eta_{e1} \Delta x_1^x(\eta_*) \partial_x \partial_y v_1^y \\
&+ \frac{2}{3}\delta\eta_{e1} \Delta x_1^x(\eta_*) \partial_x \partial_z v_1^z - \frac{1}{3}\delta\eta_{e1} \Delta x_1^y(\eta_*) \partial_y^2 v_1^y \\
&- \frac{1}{3}\delta\eta_{e1} \Delta x_1^y(\eta_*) \partial_x \partial_y v_1^x + \frac{2}{3}\delta\eta_{e1} \Delta x_1^y(\eta_*) \partial_y \partial_z v_1^z
\end{aligned}
$$

(31) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$ × time delay Δx_1^z; explicit fields of order 1 (3 terms)

$$
\begin{aligned}
&\frac{2}{3}\delta\eta_{e1} \Delta x_1^z(\eta_*) \partial_z^2 v_1^z - \frac{1}{3}\delta\eta_{e1} \Delta x_1^z(\eta_*) \partial_x \partial_z v_1^x \\
&- \frac{1}{3}\delta\eta_{e1} \Delta x_1^z(\eta_*) \partial_y \partial_z v_1^y
\end{aligned}
$$

(32) emission, at $(\eta_*, \bar{\mathbf{x}}_*)$; emission shift $\delta\eta_{e1}$ (squared); explicit fields of order 1 (6 terms)

$$
\begin{aligned}
&- \frac{1}{3}\delta\eta_{e1}^2 \partial_z^2 v_1^z - \frac{1}{6}\delta\eta_{e1}^2 \partial_\eta \partial_x v_1^x \\
&+ \frac{1}{6}\delta\eta_{e1}^2 \partial_x \partial_z v_1^x - \frac{1}{6}\delta\eta_{e1}^2 \partial_\eta \partial_y v_1^y \\
&+ \frac{1}{6}\delta\eta_{e1}^2 \partial_y \partial_z v_1^y + \frac{1}{3}\delta\eta_{e1}^2 \partial_\eta \partial_z v_1^z
\end{aligned}
$$

Total: 359 terms. The direct form of $\frac{1}{6}L_3$ has 480 terms ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$, 69 groups) and 1402 for symbolic $\hat{\mathbf{n}}$; it agrees with the shear form modulo the conservation equations (check R-4 of App. G).

D Machine-generated channel kernels of K_2

This appendix reproduces the channel kernels of $K_2(\mathbf{k}_1, \mathbf{k}_2; \hat{\mathbf{n}})$ exactly as generated by the symbolic expansion of the direct-form term list of $L_2/2$ (Sec. 8; code in App. F), with the symbol names of Sec. 11.2 substituted typographically (no algebra was changed; the script that performs the substitution also verifies symbolically that every expression below equals the compact form quoted in Sec. 11.3: residual 0 for all fifteen channels). Each expression is already the contribution to K_2, i.e. it includes the factor 2 of $K_2 = 2\,\mathrm{Sym}[K_{L_2/2}] + \dots$ and the symmetrisation $1 \leftrightarrow 2$. Notation (Sec. 11.2): $\mu = \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_2$, $\mu_a = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}}_a$, $\mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2$, $K_n = \hat{\mathbf{n}} \cdot \mathbf{K} = k_1\mu_1 + k_2\mu_2$, $P_\perp = k_1 k_2(\mu - \mu_1\mu_2)$,

$E_a = e^{ik_a\mu_a\chi}$, $E_K = E_1E_2$; first-order transfer functions $T_{X,a} \equiv T_X(k_a,\eta)$ and their η-derivatives $\dot T_{X,a}$; ray functionals $\mathcal{F}_a$, $\mathcal{D}_{\perp,a}$, $\mathcal{D}_{r,a}$ (139); observer and emission objects $X_{o,a}$, $\delta\eta_{o,a}$, $\delta\eta_{e,a}$ (134)–(136); second-order pair kernels $\Phi_2, \dot\Phi_2, \Psi_2, \dot\Psi_2, \delta_{T2}, V_2$ and $K_\omega^{(ab)} \equiv K_\omega(\mathbf{k}_a,\mathbf{k}_b)$, $K_{v^T}^{(ab)}$, $\mathcal{H}$, $\dot{\mathcal{H}}$ (140). The numerical value of every channel agrees with the independent numerical engine to $\le 1.8\times10^{-15}$. The type-B products are $K_1(\mathbf{k}_1)K_1(\mathbf{k}_2) - \Delta_0(k_1)K_1(\mathbf{k}_2) - \Delta_0(k_2)K_1(\mathbf{k}_1)$ (137).

[ray] second-order fields (C: Φ_2,Ψ_2 ISW, ω_2, h_2). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$E_K\dot\Phi_2 - E_K\dot\Psi_2 + \frac{E_K\dot{\mathcal{H}}k_1^2}{4} + 2iE_K\Psi_2K_n - \frac{E_K\dot{\mathcal{H}}k_1^2\mu_1^2}{2} - \frac{E_K\dot{\mathcal{H}}k_1^4}{4K^2} - \frac{E_K\dot{\mathcal{H}}K_n^2k_1^4}{4K^4}$$
$$- \frac{E_K\dot{\mathcal{H}}K_n^2k_1^2}{4K^2} + \frac{E_K\dot{\mathcal{H}}K_n\mu_1k_1^3}{K^2} + \frac{iE_KK_\omega^{(12)}K_nk_1\mu_1}{2} + \frac{iE_KK_\omega^{(21)}K_nk_2\mu_2}{2}$$
$$- \frac{iE_KK_\omega^{(12)}K_n^2k_1^2}{2K^2} - \frac{iE_KK_\omega^{(21)}K_n^2k_2^2}{2K^2} - \frac{E_K\dot{\mathcal{H}}k_2\mu k_1^3}{2K^2} - \frac{E_K\dot{\mathcal{H}}k_1^2k_2^2\mu^2}{4K^2} - \frac{E_K\dot{\mathcal{H}}k_2\mu K_n^2k_1^3}{2K^4}$$
$$- \frac{E_K\dot{\mathcal{H}}K_n^2k_1^2k_2^2\mu^2}{4K^4} + \frac{E_K\dot{\mathcal{H}}K_nk_2\mu\mu_1k_1^2}{K^2} - \frac{iE_KK_\omega^{(12)}k_1k_2\mu K_n^2}{2K^2} - \frac{iE_KK_\omega^{(21)}k_1k_2\mu K_n^2}{2K^2}$$

[ray] lensing, transverse $\Delta\mathbf{x}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$-2i\mathcal{D}_{\perp,1}E_2P_\perp T_{\Phi,2}k_2\mu_2 - 2i\mathcal{D}_{\perp,2}E_1P_\perp T_{\Phi,1}k_1\mu_1$$

[ray] time delay, radial $\Delta\mathbf{x}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$-2\mathcal{D}_{r,1}E_2T_{\Phi,2}k_2^2\mu_2^2 - 2\mathcal{D}_{r,2}E_1T_{\Phi,1}k_1^2\mu_1^2$$

[ray] aberration + deflection, transverse $\mathbf{k}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$2E_1\mathcal{F}_2P_\perp T_{\Phi,1} + 2E_2\mathcal{F}_1P_\perp T_{\Phi,2}$$

[ray] local 1×1 (C). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$4E_1E_2T_{\Phi,1}\dot T_{\Phi,2} + 4E_1E_2T_{\Phi,2}\dot T_{\Phi,1}$$

[obs] observer clock (D). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$E_1\dot T_{\Phi,1}\delta\eta_{o,2} + E_2\dot T_{\Phi,2}\delta\eta_{o,1} - E_1\dot T_{\delta_T,1}\delta\eta_{o,2} - E_2\dot T_{\delta_T,2}\delta\eta_{o,1} + iE_1\dot T_{V,1}\delta\eta_{o,2}k_1\mu_1$$
$$+ iE_2\dot T_{V,2}\delta\eta_{o,1}k_2\mu_2 + 2iE_1T_{\Phi,1}\delta\eta_{o,2}k_1\mu_1 + 2iE_2T_{\Phi,2}\delta\eta_{o,1}k_2\mu_2$$

[emi] emission-surface shift (D). at $\eta=\eta_*$, $\chi=\chi_*$:

$$E_1\dot T_{\delta_T,1}\delta\eta_{e,2} + E_2\dot T_{\delta_T,2}\delta\eta_{e,1} - E_1\dot T_{\Phi,1}\delta\eta_{e,2} - E_2\dot T_{\Phi,2}\delta\eta_{e,1} - E_1T_{V,1}\delta\eta_{e,2}k_1^2\mu_1^2$$
$$- E_2T_{V,2}\delta\eta_{e,1}k_2^2\mu_2^2 - iE_1T_{\Phi,1}\delta\eta_{e,2}k_1\mu_1 - iE_1\dot T_{V,1}\delta\eta_{e,2}k_1\mu_1 - iE_1T_{\delta_T,1}\delta\eta_{e,2}k_1\mu_1$$
$$- iE_2T_{\Phi,2}\delta\eta_{e,1}k_2\mu_2 - iE_2\dot T_{V,2}\delta\eta_{e,1}k_2\mu_2 - iE_2T_{\delta_T,2}\delta\eta_{e,1}k_2\mu_2$$

[obs] local 1×1 (B, endpoint). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$E_1E_2T_{\delta_T,1}T_{\delta_T,2} - 2E_1E_2T_{\Phi,1}T_{\Phi,2} + E_1E_2P_\perp T_{V,1}T_{V,2} - 2iE_1E_2T_{\Phi,1}T_{V,2}k_2\mu_2 - 2iE_1E_2T_{\Phi,2}T_{V,1}k_1\mu_1$$

[obs] second-order fields (C: $\Psi_2, \delta_{T2}, V_2, v_2^T + \omega_2$). at $\eta = \eta_o$, $\chi = 0$ ($E_a = 1$):

$$E_K\Psi_2 - E_K\delta_{T2} + iE_KV_2K_n + \frac{E_KK_\omega^{(12)}k_1\mu_1}{2} + \frac{E_KK_\omega^{(21)}k_2\mu_2}{2} + \frac{E_KK_{v^T}^{(12)}k_1\mu_1}{2} + \frac{E_KK_{v^T}^{(21)}k_2\mu_2}{2} - \frac{E_KK_\omega^{(12)}K_nk_1^2}{2K^2} - \frac{E_KK_\omega^{(21)}K_nk_2^2}{2K^2} - \frac{E_KK_{v^T}^{(12)}K_nk_1^2}{2K^2} - \frac{E_KK_{v^T}^{(21)}K_nk_2^2}{2K^2} - \frac{E_KK_\omega^{(12)}K_nk_1k_2\mu}{2K^2} - \frac{E_KK_\omega^{(21)}K_nk_1k_2\mu}{2K^2} - \frac{E_KK_{v^T}^{(12)}K_nk_1k_2\mu}{2K^2} - \frac{E_KK_{v^T}^{(21)}K_nk_1k_2\mu}{2K^2}$$

[obs] observer displacement (D). at $\eta = \eta_o$, $\chi = 0$ ($E_a = 1$):

$$E_1P_\perp T_{\delta_T,1}X_{o,2} + E_2P_\perp T_{\delta_T,2}X_{o,1} - E_1P_\perp T_{\Phi,1}X_{o,2} - E_2P_\perp T_{\Phi,2}X_{o,1} + E_1T_{\delta_T,1}X_{o,2}k_1k_2\mu_1\mu_2 + E_2T_{\delta_T,2}X_{o,1}k_1k_2\mu_1\mu_2 - iE_1P_\perp T_{V,1}X_{o,2}k_1\mu_1 - iE_2P_\perp T_{V,2}X_{o,1}k_2\mu_2 - E_1T_{\Phi,1}X_{o,2}k_1k_2\mu_1\mu_2 - E_2T_{\Phi,2}X_{o,1}k_1k_2\mu_1\mu_2 - iE_1T_{V,1}X_{o,2}k_2\mu_2k_1^2\mu_1^2 - iE_2T_{V,2}X_{o,1}k_1\mu_1k_2^2\mu_2^2$$

[emi] local 1×1 (B, endpoint). at $\eta = \eta_*$, $\chi = \chi_*$:

$$-E_1E_2T_{\delta_T,1}T_{\delta_T,2} + 2E_1E_2T_{\Phi,1}T_{\Phi,2} + E_1E_2P_\perp T_{V,1}T_{V,2} + 2iE_1E_2T_{\Phi,1}T_{V,2}k_2\mu_2 + 2iE_1E_2T_{\Phi,2}T_{V,1}k_1\mu_1$$

[emi] second-order fields (C: $\Psi_2, \delta_{T2}, V_2, v_2^T + \omega_2$). at $\eta = \eta_*$, $\chi = \chi_*$:

$$E_K\delta_{T2} - E_K\Psi_2 - iE_KV_2K_n - \frac{E_KK_\omega^{(12)}k_1\mu_1}{2} - \frac{E_KK_\omega^{(21)}k_2\mu_2}{2} - \frac{E_KK_{v^T}^{(12)}k_1\mu_1}{2} - \frac{E_KK_{v^T}^{(21)}k_2\mu_2}{2} + \frac{E_KK_\omega^{(12)}K_nk_1^2}{2K^2} + \frac{E_KK_\omega^{(21)}K_nk_2^2}{2K^2} + \frac{E_KK_{v^T}^{(12)}K_nk_1^2}{2K^2} + \frac{E_KK_{v^T}^{(21)}K_nk_2^2}{2K^2} + \frac{E_KK_\omega^{(12)}K_nk_1k_2\mu}{2K^2} + \frac{E_KK_\omega^{(21)}K_nk_1k_2\mu}{2K^2} + \frac{E_KK_{v^T}^{(12)}K_nk_1k_2\mu}{2K^2} + \frac{E_KK_{v^T}^{(21)}K_nk_1k_2\mu}{2K^2}$$

[emi] lensing at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$\mathcal{D}_{\perp,1}E_2P_\perp T_{\Phi,2} + \mathcal{D}_{\perp,2}E_1P_\perp T_{\Phi,1} - \mathcal{D}_{\perp,1}E_2P_\perp T_{\delta_T,2} - \mathcal{D}_{\perp,2}E_1P_\perp T_{\delta_T,1} + i\mathcal{D}_{\perp,1}E_2P_\perp T_{V,2}k_2\mu_2 + i\mathcal{D}_{\perp,2}E_1P_\perp T_{V,1}k_1\mu_1$$

[emi] time delay at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$\mathcal{D}_{r,1}E_2T_{V,2}k_2^2\mu_2^2 + \mathcal{D}_{r,2}E_1T_{V,1}k_1^2\mu_1^2 + i\mathcal{D}_{r,1}E_2T_{\delta_T,2}k_2\mu_2 + i\mathcal{D}_{r,2}E_1T_{\delta_T,1}k_1\mu_1 - i\mathcal{D}_{r,1}E_2T_{\Phi,2}k_2\mu_2 - i\mathcal{D}_{r,2}E_1T_{\Phi,1}k_1\mu_1$$

[emi] aberration + deflection at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$-E_1\mathcal{F}_2P_\perp T_{V,1} - E_2\mathcal{F}_1P_\perp T_{V,2}$$

E Machine-generated channels of the contracted third-order kernel

This appendix lists the term groups from which the contracted third-order kernel κ of Sec. 11.6 is assembled, exactly as produced by the symbolic expansion of S (Sec. 9, App. C.4) in the shear form, written in the frame $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ (x, y transverse, z along the observation direction; spatial components are superscripts or second subscripts, perturbative order the first subscript: v_1^x, $h_{2,xz}$, $\omega_{2,z}$, k_1^x, Δx_2^z). Only the typography has been changed (no algebra). The three lists are L_1, $\frac{1}{2}L_2$, $\frac{1}{6}L_3$ of $\Lambda = -S = \epsilon L_1 + \frac{\epsilon^2}{2}L_2 + \frac{\epsilon^3}{6}L_3 + \ldots$; "ray" groups are integrated along the background ray $\bar{\mathbf{x}} = \hat{\mathbf{n}}\chi$, "observer" groups are evaluated at $(\eta_o, \mathbf{0})$ and "emission" groups at $(\eta_*, \hat{\mathbf{n}}\chi_*)$. Ray functionals and observer quantities are those of Sec. 8.1 and Sec. 8.2 in the limit $\eta_i \to 0$: $k_1 = k_{1,\mathrm{init}} - \int_\eta^{\eta_o} G_1$, $\Delta\mathbf{x}_1 = \delta\mathbf{x}_{o1} + \hat{\mathbf{n}}\,\delta\eta_{o1} - \int_\eta^{\eta_o} \mathbf{k}_1$, $\frac{1}{2}k_2 = \frac{1}{2}k_{2,\mathrm{init}} - \int_\eta^{\eta_o}[\frac{1}{2}G_2 + \Delta\mathbf{x}_1 \cdot \nabla G_1]$ (plus the transport terms of (88)), $\frac{1}{2}\Delta\mathbf{x}_2 = \frac{1}{2}\delta\mathbf{x}_{o2} + \frac{1}{2}\hat{\mathbf{n}}\,\delta\eta_{o2} - \delta\eta_{o1}\mathbf{k}_1(\eta_o) - \int_\eta^{\eta_o}\frac{1}{2}\mathbf{k}_2$; the observer clock and displacement are (152); $\delta\eta_{e1}, \delta\eta_{e2}$ are the emission-time shifts (93).

Contraction. For $\zeta = a\,e^{i\mathbf{k}\cdot\mathbf{x}} + b\,e^{i\mathbf{q}\cdot\mathbf{x}} + c\,e^{-i\mathbf{q}\cdot\mathbf{x}}$ with $a^2 = b^2 = c^2 = 0$, coef_{abc} denotes the coefficient of abc in an expression; then $\kappa(\mathbf{k}; \hat{\mathbf{n}}) = \frac{1}{2}\int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\,\mathrm{coef}_{abc}$ (149), applied to $\Theta^{(3)} = 6\,[\frac{1}{6}L_3] + 3L_1L_2 + L_1^3 - 3\langle m_1\rangle_{\mathrm{sky}} m_2 - 3\langle m_2\rangle_{\mathrm{sky}} m_1 + 6\langle m_1\rangle_{\mathrm{sky}}^2 m_1$. In terms of kernels: a product of three first-order objects $ABC \to \sum_\pi A[\mathbf{k}_{\pi 1}]B[\mathbf{k}_{\pi 2}]C[\mathbf{k}_{\pi 3}]$ over the permutations of $(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ (class (i)); a first×second-order product $A_1B_2 \to A[\mathbf{k}]\,B_2^{\mathrm{s}}(\mathbf{q}, -\mathbf{q}) + A[\mathbf{q}]\,B_2^{\mathrm{s}}(\mathbf{k}, -\mathbf{q}) + A[-\mathbf{q}]\,B_2^{\mathrm{s}}(\mathbf{k}, \mathbf{q})$, with $B_2^{\mathrm{s}}(\mathbf{q}_1, \mathbf{q}_2) = K_B(\mathbf{q}_1, \mathbf{q}_2) + K_B(\mathbf{q}_2, \mathbf{q}_1)$ and $(\mathbf{q}, -\mathbf{q})$ the tadpole (class (ii)); a third-order field $X_3 \to K_{3c,X}(\mathbf{q}) + K_{3c,X}(-\mathbf{q})$, the contracted solution (65) (class (iii)). Every group below has been evaluated numerically by the engine of Sec. 11.6; the shear-form lists reproduce the direct-form lists at third order (Sec. 11.7).

Term list of L_1 (1 groups, 3 terms)

local; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{\partial_x v_1^x}{3} + \frac{\partial_y v_1^y}{3} - \frac{2\partial_z v_1^z}{3}$$

Term list of $\frac{1}{2}L_2$ (7 groups, 34 terms)

local; fields of order 2 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{\partial_\eta h_{2,xx}}{4} + \frac{\partial_\eta h_{2,yy}}{4} + \frac{\partial_x v_2^x}{6} + \frac{\partial_y v_2^y}{6} - \frac{\partial_z v_2^z}{3}$$

transverse direction $\mathbf{k}_1$ (aberration+deflection); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$k_1^x\partial_z v_1^x + k_1^x\partial_x v_1^z + k_1^y\partial_z v_1^y + k_1^y\partial_y v_1^z$$

local; fields of order 1+1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{v_1^x\partial_\eta v_1^x}{3} - v_1^x\partial_z v_1^x - v_1^x\partial_x v_1^z + \frac{v_1^y\partial_\eta v_1^y}{3} - v_1^y\partial_z v_1^y - v_1^y\partial_y v_1^z + \frac{v_1^z\partial_x v_1^x}{3} + \frac{v_1^z\partial_y v_1^y}{3} - \frac{2v_1^z\partial_\eta v_1^z}{3} - \frac{2v_1^z\partial_z v_1^z}{3}$$

time delay (radial $\Delta\mathbf{x}_1$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$-\frac{2\Delta x_1^z\partial_z^2 v_1^z}{3}+\frac{\Delta x_1^z\partial_z\partial_x v_1^x}{3}+\frac{\Delta x_1^z\partial_z\partial_y v_1^y}{3}$$

lensing (transverse $\Delta\mathbf{x}_1$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{\Delta x_1^x\partial_x^2 v_1^x}{3}+\frac{\Delta x_1^x\partial_y\partial_x v_1^y}{3}-\frac{2\Delta x_1^x\partial_z\partial_x v_1^z}{3}+\frac{\Delta x_1^y\partial_y^2 v_1^y}{3}+\frac{\Delta x_1^y\partial_y\partial_x v_1^x}{3}-\frac{2\Delta x_1^y\partial_z\partial_y v_1^z}{3}$$

observer clock; fields of order 1 [observer, $\eta=\eta_o$, $\chi=0$].

$$\frac{\delta\eta_{o1}\partial_x v_1^x}{3}+\frac{\delta\eta_{o1}\partial_y v_1^y}{3}-\frac{2\delta\eta_{o1}\partial_z v_1^z}{3}$$

emission-surface shift; fields of order 1 [emission, $\eta=\eta_*$].

$$-\frac{\delta\eta_{e1}\partial_x v_1^x}{3}-\frac{\delta\eta_{e1}\partial_y v_1^y}{3}+\frac{2\delta\eta_{e1}\partial_z v_1^z}{3}$$

Term list of $\frac{1}{6}L_3$ (32 groups, 359 terms)

local; fields of order 3 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{\partial_\eta h_{3,xx}}{12}+\frac{\partial_\eta h_{3,yy}}{12}+\frac{\partial_x v_3^x}{18}+\frac{\partial_y v_3^y}{18}-\frac{\partial_z v_3^z}{9}$$

local; fields of order 2+1 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\begin{aligned}
&\frac{\Phi_1\partial_\eta h_{2,xx}}{2}+\frac{\Phi_1\partial_\eta h_{2,yy}}{2}-\frac{\Phi_2\partial_x v_1^x}{3}-\frac{\Phi_2\partial_y v_1^y}{3}+\frac{2\Phi_2\partial_z v_1^z}{3}-\frac{\Psi_2\partial_x v_1^x}{3}-\frac{\Psi_2\partial_y v_1^y}{3}\\
&+\frac{2\Psi_2\partial_z v_1^z}{3}+\frac{h_{2,xx}\partial_\eta\Phi_1}{2}-\frac{h_{2,xx}\partial_x v_1^x}{6}-\frac{h_{2,xx}\partial_y v_1^y}{6}+\frac{h_{2,xx}\partial_z v_1^z}{3}-\frac{h_{2,xz}\partial_z v_1^x}{2}\\
&+\frac{h_{2,yy}\partial_\eta\Phi_1}{2}-\frac{h_{2,yy}\partial_x v_1^x}{6}-\frac{h_{2,yy}\partial_y v_1^y}{6}+\frac{h_{2,yy}\partial_z v_1^z}{3}-\frac{h_{2,yz}\partial_z v_1^y}{2}+\frac{\omega_{2,x}\partial_\eta v_1^x}{6}+\frac{\omega_{2,y}\partial_\eta v_1^y}{6}\\
&-\frac{\omega_{2,z}\partial_x v_1^x}{6}-\frac{\omega_{2,z}\partial_y v_1^y}{6}-\frac{\omega_{2,z}\partial_\eta v_1^z}{3}+\frac{\omega_{2,z}\partial_z v_1^z}{3}+\frac{v_1^x\partial_x h_{2,xx}}{4}-\frac{v_1^x\partial_\eta h_{2,xz}}{2}+\frac{v_1^x\partial_x h_{2,yy}}{4}\\
&+\frac{v_1^x\partial_\eta\omega_{2,x}}{6}+\frac{v_1^x\partial_\eta v_2^x}{6}-\frac{v_1^x\partial_z v_2^x}{2}-\frac{v_1^x\partial_x v_2^z}{2}+\frac{v_2^x\partial_\eta v_1^x}{6}-\frac{v_2^x\partial_z v_1^x}{2}-\frac{v_2^x\partial_x v_1^z}{2}+\frac{v_1^y\partial_y h_{2,xx}}{4}\\
&+\frac{v_1^y\partial_y h_{2,yy}}{4}-\frac{v_1^y\partial_\eta h_{2,yz}}{2}+\frac{v_1^y\partial_\eta\omega_{2,y}}{6}+\frac{v_1^y\partial_\eta v_2^y}{6}-\frac{v_1^y\partial_z v_2^y}{2}-\frac{v_1^y\partial_y v_2^z}{2}+\frac{v_2^y\partial_\eta v_1^y}{6}\\
&-\frac{v_2^y\partial_z v_1^y}{2}-\frac{v_2^y\partial_y v_1^z}{2}+\frac{v_1^z\partial_\eta h_{2,xx}}{4}+\frac{v_1^z\partial_z h_{2,xx}}{4}+\frac{v_1^z\partial_\eta h_{2,yy}}{4}+\frac{v_1^z\partial_z h_{2,yy}}{4}-\frac{v_1^z\partial_\eta\omega_{2,z}}{3}\\
&+\frac{v_1^z\partial_x v_2^x}{6}+\frac{v_1^z\partial_y v_2^y}{6}-\frac{v_1^z\partial_\eta v_2^z}{3}-\frac{v_1^z\partial_z v_2^z}{3}+\frac{v_2^z\partial_x v_1^x}{6}+\frac{v_2^z\partial_y v_1^y}{6}-\frac{v_2^z\partial_\eta v_1^z}{3}-\frac{v_2^z\partial_z v_1^z}{3}
\end{aligned}$$

transverse direction $\mathbf{k}_1$ (aberration+deflection); fields of order 2 [ray, $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$].

$$\frac{k_1^x\partial_\eta h_{2,xz}}{2}+\frac{k_1^x\partial_z v_2^x}{2}+\frac{k_1^x\partial_x v_2^z}{2}+\frac{k_1^y\partial_\eta h_{2,yz}}{2}+\frac{k_1^y\partial_z v_2^y}{2}+\frac{k_1^y\partial_y v_2^z}{2}$$

transverse $\mathbf{k}_2$; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\frac{k_2^x \partial_z v_1^x}{2} + \frac{k_2^x \partial_x v_1^z}{2} + \frac{k_2^y \partial_z v_1^y}{2} + \frac{k_2^y \partial_y v_1^z}{2}$$

time delay (radial $\Delta\mathbf{x}_1$); fields of order 1+1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}
&-\Delta x_1^z v_1^x \partial_z^2 v_1^x + \frac{\Delta x_1^z v_1^x \partial_z \partial_\eta v_1^x}{3} - \Delta x_1^z v_1^x \partial_z \partial_x v_1^z - \Delta x_1^z v_1^y \partial_z^2 v_1^y + \frac{\Delta x_1^z v_1^y \partial_z \partial_\eta v_1^y}{3} - \Delta x_1^z v_1^y \partial_z \partial_y v_1^z \\
&- \frac{2\Delta x_1^z v_1^z \partial_z^2 v_1^z}{3} + \frac{\Delta x_1^z v_1^z \partial_z \partial_x v_1^x}{3} + \frac{\Delta x_1^z v_1^z \partial_z \partial_y v_1^y}{3} - \frac{2\Delta x_1^z v_1^z \partial_z \partial_\eta v_1^z}{3} - \frac{2\Delta x_1^z \partial_z \Phi_1 \partial_x v_1^x}{3} \\
&- \frac{2\Delta x_1^z \partial_z \Phi_1 \partial_y v_1^y}{3} + \frac{4\Delta x_1^z \partial_z \Phi_1 \partial_z v_1^z}{3} - \frac{2\Delta x_1^z \partial_z \Psi_1 \partial_x v_1^x}{3} - \frac{2\Delta x_1^z \partial_z \Psi_1 \partial_y v_1^y}{3} + \frac{4\Delta x_1^z \partial_z \Psi_1 \partial_z v_1^z}{3} \\
&+ \frac{\Delta x_1^z \partial_\eta v_1^x \partial_z v_1^x}{3} + \frac{\Delta x_1^z \partial_x v_1^x \partial_z v_1^z}{3} - \Delta x_1^z \left(\partial_z v_1^x\right)^2 - \Delta x_1^z \partial_z v_1^x \partial_x v_1^z + \frac{\Delta x_1^z \partial_\eta v_1^y \partial_z v_1^y}{3} \\
&+ \frac{\Delta x_1^z \partial_y v_1^y \partial_z v_1^z}{3} - \Delta x_1^z \left(\partial_z v_1^y\right)^2 - \Delta x_1^z \partial_z v_1^y \partial_y v_1^z - \frac{2\Delta x_1^z \partial_\eta v_1^z \partial_z v_1^z}{3} - \frac{2\Delta x_1^z \left(\partial_z v_1^z\right)^2}{3}
\end{aligned}$$

local; fields of order 1+1+1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}
&-\Phi_1^2 \partial_x v_1^x - \Phi_1^2 \partial_y v_1^y + 2\Phi_1^2 \partial_z v_1^z - \frac{2\Phi_1 \Psi_1 \partial_x v_1^x}{3} - \frac{2\Phi_1 \Psi_1 \partial_y v_1^y}{3} + \frac{4\Phi_1 \Psi_1 \partial_z v_1^z}{3} \\
&- \frac{2\Phi_1 v_1^x \partial_\eta v_1^x}{3} + \Phi_1 v_1^x \partial_z v_1^x + \Phi_1 v_1^x \partial_x v_1^z - \frac{2\Phi_1 v_1^y \partial_\eta v_1^y}{3} + \Phi_1 v_1^y \partial_z v_1^y + \Phi_1 v_1^y \partial_y v_1^z \\
&- \frac{\Phi_1 v_1^z \partial_x v_1^x}{3} - \frac{\Phi_1 v_1^z \partial_y v_1^y}{3} + \frac{4\Phi_1 v_1^z \partial_\eta v_1^z}{3} + \frac{2\Phi_1 v_1^z \partial_z v_1^z}{3} + \frac{\Psi_1^2 \partial_x v_1^x}{3} + \frac{\Psi_1^2 \partial_y v_1^y}{3} \\
&- \frac{2\Psi_1^2 \partial_z v_1^z}{3} - \frac{2\Psi_1 v_1^x \partial_\eta v_1^x}{3} + \Psi_1 v_1^x \partial_z v_1^x + \Psi_1 v_1^x \partial_x v_1^z - \frac{2\Psi_1 v_1^y \partial_\eta v_1^y}{3} + \Psi_1 v_1^y \partial_z v_1^y \\
&+ \Psi_1 v_1^y \partial_y v_1^z - \frac{\Psi_1 v_1^z \partial_x v_1^x}{3} - \frac{\Psi_1 v_1^z \partial_y v_1^y}{3} + \frac{4\Psi_1 v_1^z \partial_\eta v_1^z}{3} + \frac{2\Psi_1 v_1^z \partial_z v_1^z}{3} - \frac{{v_1^x}^2 \partial_\eta \Phi_1}{3} \\
&- \frac{{v_1^x}^2 \partial_\eta \Psi_1}{3} - \frac{{v_1^x}^2 \partial_x v_1^x}{3} + \frac{{v_1^x}^2 \partial_y v_1^y}{3} - {v_1^x}^2 \partial_\eta v_1^z + \frac{{v_1^x}^2 \partial_z v_1^z}{3} - \frac{2 v_1^x v_1^y \partial_y v_1^x}{3} - \frac{2 v_1^x v_1^y \partial_x v_1^y}{3} \\
&- \frac{2 v_1^x v_1^z \partial_\eta v_1^x}{3} - \frac{2 v_1^x v_1^z \partial_z v_1^x}{3} - \frac{2 v_1^x v_1^z \partial_x v_1^z}{3} - \frac{{v_1^y}^2 \partial_\eta \Phi_1}{3} - \frac{{v_1^y}^2 \partial_\eta \Psi_1}{3} + \frac{{v_1^y}^2 \partial_x v_1^x}{3} \\
&- \frac{{v_1^y}^2 \partial_y v_1^y}{3} - {v_1^y}^2 \partial_\eta v_1^z + \frac{{v_1^y}^2 \partial_z v_1^z}{3} - \frac{2 v_1^y v_1^z \partial_\eta v_1^y}{3} - \frac{2 v_1^y v_1^z \partial_z v_1^y}{3} - \frac{2 v_1^y v_1^z \partial_y v_1^z}{3} \\
&+ \frac{2 {v_1^z}^2 \partial_\eta \Phi_1}{3} + \frac{2 {v_1^z}^2 \partial_\eta \Psi_1}{3} + \frac{{v_1^z}^2 \partial_x v_1^x}{3} + \frac{{v_1^z}^2 \partial_y v_1^y}{3} - \frac{2 {v_1^z}^2 \partial_\eta v_1^z}{3} - \frac{4 {v_1^z}^2 \partial_z v_1^z}{3}
\end{aligned}$$

transverse direction $\mathbf{k}_1$ (aberration+deflection)^2; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$-\frac{2k_1^{x2} \partial_x v_1^x}{3} + \frac{k_1^{x2} \partial_y v_1^y}{3} + \frac{k_1^{x2} \partial_z v_1^z}{3} - k_1^x k_1^y \partial_y v_1^x - k_1^x k_1^y \partial_x v_1^y + \frac{k_1^{y2} \partial_x v_1^x}{3} - \frac{2k_1^{y2} \partial_y v_1^y}{3} + \frac{k_1^{y2} \partial_z v_1^z}{3}$$

lensing (transverse $\Delta\mathbf{x}_1$)^2; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}
&\frac{\Delta {x_1^x}^2 \partial_x^3 v_1^x}{6} + \frac{\Delta {x_1^x}^2 \partial_y \partial_x^2 v_1^y}{6} - \frac{\Delta {x_1^x}^2 \partial_z \partial_x^2 v_1^z}{3} + \frac{\Delta x_1^x \Delta x_1^y \partial_y \partial_x^2 v_1^x}{3} + \frac{\Delta x_1^x \Delta x_1^y \partial_y^2 \partial_x v_1^y}{3} \\
&- \frac{2\Delta x_1^x \Delta x_1^y \partial_z \partial_y \partial_x v_1^z}{3} + \frac{\Delta {x_1^y}^2 \partial_y^3 v_1^y}{6} + \frac{\Delta {x_1^y}^2 \partial_y^2 \partial_x v_1^x}{6} - \frac{\Delta {x_1^y}^2 \partial_z \partial_y^2 v_1^z}{3}
\end{aligned}$$

time delay (radial $\Delta\mathbf{x}_1$)^2; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$-\frac{\Delta x_1^{z\,2}\partial_z^3 v_1^z}{3}+\frac{\Delta x_1^{z\,2}\partial_z^2\partial_x v_1^x}{6}+\frac{\Delta x_1^{z\,2}\partial_z^2\partial_y v_1^y}{6}$$

radial $\mathbf{k}_2$; fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$-\frac{k_2^z\partial_x v_1^x}{3}-\frac{k_2^z\partial_y v_1^y}{3}+\frac{2k_2^z\partial_z v_1^z}{3}$$

time delay 2nd (radial $\Delta\mathbf{x}_2$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$-\frac{\Delta x_2^z\partial_z^2 v_1^z}{3}+\frac{\Delta x_2^z\partial_z\partial_x v_1^x}{6}+\frac{\Delta x_2^z\partial_z\partial_y v_1^y}{6}$$

time delay (radial $\Delta\mathbf{x}_1$); fields of order 2 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$-\frac{\Delta x_1^z\partial_z^2 v_2^z}{3}+\frac{\Delta x_1^z\partial_z\partial_\eta h_{2,xx}}{4}+\frac{\Delta x_1^z\partial_z\partial_\eta h_{2,yy}}{4}+\frac{\Delta x_1^z\partial_z\partial_x v_2^x}{6}+\frac{\Delta x_1^z\partial_z\partial_y v_2^y}{6}$$

lensing 2nd (transverse $\Delta\mathbf{x}_2$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\frac{\Delta x_2^x\partial_x^2 v_1^x}{6}+\frac{\Delta x_2^x\partial_y\partial_x v_1^y}{6}-\frac{\Delta x_2^x\partial_z\partial_x v_1^z}{3}+\frac{\Delta x_2^y\partial_y^2 v_1^y}{6}+\frac{\Delta x_2^y\partial_y\partial_x v_1^x}{6}-\frac{\Delta x_2^y\partial_z\partial_y v_1^z}{3}$$

lensing (transverse $\Delta\mathbf{x}_1$); fields of order 2 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}&\frac{\Delta x_1^x\partial_x^2 v_2^x}{6}+\frac{\Delta x_1^x\partial_x\partial_\eta h_{2,xx}}{4}+\frac{\Delta x_1^x\partial_x\partial_\eta h_{2,yy}}{4}+\frac{\Delta x_1^x\partial_y\partial_x v_2^y}{6}-\frac{\Delta x_1^x\partial_z\partial_x v_2^z}{3}\\&+\frac{\Delta x_1^y\partial_y^2 v_2^y}{6}+\frac{\Delta x_1^y\partial_y\partial_\eta h_{2,xx}}{4}+\frac{\Delta x_1^y\partial_y\partial_\eta h_{2,yy}}{4}+\frac{\Delta x_1^y\partial_y\partial_x v_2^x}{6}-\frac{\Delta x_1^y\partial_z\partial_y v_2^z}{3}\end{aligned}$$

transverse direction $\mathbf{k}_1$ (aberration+deflection); fields of order 1+1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}&-\Phi_1 k_1^x\partial_z v_1^x-\Phi_1 k_1^x\partial_x v_1^z-\Phi_1 k_1^y\partial_z v_1^y-\Phi_1 k_1^y\partial_y v_1^z-\Psi_1 k_1^x\partial_z v_1^x-\Psi_1 k_1^x\partial_x v_1^z\\&-\Psi_1 k_1^y\partial_z v_1^y-\Psi_1 k_1^y\partial_y v_1^z+\frac{5k_1^x v_1^x\partial_x v_1^x}{3}-\frac{k_1^x v_1^x\partial_y v_1^y}{3}+k_1^x v_1^x\partial_\eta v_1^z\\&-\frac{4k_1^x v_1^x\partial_z v_1^z}{3}+k_1^x v_1^y\partial_y v_1^x+k_1^x v_1^y\partial_x v_1^y+k_1^x v_1^z\partial_\eta v_1^x+k_1^y v_1^x\partial_y v_1^x+k_1^y v_1^x\partial_x v_1^y\\&-\frac{k_1^y v_1^y\partial_x v_1^x}{3}+\frac{5k_1^y v_1^y\partial_y v_1^y}{3}+k_1^y v_1^y\partial_\eta v_1^z-\frac{4k_1^y v_1^y\partial_z v_1^z}{3}+k_1^y v_1^z\partial_\eta v_1^y\end{aligned}$$

transverse direction $\mathbf{k}_1$ (aberration+deflection); time delay (radial $\Delta\mathbf{x}_1$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\Delta x_1^z k_1^x\partial_z^2 v_1^x+\Delta x_1^z k_1^x\partial_z\partial_x v_1^z+\Delta x_1^z k_1^y\partial_z^2 v_1^y+\Delta x_1^z k_1^y\partial_z\partial_y v_1^z$$

transverse direction $\mathbf{k}_1$ (aberration+deflection); lensing (transverse $\Delta\mathbf{x}_1$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$\begin{aligned}&\Delta x_1^x k_1^x\partial_x^2 v_1^z+\Delta x_1^x k_1^x\partial_z\partial_x v_1^x+\Delta x_1^x k_1^y\partial_z\partial_x v_1^y+\Delta x_1^x k_1^y\partial_y\partial_x v_1^z\\&+\Delta x_1^y k_1^x\partial_z\partial_y v_1^x+\Delta x_1^y k_1^x\partial_y\partial_x v_1^z+\Delta x_1^y k_1^y\partial_y^2 v_1^z+\Delta x_1^y k_1^y\partial_z\partial_y v_1^y\end{aligned}$$

lensing (transverse $\Delta\mathbf{x}_1$); fields of order 1+1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$
\begin{aligned}
&-\Delta x_1^x v_1^x \partial_x^2 v_1^z + \frac{\Delta x_1^x v_1^x \partial_x \partial_\eta v_1^x}{3} - \Delta x_1^x v_1^x \partial_z \partial_x v_1^x + \frac{\Delta x_1^x v_1^y \partial_x \partial_\eta v_1^y}{3} - \Delta x_1^x v_1^y \partial_z \partial_x v_1^y \\
&- \Delta x_1^x v_1^y \partial_y \partial_x v_1^z + \frac{\Delta x_1^x v_1^z \partial_x^2 v_1^x}{3} + \frac{\Delta x_1^x v_1^z \partial_y \partial_x v_1^y}{3} - \frac{2\Delta x_1^x v_1^z \partial_x \partial_\eta v_1^z}{3} - \frac{2\Delta x_1^x v_1^z \partial_z \partial_x v_1^z}{3} \\
&- \frac{2\Delta x_1^x \partial_x \Phi_1 \partial_x v_1^x}{3} - \frac{2\Delta x_1^x \partial_x \Phi_1 \partial_y v_1^y}{3} + \frac{4\Delta x_1^x \partial_x \Phi_1 \partial_z v_1^z}{3} - \frac{2\Delta x_1^x \partial_x \Psi_1 \partial_x v_1^x}{3} \\
&- \frac{2\Delta x_1^x \partial_x \Psi_1 \partial_y v_1^y}{3} + \frac{4\Delta x_1^x \partial_x \Psi_1 \partial_z v_1^z}{3} + \frac{\Delta x_1^x \partial_\eta v_1^x \partial_x v_1^x}{3} - \Delta x_1^x \partial_x v_1^x \partial_z v_1^x \\
&- \frac{2\Delta x_1^x \partial_x v_1^x \partial_x v_1^z}{3} + \frac{\Delta x_1^x \partial_\eta v_1^y \partial_x v_1^y}{3} - \Delta x_1^x \partial_x v_1^y \partial_z v_1^y - \Delta x_1^x \partial_x v_1^y \partial_y v_1^z + \frac{\Delta x_1^x \partial_y v_1^y \partial_x v_1^z}{3} \\
&- \frac{2\Delta x_1^x \partial_\eta v_1^z \partial_x v_1^z}{3} - \frac{2\Delta x_1^x \partial_x v_1^z \partial_z v_1^z}{3} + \frac{\Delta x_1^y v_1^x \partial_x \partial_\eta v_1^x}{3} - \Delta x_1^y v_1^x \partial_z \partial_y v_1^x - \Delta x_1^y v_1^x \partial_y \partial_x v_1^z \\
&- \Delta x_1^y v_1^y \partial_y^2 v_1^z + \frac{\Delta x_1^y v_1^y \partial_y \partial_\eta v_1^y}{3} - \Delta x_1^y v_1^y \partial_z \partial_y v_1^y + \frac{\Delta x_1^y v_1^z \partial_y^2 v_1^y}{3} + \frac{\Delta x_1^y v_1^z \partial_y \partial_x v_1^x}{3} \\
&- \frac{2\Delta x_1^y v_1^z \partial_y \partial_\eta v_1^z}{3} - \frac{2\Delta x_1^y v_1^z \partial_z \partial_y v_1^z}{3} - \frac{2\Delta x_1^y \partial_y \Phi_1 \partial_x v_1^x}{3} - \frac{2\Delta x_1^y \partial_y \Phi_1 \partial_y v_1^y}{3} \\
&+ \frac{4\Delta x_1^y \partial_y \Phi_1 \partial_z v_1^z}{3} - \frac{2\Delta x_1^y \partial_y \Psi_1 \partial_x v_1^x}{3} - \frac{2\Delta x_1^y \partial_y \Psi_1 \partial_y v_1^y}{3} + \frac{4\Delta x_1^y \partial_y \Psi_1 \partial_z v_1^z}{3} \\
&+ \frac{\Delta x_1^y \partial_\eta v_1^x \partial_y v_1^x}{3} + \frac{\Delta x_1^y \partial_x v_1^x \partial_y v_1^z}{3} - \Delta x_1^y \partial_y v_1^x \partial_z v_1^x - \Delta x_1^y \partial_y v_1^x \partial_x v_1^z + \frac{\Delta x_1^y \partial_\eta v_1^y \partial_y v_1^y}{3} \\
&- \Delta x_1^y \partial_y v_1^y \partial_z v_1^y - \frac{2\Delta x_1^y \partial_y v_1^y \partial_y v_1^z}{3} - \frac{2\Delta x_1^y \partial_\eta v_1^z \partial_y v_1^z}{3} - \frac{2\Delta x_1^y \partial_y v_1^z \partial_z v_1^z}{3}
\end{aligned}
$$

lensing (transverse $\Delta\mathbf{x}_1$); time delay (radial $\Delta\mathbf{x}_1$); fields of order 1 [ray, $\int_{\eta_*}^{\eta_o} \mathrm{d}\eta$].

$$
\begin{aligned}
&\frac{\Delta x_1^x \Delta x_1^z \partial_z \partial_x^2 v_1^x}{3} - \frac{2\Delta x_1^x \Delta x_1^z \partial_z^2 \partial_x v_1^z}{3} + \frac{\Delta x_1^x \Delta x_1^z \partial_z \partial_y \partial_x v_1^y}{3} \\
&+ \frac{\Delta x_1^y \Delta x_1^z \partial_z \partial_y^2 v_1^y}{3} - \frac{2\Delta x_1^y \Delta x_1^z \partial_z^2 \partial_y v_1^z}{3} + \frac{\Delta x_1^y \Delta x_1^z \partial_z \partial_y \partial_x v_1^x}{3}
\end{aligned}
$$

observer clock[2]; fields of order 1 [observer, $\eta = \eta_o$, $\chi = 0$].

$$
-\frac{2\delta\eta_{o1}^2 \partial_z^2 v_1^z}{3} + \frac{\delta\eta_{o1}^2 \partial_z \partial_x v_1^x}{3} + \frac{\delta\eta_{o1}^2 \partial_z \partial_y v_1^y}{3}
$$

observer clock; fields of order 2 [observer, $\eta = \eta_o$, $\chi = 0$].

$$
\frac{\delta\eta_{o1} \partial_\eta h_{2,xx}}{4} + \frac{\delta\eta_{o1} \partial_\eta h_{2,yy}}{4} + \frac{\delta\eta_{o1} \partial_x v_2^x}{6} + \frac{\delta\eta_{o1} \partial_y v_2^y}{6} - \frac{\delta\eta_{o1} \partial_z v_2^z}{3}
$$

observer clock 2nd; fields of order 1 [observer, $\eta = \eta_o$, $\chi = 0$].

$$
\frac{\delta\eta_{o2} \partial_x v_1^x}{6} + \frac{\delta\eta_{o2} \partial_y v_1^y}{6} - \frac{\delta\eta_{o2} \partial_z v_1^z}{3}
$$

observer clock; observer displacement; fields of order 1 [observer, $\eta = \eta_o$, $\chi = 0$].

$$\frac{\delta\eta_{o1}\delta x^x_{o1}\partial^2_x v^x_1}{3} + \frac{\delta\eta_{o1}\delta x^x_{o1}\partial_y\partial_x v^y_1}{3} - \frac{2\delta\eta_{o1}\delta x^x_{o1}\partial_z\partial_x v^z_1}{3} + \frac{\delta\eta_{o1}\delta x^y_{o1}\partial^2_y v^y_1}{3} + \frac{\delta\eta_{o1}\delta x^y_{o1}\partial_y\partial_x v^x_1}{3} - \frac{2\delta\eta_{o1}\delta x^y_{o1}\partial_z\partial_y v^z_1}{3} - \frac{2\delta\eta_{o1}\delta x^z_{o1}\partial^2_z v^z_1}{3} + \frac{\delta\eta_{o1}\delta x^z_{o1}\partial_z\partial_x v^x_1}{3} + \frac{\delta\eta_{o1}\delta x^z_{o1}\partial_z\partial_y v^y_1}{3}$$

observer clock; fields of order 1+1 [observer, $\eta = \eta_o$, $\chi = 0$].

$$\frac{\delta\eta_{o1}v^x_1\partial_\eta v^x_1}{3} + \frac{\delta\eta_{o1}v^y_1\partial_\eta v^y_1}{3} + \frac{\delta\eta_{o1}v^z_1\partial_x v^x_1}{3} + \frac{\delta\eta_{o1}v^z_1\partial_y v^y_1}{3} - \frac{2\delta\eta_{o1}v^z_1\partial_\eta v^z_1}{3} - \frac{2\delta\eta_{o1}v^z_1\partial_z v^z_1}{3}$$

observer clock^2; fields of order 1 [observer, $\eta = \eta_o$, $\chi = 0$].

$$\frac{\delta\eta^2_{o1}\partial^2_z v^z_1}{3} + \frac{\delta\eta^2_{o1}\partial_x\partial_\eta v^x_1}{6} - \frac{\delta\eta^2_{o1}\partial_z\partial_x v^x_1}{6} + \frac{\delta\eta^2_{o1}\partial_y\partial_\eta v^y_1}{6} - \frac{\delta\eta^2_{o1}\partial_z\partial_y v^y_1}{6} - \frac{\delta\eta^2_{o1}\partial_z\partial_\eta v^z_1}{3}$$

emission-surface shift; fields of order 2 [emission, $\eta = \eta_*$].

$$-\frac{\delta\eta_{e1}\partial_\eta h_{2,xx}}{4} - \frac{\delta\eta_{e1}\partial_\eta h_{2,yy}}{4} - \frac{\delta\eta_{e1}\partial_x v^x_2}{6} - \frac{\delta\eta_{e1}\partial_y v^y_2}{6} + \frac{\delta\eta_{e1}\partial_z v^z_2}{3}$$

emission-surface shift 2nd; fields of order 1 [emission, $\eta = \eta_*$].

$$-\frac{\delta\eta_{e2}\partial_x v^x_1}{6} - \frac{\delta\eta_{e2}\partial_y v^y_1}{6} + \frac{\delta\eta_{e2}\partial_z v^z_1}{3}$$

emission-surface shift; fields of order 1+1 [emission, $\eta = \eta_*$].

$$-\frac{\delta\eta_{e1}v^x_1\partial_\eta v^x_1}{3} + \delta\eta_{e1}v^x_1\partial_z v^x_1 + \delta\eta_{e1}v^x_1\partial_x v^z_1 - \frac{\delta\eta_{e1}v^y_1\partial_\eta v^y_1}{3} + \delta\eta_{e1}v^y_1\partial_z v^y_1 + \delta\eta_{e1}v^y_1\partial_y v^z_1 - \frac{\delta\eta_{e1}v^z_1\partial_x v^x_1}{3} - \frac{\delta\eta_{e1}v^z_1\partial_y v^y_1}{3} + \frac{2\delta\eta_{e1}v^z_1\partial_\eta v^z_1}{3} + \frac{2\delta\eta_{e1}v^z_1\partial_z v^z_1}{3}$$

emission-surface shift; transverse direction $\mathbf{k}_1$ (aberration+deflection); fields of order 1 [emission, $\eta = \eta_*$].

$$-\delta\eta_{e1}k^x_1(\eta_*)\partial_z v^x_1 - \delta\eta_{e1}k^x_1(\eta_*)\partial_x v^z_1 - \delta\eta_{e1}k^y_1(\eta_*)\partial_z v^y_1 - \delta\eta_{e1}k^y_1(\eta_*)\partial_y v^z_1$$

emission-surface shift; lensing (transverse $\Delta\mathbf{x}_1$); fields of order 1 [emission, $\eta = \eta_*$].

$$-\frac{\delta\eta_{e1}\Delta x^x_1(\eta_*)\partial^2_x v^x_1}{3} - \frac{\delta\eta_{e1}\Delta x^x_1(\eta_*)\partial_y\partial_x v^y_1}{3} + \frac{2\delta\eta_{e1}\Delta x^x_1(\eta_*)\partial_z\partial_x v^z_1}{3} - \frac{\delta\eta_{e1}\Delta x^y_1(\eta_*)\partial^2_y v^y_1}{3} - \frac{\delta\eta_{e1}\Delta x^y_1(\eta_*)\partial_y\partial_x v^x_1}{3} + \frac{2\delta\eta_{e1}\Delta x^y_1(\eta_*)\partial_z\partial_y v^z_1}{3}$$

emission-surface shift; time delay (radial $\Delta\mathbf{x}_1$); fields of order 1 [emission, $\eta = \eta_*$].

$$\frac{2\delta\eta_{e1}\Delta x^z_1(\eta_*)\partial^2_z v^z_1}{3} - \frac{\delta\eta_{e1}\Delta x^z_1(\eta_*)\partial_z\partial_x v^x_1}{3} - \frac{\delta\eta_{e1}\Delta x^z_1(\eta_*)\partial_z\partial_y v^y_1}{3}$$

emission-surface shift[2]; fields of order 1 [emission, $\eta = \eta_*$].

$$-\frac{\delta\eta_{e1}^2\partial_z^2 v_1^z}{3} - \frac{\delta\eta_{e1}^2\partial_x\partial_\eta v_1^x}{6} + \frac{\delta\eta_{e1}^2\partial_z\partial_x v_1^x}{6} - \frac{\delta\eta_{e1}^2\partial_y\partial_\eta v_1^y}{6} + \frac{\delta\eta_{e1}^2\partial_z\partial_y v_1^y}{6} + \frac{\delta\eta_{e1}^2\partial_z\partial_\eta v_1^z}{3}$$

F Reproducibility

All symbolic and numerical results of this paper are produced by the following code (directory `cmb4/code/`; each directory contains its scripts, logs and machine-readable outputs). Ledger identifiers of the corresponding checks are given in brackets.

- `C1_einstein/`: Einstein–fluid equations in Poisson gauge as truncated ϵ-series through $O(\epsilon^3)$; SVT split and operators (56); first-order transfer functions; second-order kernels and Green's functions; $\zeta^{(n)}$ expansions and initial amplitudes $K^0_{\Phi_2}$, C_3; contracted third-order sources $K_{3c,X}$ (157–830 terms each) and tadpoles; position-space sources; exact dilation family [EF-1–EF-14]. Machine-readable kernels for App. A.
- `V3_referee/`: independently coded re-derivation of the field equations, kernels, Kelvin vorticity, contraction rules and C_3; literature comparisons (Kohri–Terada, Pitrou–Uzan–Bernardeau, Lu et al.) [EF-15–EF-21].
- `C4_timeintegrals/`: closed forms of $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ (integration engine with Si, Cin; cancellation of the endpoint singularities), collinear and resonant limiting forms, small- and large-η expansions, insertion into the full second-order equations [CF-1–CF-3]; source of App. B.
- `V1_referee/`: algebraic identities of Lemmas 4.1–4.2, exact-solution tests of Theorem 4.3, frame-rotation example [X-1–X-6].
- `C2_observable/`: symbolic expansion of S through third order: identities along an arbitrary ray, ray and triad, direction map, ray functionals, emission shifts, the term lists of L_1, L_2, L_3 in shear and direct form (general and $\hat{\mathbf{n}} = \hat{\mathbf{z}}$), m_n and $\Theta^{(n)}$, dilation and literature checks [X-7–X-9, R-1–R-7].
- `C5_linear_Cl/`: first-order transfer function, closed-form ISW integral (residue coefficients and elementary parts E_ℓ, $\ell \le 6$), monochromatic $C_\ell^{(11)}$, limits, sky monopole [FO-1–FO-4].
- `C3_harmonic/`: harmonic rules (A-s), (A-λ), (A-L), (B1), types C and D, (E1), lensing control, Monte Carlo tests [H-1–H-10].
- `V2_squeezed/`: adiabatic-mode transformation and squeezed-limit predictions for K_2, K_3 [SQ-1].
- `N1_planar/`: exact planar Einstein–radiation solver in comoving-orthogonal gauge, ray tracing, clock, reference maps [X-10, NP-1, NP-2].
- `A1_planewave/`: plane-wave evaluation of the complete analytic chain and comparison with the planar solver, zero-mode and label tests [PW-1–PW-7]; full $a_{\ell 0}$ lists ($\ell \le 12$) of Tables 8–9.
- `A2_second_order/`: Fourier kernel K_2 (channels of App. D, generated by `symbolic.py`), $C_\ell^{(22)}$ and its monochromatic evaluation (`cl22.py`), $K \to 0$ expansions (`k0_limits.py`) [SQ-2].
- `A3_third_order/`: contracted third-order kernel by polarisation over the term lists (`engine.py`, `third.py`, `thirteen.py`); third-order form equivalence (`test_third.py`, `test_null2.py`); residual-gauge tests (`gauge.py`); collinear residues (`residues.py`,

`diag_mu.py`); squeezed and UV probes (`ir.py`, `uv.py`); monochromatic $C_\ell^{(13)}$ and the combination (`mono.py`, `combine.py`); the channel listing `A3_channels.tex`, from which App. E is generated by `doc/gen_k3.py` [R-9–R-12, EF-22, SQ-3, H-13, L-1–L-3].

- `V4_mc/`: Monte-Carlo evaluation of $C_\ell^{(22)}$ from finite random sums of plane waves, with K_2 as a black box [L-4].
- `W3_revision/`: observer clock at finite η_i (`clock_eta_i.py`) [R-8]; harmonic rules in the observation-direction convention and the Bessel bound (`check_rules.py`) [H-11, H-12]; generator of App. C from the term lists (`gen_lterms_full.py`); printed closed forms and their re-verification (`closedforms_tex.py`) [CF-4].

G Verification ledger

Every check performed on the material of this paper, with its outcome. "Exact" means symbolic identity (computer algebra); numbers are maximal relative or absolute deviations. Identifiers: X exact observable, R line-of-sight functionals, EF Einstein–fluid solution, CF closed-form time integrals, FO first order, SQ squeezed limits, H harmonic machinery, NP non-perturbative solver, PW plane-wave end-to-end test, L loop assembly and monochromatic evaluation. The code that performs each group of checks is listed in App. F.

Exact observable (Sec. 4)

X-1 Lemma 4.1 in a generic Lorentzian metric $g = G_0 + G_{1\alpha}x^\alpha$ (random rational coefficients, no symmetry), generic unit $u(x)$, generic null p: geodesic-derived $\mathrm{d}E/\mathrm{d}\lambda$ vs $-pp\nabla u$, decomposition, $\omega_{\mu\nu}p^\mu p^\nu = 0$, final identity: exact.

X-2 Lemma 4.2: energy and Euler equations from $\nabla\cdot T = 0$ for general ρ, P; equivalence to (16) iff $\rho = 3P + c$, $P = CT^4 - c/4$; for $P = w\rho$, $\rho \propto T^4$ the identities hold iff $w = 1/3$: exact.

X-3 Off-shell identity $\mathrm{d}\ln E/\mathrm{d}l - \mathrm{d}\ln T/\mathrm{d}l + \sigma_{ee} = -[(\theta + 3u\cdot\partial\ln T)/3 + (a + h\partial\ln T)\cdot e]$ at a generic point: exact (residual -0.93 off shell in a random instance, 0 on shell).

X-4a FRW radiation: $\sigma = 0$, $T_* E_o/E_e/T_o = 1$ symbolically for all p_i.

X-4b Bianchi I $\mathrm{d}s^2 = -\mathrm{d}t^2 + \sum t^{2q_i}\mathrm{d}x_i^2$ with a radiation test fluid $u = \partial_t$: $\mathrm{d}/\mathrm{d}t[\ln E - \ln T] + \sigma_{ee} = 0$ symbolically; integrated for Kasner $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and non-Kasner $(\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$, three directions, $t_e = 1 \to t_o = 5$: $|T_{\mathrm{obs}}^{\mathrm{exact}} - T_o e^{-S}| \lesssim 2\times 10^{-31}$ with $O(1)$ direction dependence.

X-4c Milne flow in Minkowski: $\theta = 3/\tau$, $\sigma = a = 0$, $T_{\mathrm{obs}} = T(o)$ isotropic: exact.

X-5 Linear Poisson gauge: $\hat\theta, \hat a, \hat\sigma$ of (27); direct geodesic $\mathrm{d}\ln\hat E/\mathrm{d}\eta$ vs kinematic: agree; identity (29) off shell non-zero, on shell 0: exact. Sachs–Wolfe + Doppler + ISW with observer terms recovered.

X-6 Frame rotation $\mathsf{R} = \exp(\epsilon\mathsf{J})$ at $\ell = 2$ (exact spin-2 generators): $\sum_m |a_{\ell m}|^2$ invariant through $O(\epsilon^4)$ and $a^{(1)}$ invariant; $a^{(2)} \to a^{(2)} + 2\mathsf{J}a^{(1)}$; $C^{(22)}$ and $\mathrm{Re}\,C^{(13)}$ separately *not* invariant ($\sum|a_2|^2$: 321/25 vs $329107/1225 - 8798\sqrt{6}/1225$) while $\frac{1}{4}C^{(22)} + \frac{1}{3}\mathrm{Re}\,C^{(13)}$ is (107/60 both). Dipole power invariant.

Exact observable through third order and line-of-sight functionals (Secs. 4.6, 8–9)

X-7 Kinematic identity $p^a p^b \nabla_a u_b = \sigma_{pp} + \theta(p\cdot p + E^2)/3 + Ea\cdot p$ for a generic (non-null) p, no equations of motion: zero at orders 1, 2, 3.

X-8 Radiation identities (16) from $\hat{\nabla}_\mu \hat{T}^{\mu\nu} = 0$ alone (solved order by order for $\partial_\eta \delta_{T,n}$, $\partial_\eta v_n$): zero at orders 1, 2, 3.

X-9 Integrand-level comparison of route (i) $[\ln(\hat{E}_o/\hat{E}_e) = -\int \Gamma^0_{kk} \mathrm{d}\eta + [\ln(-u\cdot k)]^o_e]$ with route (ii) $[\ln(\hat{T}_o/\hat{T}_e) - S]$ along an arbitrary perturbed ray $k^i = -n^i + \epsilon k^i_1 + \ldots$ with the non-affine geodesic equation substituted: residual $R \equiv 0$ at orders 1, 2 and 3 modulo conservation (14/70/340 terms before reduction).

R-1 $-I_1 = -D[\delta_{T,1} + \Psi_1 - \hat{\mathbf{n}}\cdot v_1] + \Phi_1' + \Psi_1'$ with $I_1 = \sigma_{1,kk}$: exact.

R-2 Canonical triad (Gram–Schmidt w.r.t. $\hat{g}$, orthogonal to $\hat{u}$): $\hat{g}(e_a, e_b) = \delta_{ab}$, $\hat{g}(e_a, \hat{u}) = 0$ through order 3.

R-3 Direction map (22): $k^\mu(\eta_o)$ null through order 3 (with $|\hat{\mathbf{n}}| = 1$); first-order geodesic right-hand side $G^i_1 = -\partial_i(\Phi + \Psi) + 2n^i\, \hat{\mathbf{n}}\cdot\nabla(\Phi + \Psi) - n^i(\Phi' + \Psi')$ consistent with the null condition $\hat{\mathbf{n}}\cdot k_1 = -(\Phi + \Psi)$ at every η with the tangent integrals of (86), (88) ($-\int G$; numerically 10^{-15}, PW-4).

R-4 Line-of-sight machinery (Sec. 8.4): (T1) for the boundary function $B = \ln(-\hat{u}\cdot k) - \ln \hat{T}$, the endpoint-shifted $[B]^o_e$ equals the ray integral of its total derivative plus the endpoint corrections, identically, orders 1–3; (T2) $\Lambda^{\mathrm{dir}} - \Lambda^{\mathrm{shear}}$ of the assembled, background-ray-expanded lists vanishes piece by piece (ray/obs/emi) at orders 1, 2, 3 modulo the conservation equations and their derivatives (112/802 ray terms before reduction at orders 2/3); (T3) exact lower-index identity $-\hat{\Gamma}^0_{kk} = -\frac{1}{2}\partial_\eta \hat{g}_{kk}/(1 + 2\Psi - \omega\cdot k) - D\ln(1 + 2\Psi - \omega\cdot k)$, orders 0–3: zero.

R-5 m_n and Θ_n (99), (101) from the exponential and ratio series, including the $\ell = 0$ parts: exact.

R-6 Dilation (Sec. 13.3): for the exact family with $\lambda = 1 + \epsilon a_1 + \frac{\epsilon^2}{2} a_2$, $\beta = e^{ec}$ ($a_1 = -c/3$, a_2 free), $\hat{\sigma} \equiv 0$ through order 3, $\hat{u}^0 = \lambda^{-2}$ reproduced; assembled L_1, L_2, L_3 zero in shear and direct forms with $\delta\eta_{e1} = c\eta_*/3$, $\delta\eta_{e2} = (-a_2 + \frac{2}{9}c^2)\eta_*$; clock $\eta_o = \bar{\eta}_o/\lambda$ through second order, $T_o = \bar{T}(\bar{\eta}_o) + O(\epsilon^3)$. (The check is sensitive to the normalisation of $\delta\eta_{e2}$: a factor 2 in it is detected.)

R-7 Literature: with Boubekeur et al.'s prescriptions ($\mathbf{v} = 0$, observer at the origin, emission at fixed coordinate time) the direct form agrees with their second-order Sachs–Wolfe result exactly at order 1 and in the obs/emi parts at order 2; the ray remainder $\int[\Delta\mathbf{x}_1\cdot\nabla(\Phi_1' + \Psi_1') + 2\Phi_1\Phi_1' - 2\Psi_1\Psi_1']$ vanishes under their assumption $\Phi_1' = \Psi_1' = 0$.

Einstein–fluid equations (Secs. 5–6)

EF-1 Series arithmetic: $g\, g^{-1} = 1$ and $g_{\mu\nu}u^\mu u^\nu = -1$ to $O(\epsilon^3)$; background part of every $E_{\mu\nu}$ and of $\nabla_\mu T^{\mu\nu}$ vanishes identically.

EF-2 First-order equations reproduce the standard linearised Poisson-gauge equations (51); $\Psi_1 = \Phi_1$; $v_1^T = 0$; $\mathrm{curl}\, v_1 = 0$ identically.

EF-3 $\Phi_1''+4\Phi_1'/\eta+k^2\Phi_1/3=0$ verified; transfer functions (53) verified by substitution; conservation equations ($\nu=0..3$) vanish identically on the solution; agreement with the lead's independent linear check (T_Φ, $\delta_T\to\zeta/3$, $V\to\eta\zeta/3$).

EF-4 Second-order SVT structure (56) verified component by component ($S_{0y}=S_{yz}=S_{xy}=0$ by reflection symmetry; vector/tensor kernels free of $\sqrt{1-\mu^2}$ after projection).

EF-5 Contracted Bianchi identity $g^{\mu\alpha}\nabla_\alpha E_{\mu\nu}+g_{\nu\beta}\nabla_\mu T^{\mu\beta}\equiv 0$ to $O(\epsilon^3)$ (hence the constraints propagate at every order).

EF-6 Exact dilation family (47) solves $E_{\mu\nu}=0$ to $O(\epsilon^3)$ for arbitrary A,B; $k=0$ second-order equations fix only $\delta_2=-2\Psi_2+\frac{32}{9}c^2$; angle-averaged $k\to 0$ limit of the regular solution lies on the family.

EF-7 Vector sector: $\operatorname{curl}\hat{u}_i\neq 0$ at second order; relativistic Kelvin (57) holds and the algebraic ω_2 satisfies the ij-vector ODE identically.

EF-8 $\zeta^{(2)}$, $\zeta^{(3)}$ expansions (39)–(40); $\lim\eta^2 K_{S_{00}}=32/3$; $K_Q(0)$ and $K^0_{\Phi_2}$ (41)–(42).

EF-9 Tensor source K_h identical to Kohri–Terada (8) / Baumann et al. (11)–(12) with $w=1/3$, $\Phi=\Psi$: numerically exact.

EF-10 Superhorizon Φ_2,Ψ_2 (43)–(44) = Pitrou–Uzan–Bernardeau (3.12b,c) with $\mathcal{R}^{(2)}=-2\mathcal{R}^{(1)2}$: exact.

EF-11 Bartolo–Matarrese–Riotto II (7.30): local and gradient terms agree; velocity-quadrupole term differs ($27F$ vs our $3F$) because BMR use the late-time velocity; ours = PUB.

EF-12 Green's functions (60): $\mathcal{L}G=0$, $G(\eta,\eta)=0$, $\partial_\eta G|_{\eta'=\eta}=1$ (re-checked for the rewritten form of G_Φ in this write-up).

EF-13 K_{S_Φ} and K_{3c,S_Φ} finite as $\eta\to 0$ (small-η expansions), so particular integrals vanish at $\eta=0$; contracted vector/tensor sources vanish after angular integration.

EF-14 Squeezed limit $C_3(q\to 0)=-32/9$ equals the prediction of the exact dilation family: pass.

Independent re-derivation of the Einstein–fluid solution (Sec. 13.2)

Separately coded; arbiter exact computer algebra or 30-digit numerics at random rational points.

EF-15 Separately coded computation of $E_{\mu\nu}=G-8\pi GT$: term-by-term identical to the primary one for all ten components at orders 1, 2, 3 and for u_μ. Same method, independently coded.

EF-16 Linear solution, $\Psi_1=\Phi_1$, V_1, δ_1, T_Φ, $T_\Phi(0)=-\frac{2}{3}$: all ten $E^{(1)}$ vanish. Operator structure (56) at orders 2 and 3 from the independent E (TT and transversality imposed): exact.

EF-17 Second-order kernels $K_{S_{00}},K_{S_{0z}},K_Q,K_{S_\Phi},K_h$ from discrete plane waves vs the closed forms of App. A: 10^{-164}; $K_h=4f_{\rm RD}$ of Kohri–Terada (20): relative difference 5×10^{-29} at 6 points; Green's functions and regularity of $S^{(2)}_\Phi$ at $\eta\to 0$.

EF-18 Pitrou–Uzan–Bernardeau (A.1)–(A.8) from the raw text: all scalar sources agree (15 digits) except the trace source (A.6), whose printed form is garbled; superhorizon Φ_2 and $Q_2(0)$ vs PUB (3.12b,c): exact; $\zeta^{(2)}$, $\zeta^{(3)}$ expansions exact.

EF-19 Vector sector: the independent $0x$ constraint + relativistic Kelvin give K_ω, K_{vT} (10^{-164}); the algebraic ω_2 satisfies the ij-vector evolution equation and is its unique regular solution; Lu et al. (20) with $w=\frac{1}{3}$: kernel identical (symbolic ratio 1).

EF-20 Third order: contraction rules confirmed by brute-force lattice Wick; contracted sources K_{3c} identical to the independent $E^{(3)}$ on a three-mode realisation; C_3 re-derived fully independently (own $E^{(3)}$, small-η series, tadpoles, ζ expansion) equals (45) (symmetrised over $\mathbf{q} \to -\mathbf{q}$) to 30 digits at 3 points; squeezed limit $-\dfrac{32}{9}$ from the separate-universe argument. Zero-mode convention: shifting $\langle\Phi_2\rangle$ by $+1$ shifts the symmetrised C_3 by $+8$; shifting $\langle\Psi_2\rangle$ (with $\delta_2 \to \delta_2 - 2$) changes nothing; C_3 is a convention-dependent intermediate.

EF-21 Dilation family (47) solves the independent E to $O(\epsilon^3)$ for arbitrary $\lambda(\epsilon), \beta(\epsilon)$ and $G-8\pi GT = 0$ non-perturbatively; $\zeta_{\rm nl} = \ln\beta$.

Closed-form second-order time integrals (App. B)

CF-1 $\mathcal{K}_h$ in closed form (Si and the entire function Cin; all ϵ-singular pieces cancel symbolically): equal to $4/k^2$ times Kohri–Terada (22) [with $|\cdot|$ in the logarithm] at 12 points (10^{-15}; 25 digits with rational input), to 60-digit quadrature (25 digits), and the late-time limit equal to KT (25); exact resonant ($q + p = \sqrt{3}k$) form vs quadrature: 40 digits.

CF-2 Closed-form Φ_2 vs 60-digit quadrature at 8 random points and the $\sqrt{3}$ point: 3.5×10^{-40}; collinear limiting forms: 2.5×10^{-40}; small-η series vs Frobenius recursion and late-time asymptotics vs 80-digit evaluation: consistent.

CF-3 All second-order fields built from the closed forms inserted in the full second-order Einstein (10 components) and conservation (4) equations, three points: relative residual 5.5×10^{-50}.

CF-5 Fast evaluators of all second-order fields (`second_order_closed.py`) vs the 50-digit closed form at 40 random points ($q/k \in [0.03, 20]$, all μ, $k\eta \in [0.05, 300]$), 15 near-collinear points ($|\mu| = 1 - 10^{-6}..10^{-10}$) and the point $q + p = \sqrt{3}k$: worst relative error Φ_2 1.1×10^{-13}, Φ_2' 2.1×10^{-13}, Ψ_2 1.7×10^{-13}, δ_2 1.5×10^{-13}, $\delta_{T,2}$ 1.8×10^{-13}, V_2 1.4×10^{-13}, h_2 3.6×10^{-13}; the 50-digit reference vs direct quadrature 10^{-31}–10^{-44}; small-η series minus closed form $\propto \eta^8$; large-η expansions verified to $k\eta = 10^5$.

CF-4 The printed form of $\mathcal{K}_h$ (special-function skeleton plus nine trigonometric monomials, App. B) re-assembled from the stored closed form and compared with 30-digit quadrature of $\int_0^\eta G_h K_h$ at three points: $\leq 2.4 \times 10^{-29}$.

First-order transfer function and $C_\ell^{(11)}$ (Sec. 7)

FO-1 Transfer functions vs the first-order ODE, constraints, continuity: residual 0; Rayleigh rules in the observation-direction convention by sphere quadrature ($\ell \leq 6$): 2.9×10^{-14}; observer monopole only in $\ell = 0$, observer Doppler only in $\ell = 1$.

FO-2 $C_\ell^{(11)}$ prefactor, observer and ISW terms by a real-space computation of $\xi(\hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_2)$ at 5 angles: 6×10^{-15}.

FO-3 ISW closed form vs mpmath quadrature (5 parameter sets): 2×10^{-18}; residue formulas for the Ci/Si/Cin coefficients vs brute force: $< 10^{-169}$; $\ln w$ coefficient zero symbolically; limits $k\eta_* \to 0$ ($-\dfrac{1}{3} j_\ell$, plateau $A_s/9$), $k \to \infty$ ($k = 200, 2000$), $k \to 0$ ($\Delta_0 = O(k^4)$, $\Delta_1 = O(k^3)$).

FO-4 Sky monopole: kernel sky average $= M(k_0)$ (5×10^{-15}), $|\langle\langle m_1\rangle_{\rm sky} a_{\ell m}^*\rangle| < 10^{-14}$ ($\ell = 1..3$), $\langle\langle m_1\rangle_{\rm sky}^2\rangle = C_0/4\pi$ (4×10^{-16}).

Squeezed-limit predictions (Sec. 11.4)

SQ-1 Weinberg adiabatic mode in radiation-era Poisson gauge and the induced transformation of short-mode fields derived and checked symbolically; squeezed-limit predictions for K_2 ($K_2(\mathbf{k}_1 \to 0, \mathbf{k}_2) = -\mathcal{D}K_1 - \mathcal{S}_{\mathbf{k}_1} K_1 + O(k_1^2)$); compared with the assembled K_2 in Sec. 11.4.

Harmonic machinery (Sec. 10)

H-1 Rule (A-s), $s = 0..3$; rule (A-λ), $\lambda = \pm 1, \pm 2$, in the k-frame and in a random frame (via D-matrices and spin-weighted harmonics): 10^{-14}.

H-2 $C^{(11)}$ normalisation $2/\pi$ via the real-space route (monochromatic $\xi = \sigma^2 |f|^2 j_0(k_0\chi|\hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2|) \Rightarrow C_\ell = 4\pi\sigma^2 |f|^2 j_\ell^2$): 10^{-14}.

H-3 (B1), both pairings, plain and gradient weights, correlated maps, vs exact real-space Wick + Legendre transform: 2.6×10^{-12}; Monte Carlo (400 Gaussian realisations, exact band-limited SHT): plain product 6.6%, gradient product 4.1%, $\langle X \rangle_{\rm sky}$·map 4.5%, cross term (ii) 3.1% (statistical, excludes any factor-2 error).

H-4 Type-D Fourier $\leftrightarrow$ angular-gradient identity (two radii): 2.9×10^{-12}.

H-5 Lensing control: $\frac{1}{4} C^{(22)} = \sum F^2/(2\ell+1)[C^\phi C^\Theta + C^{\phi\Theta} C^{\Theta\phi}]$ algebraically (2.5×10^{-16}); Monte Carlo of the Taylor-expanded remapping with correlated ϕ, Θ: (22) 2.3%, (13) $= -\ell(\ell+1) R C_\ell$ 4.6%; unlensed C_ℓ recovered 2.1%.

H-6 Loop prefactors of (105) re-derived by Wick counting: $C^{(22)}$ factor 2 and $C^{(13)}$ factor 3 confirmed; $K_1^* K_3$ versus $K_1 K_3^*$ is a complex conjugation, irrelevant for $\mathrm{Re}\, C^{(13)}$.

H-7 Type C: $K_{2,\ell m}$ finite $3j$ sums vs direct 2D quadrature (random k_1, k_2, two time nodes, degree 3): 1.1×10^{-14}; isotropy reduction random frame vs $\hat{\mathbf{K}} = \hat{z}$: 8.9×10^{-15}; (C1) $\equiv$ (B1) for product kernels (monochromatic): 3.7×10^{-15}; (C2) vs (B1): 3.6×10^{-15}.

H-8 Rule (A-L): 3.2×10^{-14}; reduction (E1) with $b \le 3$, $a \le 2$, $\Delta\chi = 0$ and $\neq 0$: $\le 2.8 \times 10^{-15}$.

H-9 Monochromatic normalisation $\langle \zeta^2 \rangle = \sigma^2$: 5.9×10^{-7} (Gaussian approximation of the δ-function).

H-10 Not verified numerically (formulas only, Wick counting consistent with the verified ones): (E2), (E3), $T_3^{(\mathrm{ii})}$; the collapse of (E3) to (114)(ii) for a type-B m_2 passes analytically.

H-11 Rules (A-s) ($s = 0, 1, 2$), (A-λ) ($\lambda = \pm 1, \pm 2$, including the vanishing of all $m \neq \lambda$) and (A-L) ($L \le 3$) in the observation-direction convention (10), by Gauss–Legendre sphere quadrature at $k\chi = 3.7$, $\ell \le 6$: $\le 6.5 \times 10^{-14}$; addition theorem $\sum_\ell (2\ell+1) j_\ell^2 = 1$: 4×10^{-16}.

H-12 Bound (129): $|j_\ell(x)| \le x^\ell/(2\ell+1)!! \le (ex/(2\ell+1))^\ell$ checked on $x \in \{0.5, 3, 10, 40\}$, $\ell < 160$ (40-digit arithmetic): satisfied.

Non-perturbative solver (planar symmetry, comoving-orthogonal gauge; Sec. 13.4)

X-10 Exact planar GR + radiation solver (comoving-orthogonal gauge, lapse fixed algebraically by (16): $\alpha = e^{-\zeta}(AB^2)^{1/3}$, $T_{\rm phys} = 1/\alpha$), constraint violations $\sim 10^{-13}$: Theorem 4.3 verified non-perturbatively, $T_{\rm obs}/T_o = e^{-S}$ to 7×10^{-15} at $\epsilon = 0.1$ ($\Theta_{\rm rms} = 0.08$).

NP-1 Linear limit of the planar solver vs the Poisson-gauge transfer functions (53) through a derived first-order gauge transformation: residuals $\propto \epsilon$ (genuine second-order corrections), 1.8×10^{-8} in Φ at $\epsilon = 10^{-4}$.

NP-2 Production: $\bar{\eta}_o/\eta_* = 4, 2$, $x_o = 0, \pi/2$, 16 amplitudes $|\epsilon| \leq 0.1$ each; FRW limit 2×10^{-16}; constraints $\sim 10^{-13}$; emission event $|\alpha_e - t_*| < 3 \times 10^{-15}$; null condition 10^{-14}; symmetries ($\zeta \to -\zeta$ = half-period shift, parity) 5×10^{-12}; numerical floor of a single map $\sim 10^{-11}$ (changes of resolution, tolerance, constraint damping, t_i, initial-data order); $\Theta^{(1,2,3)}$ extracted by even/odd fits with uncertainty = spread between fit subsets (Table 9).

End-to-end plane-wave test of the analytic chain (Sec. 13.4)

PW-1 Plane-wave fields from the field equations reduced to $\zeta = \epsilon \cos x$ (closure asserted): ODE residuals $\leq 6 \times 10^{-23}$, small-η series agreement $\leq 2 \times 10^{-23}$, conservation equations at orders 2, 3: 10^{-27}, 5×10^{-21}; amplitudes from $\zeta^{(2)}_{\rm nl}(0) = \zeta^{(3)}_{\rm nl}(0) = 0$ ($C_{2,2} = -\dfrac{1}{3}$ equals $K^0_{\Phi_2}$ at $q = p = k_0$).

PW-2 Analytic chain (fields, line-of-sight lists, ray functionals, emission, clock, displacement, normalisation) vs the non-perturbative solver, four configurations, orders 1–3: agreement within the extraction uncertainty (Tables 8, 9); T_o, $\langle T_{\rm obs}\rangle_{\rm sky}$ through $O(\epsilon^2)$ agree; own convergence: Chebyshev $N = 48/64/96$ identical.

PW-3 Zero-mode convention changed ($\Psi_{2,0} = \Phi_{2,0} + c_g\eta$, $c_g = 0 \to 1$): fields and $\delta\eta_{o2}$ change at $O(1)$, all $a_{\ell 0}$, sky monopoles and T_o unchanged to $\leq 2 \times 10^{-14}$.

PW-4 Shear vs direct form (assembled lists): 4×10^{-10} (L_2), 9×10^{-7} (L_3). Sign of the tangent integrals: with $-\int G$ in k_1, k_2 (86), (88) the null condition holds to 10^{-15}; the opposite sign violates it by 2.3 and the forms then differ by 2.1 (L_2), 38 (L_3).

PW-5 Observer label at $\eta_i = 0.01$: label as $\eta \to 0$ reproduces the solver (whose comoving label is the $\eta \to 0$ position); label at the initial event (worldline integral from $\eta_{\rm init}$) shifts δx_{o1} by 1.7×10^{-5} and moves $\Theta^{(2)}$ a_{20} by 7.3×10^{-5}, $\Theta^{(3)}$ a_{10} by 7.3×10^{-4} (beyond the solver's uncertainty): an $O(\eta_i^2)$ effect that vanishes as $\eta_i \to 0$ (Sec. 8.2).

PW-6 Fit-free test: $R(\epsilon) = \Theta_{\text{non-pert}}(\epsilon) - [\epsilon\Theta^{(1)} + \dfrac{\epsilon^2}{2}\Theta^{(2)} + \dfrac{\epsilon^3}{6}\Theta^{(3)}]_{\rm analytic}$ scales as ϵ^4 for all 16 amplitudes ($\max|R|/\epsilon^4 = 11.6..14.5$).

PW-7 Zero-mode weight of the conversion $\delta \to \delta_T$ in the plane-wave fields: the conservation rules evaluated on the field jets vanish (3×10^{-11}) only with weight 1 for the $k = 0$ mode (weight 2 gives 0.27), and T_o at $O(\epsilon^2)$ then agrees with the solver.

Observer clock at finite initial time (Sec. 8.2)

R-8 Observer clock and worldline with finite η_i and the label at the initial event, (19)–(20): the expansion (90)–(92) derived symbolically; it reduces to the η_i-free form as $\eta_i \to 0$, coincides with the finite-η_i clock used in the plane-wave test, and gives $\eta_o = \bar{\eta}_o/\lambda$ through second order for the dilation family for arbitrary a_1, a_2, η_i: exact.

Contracted third-order kernel and the loop (Secs. 11.6–11.9, 12.5–12.7)

R-9 Third-order form equivalence: shear-form and direct-form $\dfrac{1}{6}L_3$, evaluated with the contracted third-order fields (65), the second-order ray functionals, clock, displacement and emission shift, give the same azimuth-averaged $\text{coef}_{abc}(\mathbf{k}, \mathbf{q}, -\mathbf{q}; \hat{\mathbf{n}})$ at $\mu = 0.31, -0.62$: 2×10^{-11} absolute, 3×10^{-14} relative.

R-10 Lower orders in the same engine: K_1 from the L_1 lists (shear and direct) vs the type-A expansion with Δ_ℓ: 2×10^{-15}; $\frac{1}{2}L_2$ shear vs direct for three generic momenta, all pairs including the $K = 0$ pair: 10^{-13}; second-order null condition for the computed $\mathbf{k}_2$: 10^{-13}. With the opposite sign of the tangent integral in $\mathbf{k}_1$, $\mathbf{k}_2$ the two forms of L_2 differ at $O(8)$ (cf. PW-4).

R-11 Residual $K = 0$ gauge invariance of the assembled κ (all tadpole routes, normalisation included) under $\eta \to \eta + \frac{\epsilon^2}{2}\tau(\eta)$: $|\delta\kappa| = 6 \times 10^{-14}$ ($\tau = 0.1\eta$), 3×10^{-13} ($\tau = 0.1\eta^3$), scale of κ 4×10^2; individual channels change by 0.15–3.6.

R-12 Triad: the direct-form observer terms written in the $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ split (per-direction Gram–Schmidt triad) combined with the canonical triad in the ray functional differ from the shear form by an observer-local term $\propto (P_3 - P_1)(\hat{\mathbf{n}} \cdot \hat{\mathbf{k}})$; the per-direction triad changes κ at the 0.3% level. The canonical global triad is used (Lemma 4.7).

EF-22 Gauge covariance of the contracted third-order solution: contracted sources with tadpoles shifted along the residual time reparametrisation, solved with the same Green's function and $\delta C_3 = \delta\Phi_3(0)$, reproduce the exact shifts of Φ_{3c}, Ψ_{3c}, V_{3c}: 10^{-13}.

SQ-2 Squeezed limit of the assembled K_2 (Theorem 11.2): $\|K_2(s\hat{\mathbf{e}}, \mathbf{k}_2) + \mathcal{D}K_1\| = 3.18 \times 10^{-1,-2,-3}$ for $s = 10^{-2,-3,-4}$ (linear in s); k_1-odd part vs $-\mathcal{S}_{\hat{\mathbf{e}}}K_1$: 1.5×10^{-5}–2.2×10^{-4} (finite-difference accuracy).

SQ-3 Squeezed limit at third order (Theorem 11.3) on the full assembled κ: $\Delta_\ell^{\tilde{\kappa}}(1, q) \to 3\mathcal{D}^2\Delta_\ell$ ($\ell = 1..4$), deviations $\propto q^2$ (factor 4.0 per halving, $q = 0.1, 0.05, 0.025$).

H-13 Reality: $\Im C_\ell^{(13)} \sim 10^{-14}$ (monochromatic, $\ell \le 12$); $C_\ell^{(11)} = 4\pi\Delta_\ell^2$ from the (13) code vs (162): 3×10^{-14}.

L-0 Monochromatic $C_\ell^{(22)}$ quadrature (Sec. 12.4): $\eta_o = 8$, settings $(n_K, n_\theta) = (24, 40)$, $(40, 48)$, $(56, 60)$ agree to $\le 1.5 \times 10^{-11}$ relative for $\ell \le 20$ ($\le 10^{-13}$ for $\ell \le 14$); $\eta_o = 4$, $(24, 40)$ vs $(40, 48)$: $\le 4 \times 10^{-11}$ ($\ell \le 12$), 10^{-9} ($\ell = 13, 14$); entries $\ell \ge 15$ at $\eta_o = 4$ (values $< 10^{-14}$) below the cancellation floor ($\sim 10^{-16}$ of the largest channel), scatter up to 2×10^{-3}. Type-B diagonal from the μ-integral vs the closed $3j$ sum (164): 4×10^{-12} ($\ell \le 12$), 5×10^{-9} ($\ell \le 20$) at $\eta_o = 8$; 3×10^{-10} ($\ell \le 12$) at $\eta_o = 4$.

L-1 Collinear residues of the monochromatic (13) integrand (Table 6): Richardson extraction (nine points $1 - \mu = 0.04 \ldots 6.25 \times 10^{-4}$, $R/(1-\mu)$+ degree-6 polynomial): six channels with non-zero residues (up to 0.355), equal to the analytic gauge responses (175) to $\le 9 \times 10^{-11}$ (2.5×10^{-10} relative); other channels $\le 5 \times 10^{-10}$; sum $\le 8 \times 10^{-10}$ (2.3×10^{-9} of the largest), $\ell = 0..8$; $(1-\mu)^{-1/2}$ coefficient $\le 2 \times 10^{-5}$ (consistent with zero); finite limits F_ℓ (168); $\mu \to -1$ equals $\mu \to +1$ to 1.5×10^{-14}. (Superseded: the earlier extraction at $1 - \mu = 10^{-6}$ included the finite part, hence its apparent sums $\sim 10^{-5}$.)

L-5 Collinear pole = gauge mode, symbolic (Sec. 12.6, code `V6_residues/`): second-order pole parts (170)–(172) with τ in closed form (171); all contracted third-order sources have only p^{-2} poles, equal to $2/p^2$ times the gauge-shift sources (173) (sympy solves for the factor: 2 in all four); C_3 pole reproduced; numerical pair profiles $P^2 \times (\cdot) \to$ pole parts with error $O(P^2)$ (4×10^{-8} at $P = 10^{-4}$): exact / pass.

L-6 Explicit residues (175) vs the engine's channels: $\le 1.2 \times 10^{-14}$ ($\ell = 0..8$); total gauge response of κ to $\tilde{\tau}$ through all routes 1.2×10^{-14} (channels $O(0.36)$).

L-2 Monochromatic $C_\ell^{(13)}$: settings $(N_\mu, N_\varphi, N_\nu) = (16, 16, 32)$ vs $(12, 20, 32)$: $\leq 8\times10^{-5}$ ($\ell \leq 10$), 3×10^{-4} ($\ell = 12$), 2×10^{-3} ($\ell = 14$); azimuthal convergence $N_\varphi = 16$ vs 32: 2×10^{-5} at $\mu = 0.5$.

L-3 UV probe of $\tilde{\kappa}(q)$ to $q = 4k$: $O(10)$, oscillatory, no growth (azimuthal resolution $N_\varphi \gtrsim 2q\chi_* + 8$ required; under-resolution by a factor 4 detected at $q = 2$). Numerical only.

L-4 Monte Carlo of the monochromatic $C_\ell^{(22)}$ (Sec. 12.8): $N = 20$ shell modes, $45 \times 4000 = 1.8 \times 10^5$ realisations, K_2 as a black box, unbiased off-diagonal estimator: agreement with the quadrature within 1.1σ at 0.3–0.7% precision for $\ell = 1..11$ ($\ell = 12$: $+2.0\sigma$, 1.4%); $C_\ell^{(11)}$ to 0.1–0.2%; $\langle\zeta^2\rangle/\sigma^2 = 0.9964(32)$; normalisation term $= -2\langle m_1\rangle_{\rm sky}\Theta^{(1)}$ to 2.3×10^{-15} per realisation. Sensitive to the Wick factor (50%) and to the $-2\langle m_1\rangle_{\rm sky} m_1$ term (6σ at $\ell = 3$). Tests the assembly, not K_2 (same evaluator).

Not yet checked / pending

Gradient-mode (Mirbabayi–Zaldarriaga) checks of the assembled K_3 beyond the squeezed limit SQ-3; an analytic proof of zero-mode invariance for general momenta and an analytic computation of the collinear residues (L-1 is numerical); non-axisymmetric and non-collinear kernel configurations, vector modes and other $\bar{\eta}_o/\eta_*$ end to end (Sec. 13.7); numerical tests of (E2), (E3), $T^{(\rm ii)}$; the UV behaviour of the loop beyond $q = 4k$ and the IR formula for tilted spectra; $C_\ell^{(13)}$ at $\eta_o = 4$ (Sec. 14).

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